

Advances in Intelligent Systems and Computing 1242

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Distributed Computing and Artificial Intelligence, Special Sessions, 17th International Conference

Advances in Intelligent Systems and Computing

Volume 1242

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Preface

Research on Intelligent Distributed Systems has matured during the last decade, and many effective applications are deployed now. Nowadays, technologies such as Internet of Things (IoT), Industrial Internet of Things (IIoT), big data, blockchain and distributed computing, in general, are changing constantly as a result of the large research and technical effort being undertaken in both universities and businesses. Most computing systems from personal laptops to edge/fog/cloud computing systems are available for parallel and distributed computing. Distributed computing performs an increasingly important role in modern signal/data processing, information fusion and electronics engineering (e.g. electronic commerce, mobile communications and wireless devices). Particularly, applying artificial intelligence in distributed environments is becoming an element of high added value and economic potential.

The 17th International Symposium on Distributed Computing and Artificial Intelligence 2020 (DCAI 2020) is a major forum for presentation of development and applications of innovative techniques in closely related areas. The exchange of ideas between scientists and technicians from both academic and business areas is essential to facilitate the development of systems that meet the demands of today's society. The technology transfer in this field is still a challenge, and for that reason, this type of contributions is specially considered in this symposium. DCAI 2020 brings in discussions and publications on development of innovative techniques to complex problems. This year's technical program covers both high quality and diversity, with contributions in well-established and evolving areas of research. Specifically, 83 papers were submitted to main track and special sessions, by authors from 26 different countries (Algeria, Angola, Brazil, Bulgaria, China, Colombia, Croatia, Denmark, Ecuador, France, Greece, India, Iran, Italy, Japan, Mexico, Nigeria, Perú, Poland, Portugal, Russia, Saudi Arabia, Spain, Taiwan, Tunisia and Venezuela), representing a truly wide area network of research activity.

Moreover, DCAI20 Special Sessions have been a very useful tool in order to complement the regular program with new or emerging topics of particular interest to the participating community. The technical program of the Special Sessions of DCAI 2020 has selected 30 papers. As in past editions of DCAI, there will be

special issues in highly ranked journals, such as Information Fusion, Neurocomputing, Electronics, IEEE Open Journal of the Communications, Smart Cities and ADCAIJ. These special issues will cover extended versions of the most highly regarded works, including from the Special Sessions of DCAI, which emphasize specialized, multi-disciplinary and transversal aspects. This year, DCAI 2020 has especially encouraged and welcomed contributions on: AI-driven methods for Multimodal Networks and Processes Modeling (AIMPM), Natural Language and Argumentation (NLA), Surveying & Maritime Internet of Things Education (SMITE), Technological Approaches to develop Sustainability of Cultural Heritage (TECTONIC) and Disruptive Information and Communication Technologies for Innovation and Digital Transformation.

This year, the symposium DCAI 2020 is organized by the University of L'Aquila (Italy). We would like to thank all the contributing authors, the members of the Program Committees, the sponsors (IBM, Armundia Group, EurAI, AEPIA, APPIA, CINI, OIT, UGR, HU, SCU, USAL, AIR Institute and UNIVAQ) and the Organizing Committee of the University of Salamanca for their hard and highly valuable work. We are especially grateful for the funding supporting by the project “Virtual-Ledgers-Tecnologías DLT/Blockchain y Cripto-IOT sobre organizaciones virtuales de agentes ligeros y su aplicación en la eficiencia en el transporte de última milla”, ID SA267P18, financed by regional government of Castilla y León and FEDER funds. And finally, we are grateful and value the Local Organization members and the Program Committee members for their hard work, which has been essential for the success of DCAI 2020.

June 2020

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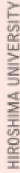
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**Special Session on AI-Driven methods
for Multimodal Networks and Processes
Modeling (AIMPM 2020)**

Special Session on AI-Driven methods for Multimodal Networks and Processes Modeling (AIMPM'20)

The special session entitled AI-driven methods for Multimodal Networks and Processes Modeling (AIMPM 2020) is a forum that will share ideas, projects, researches results, models, experiences, applications etc. associated with artificial intelligence solutions for different multimodal networks born problems (arising in transportation, telecommunication, manufacturing and other kinds of logistic systems). The session will be held in L'Aquila (Italy) as the part of the 17th International Symposium Distributed Computing and Artificial Intelligence 2020.

Recently a number of researchers involved in research on analysis and synthesis of Multimodal Networks devote their efforts to modeling different, real-life systems. The generic approaches based on the AI methods, highly developed in recent years, allow to integrate and synchronize different modes from different areas concerning: the transportation processes synchronization with concurrent manufacturing and cash ones or traffic flow congestion management in wireless mesh and ad hoc networks as well as an integration of different transportations networks (buses, rails, subway) with logistic processes of different character and nature (e.g., describing the overcrowded streams of people attending the mass sport and/or music performance events in the context of available holiday or daily traffic services routine). Due to the above mentioned reasons the aim of the workshop is to provide a platform for discussion about the new solutions (regarding models, methods, knowledge representations, etc.) that might be applied in that domain. There is a number of emerging issues with big potential for methods of artificial intelligence (evolutionary algorithms, artificial neural networks, constraint programming, constraint logic programming, data-driven programming, answer set programming, hybrid methods - AI/OR-Operation Research, fuzzy sets) like: multimodal processes management, modeling and planning production flow, production planning and scheduling, stochastic models in planning and controlling, simulation of discrete manufacturing system, supply chain management, mesh-like data network control, multimodal social networks, intelligent transport and passenger and vehicle routing, security of multimodal systems, network knowledge modeling, intelligent web mining & applications. business multimodal processes and projects planning.

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Multi-agent Path Planning Problem Under a Multi-objective Optimization Framework

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Abstract. In this study, a mixed-integer programming formulation is developed for a team of homogeneous sensing agents under a bi-objective optimization framework to solve a discrete open-loop centralized multi-agent search and rescue path planning problem. The first objective represents the maximization of probability of target detection to ensure the success of mission planning and the second objective represents minimization of the cumulative path length of all the agents to ensure resource utilization and ensure adequate area coverage. A two-phase fuzzy programming technique is used to find the Pareto optimal solution. Numerical experiments are conducted with CPLEX to evaluate the effectiveness of the solution procedure with varying number of agents, and the impact of the size of a grid-based rectangular map with a sparsely distributed non-cooperative finite number of stationary targets.

Keywords: Multi-agent path planning · Mixed-integer programming · Search and rescue

1 Introduction

Multi-agent search and rescue path planning (MASRPP) gaining significant importance and becoming an integral strategic part from the perspective of military and civilian operations such as emergency management; mountain rescue; maritime search and rescue; information collection, homeland security, etc. The MASRPP can be categorized in several ways based on the operational characteristics, such as one-sided vs. two-sided, searcher's actions action does not make any impact on the behavior of targets in one-sided search; on the contrary, target behavior can be cooperative or anti-cooperative in two-sided search. Isolated vs. continuous search based on time and location framework. Static vs moving target search, based on the movement of targets. Centralized vs decentralized

decision making or open and closed-loop decision models based on the decision making context.

Perhaps, Stone [32] offers first analytical explore the characteristics of the target search optimization problem by considering a single static target. Eagle [8] extended the model for moving targets and employed a dynamic programming technique to find a solution. Basic search and rescue path planning path problem with stationary targets is NP-Hard [35] and sequential decision inherent in the problem makes it a computationally expansive, and researchers employed several classical method like dynamic programming [9]; tree-based search techniques [12] or formulate the problem as mixed-integer programming (MIP) model and employed CPLEX [34]; A* class of algorithms [17] or other heuristic algorithm but ultimately found poor scalability even for moderate size problem. For example, Berger and Lo [2] proposed a MIP model by integrating the concept of directed acyclic graph theory. The authors employed CPLEX to find a solution. Raap et al. [25] formulated a binary integer linear program model to maximize the total area of coverage in a K-step-look-ahead planning schedule. Lo et al. [18] proposed a MIP model for a set of homogeneous agents through network representation to maximize the probability of success. The authors showed that CPLEX can provide a near-optimal solution for the medium size of problems within a reasonable amount of time. Perez-Carabaza et al. [23] used an ant colony optimization (ACO) technique to find optimal trajectories for a fleet of heterogeneous UAVs for searching a lost target. The mathematical model is formulated to minimize search time. The authors compared the solution of the problem with previously employed solution approaches such as cross-entropy optimization technique employed by Lanillos et al. [14]. The authors showed that ACO can lead to high-quality and high-level straight-segmented trajectories. Xiong et al. [37] introduced Voronoi-based ACO with Dijkstra's algorithm to find collision-free optimal trajectories. We refer to the articles [3, 7, 13, 21, 22, 24, 27, 29–31, 33], for some recent development in this direction.

However, as stated by Lanillos et al. [14], in a time-constrained scenario, maximizing the probability of finding targets may not always ensure minimization of total time. Therefore, we formulate MASRPP problem under the bi-objective framework to maximize the cumulative probability of success and minimize the cumulative number of cell visit by all the agents. Because, it is challenging to find the ideal solution due to conflicting nature of objective functions, researchers proposed several classical methods such as weighted sum and global criterion [19], goal programming [4], multi-choice goal programming [6] and others, and the subject matter to select a particular method largely depends on the decision-makers. In this article, we use a two-phase fuzzy approach proposed by Wu et al. [36] for solving the problem. Since the pioneering study by Zadeh [38], the fuzzy set theory has become an efficient methodology for solving multi-objective programming problems (Zimmermann [39]; Sakawa et al. [26]). The max-min approach [39] is extensively used to solve multi-objective linear programming problems due to computational simplicity, although the method fails to ensure a non-dominated fuzzy-efficient solution for all problems [15]. In order to overcome this problem, researchers devised a two-phase approach [11, 16]. Wu et al. [36]

analytical shown that the method guarantees a Pareto-optimal solution and it is used for solving various decision making problem [1, 5, 10, 20, 28] in different context. In this article, we also used the method proposed by Wu et al. [36] to solve the problem.

2 Mathematical Model

We assumed that a team of homogeneous agents searching stationary targets in a bounded search region. The area is divided as a set of cells N , describing possible target locations through grids. The time duration for each cell visit, with equal area, is assumed as constant and each agent is flying at slightly different altitudes to ensure collision. From prior domain knowledge, cell occupancy probabilities are generated initially and updated in subsequent visits of each agent so that the sum of cell occupancy probabilities does not violate basic axioms of the definition of probability. To maneuver its neighboring cells, any agent can move eight different possible directions $\{E, W, N, S, SE, SW, NE, NW\}$ based feasibility of movement. A graph theory-based directed acyclic network representation is employed to streamline the MASRPP problem. The entire grid is assumed as $G_k = (V_k, E_k)$, $G_k = \cup_{t \in T} V_{kt}$ and V_{kt} , the set of vertices, represent the agent location at time $t \in T$. E_k , the set of edges, represents the all possible state transition related to each agent between consecutive episodes. If we consider, O_{kt} represents one of eight feasible movement and N_{kt} represents the corresponding cell location, then $V_k = \cup_{t \in T} V_{kt} = \cup_{t \in T} N_{kt} O_{kt}$. Consequently, a binary decision variable x_{ijk} is introduced for each agent k , to represent the path in the grid.

The following notations are used to formulate mathematical model:

N	the entire search region is divided into N number of cells with equal area in the grid $\{1, 2, \dots, N \}$
T	set of time intervals with equal length defining the time horizon $\{0, 1, \dots, T - 1\}$
R	number of agents $\{1, \dots, r\}$
V_c	maximum allowed number of visits by all agents on cth cell
p_{cc}	conditional probability of actual target detection on cth cell subject to the target is located at the cth cell
p_{ct}	probability of target occupancy at cth cell during time interval t
v_{clt}	binary decision variable to track visit l , i.e. $v_{clt} = 1$ if the cth cell is included in the lth visit, and 0 otherwise
x_{ijk}	state transition binary variable; $x_{ijk} = 1$ reflects kth agent path solution includes arcs $(i, j) \in E_k$, and 0 otherwise
y_{ct}	agent position at time t in cell c . $y_{ct} = 1$ if cth cell visited by any agent at time t , and 0 otherwise

Based on the above notation, the following objective function is proposed: The first objective represents the average of cumulative probability of success

for the total number of agents to be deployed and the time horizon associated with the mission-planning as follows:

$$\text{Max } f_1 = \sum_{c \in N} \sum_{t \in T} \frac{p(z_{ct} = 1 | X_c = 1) \times p(z_{t-1} = 0 | X_c = 1) p(X_c = 1)}{RT} \quad (1)$$

As defined earlier, $p_{cc} = p(z_{ct} = 1 | X_c = 1)$, represent the probability so that a particular agent visit c where the target is present; $p_{ct} = p(z_{t-1} = 0 | X_c = 1) p(X_c = 1)$ and $p_{c't+1} = (1 - p_{cc} \delta_c^{c'}) p_{c't}$, $\delta_c^{c'} = 1, \text{ if } c = c' \text{ and } ; \delta_c^{c'} = 0, \text{ otherwise}$; represent the feedback. The second objective is maximization of total area of coverage as follows:

$$\text{Max } f_2 = \sum_{k \in R} \sum_{(i,j) \in A_k} x_{ijk} \quad (2)$$

The second objective indirectly reduce the number of repeated cell visit, and hence maximize the coverage within the time horizon.

The problem follows the following constraints:

$$\sum_{0 \leq l \leq V_c} v_{clt} = 1 \quad \forall c \in N, \forall t \in T \quad (3)$$

$$\sum_{0 \leq l \leq V_c} l v_{clt} = \sum_{t \leq t' \leq t} y_{ct} \quad \forall c \in N, \forall t \in T \quad (4)$$

Constraints (3) can help us to track the cumulative number of cth cell visit by lth visit at a time interval of $t \in T$. Constraints (4) indicates the number of all the past visits on cell c . Since, $y_{ct} \leq 1$, therefore, simultaneous visits by more than one agent on a particular cell over at time $t \in T$ is also prevented by these constraints.

$$y_{ct} = \sum_{k \in R} \sum_{i_t(c) \in V_k} \sum_{j_{t+1} \in V_k} x_{i_t(c)j_{t+1}k} \quad \forall c \in N, \quad \forall t \in T, \quad (i_t(c), j_{t+1}) \in A_k \quad (5)$$

Constraints (5) link the path for all the agents, $(i_t(c), j_{t+1})$ represents the state transition starting from the cell c . Therefore, from the initial state $i_0(k)$ and path $y_0(k)$, one can locate the position of agents at each time interval.

$$p_{c0} = p_c(t = 0) \quad \forall c \in N \quad (6)$$

$$p_{c(t+1)} = \sum_{0 \leq l \leq V_c} p_{c0} \phi(1 - P_{cc})^l v_{clt} \quad \forall c \in N, \forall t \in T \quad (7)$$

$$p_{cc} p_{ct} \leq y_{ct} \quad \forall c \in N, \forall t \in T \quad (8)$$

Constraint (6) represents the initial probability, constraint (7) update of probability under the influence of decision-maker and final, constraint (8) represent the success probability.

$$x_{0i_0(k)k} = 1, \quad \forall k \in R, i_0(k) \in V_k \quad (9)$$

$$y_{c0} = \sum_{k \in \eta} \delta_{cy_{o(k)}}, \quad \forall c \in N, \forall k \in R \quad (10)$$

$$\sum_{i \in V_k} x_{oik} = 1, \quad \forall k \in R \quad (11)$$

$$\sum_{i \in V_k} x_{idk} = 1, \quad \forall k \in R \quad (12)$$

Constraints (9) and (10) ensure initial agent location. $y_{o(k)}$ represents the origin of the agent k . $\delta_{cy_{o(k)}} = 1$, if the path start from cell c . Constraints (11) and (12), ensure the final location of k th agent defined by the decision maker.

$$\sum_{i \in V_k \cup \{o\}} x_{ijk} = \sum_{i \in V_k \cup \{d\}} x_{jik}, \quad \forall k \in R, \forall j \in V_k, (i, j) \in E_k \quad (13)$$

Constraint (13) represents the flow conservation from the origin to the destination. Finally, the following constraints represent the decision and auxiliary variables involved in the model

$$y_{ct}, v_{clt} \in \{0, 1\}, p_{ct} \in [0, 1], t \in T, \forall l \in \{0, V_c\}, x_{ijk} \in \{0, 1\} \quad (14)$$

3 Solution Procedure

In this section, we explain solution procedure and numerical illustration:

Definition 1: Multiple objective optimization problems can be represented as follows:

$$\begin{cases} \max (f_1(x), f_2(x), \dots, f_k(x)) \\ \min (g_1(x), g_2(x), \dots, g_r(x)) \\ \text{s.t. } x \in X = \{x \mid h_t(x) \leq 0, t = 1, \dots, m\} \end{cases}$$

where $x = (x_1, x_2, \dots, x_n)$ are the decision variables; $f_i(x)$, ($i = 1, \dots, k$) are maximization type objective functions; $g_j(x)$, ($j = 1, \dots, r$) are minimization type objective function; $h_t(x)$, ($t = 1, \dots, m$) are set of constraints.

If the objects are only maximization type, the following are holds:

Definition 2: A decision plan $x^0 \in X$ is said to be a Pareto optimal solution to the multiple objective optimization problems if there does not exist another $y \in X$, such that $f_k(y) \leq f_k(x^0)$ for all k and $f_s(y) < f_s(x^0)$ for at least one s (Wu et al. [36]).

Definition 3: A decision plan $x^0 \in X$ is said to be a fuzzy-efficient solution to the model if there does not exist another $y \in X$, such that $\mu_k(f_k(y)) \geq \mu_k(f_k(x^0))$ for all k and $\mu_k(f_s(y)) > \mu_s(f_s(x^0))$ for at least one s (Wu et al. [36]).

For MOLP, a Pareto-optimally not necessarily guarantees an efficient solution for all the objectives [36]. Therefore, the two-phase method is executed through the following steps [36]:

Step 1: Determine the positive and negative ideal solution for each objective function by solving single-objective optimization problems subject to a set of constraints.

Step 2: Using those, one-sided linear membership functions featuring both the continuously increasing property of the maximization objective function and decreasing property of the minimization objective function as follows:

$$\mu_{max}(f_r) = \begin{cases} \frac{f_r - f_r^{min}}{f_r^{max} - f_r^{min}} & \text{if } f_r^{min} \leq f_r \leq f_r^{max} \\ 0 & \text{if } f_r \leq f_r^{min} \end{cases}$$

$$\mu_{min}(f_r) = \begin{cases} \frac{f_r^{max} - f_r}{f_r^{max} - f_r^{min}} & \text{if } f_r^{min} \leq f_r \leq f_r^{max} \\ 0 & \text{if } f_r \geq f_r^{max} \end{cases}$$

where the possible range for the r -th objective is $[f_r^{min}, f_r^{max}]$.

Step 3: For the proposed bi-criteria problem, one needs to solve the following optimization problem in in Phase-I:

$$\text{Max } \lambda$$

$$\text{s.t. } \mu_{max}(f_1) \geq \lambda, \mu_{min}(f_2) \geq \lambda, \lambda \geq 0$$

and set of constraints defined previously.

Therefore, the decision-maker can find a solution that represents a trade-off between two objective functions. Note that the membership functions for two objective functions do not have strict upper bound for the maximization problem and lower bound for the minimization problem. Therefore, one needs to solve the following problem in Phase II to verify whether further modification is possible.

Step 4: In Phase II, the solution to the following problem leads to a final solution:

$$\text{Max } \sigma_1 + \sigma_2$$

$$\text{s.t. } \mu_{max}(f_1) - \sigma_1 \geq \lambda^*, \mu_{max}(f_2) - \sigma_2 \geq \lambda^*$$

$$\sigma_1 \geq 0, \sigma_2 \geq 0$$

where λ^* is the optimal value of in Phase I. If $\sigma_1 = \sigma_2 = 0$, then there is no better efficient solution for the model compared to Phase I and if one of $\sigma_1 > 0$ and $\sigma_2 > 0$ or both, the solution obtained in Phase II is more efficient [36]. From the perspective of the search and rescue problem, it is difficult to define the strict upper or lower bound for each objective function. Therefore, the method can provide flexibility for the decision-maker to overcome the impact of imprecise knowledge about the context.

For performance evaluation, all numerical experiments were executed with Intel Core i7-6600 CPU with 2.81 GHz processors and 16.0 GB RAM. For numerical verification, we assume the use of 8×8 with the sum of target cell occupancy probability up to one. The number of targets is 2 and the time horizon is $T = 12$ and the multi-agent team size $R = 4$. The search pattern and numerical results are presented in Fig. 1(a) and 1(b).

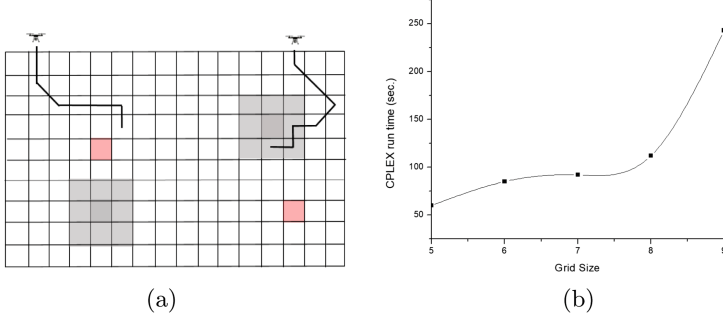


Fig. 1. (a) The overview of a general grid map, targets and search patterns scheme (b) Computational time vs increasing number of grid size n ($n \times n$).

Figure 1(a) reflects an overview of a conceptual grid map that represents a virtual search and rescue area divided into equal subareas, belief distribution, agent path location and orientation through time. Although, for numerical validation, we use a simple scenario, Fig. 1(b) demonstrates that the computational time increases exponentially with an increasing number of grid sizes. For this study, we consider a simple problem due to this time complexity, without heuristic it is difficult to find a solution for the problem with a large number of grid sizes.

4 Conclusion

This study proposed a multi-objective mixed-integer programming (MIP) model for path planning for a set of homogeneous agents searching for static objects. Although MASRPP is studied in various perspectives, there is a limited study under multi-objective formulation. We employed CPLEX and two-phase approaches to find the paths of agents. Numerical experiments are conducted to illustrate the formulation. A graph-theoretic representation is employed to reduce complexities inherent in operational constraints; speed-up computation with the minimum number of cell visits with a reasonable number of repetitions. The fixed time horizon is discretized into an equal number of episodes and the entire problem is solved sequentially by dividing it into two parts representing abstract and movement planning that ensure flow conservation among all the agents; track inward and outward movement of each agent from their respective present cell location, and reflect updated target occupancy information.

In the present article, we try to provide a foundation for a small problem and the study is based on several assumptions. Therefore, the present study needs to be extended in several directions. First, each solution method for multi-objective linear optimization has some advantages and drawbacks. Therefore, it is necessary to verify the trend of a solution with other methods also. Second, the up-gradation of the probability map in each cycle itself a broad area of research. The present study is based on a simple assumption to update the probability of occupancy cells. One may employ machine-learning tool by considering environmental or system parameters based on the expert knowledge to update probability. Third, due to the growing number of binary variables as the number of grids increases, we found that computational time increases exponentially, even for small problems illustrated in this study. Therefore, as a future extension, one needs to develop heuristic algorithms to solve the problem with a higher number of roads or scheduled for consecutive days. Finally, but not the last, we assume altitude differentiation from the perspective of collision avoidance and neglected the several constraints associated with each agent, fuel constraints, sensor capacity, search pattern, etc. those need to be integrated to formulate a robust path planning model.

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Rules in Detection of Deadlocks in Multithreaded Applications

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Abstract. This study presents an original approach to the detection of deadlocks in multithreaded applications. The solution proposed takes into consideration the order of introducing and removing locks in the programme code, which was used in three exemplary applications. The solution proposed constitutes a continuation of research into a model of multithreaded applications.

Keywords: Multithreaded applications · Eligibility criteria · Deadlock

1 Introduction

When every multithreaded application is being created that uses the *pthread* library, there is a possibility that the programmer's error will lead to resource conflicts that cause undesirable phenomena, i.e. race condition, deadlock, atomicity violation and order violation. Therefore, applications that are created need to fulfil a number of criteria to guarantee that the aforementioned phenomena will not occur.

1.1 Definition of Deadlock and Necessary Conditions

In the book [3], the deadlock phenomenon was defined as a situation where several processes are waiting infinitely for an event that may be commenced by only one of the processes waiting. The deadlock phenomenon occurs in many fields, not only information technology, and it can be detected in multithreaded applications among others owing to Petri Nets (PN) [1], the directed graph [3] and Gadara Nets [13]. In the study [2], deadlocks were divided into resource deadlocks and communication deadlocks.

Resource deadlocks occur only when processes or threads need for their work a certain group of shared resources, and each of them possesses only selected processes, the effect being waiting one for another indefinitely [11]. A thread in multithreaded applications may wait for an access to the resources it already possesses. An example of this phenomenon was described in Sect. 4.3.

In the present article, detection of the deadlock phenomenon will be possible only in the use of an original model presented in Sect. 2, which was preceded

by a section about the mechanisms of the synchronization of threads. Section 3 constitutes a formulation of the problem. Section 4 contains a detailed description of the deadlock phenomenon taking into consideration the division into those conflicts that cause this phenomenon to occur. Section 5 covers the sufficient condition that precedes conclusions in the study.

1.2 Thread Synchronization Mechanisms

The *pthread* library provides mechanisms that allow one to create safe applications provided that the programmer uses these correctly. These mechanisms include mutexes and condition variables. In practice, multithreaded application always use synchronization mechanisms.

Mutexes serve the purpose of the execution of critical sections, where a critical section is to be understood as a fragment of the application's code found between the call of the function that starts a mutex, where a shared resource is used.

Condition variables is the second mechanism that is provided with the *pthread* library. It makes it possible to force a sequence of operations by using signals. Owing to this mechanism, the waiting thread may go into a state that does not put a strain on the processor and it may resume its operation once it has received a signal from the thread whose operation had to be performed earlier.

Condition variables always use mutexes in their operation. These variables are initiated by using the `pthread_cond_init` function or the `PTHREAD_COND_INITIALIZER` macro, and the `pthread_cond_destroy` function is to be used to unlock them. The programmer is obliged to unlock the condition variable as in any other case dangerous memory leaks may occur in the application.

2 Model

In study [7], an original model of multithreaded applications was presented with the aid of which a theorem was developed whose fulfilment means that detrimental competition occurs in the application. The model presents as follows:

$$C_P = (T_P, U_P, R_P, O_P, S_P, \mathbb{M}_P, F_P) \quad (1)$$

where:

1. P is the index of the application,
2. $T_P = \{t_i | i = 0 \dots \alpha\}$, ($\alpha \in \mathbb{N}$) is the set of threads t_i and application C_P , where t_0 the main thread, $|T_P| > 1$,
3. $U_P = (u_b | b = 1 \dots \beta)$, ($\beta \in \mathbb{N}^+$) is a sequence of sets u_b , which are subsets of a set of T_P that contain threads working in the same time interval in application C_P , with $|U_P| > 2$, $u_1 = \{t_0\}$ i $u_\beta = \{t_0\}$,
4. $R_P = \{r_c | c = 1 \dots \gamma\}$, ($\gamma \in \mathbb{N}^+$), $r_c = (v_c, w_c)$ is a set of shared resources in application C_P , defined as a pair (*variable*, *value*),

5. $O_P = \{o_{i,j} | i = 1 \dots \delta, j = 1 \dots \epsilon\}$, $(\delta, \epsilon, \in \mathbb{N}^+)$ is a set of all application operations C_P , which at a certain level of abstraction are atomic operations. The index i indicates the number of the thread in which the operation occurs, and the index j is the ordinal number of operations working within the same thread.
6. $S_P = \{s_p | p = 1 \dots \kappa\}$, $s_p = (o_{i,j} | i = 1 \dots \delta, j = 1 \dots \epsilon)$, $(\delta, \epsilon, \kappa, \in \mathbb{N}^+)$, $S_P \subseteq O_P^*$ - is a set of the sequences of the operations that are protected with a mutex,
7. $F_P = \{f_n | n = 1 \dots \iota\}$ i $F \subseteq (O_P \times O_P) \cup (O_P \times R_P) \cup (O_P \times \mathbb{M}_P)$, $(\iota \in \mathbb{N}^+)$ - is a set of edges that comprises the following:
 - (a) transition edges - these determine the order of the execution of the operation. These edges are pairs $f_n = (o_{i,j}, o_{i,k})$, where elements describe two operations that are performed one after another $o_{i,j} \in O_P$. The transition edge may also lead from the operation to the mutex and from the mutex to the operation, i.e. $f_n = (o_{i,j}, m_l^b)$ or $f_n = (m_l^b, o_{i,j})$. In the application code, locking and unlocking of the mutex is done with the use of usual functions from the *pthread* library.
 - (b) usage edges - these indicate those resources that are subject to change while the operation is being performed. The edges are pairs $f_n = (o_{i,j}, r_c)$, where the o is operation $o_{i,j} \in O_P$, and the r is resource $r_c \in R_P$,
 - (c) dependency edges - these indicate operations that depend from the current value of one of the resources so that the operation could be correctly executed. These edges are pairs $f_n = (r_c, o_{i,j})$, where the first element is resource $r_c \in R_P$, and the other element is operation $o_{i,j} \in O_P$,
8. $\mathbb{M}_P = (MU^b | b = 1 \dots \beta)$ - a sequence of sets MU_P^b ; $MU_P^b = \{m_l^b | l = 1 \dots \theta^b\}$ a set of mutexes m_l^b that are obligatory for threads from set u_b ; $m_l^b = \{s_p \in S_P\}$ - a set of excluding operations (hereinafter referred to as the “mutex”). Additionally, for every mutex m_l^b functions are defined whose result are matrices with the ordinal numbers of mutexes. Each subsequent line of the matrix is a possible variant of locking and unlocking subsequent mutexes. These functions are as follows:
 - (a) $\phi_l^b(s_p) = \begin{bmatrix} k, \dots, \beta \\ \dots, \dots, \dots \\ q, \dots, \beta \end{bmatrix}$, where k, q, $\beta \in \mathbb{N}$ - subsequent values in line mean mutexes from set $\mathbb{M}_P$, place in the line means the call of the lock of the l^{th} mutex in b^{th} time interval on the set of operations s_p . This function performs a transition from the graph of the operation and, as a result, it returns the matrix, which includes the ordinal numbers of the mutexes from set MU_P^b in the order in which they are started and from zeros for every encountered operation of the mutex unlocking. The matrix lines reflect the subsequent paths that are formed as a result of the use of the control statement in the application's code.
 - (b) $\psi_l^b(s_p) = \begin{bmatrix} k, \dots, \beta \\ \dots, \dots, \dots \\ q, \dots, \beta \end{bmatrix}$, where k, q, $\beta \in \mathbb{N}$ - subsequent values in line mean mutexes from set $\mathbb{M}_P$, place in the line means the call of the lock of the l^{th} mutex in b^{th} time interval on the set of operations s_p . This function performs a transition through the graph of the operation, and its effect

is a matrix that is composed of zeros for every operation encountered related to starting a mutex and the ordinal number of the mutex when the mutex unlocking operation is being performed. The matrix lines reflect the subsequent paths that are formed as a result of the use of control statements in the application's code.

3 Formulation of the Problem

A multithreaded C_P application is given written in C language with the use of the *pthread* library. In this application, a deadlock phenomenon occurs. Is it possible to detect a conflict that causes the deadlock of the application by using the model from Sect. 2?

The applications described in Sect. 4 will serve as an example as each of them possesses one of the three different conflicts that cause the deadlock phenomenon. Section 5 includes a description of the sufficient condition.

4 Deadlock

With an increasing number of deadlocks, the risk is growing of the occurrence of the deadlock phenomenon. In order to minimize its occurrence, the number of mutexes locked is to be minimized; however, too small an amount of mutexes may lead to the application being slowed down or to occur the race condition.

A communication deadlock in multithreaded applications may occur at the moment condition variables are being incorrectly used. Any operations on a condition variables need to be protected by a mutex at all times; hence, in this study, a condition variable will be treated as remaining resources.

A deadlock phenomenon frequently occurs at the moment when the programmer is correcting an error or is trying to prevent conflicts that cause the phenomena of race condition or a atomicity violation ([5, 9]).

The term of deadlock is most frequently understood to be a resource deadlock, and this may occur in three variants:

1. through a mutual exclusion of mutex pairs (an example code: <https://goo.gl/o9VVhu>, called as EC0),
2. through an omission of the mutex unlocking e.g. as a result of a control statement (an example code: <https://goo.gl/ju1Rnx>, called as EC1),
3. through a renewed attempt to lock a mutex as a result of operation of a loop (an example code: <https://goo.gl/Q5WyTg>, called as EC2).

Figure 1 presents three diagrams that correspond to those fragments of a pseudocode where conflicts occur that cause the deadlock phenomenon. These variants were considered in several studies ([5, 9]) including those related to an automatic fixing of errors connected with the atomicity violation. The AFix and Axis tools described therein can automatically remove a atomicity violation while not introducing any new conflicts that might lead to a deadlock.

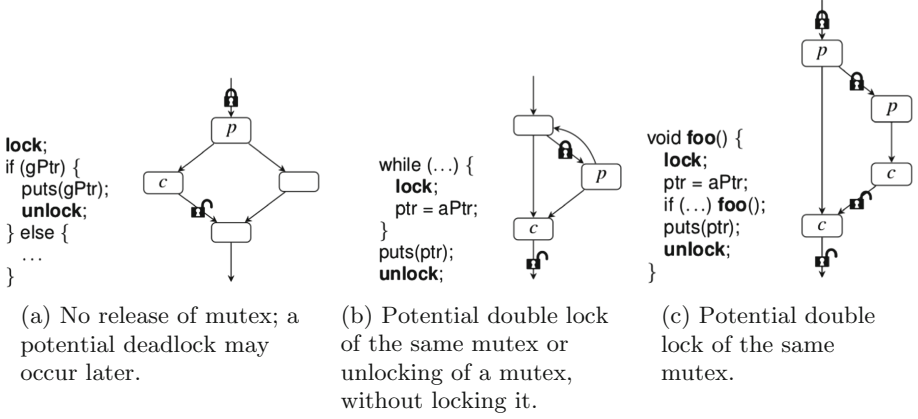


Fig. 1. Possible errors that cause the deadlock phenomenon. Source [5].

The model proposed of multithreaded applications is to make it possible to detect resource conflicts that cause the phenomena discussed herein faster as compared to other methods, by making an analysis of the source code only. This approach should be faster as:

- there is no need to carry out a complex process of isolating fragments of the code, as in the case in the CTrigger tool [8],
- there is no phase of multiple testing of the code with various parameters.

4.1 Model of EC0 Application

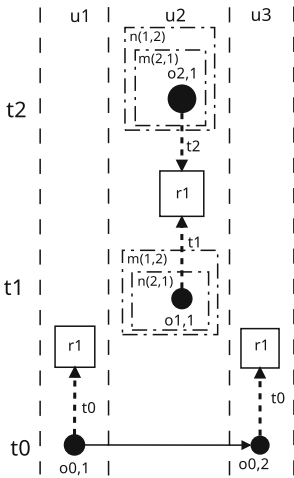
In accordance with the model from Sect. 3, the graphical presentation of the EC0 application is found in Fig. 2, and its model is as follows:

$$\begin{aligned}
 T_{EC0} &= (t_0, t_1, t_2), & S_{EC0} &= \{\{o_{1,1}\}, \{o_{2,1}\}\}, \\
 U_{EC0} &= (\{t_0\}, \{t_1, t_2\}, \{t_0\}), & MU_{EC0}^2 &= \{\{s_1, s_2\}\}, \{s_1, s_2\}\}, \\
 R_{EC0} &= \{(counter, 0)\}, & F_{EC0} &= \{(o_{0,1}, r_1), (o_{1,1}, r_1), (o_{2,1}, r_1), \\
 O_{EC0} &= \{o_{0,1}, o_{0,2}, o_{1,1}, o_{2,1}\}, & & (o_{0,1}, o_{0,2}), (o_{0,2}, r_1)\}.
 \end{aligned}$$

In a situation where mutexes can be excluded mutually (the threads do not perform any pre-emption) and the waiting time for the mutex to be locked is unlimited, a deadlock will occur. From among the variants listed, this one is dependent from time only. It is possible, however, that the programmer, by using other structures of the C language, will obfuscate the code and this phenomenon will be hidden from the programmer [14]. Obfuscating the code is to be understood as not using the rules of the application development standard, e.g. MISRA C, Netrino or GNU Coding standards. This type deadlocks can be found very easily owing to the ϕ_l^b function as proposed in the model.

This function determines the order of locking mutexes in selected sets of operations. The order of locking mutexes in the EC0 application can be presented in the form of a matrix:

$$\begin{array}{l} \phi_1^2(s_1) \quad [1, 2, 0, 0] \\ \phi_2^2(s_2) \quad [2, 1, 0, 0] \end{array} \quad (2)$$



Listing 2 presents the results of the ϕ_l^b function for the sets of operations s_1 and s_2 . By analysing data for set s_1 , it is evident that mutex m_1^2 is locked first followed by mutex m_2^2 . For the set of operations s_2 , these mutexes are locked in the reverse order; hence, the threads are mutually exclusive by taking one mutex each, and they are infinitely waiting for one another.

To conclude: when in at least two threads, a group of mutexes is locked on a selected set of operations, one needs to take care to have the order in which they are locked to be identical in each of the thread. In any other case, there is a probability of a deadlock, whose only condition is time.

4.2 Model of EC1 Application

Fig. 2. Graphical representation of EC0 application model.

This case is the result of the use of a control statement, as the logical expressions used in it may work in a different way than expected by the programmer.

Figure 3 includes a graphical presentation of the EC1 application, whose model is as follows:

$$\begin{aligned} T_{EC1} &= (t_0, t_1, t_2), & MU_{EC1}^2 &= \{\{s_1, s_2\}\}, \\ U_{EC1} &= (\{t_0\}, \{t_1, t_2\}, \{t_0\}), \\ R_{EC1} &= \{(counter, 0)\}, & F_{EC1} &= \{(o_{0,1}, r_1), (o_{0,1}, o_{0,2}), \\ O_{EC1} &= \{o_{0,1}, o_{0,2}, o_{1,1}, o_{2,1}, o_{2,2}, o_{2,3}\}, & & (o_{0,2}, r_1), (o_{1,1}, r_1), (o_{2,1}, o_{2,2}), \\ S_{EC1} &= \{\{o_{1,1}\}, \{o_{2,1}, o_{2,2}, o_{2,3}\}\}, & & (o_{2,2}, r_1), (o_{2,2}, m_1), (o_{2,1}, o_{2,3})\}. \end{aligned}$$

Edge $f_7 = (o_{2,2}, m_1)$ that is present in this model that is led from operation $o_{2,2}$ to the mutex provides information on the fact that right after the performance of the operation, mutex m_1 is unlocked. There is no analogical edge from operation $o_{2,3}$ to the mutex; therefore, it needs to be interpreted as a lack of unlocking of this mutex after the performance of this operation.

If edge f_7 were non-present, it would have to be considered that both after operations $o_{2,2}$ and $o_{2,3}$, the mutex is unlocked. This is the first suggestion that there is a resource conflict in this thread, which may lead to a deadlock phenomenon. The occurrence of the deadlock in the EC1 application needs to be confirmed

$$\begin{array}{l} \psi_1^2(s_1) \quad [0, 1] \\ \psi_1^2(s_2) \quad \begin{bmatrix} 0, 1 \\ 0, \infty \end{bmatrix} \end{array} \quad (3)$$

through a verification of the order in which the mutexes are unlocked with the use of function ψ_l^b .

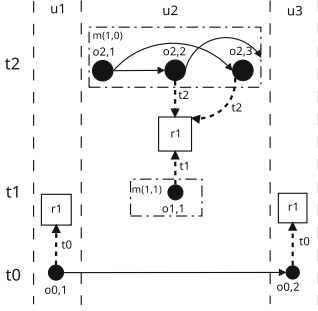


Fig. 3. Graphical representation of EC1 application model.

locking operation in the other block, any other thread that will want to lock mutex m_1^2 will be waiting infinitely. Contrary to the EC0 application, in which the deadlock phenomenon depends from time, this phenomenon depends from input data in the EC1 application.

4.3 Model of EC2 Application

The cause of a deadlock phenomenon in the EC2 application is very similar to the one in the EC1 application, namely the mutex locking operation is found in a different code block than the mutex unlocking operation. A graphical presentation of the EC2 application is found in Fig. 4, and its model is as follows:

$$\begin{aligned}
 T_{EC2} &= (t_0, t_1), & MU_{EC2}^2 &= \{\{s_1\}\}, \\
 U_{EC2} &= (\{t_0\}, \{t_1\}, \{t_0\}), \\
 R_{EC2} &= \{(counter, 0)\}, & F_{EC2} &= \{(o_{0,1}, r_1), (o_{0,2}, r_1), (o_{0,1}, o_{0,2}) \\
 O_{EC2} &= \{o_{0,1}, o_{0,2}, o_{1,1}, o_{1,2}, o_{1,3}\}, & & (o_{1,1}, o_{1,2}), (o_{1,2}, m_1), (m_1, o_{1,3}), \\
 S_{EC2} &= \{\{o_{1,3}\}\}, & & (o_{1,3}, r_1), (o_{1,3}, o_{1,2})\}.
 \end{aligned}$$

In the EC2 application, unlike the previous applications, only one thread will be enough for a deadlock phenomenon to occur. An adequate input parameter is a condition for its occurrence. Edges f_5 and f_6 , that lead one by one to mutex m_1^2 and from mutex m_1^2 to operation $o_{1,3}$ emphasize only that a mutex is locked between operation $o_{1,2}$ and operation $o_{1,3}$. Both these edges could be substituted by edge $(o_{1,2}, o_{1,3})$; however, it is edge $f_8 = (o_{1,3}, o_{1,2})$ that is also present in the model, which points to the cyclical nature of the operation.

$$\begin{aligned}
 \phi_1^2(s_1) & \begin{bmatrix} 1, 0 \\ 1, \infty \end{bmatrix} \\
 \psi_1^2(s_1) & \begin{bmatrix} 0, 1 \\ \infty, 1 \end{bmatrix}
 \end{aligned} \tag{4}$$

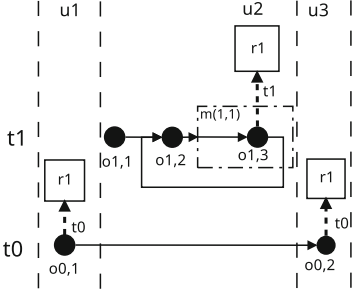


Fig. 4. Graphical representation of EC2 application model.

A repeated call of the mutex locking operation with the omission of the mutex unlocking operation constitutes an important element of this cyclical nature. In this case, this leads to a deadlock phenomenon. In order to detect this deadlock, one needs to verify whether there is a cycle of the calls of operations, where an edge will be found that leads from an external operation to the mutex, e.g. $f_5 = (o_{1,2}, m_1^2)$ and, at the same time, no such edge exists that would lead from an operation inside the mutex to the mutex.

In order to detect a conflict that causes a deadlock phenomenon in the EC2 application, one needs to examine whether there exists a cycle of operations where the mutex unlocking operation is missing. Therefore, in order to detect a deadlock, one needs to check whether there exist any edges that lead from those operations that are not protected with a mutex to mutex locking operations and, at the same time, whether an edge exists in the whole cycle that leads from a mutex protected operation to a mutex unlocking operation.

5 Sufficient Condition

The examples described above of applications present three various reasons for the occurrence of the deadlock phenomenon. Based on the examples described in Sect. 4, it may be stated that the deadlock phenomenon occurs if:

1. the groups of the mutexes locked possess a non-uniform order of their locking;
2. the mutex unlocking operation may be omitted as it is found in the body of the control statement;
3. the mutex locking operation is found in a loop, and its unlocking operation is found outside of the loop.

The reasons presented above that cause the deadlock phenomenon in multi-threaded applications can be generalized in the form of the following theorem:

Theorem 1. Let $O_P = \{o_{m,j}, \dots, o_{n,q}\}$ mean a set of operations performed within the threads of set $u_b \in U_P$, a $S_P = \{s_i, \dots, s_p\}$ mean a set of operations protected with a mutex.

The deadlock phenomenon will occur if:

$$\exists(a, b) \in \phi_l^b\{s_i\} \bigwedge \exists(b, a) \in \phi_l^b\{s_k\} \quad (5)$$

$$\infty \in \psi_l^b(s_p) \quad (6)$$

$$\exists(o_{i,j}, \dots, o_{m,j}) \in S_P \bigwedge \nexists f_n = (o_{k,l}, m_l^b) \text{ where } o_{k,l} \in S_P \quad (7)$$

Proof. According to Theorem 1 and mentioned definitions, the necessary condition for the occurrence of a deadlock in EC0 application (exclusion of a pair of mutexes) is the existence a pair of operations (a, b) to establish q_c and q_d mutexes, which order of execution are different in threads, it is caused by the work of a pair of threads using a pair of locks to exclude each other. In case of EC1 and EC2 applications, the necessary condition is the existence of a cyclic operations, which start from establish q_c mutex and resulting in the failure to unlock a pre-established q_c mutex - more precisely, condition will be met if matrices (received from $\phi_l^b(s_p)$ and $\psi_l^b(s_p)$ functions) contain ones in the same places.

6 Conclusions

The model presented in this study has made it possible to explicitly determine the reason for the mutex in three exemplary multithreaded applications. These cases are not all possible errors, but this model provides information on the place where such an analysis is to be performed. This allows model to detect other types of errors in multithreaded applications.

Owing to the theoretical nature of the investigations carried out for the needs of this article, it is justifiable to claim that this solution may not be completely effective in the case of the deadlock phenomenon. However, the main contribution includes the development of rules for detecting deadlocks in multithreaded applications. The rules were pre-verified in other researches, which allowed to indicate it may claim based on this that it is sufficient to indicate the place of all and any possible resource conflicts that cause deadlocks.

Further work related to the development of this model will include its use to detect the phenomena of a atomicity violation and an order violation. Work is to be commenced related to an automation of the model building process based on the application code. It is possible to use the model for other applications, but an element similar to mutexes is required.

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Optimization of Customer Order Processing for the Pizza Chains

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Abstract. The pizza chain is one of the examples of modern chain business. Each of the pizza parlors in the chain has the same decor, organization, menu and take-away delivery method. Such chains usually cover large urban agglomerations. In the era of universal access to the Internet and mobile devices, most customers order pizzas on-line (via an application) with the option of delivery. The following key questions arise in relation to the customer order processing for the pizza chain owner: *How to allocate individual customer orders to selected pizza parlors so that the cost of their processing (production and delivery) is the lowest? How to deliver on time? etc.* To answer these questions, an optimization model has been developed, which combines routing, allocation and planning. The model has been implemented using the Gurobi solver. To verify the correctness of the model and the effectiveness of the proposed method of its solution, numerous computational experiments have been performed.

Keywords: Vehicle routing problem · Mathematical programming · Optimization · Network structure · Decision support

1 Introduction

With the development of the Internet and the spread of mobile devices, the existing chain businesses are back in fashion and many new ones are emerging. This applies to retail, restaurant and service chains. This is for two reasons. First of all, IT development has enabled easier order placement by customers, faster communication between supplier and customer (B2C) and between suppliers (B2B), shortened supply chains, standardized offers, etc. [1] Secondly, logistics services with new technologies, such as e-mobility, drones, etc. have developed [2].

One example of chain business is the pizza chain [3]. Usually such chains have several or more points (pizza parlors) in a given location. Locations are usually major cities or urban agglomerations. Each pizza parlor has a similar decor, menu, etc. and usually serves its products in two ways, i.e. on site and delivery. The latter is becoming increasingly popular, which is due, on the one hand, to the current way of life of many societies and, on the other hand, to the on-line order processing, which significantly reduces costs. The pizza chains also react to changing market, customer needs, competition, shortening the cycle from order placement to delivery, etc. through the

widespread use of IT solutions. It is generally about reducing costs and improving the quality of business. One of the innovative ideas for such changes is the introduction of customer order processing optimization. Such optimization covers, both, production planning and delivery. There are many problems in this area that need to be properly modeled and solved, such as optimizing routes [4], allocating orders to specific pizza parlors, allocating products to couriers, etc. Each of these problems is nontrivial and is linked to other problems to a different extent and scope, which makes the search for a solution even more complicated.

The article proposes an integrated model for optimizing the customer order processing of pizza chains, which combines the issues of routing [4, 5], production planning, resource allocation, etc. The model was formulated in the form of the mixed integer linear programming (MILP) [6] and it has been implemented using the GUROBI solver environment [7].

2 Problem Statement – Illustrative Example

The pizza chain is given, which has its branches (pickup points) in specific locations of a city/urban agglomeration. Each of the branches produces products (pizzas, pasta, etc.) from the same menu and provides delivery service. Customers place orders via a central Internet application, a mobile app or via a hotline, choosing the assortment, quantity and providing the delivery address. When ordering, they do not specify the pick-up point (pizza parlor); they are not interested in where the product will be delivered from.

Orders are collected within given periods of time and forwarded to individual pick-up points. The general schematic diagram of such chain is shown in Fig. 1.

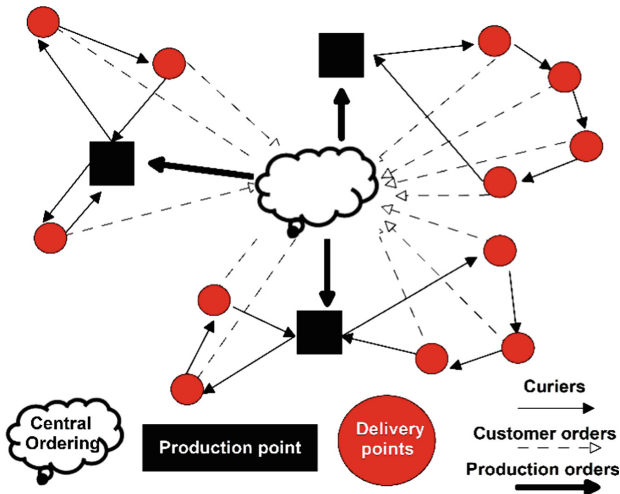


Fig. 1. Order processing in an exemplar pizza chain.

The problem formulated in this way raises a fundamental question: *How to plan production and delivery to keep costs to a minimum?*

To answer this question, answers to the following more detailed questions should be found.

- a) How to allocate individual customer orders among individual pickup points?
- b) How to arrange deliveries so that the time and delivery cost are as short and low as possible?

When looking for answers to these questions, it is important to take into account many constraints that arise when analyzing this problem. They cover the production capacity of individual pizza parlors, the number of couriers in a given parlor, the delivery time deadline, the capacity of couriers (usually deliver using scooters, e-mobile means, etc.), route lengths, etc. In addition, it should be noted that the two questions are correlated with each other. The order allocation to a particular pick-up point depends to some extent on the delivery point.

3 Mathematical Model

The mathematical model of the problem has been formulated as a MILP task. The MILP models belong to a well-known and widespread class of mathematical programming (MP) models, which are often used for planning, scheduling, routing, resource allocation, etc. [8].

Table 2 presents indexes, parameters and decision variables of the proposed model. Table 1, on the other hand, describes the significance of particular constraints of the model (1)..(11). As an objection function (1) in this model version, the total distance traveled by couriers is assumed. This path is minimized during model optimization. This objective function seems justified as the route length affects both, the order processing cost and time, which is an important quality factor of the service provided.

Table 1. Description of the constraints

Constraints	Description
(1)	Objective function – minimizing the route traveled
(2)	If the courier arrived at the point, courier must also leave it
(3)	Only one courier for each customer
(4)	Couriers start and stop in production points
(5)	Processing of all customer orders through courier deliveries
(6)	Combining variables $X_{c,d,p,j}$ and $Y_{c,d,p,j}$ - no delivery if the courier does not go
(7)	Courier capacity constraint c
(8)	Point production capacity constraint d
(9)	Courier only takes one delivery during the planning period
(10)	Deliveries must be performed within time no longer than t
(11)	Binarity

Table 2. Indices, sets, parameters and decision variables

Symbol	Description
<i>Indices and Sets</i>	
P	A set of delivery points (customers)
D	A set of production points (pizzerias)
C	A set of all couriers
p, j	Point index ($p, j \in P \cup D$)
c	Couriers index ($c \in C$)
d	Production point index ($d \in D$)
<i>Parameters</i>	
vi_p	Customer's order volume (volumetric weight) p ($p \in P$)
vc_c	Courier's c payload ($c \in C$)
$di_{p,j}$	Distance between points p, j ($p, j \in P \cup D$)
$ti_{p,j}$	Transfer time between points p, j ($p, j \in P \cup D$)
$w_{c,d}$	If courier c is assigned to production points d then $w_{c,d} = 1$ otherwise $w_{c,d} = 0$ ($d \in P, c \in C$)
t	Deadline for delivery
st	Very large constant
<i>Decision variables</i>	
$X_{c,d,p,j}$	If courier c starting from production point d and travels from point p to point j , then $X_{c,d,p,j} = 1$, otherwise $X_{c,d,p,j} = 0$, ($c \in C, p, j \in P \cup D, d \in D$)
$Y_{c,d,p,j}$	The order value for courier c , who starting delivery from production point d and travels from point p to point j ($c \in C, p, j \in P \cup D, d \in D$)

In the future, the objective function can be modified and can only cover, for example, delivery time, courier count, number of points, etc.

$$\min \sum_{c \in C} \sum_{d \in D} \sum_{j \in P \cup D} \sum_{p \in P \cup D} (X_{c,d,p,j} \cdot di_{p,j}) \quad (1)$$

$$\sum_{j \in P \cup D} X_{c,d,p,j} = \sum_{j \in P \cup D} X_{c,d,j,p} \quad \forall c \in C, p \in P \cup D, d \in D \quad (2)$$

$$\sum_{c \in C} \sum_{d \in D} \sum_{p \in P} X_{c,d,j,p} = 1 \quad \forall j \in P \quad (3)$$

$$st \cdot \sum_{j \in P} X_{c,d,d,p} \geq \sum_{j \in P} \sum_{p \in P} X_{c,d,p,j} \quad \forall c \in C, d \in D \quad (4)$$

$$\sum_{c \in C} \sum_{d \in D} \sum_{j \in P \cup D} Y_{c,d,j,p} - \sum_{c \in C} \sum_{d \in D} \sum_{j \in P} Y_{c,d,p,j} = vi_p \quad \forall p \in P \quad (5)$$

$$st \cdot X_{c,d,j,p} \geq Y_{c,d,j,p} \quad \forall c \in C, d \in D, p, j \in P \cup D \quad (6)$$

$$vc_c \geq Y_{c,d,j,p} \forall c \in C, d \in D, p, j \in P \cup D \quad (7)$$

$$vd_d \geq \sum_{c \in C} \sum_{p \in P} Y_{c,d,d,p} \forall d \in D \quad (8)$$

$$\sum_{d \in D} \sum_{p \in P} X_{c,d,d,p} \leq 1 \forall c \in C \quad (9)$$

$$\sum_{d \in D} \sum_{p \in P \cup D} \sum_{j \in P \cup D} (ti_{p,j} \cdot X_{c,d,p,j}) \leq t \forall c \in C \quad (10)$$

$$X_{c,d,j,p} \in \{0, 1\} \forall c \in C, d \in D, p, j \in P \cup D \quad (11)$$

To implement the model (1)..(11) the AMPL modeling language and solver Gurobi have been used [9]. AMPL is an algebraic modeling language to describe and solve high-complexity problems for large-scale mathematical computing. AMPL supports dozens of solvers, both open source and commercial software, including CHIP, CPLEX, Xpress, FortMP, Gurobi, MINOS, IPOPT, CPLEX CP Optimizer, etc.

The Gurobi is a commercial optimization solver for linear programming (LP), mixed integer linear programming (MILP), quadratic programming (QP), quadratically constrained programming (QCP), mixed-integer quadratic programming (MIQP) etc. [10]. The implementation of the modeled problem using AMPL and Gurobi solver optimization is shown in Fig. 2. The model in AMPL is shown in Appendix A.

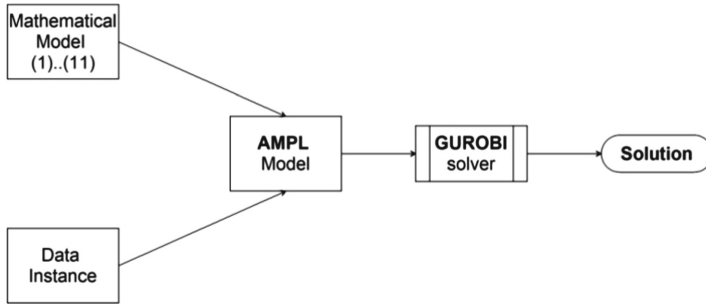


Fig. 2. Schematic diagram of how to implement the modeled problem using AMPL and optimization with the Gurobi solver.

4 Computational Experiments

To verify the correctness of the proposed model and the scope of supported decisions (answers to decision questions (a), (b)), numerous computational experiments have been conducted. The calculations have been made for four data instance sets, p1..p4. The different sets differ in the number of production points and delivery points. The data instances of each set differ in the number of couriers. The same processing time

Table 3. Results of computational experiments

Np.	Production points	Delivery points	Number of couriers	Non zeros	Fc	T
p1a	2	10	1	2362	NFSF	1
p1b	2	10	2	4700	296	1
p1c	2	10	3	7050	296	2
p2a	2	20	1	8668	598	1
p2b	2	20	2	17372	570	245
p2c	2	20	3	26058	570	345
p3a	2	40	2	66404	741	423
p3b	2	40	3	99686	708	546
p3c	2	40	4	132808	684	734
p4a	3	40	2	101788	676	546
p4b	3	40	3	152682	651	789
p4c	3	40	4	203576	634	1243
p5a	3	60	2	247005	NFSF	287
p5b	3	60	3	370507	946	3487
p5c	3	60	4	494009	867	4768
p6a	5	100	2	1143453	2131	11234
p6b	5	100	3	1715179	2072	11975
p6c	5	100	4	2286905	1946	12345
p6d	5	100	6	3430357	1846	13491
p6d	5	100	8	4573809	1846	14127

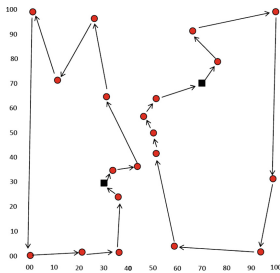


Fig. 3. Exemplary distribution network for the example 2pa (FC = 642)

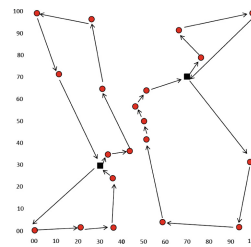


Fig. 4. Exemplary distribution network for the example 2pb (FC = 610)

$t = 25$ is assumed for all data instances. The calculations (optimization) have been made for each data instance. The optimal value of the objective function (Fc) and the calculation time (T) for each instance are shown in Table 3. Additionally, for data instances p2a and p2b, the obtained results are illustrated in the form of optimal delivery routes for couriers (in Fig. 5 and Fig. 6 respectively). For comparison, Fig. 3 and Fig. 4 show a non-optimal solution for the same data instances (p2a and p2b).

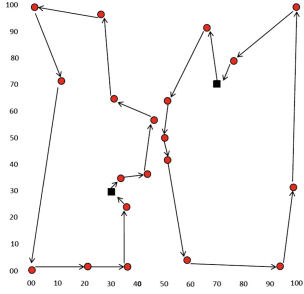


Fig. 5. Exemplary distribution network for the example 2pa ($FC^{OPT} = 598$)

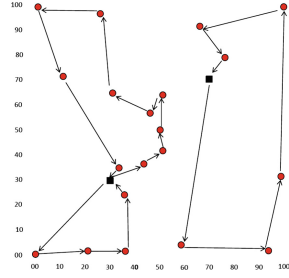


Fig. 6. Exemplary distribution network for the example 2pb ($FC^{OPT} = 570$)

A detailed analysis of the results obtained (Table 3), including the values of decision variables led to the following observations. The optimal value of the objective function usually depends on the number of couriers (with the same number of orders and production points). In some cases, increasing the number of couriers does not result in a better (smaller) value of the objective function (see p2b, p2c, p1b, p1c, p6d, p6e). Allocation of orders among a larger number of production points (pizza parlors) results in a better value of the objective function. For some data instances (p1a, p5a) no solution could be found. This means that with the assumed values of parameters (number of orders, couriers and production points), it is not possible to process customer orders.

A separate category for analysis is the time to achieve an optimal solution. No surprises here. For greater tasks, time increases significantly. Therefore, the use of the MP solver is insufficient for bigger problems.

5 Conclusion

The model proposed in the article allows for comprehensive optimization of customer order processing for an exemplary pizza chain. The proposed set of constraints and the objective function ensure both, the optimization of the order allocation to individual pizza parlors and the delivery method optimization (optimal selection of routes and couriers). The proposed set of constraints with minor modifications also supports the decision maker in many other decision-making and optimization problems, which can be formulated with the following questions

- Is it possible for a given set of customer orders to be processed within a certain period of time by a given pizza chain
- What is the minimum number of couriers for a given pizza chain to process a given set of orders within a certain period of time
- Whether a given set of orders will be processed in the specified period of time if a number of couriers is absent

These and other questions will be introduced in subsequent versions of the model to be developed in future research. Both, the current version of the model and its development due to the high computational complexity of modeled problems (NP-hard problems) for larger data instances will be implemented and solved using the authors' hybrid approach. In the currently developed version of the hybrid approach [11–13] three MP, CLP (Constraint Logic Programming) [14] and GA (Genetic Algorithm) [15] environments are integrated. The authors have been developing the hybrid approach for many years and applied to problems related to transport, routing, resource allocation, planning, etc. Its application to the modeling and implementation of optimization and decision-making problems provides greater modeling flexibility and efficiency in finding a solution. This results from a significant narrowing of the solution search with CLP mechanisms, appropriate problem representation and the use of data instance features.

It is also planned to include fuzzy logic [16, 17] in future models and to use variants of the proposed model to optimize orders in supply chains [18], production and distribution problems [19] and multimodal transportation networks [20].

Appendix AMPL Model

```

set Delivery_points;
set Production_points;
set Couriers;
set Points;
param vi{Delivery_points};
param vd{Production_points};
param vc{Couriers};
param di{Points,Points};
param ti{Points,Points};
param w{Couriers,Production_points};
param lpp;
param st;
param t;
var X{Couriers,Production_points,Points,Points} >=0, binary;
var Y{Couriers,Production_points,Points,Points} >=0, integer;
Minimize cost:
    sum{c in Couriers, d in Production_points, p in Points, j in Points}
        di[p,j]*X[c,d,p,j];
subject to C2 {c in Couriers, d in Production_points, p in Points}:
    sum{j in Points} X[c,d,p,j] = sum{j in Points} X[c,d,j,p];
subject to C3 {p in Delivery_points}:
    sum{c in Couriers, d in Production_points, j in Points}
        X[c,d,p + lpp, j] = 1;
subject to C4 {c in Couriers, d in Production_points}:

```

```

    st*sum{j in Points} X[c,d,d,j] >=sum{j in Points,p in Points}
    X[c,d,p,j];
subject to C5 {p in Delivery_points}:
    sum{c in Couriers, d in Production_points, j in Points}
Y[c,d,j,p+lpp]-
    sum{c in Couriers, d in Production_points, j in Points}
    Y[c,d,p+lpp,j] =vi[p];
subject to C6 {c in Couriers, d in Production_points, p in Points, j in
Points}: st*X[c,d,p,j] >=Y[c,d,p,j];
subject to C7 {c in Couriers, d in Production_points, p in Points,
j in Points}:vc[c] >=Y[c,d,p,j];
subject to C8 {d in Production_points}:
    sum{c in Couriers,j in Points} Y[c,d,d,j] <=vd[d];
subject to C9 {c in Couriers}:
    sum{d in Production_points, j in Points} X[c,d,d,j] <=1;
subject to C10 {c in Couriers}:
    sum{d in Production_points, p in Points, j in Points}
ti[p,j]*X[c,d,p,j] <=t;

```

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UAV Fleet Mission Planning Subject to Robustness Constraints

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Abstract. Fleet mission planning for Unmanned Aerial Vehicles (UAVs) involves creating flight plans for a specific set of objectives, which typically, have to be achieved over a specific time period. The key challenge is to develop methods allowing to prototype mission plans, encompassing UAV routes and schedules, that are robust to changing weather conditions and energy constraints. This paper presents a declarative approach to solving UAV mission planning problems subject to weather uncertainty. The approach was tested using several examples, for which we analyzed how the achievement of mission goals depended on parameters, such as UAV fleet size, UAV energy capacity, weather changes, including wind direction and wind speed, as well as the structure of the distribution network and the time horizon.

Keywords: UAV routing and scheduling · UAV fleet mission planning

1 Introduction

Mission planning is essential for any operation of an UAV fleet. UAV fleet mission planning problems are an extension of the well-known Vehicle Routing Problem (VRP), but with the added complexity of three-dimensional operations and combined routing and scheduling [20]. While the classical VRP is well-studied, the methods and approaches found in the field of UAV mission planning are still at a nascent stage and are mostly focused on the advancement of new technologies in the area of UAV operations [2]. Typically, mission planning for UAVs must take into account constraints on UAV range (dependent on UAV characteristics and weather conditions), airspace regulations and restrictions, as well as congestion (collision avoidance, safety distance, etc.) and UAV characteristics (air speed, maximum payload, energy capacity, physical dimensions, etc.) [15].

In outdoor UAV mission planning, it is necessary to address changes in weather conditions that influence the energy consumption and energy capacity of UAVs [10]. Some weather parameters can strongly influence the choice of strategy for UAV

mission planning. This is especially true of wind direction and speed, which impact energy consumption through either headwind or tailwind effects [18].

This study addresses the current gap in state-of-the-art mission planning for UAV fleets, by proposing a model that takes into account changing weather (wind speed and directions) conditions and allows to generate alternative robust mission plans. Another aim of this paper is to propose solutions that will ensure maximum customer satisfaction (completion of a maximum number of orders from all customers).

This study elaborates on the result of our previous works [1, 15, 16], and presents an expanded approach to solving mission planning problems, which takes into account UAV energy capacity and flight characteristics, variability of weather conditions, structure of the distribution network, and time horizon. Special attention is paid to prototype UAV flying missions robust to changing weather conditions e.g. missions resistant to weather uncertainty.

The remainder of this paper is structured as follows: Sect. 2 provides an overview of literature. A motivation example introducing the problem considered is presented in Sect. 3. A reference model for a UAV fleet routing and scheduling problem is presented in Sect. 4. The problem is formulated in Sect. 5, which also presents a Constraint Satisfaction Problem-based method for planning of UAV fleet delivery missions. An example illustrating the approach proposed is given in Sect. 6. Conclusions are formulated and directions of future research are suggested in Sect. 7.

2 Literature Review

The aim of mission planning is to find a sequence of points that connect a starting location to a destination. It differs from trajectory planning, in which the solution path is expressed in degrees of freedom of the vehicle [4, 6, 12]. In general VRPs, the standard objective function is typically to minimize the time for visiting a set number of nodes. In UAV fleet mission planning, several variants of objective functions, such as reducing individual UAV costs, increasing safety in operations, reducing lead time, and increasing the load capacity of the entire system, are considered [3, 4, 11, 14, 17]. In this context, the problems mentioned above can be seen as extensions of the vehicle routing and scheduling problems belonging to the class of NP-hard problems [4, 5].

The literature clearly shows that the decision criteria in UAV mission planning are many and complex [4, 16]. Specifically, the decision space comprises aspects related to routing and scheduling [1], changing weather conditions (wind speed, wind direction) [16], UAV weight and shape [8, 15], energy consumption affected by weather conditions [8], and the payload carried by the UAVs [5]. The multiplicity of these elements suggests the potential intractability of mission planning, as it is highly challenging to develop models considering all these aspects concurrently [4].

A number of contributions have proposed to subdivide the mission area, taking into account UAVs' relative capabilities, and to cluster the subsequent smaller areas to reduce problem size [4, 7, 9, 19]. Certain recent studies have used strategies to decompose the distribution network to reduce problem complexity [7, 9, 19]. According to this concept, a set of feasible weather resistant UAV fleet routes and schedules are created for all arbitrarily distinguished clusters of customer nodes.

Also, it is worth emphasizing that the possibility of taking into account the influence of weather conditions on energy consumption, and hence on the customer-servicing route and schedule, provides the basis for the construction of a model that allows to identify missions robust to specific weather changes. However, the literature offers no reports of results regarding the planning of UAV flight missions that would guarantee robustness to given variable weather conditions. This research gap has become an inspiration for the present study, which investigates the conditions that must be met for a planned UAV mission to achieve an assumed level of robustness.

3 A Motivational Example

Consider a company which provides goods transport services using a fleet of UAVs. The transport network covers an area of 100 km^2 and consists of 6 nodes: a base N_1 and 5 customers $N_2 - N_6$ waiting for the ordered quantity of goods $z_2 - z_6$, see Fig. 1. The supplier's fleet consists of 3 UAVs D_1, D_2, D_3 , the technical parameters of which are given in the table in Fig. 1b. The quantities demanded by the customers are known: $z_2 = 4 \text{ [kg]}$, $z_3 = 11 \text{ [kg]}$, $z_4 = 14 \text{ [kg]}$, $z_5 = 17 \text{ [kg]}$, $z_6 = 22 \text{ [kg]}$ and so is the time horizon during which the deliveries are to be made $t_{\max} = 2500 \text{ [s]}$. Deliveries are made in various weather conditions: a forecast wind speed $v_w = 10 \frac{\text{m}}{\text{s}}$ and a forecast wind direction $\theta = 180^\circ$ are also known. For these weather conditions, an answer is sought to the following question: *Can the given fleet of UAVs $\{D_1, D_2, D_3\}$ guarantee that the quantities demanded ($z_2 - z_6$) will be delivered in the given time horizon ($t_{\max}$) and under the forecast weather conditions (v_w, θ)?*

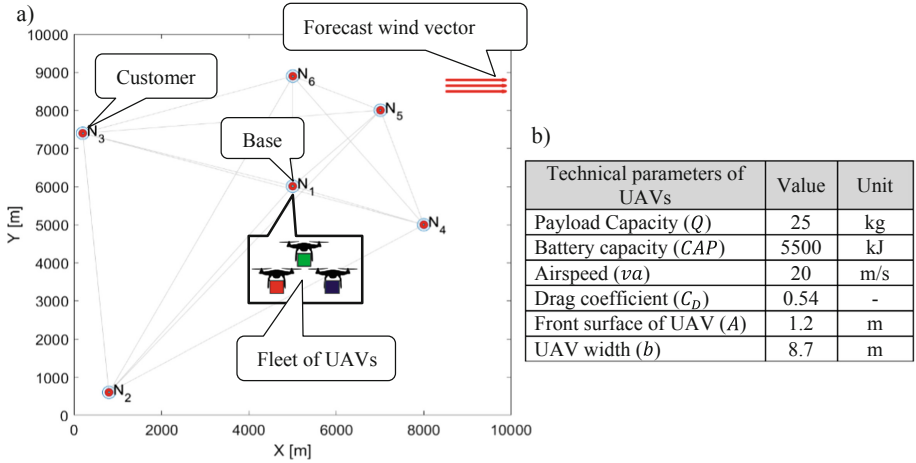


Fig. 1. Transportation network (a), technical parameters of UAVs (b)

In other words, a proactive flight mission plan is being sought, covering UAV flight routes and schedule, which will ensure that the expected amount of goods is delivered

in the given weather conditions. An example of such a plan is shown in Fig. 2a. In this solution, UAV routes (sequences of nodes visited one by one by the UAVs): $\pi_1 = (N_1, N_6, N_3, N_2, N_1)$, $\pi_2 = (N_1, N_5, N_6, N_1)$, $\pi_3 = (N_1, N_5, N_4, N_1)$, allow to deliver the entire demanded quantity of goods to customers, without the minimum battery level of each UAV being exceeded. Energy consumption represents 98.5%, 36.9%, and 43.3%, respectively, of the adopted battery capacity value $CAP = 5500$ kJ. By implementing this flight mission plan, the supplier can meet the customers' demand for goods in the forecast weather conditions.

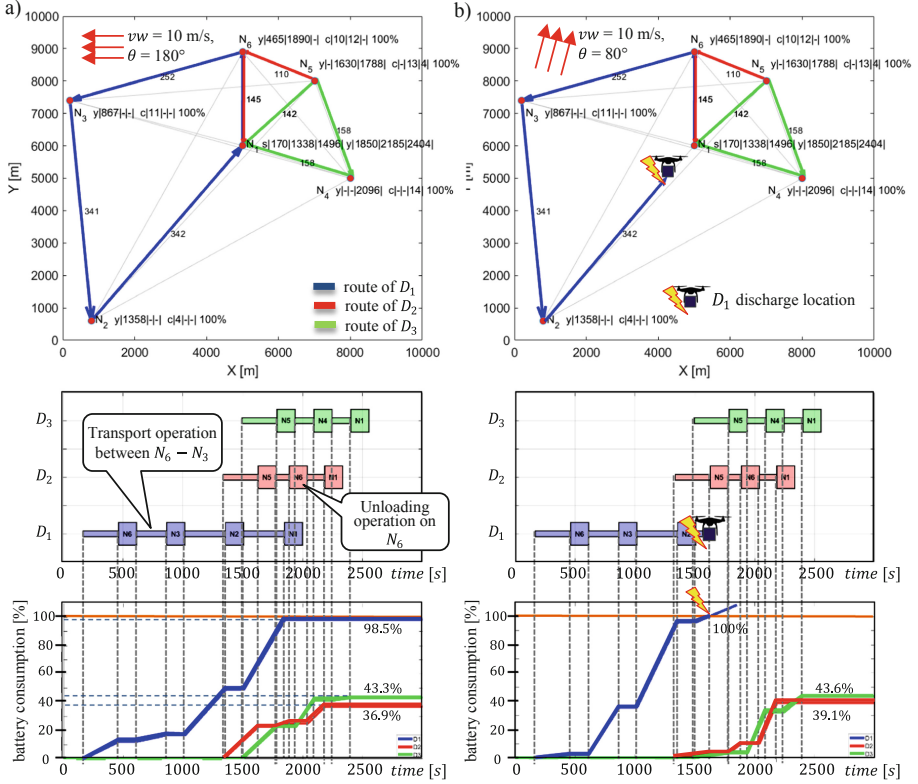


Fig. 2. A UAV fleet mission specified by routings, delivery flow charts, and energy consumption time diagrams under different wind directions: (a) $vw = 10$ m/s, $\theta = 180^\circ$ (b) $vw = 10$ m/s, $\theta = 80^\circ$

Let us consider a situation (Fig. 2b), in which the flight mission described above is conducted in different weather conditions, e.g. at a different wind direction: $\theta' = 80^\circ$. It turns out that the previously planned acceptable energy consumption level D_1 (for $\theta = 180^\circ$) has been exceeded. The energy consumption levels for the case under consideration are 108.8%, 39.1%, and 43.6%, respectively. This means that the change in weather conditions makes it impossible for one of the three UAVs used to complete

its flight. Mission D_1 will end in the middle of the flight between nodes $N_2 - N_1$ (battery depletion – see Fig. 2b).

The example shows that proactive mission plans generated in this way are very sensitive to changes in weather conditions. This indicates that changes in weather can significantly affect the implementation of a given flight plan, and so an answer to the following question must be sought: *Does there exist, for the given fleet of UAVs which deliver supplies in the distribution network under consideration, a delivery plan (delivery routes and schedule) robust to given changes in weather conditions?*

The essence of this problem is illustrated in Fig. 3. To seek flight missions robust to given changes in weather conditions is to designate UAV routes (and the corresponding delivery schedules) which will guarantee timely delivery of goods, despite changes in weather conditions occurring over the time horizon under consideration. The weather conditions for which routes guaranteeing timely delivery are sought, are presented in the form of a radar chart (Fig. 3) which defines the upper value of wind speed vw for a given wind direction θ (given by function $vw(\Theta)$).

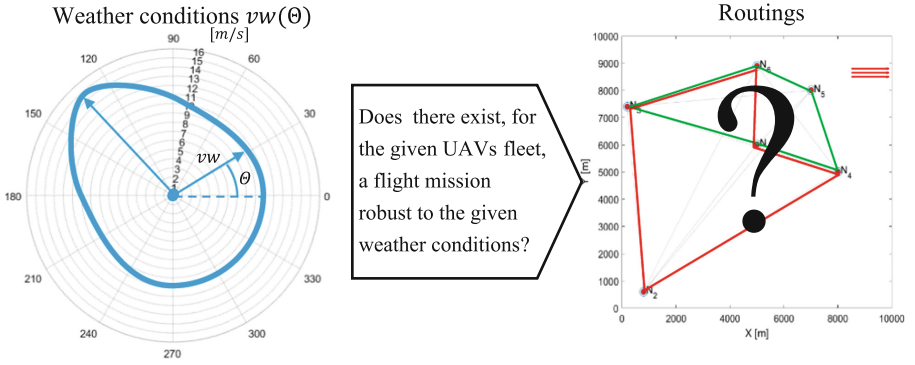


Fig. 3. Illustration of robust UAV fleet mission planning

Determination/planning of UAV flight missions that guarantee robustness to given weather conditions is a combinatorial optimization problem belonging to the class of synthesis problems [1]. This type of problems are NP-hard. The search for robust UAV fleet missions can be modeled using the CSP formalism, which allows to implement the proposed model directly in commercially available constraint programming environments, such as IBM ILOG CPLEX, Gurobi, ECL^{PS}^c, Oz Mozart, and others, which are a subclass of declarative programming environments.

4 Model

The mathematical formulation of the model of Robust Mission Planning employs the following parameters, variables, sets, and constraints:

Parameters

Network

$G = (N, E)$	graph of a transportation network: $N = \{1..n\}$ is a set of nodes, $E = \{\{i, j\} i, j \in N, i \neq j\}$ is a set of edges
z_i	demand at node $i \in N$, $z_1 = 0$
$d_{i,j}$	travel distance from node i to node j
$t_{i,j}$	travel time from node i to node j
w	time spent on take-off and landing operations
ts	time interval at which UAVs can take off from the base

Technical Parameters of UAVs

K	size of the fleet
Q	maximum loading capacity
C_D	aerodynamic drag coefficient
A	front facing area
Ep	empty weight

Environmental Parameters

H	time horizon $H = [0, t_{max}]$
$vw(\theta)$	function determining the upper value of wind speed for wind direction θ
$va_{i,j}$	airspeed of a UAV traveling from node i to node j
$\varphi_{i,j}$	heading angle, angle of the airspeed vector when the UAV travels from node i to node j
$vg_{i,j}$	ground speed of a UAV travelling from node i to node j
$\vartheta_{i,j}$	course angle, angle of the ground speed vector when the UAV travels from node i to node j

Decision Variables

$x_{i,j}^k$	binary variable which indicates if the k -th UAV travels from node i to node j
-------------	--

$$x_{i,j}^k = \begin{cases} 1 & \text{if } k\text{-th UAV travels from node } i \text{ to node } j \\ 0 & \text{otherwise.} \end{cases}$$

y_i^k	time at which the k -th UAV arrives at node i
c_i^k	weight of freight delivered to node i by the k -th UAV
$f_{i,j}^k$	weight of freight carried from node i to node j by the k -th UAV
$P_{i,j}^k$	energy per unit of time, consumed by k -th UAV during a flight from node i to node j
s^k	take-off time of the k -th UAV

cp_i total weight of freight delivered to node i
 π_k route of the k -th UAV $\pi_k = (v_1, \dots, v_i, v_{i+1}, \dots, v_\mu)$, $v_i \in N$, $x_{v_i, v_{i+1}}^k = 1$

Sets

Y^k set of times y_i^k – schedule of the k -th UAV
 Y family of Y^k – schedule of the UAV fleet
 C^k set of c_i^k – payload weight delivered by the k -th UAV
 C family of C^k
 Π set of UAV routes $\pi_{m,l}^k$
 S flying mission $S = (\Pi, Y, C)$

Constraints

Routes. Relationships between the variables describing UAV take-off times/mission start times and task order:

$$s^k \geq 0, k = 1 \dots K, \quad (1)$$

$$(k \neq q) \Rightarrow (|s^k - s^q| \geq ts), k, q = 1 \dots K, \quad (2)$$

$$\sum_{j=1}^n x_{1,j}^k = 1, k = 1 \dots K, \quad (3)$$

$$(x_{1,j}^k = 1) \Rightarrow (y_j^k = s^k + t_{1,j}), j = 1 \dots n; k = 1 \dots K, \quad (4)$$

$$(k \neq q \wedge y_i^k \neq 0 \wedge y_i^q \neq 0) \Rightarrow (|y_i^k - y_i^q| \geq w), i = 1 \dots n; k, q = 1 \dots K, \quad (5)$$

$$(x_{i,j}^k = 1) \Rightarrow (y_j^k = y_i^k + t_{i,j} + w), j = 1 \dots n; i = 2 \dots n; k = 1 \dots K, \quad (6)$$

$$y_i^k \geq 0, i = 1 \dots n; k = 1 \dots K, \quad (7)$$

$$\sum_{j=1}^n x_{i,j}^k = \sum_{j=1}^n x_{j,i}^k, i = 1 \dots n; k = 1 \dots K, \quad (8)$$

$$y_i^k \leq t_{max} \times \sum_{j=1}^n x_{i,j}^k, i = 1 \dots n; k = 1 \dots K, \quad (9)$$

$$x_{i,i}^k = 0, i = 1 \dots n; k = 1 \dots K. \quad (10)$$

Delivery of freight. Relationships between the variables describing the amount of freight delivered to nodes by UAVs and the demand for goods at a given node:

$$c_i^k \geq 0, i = 1 \dots n; k = 1 \dots K, \quad (11)$$

$$c_i^k \leq Q \times \sum_{j=1}^n x_{i,j}^k, i = 1 \dots n; k = 1 \dots K, \quad (12)$$

$$\sum_{i=1}^n c_i^k \leq Q, k = 1 \dots K, \quad (13)$$

$$(x_{i,j}^k = 1) \Rightarrow c_j^k \geq 1, k = 1 \dots K; i = 1 \dots n; j = 2 \dots n, \quad (14)$$

$$\sum_{k=1}^K c_i^k = cp_i, i = 1 \dots n, \quad (15)$$

$$cp_i \leq z_i, i = 1 \dots n, \quad (16)$$

$$\sum_{i=1}^n c_i^k = cs^k, k = 1 \dots K, \quad (17)$$

$$(x_{1,j}^k = 1) \Rightarrow (fc_j^k = cs^k), j = 1 \dots n; k = 1 \dots K, \quad (18)$$

$$(x_{i,j}^k = 1) \Rightarrow (fc_j^k = fc_i^k - c_i^k), i, j = 1 \dots n; k = 1 \dots K, \quad (19)$$

$$(x_{1,j}^k = 1) \Rightarrow (f_{1,j}^k = cs^k), j = 1 \dots n; k = 1 \dots K, \quad (20)$$

$$(x_{i,j}^k = 1) \Rightarrow (f_{i,j}^k = fc_j^k), i, j = 1 \dots n; k = 1 \dots K, \quad (21)$$

Energy consumption. The amount of energy needed to complete tasks performed by an UAV cannot exceed the maximum capacity of its battery.

$$bat^k(\theta) \leq CAP, k = 1 \dots K, \forall \theta \in [0^\circ, 360^\circ], \quad (22)$$

$$\sum_{i=1}^n \sum_{j=1}^n x_{i,j}^k \times t_{i,j} \times P_{i,j}^k(\theta) = bat^k(\theta), k = 1 \dots K, \forall \theta \in [0^\circ, 360^\circ], \quad (23)$$

$$P_{i,j}^k(\theta) = \frac{1}{2} C_D \times A \times D \times (va_{i,j}(\theta))^3 + \frac{(ep + f_{i,j}^k)^2}{D \times b^2 \times va_{i,j}(\theta)}, \forall \theta \in [0^\circ, 360^\circ], \quad (24)$$

where $va_{i,j}$ and $t_{i,j}$ depend on the assumed strategy for delivering goods – ground speed $vg_{i,j}$ is constant and airspeed $va_{i,j}$ is calculated from $\theta \in [0^\circ, 360^\circ]$:

$$va_{i,j}(\theta) = \sqrt{(vg_{i,j} \times \cos \vartheta_{i,j} - vw(\theta) \times \cos \theta)^2 + (vg_{i,j} \times \sin \vartheta_{i,j} - vw(\theta) \times \sin \theta)^2}, \quad (25)$$

$$t_{i,j} = \frac{d_{i,j}}{vg_{i,j}}, \quad (26)$$

Customer satisfaction. Customer satisfaction should be equal to or higher than an arbitrarily assumed value CSL . Customer satisfaction is described by the following formula expressing an average customer satisfaction level (the volume of planned deliveries expressed as a percentage):

$$\frac{\sum_{i=1}^n cp_i}{\sum_{i=1}^n z_i} \times 100\% \geq CSL. \quad (27)$$

5 Problem Formulation

The model we propose allows to define the problem of synthesis of robust UAV fleet missions, discussed in further part of this paper, in the following way: Consider a UAV fleet of size K servicing customers in a delivery distribution network G . Does there exist a mission S (determined by variables Π, Y, C) that ensures customer satisfaction CSL (27) under the constraints (22)–(26) related to energy consumption (resulting from the function $\nu w(\theta)$ which determine the upper values of relevant weather parameters – see example in Fig. 3)? The investigated problem can be viewed as a Constraint Satisfaction Problem (CSP) [13, 16] given by Formula (28):

$$CP = (\mathcal{V}, \mathcal{D}, \mathcal{C}), \quad (28)$$

where: $\mathcal{V} = \{\Pi, Y, C\}$ —a set of decision variables determining a mission S : Π —a set of UAV routes, Y —a schedule of a UAV fleet, C —a set of payload weights delivered by the UAVs, $\mathcal{D}$ —a finite set of decision variable-domain descriptions, $\mathcal{C}$ —a set of constraints specifying the relationships between UAV routes, UAV schedules, and transported materials (Formulas (1)–(27)).

To solve the CP given by Formula (28), one has to determine the values of the decision variables for which all the constraints are satisfied. By implementing CP (Formula (28)) in a constraint programming environment, such as IBM ILOG, one can answer the question formulated above.

6 Computational Experiments

In Sect. 3, it has been noted that the solution shown in Fig. 2a is unacceptable for the supply network under consideration if the following weather conditions exist: $\nu w = 10 \frac{\text{m}}{\text{s}}$; $\Theta = 80^\circ$ (the delivery plan will result in the depletion of battery D_1 before end of mission). The model developed in this study can be used to determine proactive flight missions that will ensure timely delivery of the demanded quantity of goods under variable weather conditions (within a specific range of values). In particular, an answer is sought to the following question: *Do there exist routes Π for the fleet of 3 UAVs (D_1, D_2, D_3 – Fig. 1b) and the transport network shown in Fig. 1 such that the demanded quantity of goods can be delivered on time ($CSL = 100\%$) under*

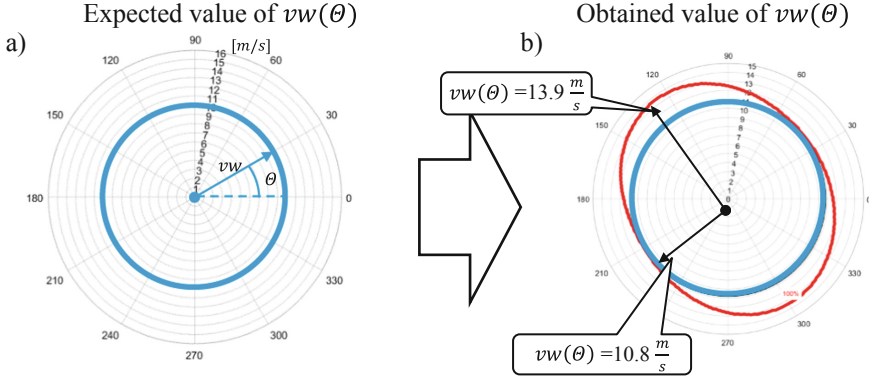


Fig. 4. Function $vw(\theta)$ determining the upper value of wind speed for a changing wind direction

the given variability of weather conditions: $vw = 0 \frac{m}{s} - 10 \frac{m}{s}$; $\theta = 0^\circ - 360^\circ$? The weather conditions are represented in the radar chart in Fig. 4a, whose outer perimeter defines the upper value of wind speed vw for angle $\theta \in [0^\circ, 360^\circ]$.

To answer this question one must solve problem (28). This problem was implemented and solved in the declarative programming environment IBM ILOG (Intel Core i7-M4800MQ 2.7 GHz, 32 GB RAM). The first acceptable solution was obtained in 7.55 s. Figure 5 shows the routes obtained: $\pi_1 = (N_1, N_6, N_3, N_1)$, $\pi_2 = (N_1, N_2, N_4, N_1)$, $\pi_3 = (N_1, N_5, N_6, N_1)$ and the corresponding delivery schedule. The adoption of such a solution guarantees that the demanded quantity of goods will be delivered to each node ($CSL = 100\%$) on time under variable weather conditions ($vw = 0 \frac{m}{s} - 10 \frac{m}{s}$; $\theta = 0^\circ - 360^\circ$). Battery consumption levels for $\theta = 180^\circ$ are: 64.8%, 37.2%, and 90.2%, respectively, and for $\theta = 80^\circ$: 55.4%, 38.5%, and 87.1%. Figure 4b shows a radar chart that defines the upper wind speed values for which the mission guarantees timely delivery of goods. As it can be seen, for each wind direction θ , the upper wind speed is $vw(\theta) \geq 10.8 \frac{m}{s}$, which means that the solution obtained is robust to the given (see Fig. 4a) weather conditions. The model was used in a series of experiments carried out to determine the parameters of the distribution network (the number of UAVs and the number of customers) for which the problem can be solved in under 600 s [13]. Table 1 shows the results of the experiments. The cells marked in yellow mean that for this set of input data (number of customers n , UAV fleet size K , wind speed vw), there is no solution that guarantees timely delivery of goods to customers. The experiments show that it is possible to synthesize flight missions robust to variable weather conditions for problems in which the transport network is not larger than 10 nodes and 3 UAVs.

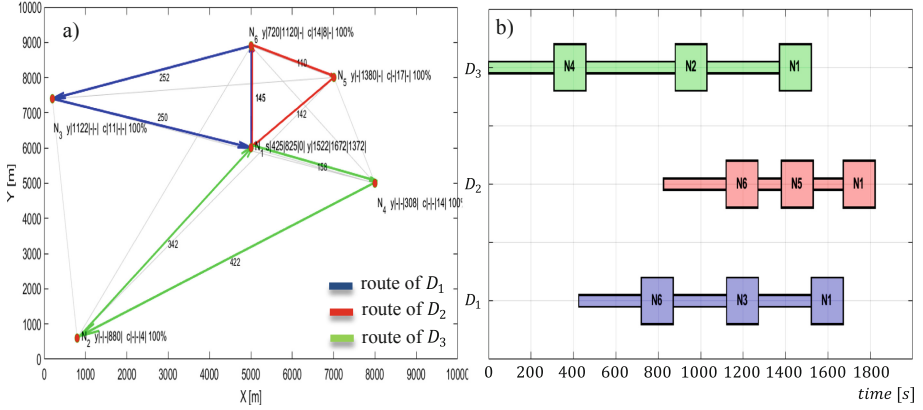


Fig. 5. Routes (a) delivery flow chart (b) guaranteeing delivery of ordered goods under changing weather conditions ($vw = 0 \frac{m}{s} - 10 \frac{m}{s}$; $\Theta = 0^\circ - 360^\circ$)

Table 1. Results of selected experiments for different $vw(\Theta)$, where $\Theta \in [0^\circ, 360^\circ]$

n	K	$vw(\Theta) = 0 - 9$	$vw(\Theta) = 0 - 10$	$vw(\Theta) = 0 - 11$	NC	NDV
		TC [s]	TC [s]	TC [s]		
4	2	5.11	4.06	4.04	454	1110
	3	4.48	4.52	4.39	825	2341
	4	4.9	4.77	4.9	1296	4024
6	2	4.71	4.72	5.6	1084	3360
	3	6.88	6.52	6.78	2115	7707
	4	16.34	47.51	13.7	3478	13834
8	2	10.29	11.1	12.48	2546	10042
	3	119.7	34.98	134.8	5561	25769
	4	$\tau > 600$	$\tau > 600$	$\tau > 600$	10012	48636
10	2	30.02	30.12	39.87	4920	21636
	3	$\tau > 600$	$\tau > 600$	$\tau > 600$	11703	57967
	4	$\tau > 600$	$\tau > 600$	$\tau > 600$	21378	111310

n – number of nodes; K – size of the UAV fleet; TC – time of computation (s); NC – number of constraints;

7 Conclusions

The declarative model proposed here (implemented in the ILGO IBM environment) allows to determine UAV missions robust to forecast whether changes. The permissible size of the distribution network (10 nodes and 3 UAVs) for which such missions can be determined, makes the proposed model particularly suitable for online application. It is worth emphasizing that the possibility of taking into account the influence of weather conditions on energy consumption, and hence on the customer-servicing route and schedule, provides the basis for the construction of a model that allows to identify missions robust to specific weather changes.

In our future research on robust UAV mission planning we want to explore the relationships linking fleet size with the assumed amount of deliveries subject to changes in forecast weather during mission execution as well as planning missions guaranteeing congestion-free execution of delivery process. Particular attention will be paid to the pick-up delivery problem with time windows and to planning the size of fleets composed of heterogeneous UAVs.

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**Special Session on Natural Language
and Argumentation 2020 (NLA 2020)**

Special Session on Natural Language and Argumentation 2020 (NLA'20)

We are in the reality of natural and computational systems of argumentation provided by reasoning, with natural and artificial languages. Intelligent systems of argumentation target advanced methods for exchanging, saving, reasoning, accessing, and updating information in memory. The special session on Natural Language and Argumentation (NLA) covers theories and applications. Formal models of argumentation like the Dung framework assume that natural language arguments have properly been mapped to logical formulas or partial proofs. Argument mining, when mainly working with existing machine learning methods, encounters difficulties to properly analyse arguments and relations between arguments, over general data, and especially when natural language expressions involve logical constructions. On the other side, traditional methods map sentences to logical formulas, which can be available after having been handled by a theorem prover. E.g., categorial analyses yield discourse representation structures, by using a parser (like Boxer, or Grail), and theorem provers (e.g., Coq) handle corresponding logical representations. The first two approaches (the Dung framework, and typical argument mining) suffer from the lack of development of the relations between natural language texts and dialogues, and do not handle the logical structure of meanings, while the third one (the predominant, traditional logical approach) is limited by the lack of sophisticated semantic lexicon for encompassing the logical structure carried by some words, and interconnections with other methods.

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Approximation Spaces of Temporal Processes and Effectiveness of Interval Semantics

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Abstract. A series of positive results related to the generalized problem of Yu.L. Ershov on the structure of Σ -degrees of dense linear orders is obtained. In particular, we prove that interval models of temporal logic, as well as finite fragments of approximation spaces generated by interval Boolean algebras, are Σ -definable (effectively interpretable) in hereditarily finite superstructures over dense linear orders. These results are used in the analysis of semantics of verbs in natural languages within the approach in formal semantics proposed by R. Montague.

Keywords: Effective model theory · Approximation spaces · Montague semantics for natural languages

1 Introduction

In this short paper, we consider three different topics of effective model theory [9, 15] and its applications. First, a natural example of approximation spaces [17, 18] (so called temporal approximation spaces over linear orders) is introduced and studied. Second, a number of positive results is obtained on the generalized problem of Yu.L. Ershov [8, 9, 14] concerning the structure of Σ -degrees of dense linear orders. In particular, we establish Σ -definability (effective interpretability) of interval models of temporal logic, as well as finite fragments of temporal approximation spaces generated by interval Boolean algebras, in hereditarily finite superstructures over dense linear orders. Third, we present some applications of these results in mathematical linguistics, in particular, in the analysis of semantics of verbs in natural languages (for example, English and Russian). In mathematical linguistics, for the analysis of the information expressed by a sentence in a natural language, one of the central tasks is to determine the semantic meanings of verbs that are the members of this sentence. We discuss some computational aspects of these questions within the approach in formal semantics proposed by R. Montague [11, 12] (see also [3, 4]).

The motivation of this paper should be explained. First of all, the author is not a linguist. He is a mathematician, namely a specialist in mathematical logic, effective model theory and theoretical computability. There are some properties of dense linear orders (e.g., elimination of quantifiers and decidability), which are well-known for logicians and which could be useful in the analysis of algorithmic aspects of interval semantics of verbs in natural languages. As far as it is known for the author, there are no examples of such analysis in the literature (at least, in the works of R. Montague [11, 12], M. Bennett and B. Partee [2]). So, this paper can be considered as a first (small) step in this direction.

We deliberately restrict ourselves to the case of linear orders, and hence of linear time, since the computational properties of dense linear orders are extremely nice. The models dealing with branching time should be considered differently. Linear time is sufficient to describe formally (and hence analyse effectively) such important features of verbs as tense and aspect.

2 Interval Extensions of Dense Linear Orders and Temporal Processes

For an arbitrary dense linear order $\mathbb{L} = \langle L, \leq \rangle$, define its interval extension $\mathcal{I}(\mathbb{L}) = \langle I, \leq, \subseteq \rangle$ as follows. As usual, a nonempty set $i \subseteq L$ is called an *interval* in $\mathbb{L}$ if, for any $l_1, l_2, l_3 \in L$ such that $l_1, l_3 \in i$ and $l_1 \leq l_3$, from $l_1 \leq l_2 \leq l_3$ it follows that $l_2 \in i$.

Let I be the set of all intervals in $\mathbb{L}$. Elements of L can be considered as intervals of the form $[l, l]$, $l \in L$. Thus, $L \subseteq I$.

The relation $\leq$ of structure $\mathbb{L}$ induces a partial order relation $\leq$ on set I . Namely, for elements $i_1, i_2 \in I$, we set $i_1 \leq i_2$ if and only if, for any $l_1 \in i_1$ and any $l_2 \in i_2$, the condition $l_1 \leq l_2$ is satisfied. Similarly, we can define the relation $<$ on I . The relation $\subseteq$ on I is interpreted as the standard set-theoretic inclusion relation.

Thus, the interval extension of a dense linear order is an extension in the model-theoretical sense.

The following notions are standard.

Definition 1. Let $i_1, i_2 \in I$ be arbitrary elements (i.e., intervals) of interval extension $\mathcal{I}(\mathbb{L})$.

1. i_1 is a *subinterval* of i_2 if
 $\mathcal{I}(\mathbb{L}) \models (i_1 \subseteq i_2)$;
2. i_1 is a *proper subinterval* of i_2 if
 $\mathcal{I}(\mathbb{L}) \models (i_1 \subseteq i_2) \wedge (i_1 \neq i_2)$;
3. i_1 is an *initial subinterval* of i_2 if
 $\mathcal{I}(\mathbb{L}) \models (i_1 \subseteq i_2) \wedge \neg \exists i_3 ((i_3 \subseteq i_2) \wedge (i_3 < i_1))$;
4. i_1 is a *final subinterval* of i_2 if
 $\mathcal{I}(\mathbb{L}) \models (i_1 \subseteq i_2) \wedge \neg \exists i_3 (((i_3 \subseteq i_2) \wedge (i_1 < i_3))$;
5. i_1 is a *point interval* if
 $\mathcal{I}(\mathbb{L}) \models \forall i_0 ((i_0 \subseteq i_1) \rightarrow (i_0 = i_1))$.

The definition of an approximation space [5, 7] is given below in the most general form. However, in this paper we will consider only very special examples of such spaces, generated by interval extensions.

Definition 2. An *approximation space* is an ordered triple

$$\mathcal{X} = \langle X, F, \leq \rangle,$$

where X is a topological T_0 -space, which is an A -space in terms of [5, 7], $F \subseteq X$ is a basic subset of *finite elements* and $\leq$ is a specialization order on X .

In the same way as in [5, 7], we denote by $a \prec x$ the fact that $a \in F$ and $a \leq x$.

It is known (see [7]) that any A -space has the following important property: every element $x \in X$ is a limit of its F -approximations:

$$x = \sup\{a \mid a \prec x\}.$$

Also, we will consider so called *structured* approximation spaces [17, 18], i.e., we assume F to be the domain of some structure $\mathcal{F}$.

One of the examples of structured approximation spaces is the approximation space of temporal processes over interval extension defined below. The elements of this extension (i.e., intervals) correspond to “finite” elements which approximate temporal “processes” considered in the broadest possible sense.

Definition 3. Let $\mathbb{L}$ be a dense linear order. *The space of temporal processes* over $\mathbb{L}$ is the approximation space $\mathcal{T} = (P(L) \setminus \{\emptyset\}, \mathcal{I}(\mathbb{L}), \subseteq)$, where $\mathcal{I}(\mathbb{L})$ is the interval extension of $\mathbb{L}$, $P(L)$ is the set of all subsets of L , and $\subseteq$ is the standard set-theoretic inclusion relation on $P(L)$.

Remark 1. Strictly speaking, $\mathcal{T}$ can be considered as an approximation space only if, instead of $\mathcal{I}(\mathbb{L})$, we define the set F of “finite” elements to be the set of all finite unions of elements from $\mathcal{I}(\mathbb{L})$. We restrict ourselves to the *basis* of F since it is the most natural formalization of a set of “simple” (not atomic) elements.

As a dense linear order $\mathbb{L}$, which generates the space of temporal processes, in this paper we consider an arbitrary dense linear order without endpoints. In formal semantics of natural languages, the ordered set $\mathbb{R}$ of real numbers, treated as an axis (scale) of time, is considered as a dense linear order in an abstract sense, hence its cardinality is not essential.

In applications, it is usually enough to consider only finitely generated fragments of such spaces. Again, Remark 1 is necessary to make precise the

Definition 4. Let $\mathbb{L}$ be a dense linear order and let $P_1, \dots, P_n \subseteq L$, $n \geq 1$ (we assume all P_i are nonempty). By $\mathcal{T}(P_1, \dots, P_n)$ we denote the approximation space $(\{P_1, \dots, P_n\} \cup \mathcal{I}(\mathbb{L}), \mathcal{I}(\mathbb{L}), \subseteq)$.

One of the main results obtained in this paper is Σ -definability of the satisfiability relation for Δ_0^{DL} -formulas over such spaces in hereditarily finite superstructure over the Morley expansion of $(\mathbb{L}, P_1, \dots, P_n)$, where $P_1, \dots, P_n \subseteq L$. The formalism of the version of dynamic logic DL with bounded modalities is used here for reasoning about temporal processes (or states), in particular, expressed by verbs in natural languages.

Let σ be a finite predicate signature containing, among other symbols, a binary predicate symbol $\subseteq$. We recall the definition of a formula of dynamic logic DL_σ from [17] (see also [16, 18]), which is essentially different from that in [6, 10]. Namely, formulas of logic DL_σ have variables of two types — for finite objects and for arbitrary, potentially infinite, objects that can only be accessed with the help of their finite fragments (approximations). We denote these sets by FV and SV , respectively. For the formula θ , the sets of its free variables of these two types are denoted by $FV(\theta)$ and $SV(\theta)$, respectively. If θ is a first-order logic formula of signature σ , then all its variables, including free ones, are considered to be finite. Variables denoted by uppercase letters ($S, P, \dots$) are by default considered as variables of type SV .

Definition 5. The set of Δ_0^{DL} -formulas of logic DL_σ is defined as the least set R such that

- 1) if θ is a first-order logic formula of signature σ , then $\theta \in R$ ($SV(\theta) = \emptyset$, $FV(\theta)$ is the set of all free variables of formula θ);
- 2) if $\theta \in R$, $S \in SV$, $a \in FV$, then

$$[a|S]\theta \in R, \langle a|S \rangle \theta \in R$$

($SV([a|S]\theta) = SV(\theta) \cup \{S\}$, $FV([a|S]\theta) = FV(\theta) \setminus \{a\}$), similarly for $\langle a|S \rangle \theta$;

- 3) if $\theta \in R$, $a, s \in FV$, then

$$[a|s]\theta \in R, \langle a|s \rangle \theta \in R$$

($SV([a|s]\theta) = SV(\theta)$, $FV([a|s]\theta) = (FV(\theta) \setminus \{a\}) \cup \{s\}$), similarly for $\langle a|s \rangle \theta$;

- 4) if $\theta_0, \theta_1 \in R$, then $\neg\theta_0 \in R$, $(\theta_0 \wedge \theta_1) \in R$, $(\theta_0 \vee \theta_1) \in R$ and $(\theta_0 \rightarrow \theta_1) \in R$.

The expressions $[a|S]$, $\langle a|S \rangle$, $[a|s]$ and $\langle a|s \rangle$ will be called *bounded modalities*.

Definition 6. Let $\mathcal{X} = (X, F, \leq)$ be a structured approximation space over the structure $\mathcal{F} = (F, \sigma^\mathcal{F})$ of signature σ . We define the *satisfiability relation* on $\mathcal{X}$ for a formula φ of logic DL_σ and an evaluation $\gamma : SV(\varphi) \cup FV(\varphi) \rightarrow X$ with $\gamma(x) \in F$ for any $x \in FV(\varphi)$, denoted by

$$\mathcal{X} \models \varphi \upharpoonright \gamma,$$

by induction on the complexity of φ . Let, for $x \in FV$ and $a \in F$, $\gamma_a^x = (\gamma \setminus (\{x\} \times F)) \cup \{\langle x, a \rangle\}$, similarly, for $S \in SV$ and $S_0 \in X$, $\gamma_{S_0}^S = (\gamma \setminus (\{S\} \times X)) \cup \{\langle S, S_0 \rangle\}$.

- 1) $\mathcal{X} \models [x|S]\theta(x) \uparrow \gamma$ if, for all $a \prec \gamma(S)$, $\mathcal{X} \models \theta \uparrow \gamma_a^x$;
- 2) $\mathcal{X} \models \langle x|S\rangle\theta(x) \uparrow \gamma$ if there exists $a \prec \gamma(S)$ such that $\mathcal{X} \models \theta \uparrow \gamma_a^x$;
- 3) $\mathcal{X} \models [x|s]\theta(x) \uparrow \gamma$ if, for all $a \prec \gamma(s)$, $\mathcal{X} \models \theta \uparrow \gamma_a^x$;
- 4) $\mathcal{X} \models \langle x|s\rangle\theta(x) \uparrow \gamma$ if there exists $a \prec \gamma(s)$ such that $\mathcal{X} \models \theta \uparrow \gamma_a^x$;
- 5) $\mathcal{X} \models (\exists S)\theta(S) \uparrow \gamma$ if there exists $S_0 \in X$ such that $\mathcal{X} \models \theta \uparrow \gamma_{S_0}^x$;
- 6) $\mathcal{X} \models (\forall x)\theta(x) \uparrow \gamma$ if, for all $a \in F$, $\mathcal{X} \models \theta \uparrow \gamma_a^x$

and so on (we assume that the predicate symbol $\leq$ is interpreted in X in the standard way).

In case of approximation spaces of temporal processes, bounded modalities of form $[I|P]$ and $\langle I|P\rangle$, together with the unbounded modalities of form $[I] = [I|L]$ and $\langle I\rangle = \langle I|L\rangle$, allow us to formulate judgments about “possible worlds”, understood as subintervals of a process (other processes on a selected interval can be arranged with arbitrary complexity, but the “access” to them, in turn, is carried out only by subintervals). Atomic intervals (that is, intervals of the form $\{l\}$) give the exact and complete information about all the processes taking place at the moment. For these reasons and for connecting definability by Δ_0^{DL} -formulas with the first-order definability, we consider the following notion.

Definition 7. Let $\mathbb{L}$ be a dense linear order. *The atomic space of temporal processes* over $\mathbb{L}$ is the approximation space $\mathcal{T}_0 = (P(L) \setminus \{\emptyset\}, P_{fin}(L) \setminus \{\emptyset\}, \subseteq)$, where $P(L)$ is the set of all subsets of L , $P_{fin}(L)$ is the set of all finite subsets of L (treated as an extension of $\mathbb{L}$), and $\subseteq$ is the standard set-theoretic inclusion relation on $P(L)$.

3 Effectiveness and Computational Reasoning About Temporal Processes

The notion of Σ -definability, or effective interpretability, is one of the central notions studied in effective model theory [15]. We will use some modifications of the standard definition from [8].

Definition 8. Let $\mathfrak{M}$ be a structure of relational signature $\langle P_0^{n_0}, \dots, P_k^{n_k} \rangle$, and let $\mathbb{A}$ be an admissible set. $\mathfrak{M}$ is called Σ -*definable* in $\mathbb{A}$, if there are Σ -formulas

$$\Phi(x_0, y), \Psi(x_0, x_1, y), \Psi^*(x_0, x_1, y), \Phi_0(x_0, \dots, x_{n_0-1}, y),$$

$$\Phi_0^*(x_0, \dots, x_{n_0-1}, y), \dots, \Phi_k(x_0, \dots, x_{n_k-1}, y), \Phi_k^*(x_0, \dots, x_{n_k-1}, y)$$

in the signature of $\sigma_{\mathbb{A}}$ and $a \in A$ such that for $M_0 \models \Phi^{\mathbb{A}}(x_0, a)$ and $\eta \models \Psi^{\mathbb{A}}(x_0, x_1, a) \cap M_0^2$ the following holds: $M_0 \neq \emptyset$, η is a congruence relation on structure

$$\mathfrak{M}_0 \models \langle M_0; P_0^{\mathfrak{M}_0}, \dots, P_k^{\mathfrak{M}_0} \rangle,$$

there $P_k^{\mathfrak{M}_0} \models \Phi_k^{\mathbb{A}}(x_0, \dots, x_{n_k-1}) \cap M_0^{n_k}$ for any k , $\Psi^{\mathbb{A}}(x_0, x_1, a) \cap M_0^2 = M_0^2 \setminus \Psi^{\mathbb{A}}(x_0, x_1, a)$, $\Phi_k^{\mathbb{A}}(x_0, \dots, x_{n_k-1}, a) \cap M_0^{n_k} = M_0^{n_k} \setminus \Phi_k^{\mathbb{A}}(x_0, \dots, x_{n_k-1})$ for all k , and $\mathfrak{M}$ is isomorphic to the quotient structure $\mathfrak{M}_0/\eta$.

For structures $\mathfrak{A}$ and $\mathfrak{B}$, we denote by $\mathfrak{A} \leq_{\Sigma} \mathfrak{B}$ the fact that $\mathfrak{A}$ is Σ -definable in $\mathbb{HF}(\mathfrak{B})$, the least admissible set over $\mathfrak{B}$. We denote by $\mathfrak{A} \equiv_{\Sigma} \mathfrak{B}$ the fact that $\mathfrak{A} \leq_{\Sigma} \mathfrak{B}$ and $\mathfrak{B} \leq_{\Sigma} \mathfrak{A}$.

Definition 9. An approximation space $\mathcal{X}_1$ is Δ^{DL} -reducible to an approximation space $\mathcal{X}_2$ (denoted by $\mathcal{X}_1 \leq_{DL} \mathcal{X}_2$), if $\mathcal{X}_1$ as a structure is Δ_0^{DL} -definable in the approximation space $\mathcal{X}_2$, and

- 1) the structure of finite elements $\mathcal{F}_1$ is Δ_0^{DL} -definable in $\mathcal{X}_2$ inside $\mathcal{F}_2$,
- 2) there is an effective procedure that associates with every Δ_0^{DL} -formula of space $\mathcal{X}_1$ a Δ_0^{DL} -formula of space $\mathcal{X}_2$, which defines the corresponding predicate in this presentation space $\mathcal{X}_1$ in space $\mathcal{X}_2$.

Theorem 1. *Approximation spaces $\mathcal{T}$ and $\mathcal{T}_0$ are effectively DL-equivalent:*

$$\mathcal{T} \equiv_{DL} \mathcal{T}_0.$$

Proof. The sketch of proof is as follows. It is enough to note that for arbitrary processes $P_1, P_2 \subseteq L$, the relation $P_1 \subseteq P_2$ is expressed in $\mathcal{T}_0$ by the formula $[t_1|P_1][t_2|P_2](t_1 = t_2)$, and the relation $P_1 \leq P_2$ — by the formula $[t_1|P_1][t_2|P_2](t_1 \leq t_2)$, where in both cases $t_1, t_2 \in L$. Moreover, $P \subseteq L$ is an interval (with the endpoints) if and only if $\mathcal{T}_0 \models \langle t_1|P \rangle \langle t_2|P \rangle [t|P](t_1 \leq t \leq t_2)$. Similar formulas defining different kinds of intervals (open, etc.) can be written.

In the same way can be proved the following statement.

Proposition 1. *For any $n \geq 1$ and any $P_1, \dots, P_n \subseteq L$,*

$$\mathcal{T}(P_1, \dots, P_n)_{\Delta_0^{DL}} \equiv_{\Sigma} (\mathbb{L}, P_1, \dots, P_n)_{Morley},$$

there $(\mathbb{L}, P_1, \dots, P_n)_{Morley}$ is the Morley expansion of $(\mathbb{L}, P_1, \dots, P_n)$, i.e., expansion obtained by adding to the signature new symbols for all first-order definable relations, and $\mathcal{T}(P_1, \dots, P_n)_{\Delta_0^{DL}}$ is the expansion obtained by adding to the signature new symbols for all Δ_0^{DL} -definable relations.

Corollary 1. *If the elementary theory of $(\mathbb{L}, P_1, \dots, P_n)$ is c-simple (i.e., sub-model complete and decidable) then there exists a uniform effective procedure for checking the truth of Δ_0^{DL} -formulas in $\mathcal{T}(P_1, \dots, P_n)$.*

In particular, if all processes $P_1, \dots, P_n$ are elements of the interval Boolean algebra generated by $\mathcal{I}(\mathbb{L})$ (i.e., each process is a finite union of intervals), then condition of the corollary above is satisfied. This special case is usually the only one used in mathematical linguistics.

4 Application to Temporal Logic and Formal Semantics of Natural Languages

Generally speaking, the language of dynamic logic over interval temporal spaces is a convenient “high level” language for working with temporal logic and analyzing the interval semantics of natural languages. This language is effectively

interpreted in the language of first-order logic over the corresponding linear order, expanded by unary predicates for denoting processes. However, the formulas obtained in this case are difficult to analyze and are purely technical in nature.

The basic relations of the temporal logic of J.F. Allen [1] are formalized in dynamic logic as follows: for arbitrary temporal processes $P_1, P_2 \subseteq T$,

P_1 **before** P_2 corresponds to the relation $[I_1|P_1][I_2|P_2](I_1 \leq I_2)$;
 P_1 **after** P_2 corresponds to the relation $[I_1|P_1][I_2|P_2](I_2 \leq I_1)$;
 P_1 **while** P_2 corresponds to the relation $[I_1|P_1]\langle I_2|P_2 \rangle(I_1 = I_2)$;
 P_1 **overlaps** P_2 corresponds to the relation $\langle I_1|P_1 \rangle \langle I_2|P_2 \rangle (I_1 = I_2)$
(or, in the different interpretation, to the relation $\langle I_1|P_1 \rangle \langle I_2|P_2 \rangle ((I_1 = I_2)) \wedge \wedge$
("I₁ is a final subinterval of P_1) $\wedge$ ("I₂ is an initial subinterval of P_2)),
etc.

Much more interesting example of application of spaces of temporal processes and their effectiveness properties can be found in the area of mathematical linguistics related to interval semantics of verbs in natural languages. The papers by R. Montague [11, 12], an American specialist in mathematical logic, gave rise to a new direction in linguistics, later called formal semantics. It combines various approaches taken from logic and philosophy of language, and its main task is to describe natural language semantics within the framework of a solid theory. One of the methods used in this direction is the analysis of the grammatical categories of tense and aspect. From this viewpoint, R. Montague formalized the semantic meaning of verbs in English. We recall some examples of such formalization. First, here is his analysis of tense *Present Progressive* (as reformulated in [2]).

The sentence (i.e., state) **John is walking** is true at time p if and only if there is an open interval I such that p is a subinterval of I and for all $t \in I$ state **John walks** is true in moment t .

Interval extensions for the first time were essentially used by American linguists M. Bennett and B. Partee [2]. As an example, we consider a formal description of tense *Past Simple*.

The sentence (i.e., state) **John ate the fish** ($= \alpha$) is true on interval I , if I is a point interval, α refers to the interval I' , and there exists an interval $I'' < I'$ such that $I'' < I$ and the state **John eats the fish** is true on I'' .

For another example, consider the formal description of tense *Present Perfect*.

The sentence (i.e., state) **John has eaten the fish** ($= \alpha$) is true on interval I , if I is a point interval, α refers to the interval I' , I is a subinterval of I' and there is an interval $I'' < I'$ such that either I is the final point of I'' , or $I'' < I$ and the state **John eats the fish** is true on I'' .

It is easy to construct Δ_0^{DL} -formulas of signature $\langle \leq, \subseteq \rangle$ describing the corresponding relations between these processes (or states) in the space of temporal processes $\mathcal{T}$. Namely, if

1) “ I is an open interval” denotes the formula

$$[I'|I](\langle I_0|I'\rangle(I_0 \neq I') \vee \langle I_1|I\rangle\langle I_2|I\rangle(I_1 < I' < I_2));$$

2) “ I is a bounded interval” denotes the formula

$$\langle I_1\rangle\langle I_2\rangle[I'|I](I_1 \leq I \leq I_2);$$

3) “ I_1 is the left endpoint of I_2 ” denotes the formula

$$(I_1 \leq I_2) \wedge \neg\langle I\rangle(I_1 < I < I_2);$$

4) “ I_1 is the right endpoint of I_2 ” denotes the formula

$$(I_1 \leq I_2) \wedge (I_1 \subseteq I_2) \wedge ([I|I_1](I = I_1))$$

etc., we get

$$p \subseteq \text{“John is walking”} \iff$$

$$\iff \langle I|\text{“John walks”}\rangle((p \subseteq I) \wedge (\text{“}I \text{ is an open interval”})),$$

$$p \subseteq \text{“John ate the fish”} \iff [I|\text{“John eats the fish”}](I < p),$$

$$p \subseteq \text{“John has eaten the fish”} \iff [I|\text{“John eats the fish”}](I \leq p).$$

In the examples above we consider the states **John walks**, **John is walking**, **John ate the fish**, **John ate the fish** and **John has eaten the fish**, together with the point interval treated as the “present moment”. Actually, in these examples it is shown how to define from *Present Simple* more complex tenses. Hence, by the results obtained above, the reasoning about the statements expressed by various combinations of tenses and aspects of English can be carried using some uniform and effective procedure.

5 Future Work

The structure of tenses and aspects of verbs in Russian is rather different than that in English. Namely, with three tenses (*Present*, *Past* and *Future*), there are two aspects: *Perfect* and *Imperfect*. The main difficulty for the analysis of Russian verbs is that these two aspects are *independent* in sense there is no basic and no derivable one. These issues are discussed in [13].

Another natural extension of the results above considers as the time scale, instead of dense linear orders, the ordered field of real numbers. In this case it is possible to argue about sentences containing phrases like “Tomorrow”, “Yesterday”, “Two days ago”, etc. It is known that the elementary theory of reals is regular (i.e., model complete and decidable) and from ω -minimality it follows that similar effectiveness results are also true. However, this is so only if it is not allowed to use natural numbers, and hence the space is not rich enough for our purposes.

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Dynamic Multi-level Attention Models for Dialogue Response Generation

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Abstract. One of the key challenges for creating a successful chat bot is to find an effective way to learn from human-human conversation data. Recently, a few neural network based dialog models, including the RNN language model (RNNLM) and the hierarchical recurrent encoder-decoder (HRED) model have shown promising results on dialog response generation. However, there is a critical challenge that the responses generated by these models incline to chit-chat style instead of being informative. In this paper, we empirically investigate this problem and also propose multilevel attention models to extend HRED with a hope that the attention mechanism can capture more informative content. The experiment studies on two multi-turn dialogue Datasets have shown the model's potential.

Keywords: Dialog generation · Multi-level attention · Encoder-decoder

1 Introduction

Recently, with development of neural network techniques, especially sequence-to-sequence (SEQ2SEQ) framework [19], the neural network based approaches have been widely applied to generate answers in question/answering services and multiple-turn dialogue. Currently the concept of Hierarchical Recurrent Encoder-Decoder (HRED) [14] has shown promising results on dialogue response generation. HRED has been widely lauded as an effective mechanism as a dialogue encoder framework [9]. Although HRED and its variants have shown promising potential, there are still a lot of challenges in dialogue applications and a notable one is informative dialogue response. Existing approaches to automatic response generation tend to adopt the chit-chat style, e.g., *Yes, it's OK*.

There are two possible reasons behind this. First, training data is unbalanced and insufficient and hence the trained model is not capable to abstract the complexity nature of human-human dialogues. The high frequency of generic responses found in the training set makes the model focus more on non-informative generalization abilities. Another possible reason is the deficiency

of the model itself. Inspired by the success of neural attention (alignment) models for sequence-to-sequence tasks such as machine translation [1]. The previous study has demonstrated that the neural attention mechanism can improve the effectiveness of a neural language model [18]. Similarly Mei et al. proposed a way to add attention to RNNLM by introducing a dynamic attention mechanism in which the scope of attention increases as the recurrence operation progresses from the start through the end of the conversation [11]. The dynamic attention model promotes coherence of the generated dialogue responses by favouring the generation of words that have syntactic or semantic associations with salient words in the conversation history.

However, the currently widely adopted attention models need to be extended to model the hierarchy of the context since they usually pay attention to word-level hidden states and cannot further combine word-level attention with session-level attention. In our work, the dynamic attention model is on session level and word level hidden states to learn certain context and capture more informative content. The experimental study on Ubuntu Dataset and Weibo dialogue datasets shows that our proposed model’s potential.

2 Related Work

Over the past ten years, dialogue generation models has been attracting considerable research interest. Traditional methods often rely on hand-coded rules and templates to generate proper responses, and this process is laborious and domain migration is less effective [16]. An alternative approach is to use retrieval model based dialogue system. However, such systems cannot generate new discourse when answering an unknown question [5].

Gu et al. argued that it would be beneficial that SEQ2SEQ system can accommodate both understanding and copy mechanism in case that system needs to refer to some words of target side sentence [3]. It is difficult for SEQ2SEQ system to learn the meaning of the rare words such as proper nouns and to generate them with standard RNN model. Therefore, several researchers extended attention-based encoder-decoder with CopyNet (Pointer network), which predicts words based on combined probability distribution [4].

Serban et al. further extend SEQ2SEQ framework with hierarchical encoders [14]. The word-level RNN encodes all tokens in each dialogue turn into utterance vectors. The context-level encoders recursively summarizes the dialogue turns into a hidden states as representation of previous dialogue context, which feeded into decoder as condition to predict next turn utterance.

3 Methodology

3.1 Task Definition and Model Overview

Follow the definition of [14], we also consider a dialogue as a sequence of M turns of utterances, i.e., $D = \{U_1, \dots, U_M\}$, and the dialogue involves two speakers.

Each U_t contains a sequence of N_t words, i.e., $U_t = \{w_{t,1}, \dots, w_{t,N_t}\}$, where $w_{t,i}$ is the word embedding from vocabulary V and representing the i th word in turn t . For a given multi-turn dialogue context $\{U_1, \dots, U_t\}$, our goal is to generate a proper and informative next turn utterance U_{t+1} .

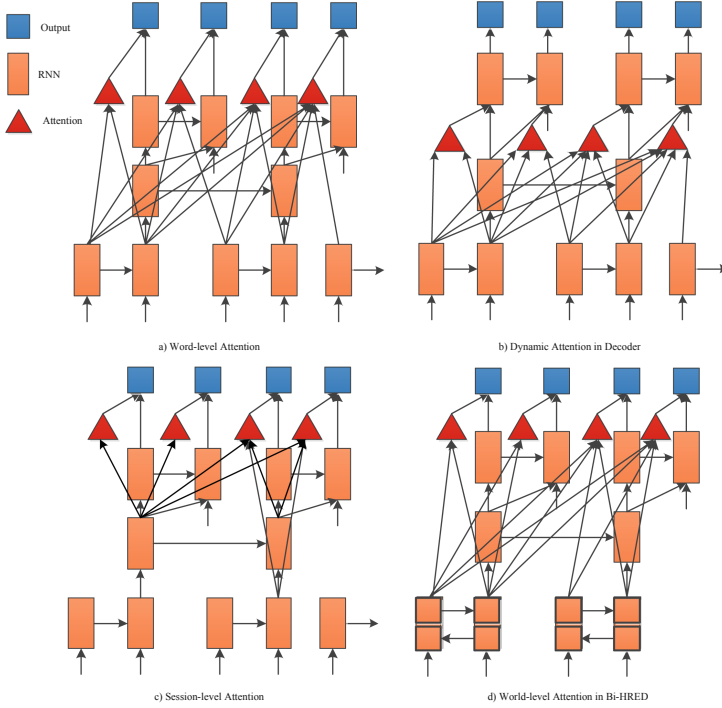


Fig. 1. HRED attention models

3.2 Hierarchical Encoder

The seminal encoder-decoder framework [17] predicts a sequence of words for the next dialog turn $d_{t+1} = \{w_{t+1,1}, \dots, w_{t+1,i}, \dots, w_{t+1,N_t}\}$ as the target in the encoder-decoder model. The source is the previous dialog turns $\{d_1, \dots, d_t\}$. Each turn d_t contains a sequence of words. HRED consists of three RNNs, an encoder RNN, a context RNN, and a decoder RNN, which correspond to the bottom, the middle, and the top layer of Fig. 1, respectively. The top layer is the output layer. The encoder RNN encodes all tokens up to i in the dialog turn d_t into a hidden vector $h_{t,i}^e$ with the recurrence function. The last hidden vector h_{t,N_t}^e is considered as the encoder vector represented as h_t^e . **The context RNN recursively summarizes the dialog turns up to the t turn into a hidden vector $h_t^c = f(h_{t-1}^c, h_t^e)$.** The context vector h_t^c is then fed to the decoder RNN to generate the next response d_{t+1} word by word. The decoder RNN works

similar to that of RNNLM except that the context vector h_t^c is used as extra input for each word generation in addition to the previous local hidden vector $h_{t+1,i-1}^d$ and the word before $w_{t+1,i-1}$

$$h_{t+1,i}^d = f(h_t^c, h_{t+1,i-1}^d, w_{t+1,i-1}) \quad (1)$$

where h_t^c is the session-level hidden states from turn t , $h_{t+1,i-1}^d$ is the decoder hidden states of previous step in turn $t + 1$.

3.3 Dynamic Multilevel Attention

The standard attention model has been proven to be effective for sequence-to-sequence modeling [11]. The attention module computes a weighted average vector of the encoder hidden states. The typical attention mechanism is set between a pair of source and target sentences. The scope of attention is always the source side sentence. In dialogue, source side sentence and target side sentence are consecutive dialogue turns. When human converse, we not only pay attention to the last turn, but also any part of the conversation that we are interested in previous turns. Therefore, in this research we propose to expand the attention range to be across turn boundaries. Figure 1-a) shows how dynamic attention works. We propose to compute attention as the weighted average vector of all previous encoder hidden states from turn 1 to the current turn $t + 1$.

$$z_{t+1,j} = \sum_{\tau=1}^t \sum_{i=0}^{N_t} \alpha_{ij}^{\tau} h_{\tau,i}^e \quad (2)$$

$$\alpha_{ij}^{\tau} = \exp(\beta_{ij}^{\tau}) / \sum_{\tau=1}^t \sum_{m=0}^{N_t} \exp(\beta_{mj}^{\tau}) \quad (3)$$

$$\beta_{ij}^{\tau} = b^T \tanh(W_{att} h_{t+1,j}^d + U_{att} h_{\tau,i}^e) \quad (4)$$

As showed in Eqs. 5 and 6, we take attention directly as input to the output layer [10]. When the decoder is generating the j word in turn $t + 1$, w_j^{t+1} , the attention $z_{t+1,j}$ is computed with Eq. 2. α_{ij}^{τ} is the attention weight between $h_{t+1,j}^d$ and the encoder hidden state $h_{\tau,i}^e$. These weights are determined by Eqs. 3 and 4.

$$g(h_{t+1,j}^d, z_{t+1,j}, v_k) = O_{v_k}^{\top} (O_h h_{t+1,j}^d + O_z z_{t+1,j}) \quad (5)$$

$$P(w_j^{t+1} = v_k | W_{1:t}, w_{0:j-1}^{t+1}) = \frac{\exp(g(h_{t+1,j}^d, z_{t+1,j}, v_k))}{\sum_l \exp(g(h_{t+1,j}^d, z_{t+1,j}, v_l))} \quad (6)$$

where $O_{v_k}^{\top} \in \mathbb{R}^d$ is the projection vector corresponding to the word v_k , $O_h \in \mathbb{R}^{d \times d}$ and $O_z \in \mathbb{R}^{d_z \times d}$ respectively project $h_{t+1,j}^d$ and $z_{t+1,j}$ into the same d dimensional space for the hidden vector, $W_{1:t}$ represents all words in previous turns, and $w_{0:j-1}^{t+1}$ represents all previous words in the current turn. As showed in Fig. 1-b), attention can be taken to the decoder RNN as input [1]. Session

attention is shown in Fig. 1-c). To make the model simpler, the session attention directly contributes to the Output layer. The Eq. 2 is extended as:

$$z_{t+1,j} = \sum_{\tau=1}^t \beta_{\tau,j} h_{\tau}^c \quad (7)$$

where h_{τ}^c represents all previous session-level RNN hidden states before current turn. We add both word-level attention and session attention to leverage different context information at same time. The combination of context vector directly contributes to the output layer and hence resulting in Eq. 8.

$$z_{t+1,j} = \sum_{\tau=1}^t \alpha_{\tau,j} h_{\tau}^c + \sum_{\tau=1}^t \sum_{i=0}^{N_t} \alpha_{ij}^{\tau} h_{\tau,i}^e \quad (8)$$

where $\alpha_{\tau,j}$ and α_{ij}^{τ} denotes the attention weights of session level and word level hidden states. Clearly, we have $\sum_{\tau=1}^t \alpha_{\tau,j} + \sum_{\tau=1}^t \sum_{i=0}^{I_t} \alpha_{ij}^{\tau} = 1$.

Given the success of Bidirectional HRED (Bi-HRED) [14,15], we finally extend the dynamic attention model and the dynamic output attention model to Bi-HRED. Bi-Directional HRED is a variant of HRED. The encoder RNN runs two chains: one forward and the other backward. As showed in Fig. 1-d), the last hidden state vector of each chain is concatenated as input to the context RNN. Although the last encoder hidden state summarizes the entire turn, it emphasizes more on the words close to the end and may not represent semantic dependency at the beginning of sentence. Bi-Directional HRED eases this problem by adding a backward chain.

4 Experiment Study

4.1 Dataset and Evaluation Configuration

We evaluate our proposed attention model on two datasets (preprocessed version of the Ubuntu Dialogue Corpus [15] and multi-turn dialogue datasets¹ collected from Sina Weibo). Ubuntu dialogues are highly related to technical issues of Ubuntu system. The dialogue involves two participants. one describes the technical issue and another provides suggestion. The Weibo dialogue dataset contains 5M multi-turn dialogues with a total of 25 million dialog turns and around 160 million tokens. On average, there are 5 turns per dialogue.

Following previous works [11,15], we use BLEU and Embedding Average (EACosine) as metrics to measure the response generation accuracy. BLEU [12] compares n-grams of the generated responses with the n-grams of the reference responses and count the number of matches. The matches are position-independent. Embedding Average [15] is an embedding-based textual similarity metric, which computes the cosine distance of sentence embedding between

¹ <http://tcci.ccf.org.cn/conference/2018/dldoc/taskgline05.pdf>.

generated response and the golden truth response. The sentence embedding of response is calculated as the averaging of all word embedding in the response. The metrics mentioned above have been commonly accepted for the measurement of sentence similarity. Adapted from [15], evaluations were also conducted using average entropy [15] to measure the performance of our proposed model that can provide more information content in response generation. In this paper, we also perform human evaluation, since accurate automatic evaluation of response generation is still a unsolved problem [13].

On both Ubuntu and Weibo datasets, we compare to 3 baseline models, including SEQ2SEQ model with attention (SEQ2SEQ ATT) [1], LSTM language model (LSTM LM) [11], and Hierarchical Recurrent Encoder-Decoder (HRED) [14]. SEQ2SEQ model is commonly adopted in dialogue Generation [7] and QA [8]. As mentioned in Sect. 3, HRED+ wd Att and sess Att stands for world-level and session-level dynamic attention respectively. HRED+wd sess att stands for word level and session level attention mechanism. On both Ubuntu and Weibo datasets, we set mini-batch size as 80. The encoder RNN is a standard GRU. Bi-HRED extends the HRED model with a bi-directions word level encoder. All the models are trained using Adam [6] for optimization in RNN back-propagation.

4.2 Result and Analysis

Table 1. Model performance

	Ubuntu		Weibo	
	BLEU	EACosine	BLEU	EACosine
SEQ2SEQ ATT	0.0463	0.471508	0.430400	0.323804
LSTMLM	0.0940	0.216193	0.707549	0.731520
HRED	0.9104	0.544941	1.168589	0.758465
Bi-HRED	0.8791	0.554000	1.169589	0.755272
HRED+wd Att	0.7567	0.558373	1.088861	0.758468
HRED+sess Att	0.9018	0.557808	1.108788	0.753484
HRED+wd sess Att	0.9320	0.570629	1.062517	0.759513
Bi-HRED+wd Att	0.9698	0.566351	1.155858	0.825979
Bi-HRED+sess Att	0.9842	0.563385	1.169626	0.752489
Bi-HRED+wd sess Att	1.0276	0.57163	1.219779	0.762425

Table 1 provides comparison of BLEU and Embedding metric evaluation results for our models and the baselines in Ubuntu and Weibo datasets. On Ubuntu datasets, HRED achieves a best BLUE score of 0.9104 and Bi-HRED achieves a best EACosine (Embedding Average Cosine) score of 0.554. The similar performance comparison is reported in [14]. On Weibo datasets, similar in the right

column of Table 1. HRED and Bi-HRED baseline achieves a best BLEU (1.1696) and EACosine (0.758) respectively. In the performance of the SEQ2SEQ and LSTM language model perform is unsatisfactory, suggesting that the SEQ2SEQ and LSTM language model is inadequate to model the complexity of Ubuntu dialogues or Weibo. The LSTM baseline also tends to output much more generic responses (see Table 2).

Table 2. Examples of generated dialog responses

Context	Responses
T1: having permissions problems with ubuntu server, trying to get /var/www to let me create files ... or at least move them to that dir but it 's not letting me T2: have you tried it with sudo mv?	HRED+wd sess att: yes, but i still get permission denied HRED: yes, but it doesn't work LSTM: ?
T1: hello all. is there a way to use legacy video drivers in a new ubuntu version? I have 10.10 right now, but I would like to upgrade to 12.04 and my video card is not supported in the new version T2: Which card very exactly do you have? T3: working on it ... is there an easy way to list installed hardware?	HRED+wd sess Att: lspci grep VGA HRED: What do you want to do? LSTM: what?
T1: i'm right now, no problem whatsoever booting in, but i need to do it MANUALLY T2: Can you give more detail what error are you seeing and when? T3: nothing! just the minimal console	HRED+wd sess Att: Can you pastebin the output of "sudo fdisk -l"? HRED: What is the exact error message? LSTM: i don't know

Our proposed models, namely, HRED+wd att, HRED+sess att, and HRED+wd sess att perform substantially better than the baseline model in BLEU score and EACosine on both Ubuntu and Weibo datasets. For ubuntu dataset, HRED+wd sess att outperform the best baseline by 2.4% and 4.7% in BLEU score and Embedding Average respectively. This confirms our hypothesis that generating responses with word-level and session-level attention can capture more informative content in context and lead to longer and more meaningful responses compared to the LSTM model and HRED model. Bi-HRED + wd, Bi-HRED + sess, Bi-HRED + wd sess att achieve further improvement in both BLEU and Embedding Average Cosine. Bi-HRED + wd sess att achieves the best performance in both BLEU and Embedding Average Cosine, i.e., 16.9% and 2.5% over the Bi-HRED baseline on Ubuntu dataset.

We compare the output from the LSTM language model, HRED baseline, and our best model (Bi-HRED with wd sess level attention). As can be seen in Table 2, word level attention model captures proper topic with more informative content generated in certain scenarios. Following experiment in [15], we measure the ability of our proposed models to promote information-richness in response generation using the average response length and average entropy [15] w.r.t. the maximum likelihood unigram model over the generated responses (Table 3). The $|U|$ was computed as the mean over the generated response length. We calculated the H_w (information entropy per word) [15] as $H_w = -\sum_{w \in U} p(w) \log p(w)$.

The H_U (information entropy per response) [15] were computed as the maximum-likelihood unigram distribution of the training corpus.

Table 3. Response information entropy

	Ubuntu			Weibo		
	$ U $	$\mathbf{H}_w$	$\mathbf{H}_U$	$ U $	$\mathbf{H}_w$	$\mathbf{H}_U$
SEQ2SEQ ATT	4.06	4.73	19.22	2.20	4.05	9.25
LSTMLM	4.27	6.15	26.25	4.23	4.87	19.49
HRED	11.06	6.49	71.79	4.514	5.40	23.11
Bi-HRED	11.47	6.67	76.50	4.69	5.40	24.00
HRED+wd Att	9.62	6.43	61.92	4.50	5.33	22.71
HRED+sess Att	10.86	6.41	76.80	4.62	5.47	23.90
HRED+wd sess Att	14.25	6.14	87.49	4.63	5.27	23.07
Bi-HRED+wd Att	11.78	6.49	76.46	6.34	4.95	29.67
Bi-HRED+sess Att	12.76	6.35	81.01	4.62	5.43	23.74
Bi-HRED+wd sess Att	12.97	6.73	87.33	4.66	5.78	25.49
Human	18.30	8.90	162.88	6.26	7.51	47.04

As illustrated in Table 3, our models outperform baseline models (SEQ2SEQ ATT, LSTMLM, and HRED) on both Ubuntu and Weibo datasets, which is consistent with previous literature [15]. Our proposed models also produce longer responses over Ubuntu and Weibo datasets. It is seen from Table 3, the responses generated by human even outperform all neural generative models on response length and information entropy, it appears suggest that a higher entropy is desirable [15], which is in good agreement with our previous experiment about BLEU and embedding average scores.

Table 4. Human evaluation on different attention strategies

	Win	Loss	Tie	Kappa
wd vs HRED	18.3%	14.6%	67.1%	0.36
sess vs HRED	23.2%	12.2%	64.6%	0.35
wd+sess vs HRED	24.2%	12.4%	63.4%	0.37

We also perform a human evaluation. Three human annotators with ubuntu system experiments were recruited. Each annotator judged 300 randomly chosen examples from each of our proposed model, including word-level, session-level, and wd sess level attentions. The baseline is Bi-HRED. For each sample, the generated responses from our proposed model and baseline were showed with

context and we anonymized the model identities. We ask human annotators to choose which response generation is more relevant, informative and fluent according to the dialog context and tie was allowed. Agreements among the annotators were calculated using Fleiss kappa [2]. The averaged human evaluation result is showed in Table 4, all our attention models outperform Bi-HRED baseline in pair-wise comparisons and high kappa scores indicate that annotators reached agreements in most case, which further verifies that our proposed multi-level attention model can generate more coherent and informative responses.

5 Conclusion and Future Work

To deal with the challenge of general response in automatic dialogue, in this research we propose a dynamic multilevel attention model, which can selectively focus on semantically more important part of the dialogue while keeping global features at session-level via attention mechanism. The proposed model is capable of generating responses towards certain words in context, towards the semantic representations of multiple sentences in previous dialogue, or towards both of them at same time. We evaluate the model with on Ubuntu Dataset and Weibo dialogue dataset. The experimental result has proven the effectiveness. For future work, we need to explore how to better incorporate external knowledge into technical-domain dialogue generation such as Ubuntu Troubleshoot scenario.

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Inferential Semantics as Argumentative Dialogues

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Abstract. This paper is at the same time a first step towards an “implementation” of the inferentialist view of meaning and a first proposal for a logical structure which describes an argumentation. According to inferentialism the meaning of a statement lies in its argumentative use, its justifications, its refutations and more generally its deductive relation to other statements. In this first step we design a simple notion of argumentative dialogue. Such dialogues can be either carried in purely logical terms or in natural language. Indeed, a sentence can be mapped to logical formulas representing the possible meanings of the sentence, as implemented with some categorial parsers. We then present our version of dialogical logic, which we recently proved complete for first order classical logic. Next we explain, through examples, how argumentative dialogues can be modeled within our version of dialogical logic. Finally, we discuss how this framework can be extended to real argumentative dialogues, in particular with a proper treatment of axioms.

Keywords: Proof theory · Dialogical logic · Argumentation · Natural language semantics

1 Introduction

A problem with the standard view of both natural language semantics and of logical interpretations of formulas is that the models or possible worlds in which a sentence is true cannot be computed or even enumerated [16]. As far as pure logic is concerned there is an alternative view of meaning called *inferentialism* [4–7, 17]. Although initially inferentialism took place within a constructivist view of logic [6], there is no necessary conceptual connection between accepting an inferentialist position and refusing classical logic as explained in [5].

Inferentialisms takes the inferential activity of agents to be the primary semantic notion rather than denotation or reference in a given situation. According to inferentialism, knowing the meaning of a sentence means being able to recognize what are the conditions allowing the sentence to be correctly asserted:

roughly speaking being able to recognize what counts as a justification of the sentence. From this it follows that there is a very tight link between inferentialism and argumentation theory: the study of arguments – reasonings, proofs, and clues which make it possible to demonstrate or support an affirmation, a fact or a proposition. Thus for an inferentialist the study of the “good” arguments for a proposition is the study of its semantics. A very restricted view of inferentialism is proof theoretical semantics (see e.g. [9]); for instance, the proof-theoretic semantics for intuitionistic logic defines the interpretation of a formula by the set of its proofs: while it is satisfactory for provable formulas, it leaves unprovable sentences without meaning, and so is clearly inappropriate for natural language semantics. Another reason for the initial connection between inferentialism and intuitionistic logic is that proof theoretic semantics only makes sense for constructive logics, including intuitionistic logic, and not for classical logic.

An argument in favor of a statement is often developed when a critical audience, real or imaginary, doubts the truth, or the plausibility of the proposition. In this case, in order to successfully assert the statement, a speaker or proponent of it must be capable of providing all the justifications that the audience is entitled to demand. Taking this idea seriously, an approximation of the meaning of a sentence in a given situation can be obtained by studying the *argumentative dialogues* that arise once the sentence is asserted in front of such a critical audience. The aim of this paper is to sketch a frame in which this study can be carried forward. We use the dialogical logic paradigm [8, 12] as a starting point: dialogical logic analyzes the concept of validity of a formula A through the concept of *winning strategy* in a particular type of two-player game. This type of game is nothing more than an argumentative dialogue between a player, called Proponent, which affirms a certain formula A and another player, called Opponent (called “teacher” in our version of dialogical logic), which contends its affirmation. The argumentative dialogue starts by Proponent affirming a certain formula A . Opponent takes its turn and attacks the claim made by Proponent according to the logical form of A . Proponent can, depending on his previous assertion and on the form of the attack made by the opponent, either defend his previous claim or counter-attack. The debate evolves following this back-and-forth pattern. Proponent wins the debate if he has the last word. An important characteristic of dialogic logic is that the concept of validity of a formula does not depend on a notion of model external to the dialogic game, contrary to what happens in his cousin theory game theoretical semantic [3, 11]. We propose a new variant of dialogical logic for first order logic, for which we were able to give a clear and simple proof of the following completeness result: a first order statement F is provable in classical logic (e.g. in Gentzen **LK**) if and only if Proponent has a winning strategy argumentative dialogues starting with the formula F .

The link to natural language semantics and to argumentation in natural language is crystal clear in type logical grammars. Such grammars compute the possible meanings of a sentence viewed as logical formulae, as in the **Graïl** platform a wide-scale categorial parser which maps French sentences to logical formulas [13–15]. Because of this close and computable connection between sentences and

logical formulae [16], we can extend the logical notion of argumentative dialogue into an argumentative dialogue starting with a natural language sentence, and even to an argumentative dialogue with natural language questions and answers, instead of logical formulas.

Admittedly, our first step in relating natural language semantics and argumentation is limited. In particular, argumentative dialogues in real-life situations can end without any of the two players winning: indeed, dialogues are never rooted in pure logic, but the players also have *axioms*, i.e. word meaning and assumptions about the world. These assumptions, especially the second ones, are not necessarily shared. For this reason, we focus our approach on situations where the definitions and postulates are known and precise, like in arguments between teacher and pupils in maths lessons (thanks to the data provided by maths education specialists) or in a future project on trial minutes.

Among the axioms some correspond to beliefs or postulates and are subjective and dependent on the speaker, but other axioms implement word meaning, which is, in a first approximation, objective and independent of the speaker in the language community. These latter axioms are fundamental to language understanding, natural language semantics, and argumentation. This is the lexical component of natural language semantics, and it is not so easy to properly integrate this aspect into compositional semantics which produces logical formulas describing the logical structure of the sentence without connecting the various predicates in the formulas like *book*(x) and *read*(u, v). But by now there exist several computable type theoretical formalisms that properly handle lexical semantics [1, 18] parts of the later one being integrated in the Grail platform [15]. Our argumentative dialogues already handle lexical meaning as formulas, on a par with the main sentence under discussion which is also viewed as a formula. Nevertheless, we plan to have a specific hence more accurate treatment of those common axioms describing lexical semantics.

Another expected extension of argumentative dialogues is to extend our dialogues and their completeness result to first order modal logic, whose expressive power is needed for interpreting, e.g. belief verbs.

2 Dialogical Logic

We present here our variant of dialogical logic that deals with first order formulas in negation normal form i.e. in which the symbol of negation only appears as a prefix of atomic formulas—every first order formula is classically equivalent to a formula in negation normal form. As negation amounts to swapping the two players, by adopting this choice we obtain dialogical games in which the role of the two participants never change during the course of the game: the Proponent always asserts formulas while the Opponent always attacks the formulas. By defining dialogic games in this way, we are able to obtain a simpler definition of the notion of strategy than, for example, that given by Felscher [8]. As we will see, a strategy is a tree of dialogic games with binary branching, in which a branching is the consequence of the assertion of a conjunction.

The dialogue games that we are going to define have a reminiscent flavor of Socratic dialogues. Two characters participate in the dialogue-games: the pupil and the teacher. The pupil claims that a certain proposition is true and therefore asserts it. The task of the teacher is to guide the pupil in the discovery of a *justification* for the asserted proposition. The teacher is a logic expert and whenever the student asserts a proposition, the teacher, in a somewhat pedantic way, reacts by questioning it by the following scheme.

conjunction. If the student claims that “ A_1 and A_2 ” the teacher will say “suppose that I claim that non A_i is justified, could you show me that I’m irrational by continuing the debate on A_i ?”

disjunction. If the student claims that “ A_1 or A_2 ” the teacher will say “Suppose that I claim instead that both non A_1 and not A_2 are justified, could you show me that I’m irrational by choosing one of either A_1 or A_2 and continuing our debate on the proposition of your choice?”

The teacher will have to concede that the pupil is able to justify the statement when the dubitative rational process that he has undergone, together with the answers offered by the student, force him to bet on asserting both a simple primitive proposition and its negation. Given the pupil’s inexperience, the teacher is forgiving of him. If he realizes that the fact of *choosing* to assert a certain proposition A leads him to a deadlock, he can always retrace his step and choose to assert a different proposition B . This corresponds to the choice of a different subformula in an asserted disjunction or to the choice of a different term in an asserted existential formula. We now present the system more formally.

Formulas. The set of first order terms and first order formulas are defined as usual. In our dialogical games each formula will be in negation normal form. In other words the negation symbol $\neg$ will appear only as a prefix of an atomic formula and the only binary connective appearing in the formula will be $\wedge$ and $\vee$. This is not a limitation since each formula is equivalent to at least one formula in negation normal form—although admittedly the dialogue on the negation normal form does not match the structure of the non negation normal form.

Moves. Here we define the rules that permits to attack a formula and what counts as a defence against the attack. The symbol prefixed by ? in the central column of the table are called **attack symbols**.

Assertion	Attack	Defence
$A_1 \wedge A_2$	$?\wedge_i$	A_i
$A_1 \vee A_2$	$?\vee$	A_i
$\forall xA$	$?\forall[y/x]$	$A(y)$
$\exists xA$	$?\exists$	$A(t)$

In the table $i \in \{1, 2\}$, x, y stand for arbitrary variables and t stands for an arbitrary term.

A **move** is a couple (i, s) where $i \in \mathbb{N} \cup \{\star\}$ and s is either a formula or an attack symbol. Given M_n one of the moves in the sequence of moves

$$M_0, M_1 \dots M_j \dots$$

we say that:

- M_n is an **assertion** if it is of the form (i, F) and we call F the formula asserted by the move, or the asserted formula;
- M_n is a question if it is of the form (i, s) where s is one of the attack symbol.

A question M_n of the form (i, s) is **justified** if $i < n$ and $M_i := (i', F)$ and s is an attack on F , and we say that M_i is the **enabler** of M_n . An assertion M_n of the form (i, F) is **justified** if $i < n$ and $M_i := (i', s)$ and M_i is a justified attack on $M_j := (j', F')$ where s attacks F' and F is a defence of s . M_i is the **enabler** of M_n .

Games. A **dialogical game** for a formula F is a sequence of moves

$$M_o, \dots M_j \dots$$

such that

1. M_o is $(\star, F)$ where F is a formula
2. $M_1 \dots M_j \dots$ is a strictly alternated sequence of justified questions (odd index moves) and justified assertions (even index moves). Moreover the enabler of a question is the preceding assertion.
3. if M_n and M'_n asserts the same sub-formula occurrence of F and have the same enabler then $n = n'$
4. if the formula asserted by a move M_n is a literal then there is a question M_j ($j < n$) such that M_j is of the form $(j - 1, s)$, the enabler of M_j is (k, F) , and the negation of the literal asserted by M_n is a direct subformula of F that is a defence against s .

Even numbered moves are called pupil moves or **P** moves while odd numbered moves are called teacher moves or **T** moves. A move M is **legal** for a dialogical game D if and only if the sequence D, M is a dialogical game. We say that a dialogical game D is won by the pupil when

1. D is finite and ends in a **P**-move
2. there is no **T**-move that is legal for D

It is easy to see that if D is won by **P** the last move of D asserts a literal.

Strategies. We now define the concept of **strategy**. Strategies are the dialogical counterpart of formal derivations in a proof system. Informally speaking a strategy is something that tells the pupil how to react against any possible attack of the teacher. Formally, a strategy $\mathfrak{S}$ for a formula A is a tree of dialogical games that is rooted on $(\star, A)$ in which each **T** move has at most one **P** move as a child and where a **P** move has two children if and only if it is of the form $(j, A \wedge B)$. A strategy is **P**-winning if and only if each of its branches is a dialogical game won by **P**. We say that a formula is dialogically valid if and only if there is a winning strategy for the formula.

3 Meaning, Proofs and Argumentation

We recently proved completeness of our argumentative dialogues for first order classical logic:

A formula is provable if and only if any of its prenex negation normal forms admit a winning strategy.

Proof (sketch). The difficult direction is from Strategy to Proof. To do so, we first compute a derivation in the sequent calculus $G3$ [19] from a winning strategy for a quantifier free formula F (*). We then show that a strategy for a prenex formula can be divided into two parts (1) a part dealing only with (head) quantifiers (2) a part dealing with quantifier free formulas, solved by (*). Regarding (1) the difficulty (already noticed in [8] in the simpler intuitionistic case) is that some winning strategies do not respect the eigen-variable condition enjoyed by correct proofs. So we first show that (1.1) each quantifier part of a winning strategy for a formula F can be mapped to a correct quantifier structure (a kind of proof net for the quantifiers), (1.2) each correct quantifier structure for a formula F can be mapped to the quantifier part π' (the part below the *mid-sequent* obtained from the mid-sequent theorem) of a normal derivation π of F .

Such a result means that we can consider our dialogical system as a proof system for classical logic.

Consequently we can define the semantics of a sentence as the set of the argumentative dialogues starting with this formula. This set may or not include a winning strategy for the proponent—a tree may be viewed as the set of the prefixes of its branches. So this is a kind of proof theoretical semantics (see e.g. [9]) where formulas are interpreted by the set of their proofs. But there is a major difference with proof theoretical semantics: even non provable formulas have a non trivial interpretation (as in Ludics see e.g. [10]), which is satisfying, since for most formulas neither F nor $\neg F$ are provable.

This also has an important consequence for computational semantics of natural language. While there is no way to compute or enumerate the possible worlds in which a given sentence is true, one can start argumentative dialogues from any formula, just by following the dialogue rules, and this actually produces some kind of enumeration. It is too early to elaborate on this aspect of the project, but the prospects seem promising.

4 A Toy Example of Natural Language Argumentation

Let us consider the following argumentative dialogue between **P** and **T**:

P David is a murderer

T Can you justify you assertion?

P Indeed, David killed Paul

T I see.

In the first step of the dialogue the student asserts a proposition. At the professor's request to justify his assertion, the student asserts another proposition i.e. "David killed Paul". This seems to indicate that the professor *believes* that it is sufficient to be able to assert "David killed Paul" to justify the claim "David is a murderer". If we were to model this mini argumentative dialogue faithfully through dialogical logic we would be embarrassed. Once the first statement has been translated into a logical formula, we found ourselves with a literal i.e., $\text{Murderer}(d)$ where d is a first order constant representing the individual David. Now condition 4 of the dialogical game definition prevents the pupil from asserting a literal unless there is a preceding question triggering the assertion of the literal negation. Moreover we do not see the deductive link between the proposition "David killed Paul" and the proposition "David is a murderer".

As the definition of dialogical games ensures logical correctness, we want to model this argumentative dialogue without a single change in the definition of dialogical games. The dialogue seems to suggest that the professor believes in the truth of a proposition, or of a set of propositions Γ , which, together with the proposition "David killed Paul" logically imply the proposition "David is a murderer". i.e. that there is a winning strategy $\mathfrak{S}$ for the formula $\Gamma \wedge \text{Killed}(d, p) \Rightarrow \text{Murderer}(d)$. Suppose that *in this case* Γ is $\forall x(\text{Murderer}(x) \Leftrightarrow \exists z(\text{Killed}(x, z)))$.

Figure 1, where the blue dotted arrow points back from **P** moves to the **T** move that enables them, shows a winning strategy for negation normal form of the formula:

$$F \equiv \forall x(\text{Murderer}(x) \Leftrightarrow \exists z(\text{Killed}(x, z))) \wedge \text{Killed}(d, p) \Rightarrow \text{Murderer}(d).$$

In the previous dialogue it is reasonable to suppose that the implicit proposition that allows the deductive link between the first and second statements made by the student is $\forall x(\text{Murderer}(x) \Leftrightarrow \exists z(\text{Killed}(x, z)))$. However, if we were to model more complex argumentative dialogues, it would be much more difficult to trace the set of hidden hypotheses that create the deductive link between the asserted propositions. To make the process more realistic we can impose that Γ contains *definitions* of words of open classes like nouns, verbs, adjectives, adverbs, etc. Those definitions are stipulated to be components of the sentences which are uttered by the dialogue proponent. In other words in order to check whether the thesis of an argumentative dialogue is implied by the sentence uttered by the person **P** who is asked for a justification one could use the following general procedure.

- Translate each sentence uttered by **P** into a logical formula F obtaining a set of formulas Δ
- For each relational symbol R such that $R(\mathbf{t})$ is a subformula of some formula in Δ find a definition of the type $\forall x(R \Leftrightarrow B)$, obtaining a set of formulas Γ .
- Try to find a winning strategy for $\Gamma \wedge \Delta \Rightarrow A$ where A is the thesis of the dialogue.

It is reasonable to ask if there are *fragments of natural language* in which the meaning of words that belong to open classes is defined with sufficient precision.

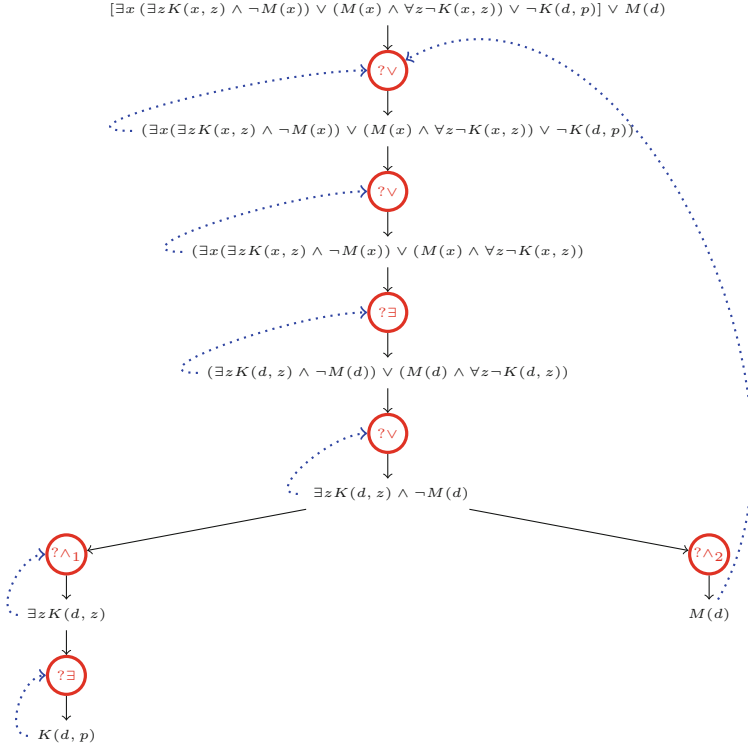


Fig. 1. A winning strategy for F . M stands for Murderer and K for Killed. Instead of integers we use blue dotted arrow that points back from **P** moves to the **T** move that enables them

Dialogical logic has already been applied in the area of mathematics pedagogy to model argumentative dialogues between pupils looking for a proof of a particular mathematical theorem (see e.g. [2]), and we think that another possible domain of application are (parts of) the minutes of a trial.

5 Conclusion and Perspective

In this first attempt to connect natural language semantics with argumentative dialogues, we proposed a notion of argumentative dialogues which enjoys a completeness theorem for first order classical logic. This notion can be used for defining the semantics of a sentences as the sets of all possible argumentative dialogues starting with this sentence.

When modelling natural language argumentation, we observe at least axioms representing lexical semantics are needed. Although those lexical axioms can be treated as ordinary formulas as done in this paper, but it would be much more convenient to have a specific treatment of axioms. It would also open the door to a treatment of axioms that are not shared by the opponent and the proponent.

An even more interesting proposal would be to view unclosed dialogues as a way to trigger the discovery of axioms: what formula would close this dialogue? what other dialogue does this formula solves?

Regarding the purely logical aspects of our work, we would like to extend the transformation of a winning strategy to a proof to non necessarily prenex formulas (this is nearly done), to non necessary negation normal form formulas. Later on, we would like to encompass some standard modal logics (e.g. S4 which is proof theoretically well behaved) because modalities are needed when interpreting natural language sentences.

We also want to further explore the semantics in terms of sets of argumentative dialogues, both for formulas and for natural languages sentences—in this direction, a connection with ordinary models which have been more studied in linguistics would be much welcome.

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**Special Session on Surveying and
Maritime Internet of Things Education
(SMITE 2020) and Special Session on
TEChnological Approaches To Develop
SustaiNability of Cultural Heritage
(TECTONIC 2020)**

Special Session on Surveying and Maritime Internet of Things Education (SMITE'20)

The special session entitled Surveying & Maritime Internet of Things Education (SMITE 2020) is a forum that will share ideas, projects, researches results, models, experiences, etc. associated with Maritime & Surveyor IoT applications. The session will be held in L'Aquila (Italy) as the part of the 17th International Symposium Distributed Computing and Artificial Intelligence 2020 (<http://www.dcai-conference.net/>).

The Internet of Things (IoT) is a network of physical “smart” devices (applicable in vessels, vehicles, buildings, factories etc.) embedded with electronics, software, sensors, actuators, that allow interconnectivity between these devices & data exchange. Over the past 5 years this new technology has grown rapidly and has found applications ranging from people whose devices monitor health and wellness to manufacturers that utilize sensors to optimize the maintenance of equipment and protect the safety of workers. It is expected that by 2025, IoT may reach a total potential market impact of up to \$11.1 trillion. Large shipping corporations are already investing heavily in enabling IoT technology solutions in their fleet to improve transparency, safety, and cost efficiency by optimising procedures, maintenance & energy efficiency.

Shipowners are set to spend an average of \$2.5m, each, on Internet of Things (IoT) solutions over the next three years. In addition, the marine surveying infrastructure transformation through IoT technologies is expected to enable the shipping industry, port authorities or environmental agencies, to inspect shipping assets, offshore structures, waterways, and ensure compliance with various standards or specifications. In contrast, formal Maritime & Surveying IoT training on development, installation, service, maintenance & sustainability awareness is at its infancy, especially in the European Union.

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Special Session on TEChnological Approaches TO Develop DustaiNability of Cultural Heritage (TECTONIC'20)

The special session entitled TEChnological Approaches TO develop sustaiNability of Cultural heritage (TECTONIC 2020) is a forum that will share ideas, projects, researches results, models, experiences, applications etc. focus on Preventive Conservation of marine and aerial Cultural Heritage. New technological proposals, many of them based on Artificial Intelligence, are currently being applied as measures and processes for the conservation of cultural heritage. This special session has its main objectives:

To maximize the value of research outcomes by promoting knowledge exchange, interactions, partnerships and inclusive engagement between cultural heritage researchers, individuals and organizations outside the immediate research community.

To encourage the implementation and transmission of research outcomes and to communicate them and the knowledge acquired among researchers and stakeholder sectors.

Nowadays Preventive Conservation of Cultural Heritage (CH) and, in particular, Underwater Cultural Heritage (UCH) represents a crucial issue to safeguard tangible testimonies of past human life preserved on the seabed. Solutions to preventive conservation in the UCH sector are still costly and sometimes very difficult to develop and implement. This is because the research activities in the marine environments imply greater difficulties than the aerial environment in terms of methods as well as in the use of advanced technologies. Underwater sites are generally difficult to access, and more dangerous, compared with working on dry land. Also, working in marine environment, and use specific equipment requires a lot of experience, highly skilled operators and the coordination of different “technical actors”, thus keeping such solutions out of the reach of “common operators”. In this context, the whole management of an underwater site, starting from the evaluation of its state of conservation until the restoration works, is often carried out by “external experts”, implying enormous costs, not always manageable and applicable.

In this framework, the main objective of the session is to share research solutions tailored on the technical needs of all the actors and operators working in the Preventive Conservation of Cultural Heritage, and in particular, but not limited to this, in the UCH field (i.e. underwater site manager, managing institution, managing authority, etc.).

The session tries to joint experts and researchers that recently focus their work on intelligent approaches for a more sustainable management of aerial and UCHs, facilitating monitoring operations, encouraging the conservation in situ and providing accurate guidelines for a correct maintenance.

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Representation of the Knowledge and Fuzzy Reasoning

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Abstract. In this paper, we have described with certain detail some theories of the fuzzy logic, and their application to the artificial intelligence like scheme of representation of the knowledge, like base of new reasoning models, and also like effective means of approaching the problem of the linguistic classification of variables. After a fleeting representation of the nature and reach of the fuzzy sets, we enter of full in their characterization and nomenclature. Subsequently we approach the problems of the fuzzy relationships, and we propose the formulation of Zadeh for the representation of knowledge of the type: If x is A , Then y is B . This allows us to define the generalized modus ponens as inferential mechanism of the fuzzy systems. Finally, it is mentioned some of the reasoning ways that can be found in the fuzzy systems.

Keywords: Representation of the knowledge · Fuzzy reasoning · Artificial intelligence

1 Introduction

Most of the human declarations are ambiguous, and this ambiguity is an essential characteristic, not only of the language, but also of the classification processes, of the taxonomy establishment and hierarchies, and of the reasoning processes in themselves. The mathematics and the artificial intelligence, they attempt as always to interesting problems related with the cognitive sciences, they could not be to the margin of this peculiarity, and in 1965 Lotfi Zadeh made public works related with this topic in its famous article Fuzzy Sets. Fuzzy logic is a form of many-valued logic in which the truth values of variables may be any real number between 0 and 1 both inclusive. It is employed to handle the concept of partial truth, where the truth value may range between completely true and completely false. [1] By contrast, in Boolean logic, the truth values of variables may only be the integer values 0 or 1.

The term fuzzy logic was introduced with the 1965 proposal of fuzzy set theory by Lotfi Zadeh. [2, 3] Fuzzy logic had however been studied since the 1920s, as infinite-valued logic notably by Łukasiewicz and Tarski. [4] Fuzzy logic is based on the observation that people make decisions based on imprecise and non-numerical information. Fuzzy models or sets are mathematical means of representing vagueness and imprecise information (hence the term fuzzy). These models have the capability of recognising, representing, manipulating, interpreting, and utilising data and information

that are vague and lack certainty. [5] Fuzzy logic has been applied to many fields, from control theory to artificial intelligence.

Classical logic only permits conclusions which are either true or false. However, there are also propositions with variable answers, such as one might find when asking a group of people to identify a color. In such instances, the truth appears as the result of reasoning from inexact or partial knowledge in which the sampled answers are mapped on a spectrum. Both degrees of truth and probabilities range between 0 and 1 and hence may seem similar at first, but fuzzy logic uses degrees of truth as a mathematical model of vagueness, while probability is a mathematical model of ignorance. Applying truth values [6]: A basic application might characterize various sub-ranges of a continuous variable. For instance, a temperature measurement for anti-lock brakes might have several separate membership functions defining particular temperature ranges needed to control the brakes properly. Each function maps the same temperature value to a truth value in the 0 to 1 range. These truth values can then be used to determine how the brakes should be controlled. Linguistic variables [7]: While variables in mathematics usually take numerical values, in fuzzy logic applications, non-numeric values are often used to facilitate the expression of rules and facts. A linguistic variable such as age may accept values such as young and its antonym old. Because natural languages do not always contain enough value terms to express a fuzzy value scale, it is common practice to modify linguistic values with adjectives or adverbs.

2 General Aspects of the Fuzzy Sets

Considering an universe randomly. For example the universe formed by the set N of the natural numbers. Defining a subset A of N characterized by the following description: “ A is the set formed by the natural pairs numbers smaller than ten”. The subset A of N is perfectly defined in the following way: $A = \{2, 4, 6, 8\}$ clearly: $2 \in A$; $3 \notin A$; $10 \notin A$. In this case we don’t have any problem to establish the degrees of belongs of an element of the speech universe with regard to the considered subset. Still, considering now the universe C characterized by the following description: “ C is the set formed by all the alive human beings”, and B a subset of C characterized by the description: “ B is the subset C of the black and tall men”. In this case, we have problems in establishing the degrees of belongs of an element from the universe to the subset B considered. Clearly, an ordinary set can be defined as a collection of elements. If an element of the universe is represented in the collection, the element in question belongs to this set. In these cases, one can say that the degrees of belongs of an element chosen of the referential (referential is the synonym of Universe of Speech traditionally employee when one speaks of fuzzy sets) has a Boolean value, so that: If the element belongs to the set, the Boolean value is “1”; If the element doesn’t belong to the set, the Boolean value is “0”. This way, we can build a function “ f ”, that for ordinary sets is a Boolean function, so that given an element “ x ” of the referential U , and given a subset A of U [7]:

$$f_A(x) = 1 \Leftrightarrow x \in A \quad (1)$$

$$f_A(x) = 0 \Leftrightarrow x \notin A \quad (2)$$

We will enlarge the question now to that special type of sets that we have denominated fuzzy sets. In this case, we said that linguistic, subjective character, they prevent us from establishing with clarity the degree of belongs of some elements from the speech universe to the considered fuzzy set. Thus, there will be elements of the referential that clearly belong to the set, there will be others that clearly they don't belong, and there will be others that belong in certain degree - although not totally. Continuing with the previously position, the problem is very easy of solving if we consider that the function “f” adopts the following value, given an element “x” of the referential U, and a fuzzy subset A of U [10]:

$$f_A(x) = 1 \Leftrightarrow x \in A \quad (3)$$

$$f_A(x) = 0 \Leftrightarrow x \notin A \quad (4)$$

- $0 < f_A(x) < 1$ “x belongs in certain degree to A”

The function “f” somehow quantifies the degree of belongs of an element of referential of a considered set. Thus, a fuzzy set is one that doesn't have any clear border between the belongs and don't belongs of certain elements of the referential. Therefore, in order to establish the fuzzy limits of the corresponding set, we will need a criterion that will almost always be arbitrary. Let us analyze the following example: Considering the referential U of alive people, and also considering the fuzzy subset A of U defined by the label “A is the set of alive young people”. A property that we seems appropriate to characterize to the fuzzy subset A is the age of the elements of the referential, but to what criterion? ...We are before the non trivial problem of the definition of criterions for the fuzzification of sets. In our case, and continuing with the example, we will consider “young” all those elements of the referential whose age allow them easily catch the train and “not young” all those elements of the referential that can also legally benefit from card of the third age (respectively, ≤ 25 years and ≥ 65 years). This way:

$$f_A(x) = 1 \forall x / \text{Age}(x) \leq 25 \quad (5)$$

$$f_A(x) = 0 \forall x / \text{Age}(x) \geq 65 \quad (6)$$

But what happen with all those elements of the referential with ages located between 25 and 65 years? Which is their youth degree based on the criterion of age?... We are before a new problem: the characterization of the fuzzy area. In order to leave this problem, we will build a lineal function in the following way:

$$f_A(x) = \frac{65 - \text{Age}(x)}{65 - 25} = \frac{65 - \text{Age}(x)}{40} \quad \forall x / \text{Age}(x) \in [25, 65] \quad (7)$$

With this approach we have been able to segment our numeric space [0–1] in to three areas; two of them are not fuzzy and they refer to those elements of the referential they do or don't belong, to the considered fuzzy subset; and a third area, of fuzzy character that refers to those elements of the referential that belong in certain degree to the considered fuzzy subset [8].

3 Characterization and Nomenclature of the Fuzzy Sets

Any set, may it be fuzzy or ordinary, it has to be able to be described in a convenient way. In the case of the ordinary sets, and since it can be established without ambiguities, the corresponding relationship of belongs from the referential elements to the considered set. It is equivalent to characterize to the set in question in function to their domain (for example A is the set of the natural pairs numbers smaller than ten), or making explicit the elements that constitute them (for example $A = \{2, 4, 6, 8\}$). [8] On the other hand, we have already seen that, for each element of given referential, we can define a function “f” that will be of Boolean character in the case of ordinary sets, so that will assign their corresponding logical value to each element of the referential 0 or 1, according to the element in question belong, or don't belong, to the set. Therefore: Given a referential U, and be $A \subset U$, $\exists f_A(x) = 1 \Leftrightarrow x \in A$ and $f_A(x) = 0 \Leftrightarrow x \notin A$. Applying this approach to the ordinary set A that serves us as example, A will be perfectly certain with the following expression [6]:

$$f_A(x) = \left\{ \begin{array}{l} f_A(1) = 0 + f_A(2) = 1 + f_A(3) = 0 + f_A(4) = 1 + f_A(5) = \\ 0 + f_A(6) = 1 + f_A(7) = 0 + f_A(8) = 1 + f_A(9) = 0 + f_A(10) = \\ = 0 + \dots \end{array} \right\} \quad (8)$$

expression in the one which the sign “+” is read “and”

An equivalent expression to the previous, but something more simplified, it is the following:

$$f_A(x) = \left\{ \frac{0}{1} + \frac{1}{2} + \frac{0}{3} + \frac{1}{4} + \frac{0}{5} + \frac{1}{6} + \frac{0}{7} + \frac{1}{8} + \frac{0}{9} + \frac{0}{10} + \dots \right\} \quad (9)$$

where the numerators of the fractions represent the value of the function of degree of belongs, and the denominators are the elements of the considered referential. Thus, ordinary subset A of a referential U can be described:

- Implicitly
- Explicitly
- By means of a boolean function $f_A(x)$, $\forall x \in U$

For obvious reasons when we work with fuzzy sets we will prefer to use implicit descriptions or to use the functions of degree of belongs. In this last case we will keep

in mind that it gets lost the Boolean character of the aforementioned function. This way: $\forall A \subset U / \text{fuzzy} A \rightarrow \exists f_A(x) / f_A(x) : U \rightarrow [0, 1] \forall x$ that is to say, the function $f_A(x)$ can take any value in the interval $[0,1]$. This way of naming the fuzzy sets allows us directly to establish that the ordinary sets are a case particular of the fuzzy sets.

4 Representation of the Knowledge and Fuzzy Reasoning

Although a wide treatment of the derivative problems of the representation of the knowledge and of the fuzzy reasoning it surpasses the pretenses of this paper, it seems convenient to begin an approach to both questions. All the concepts seen up to now they have allowed us to characterize to the fuzzy sets and also to distinguish them of the ordinary sets. Let us remember that from the perspective of the artificial intelligence, the fuzzy model should allow us the representation of declarations like the following [7]:

- It usually takes around 45 min in arriving from Luanda to Catete, for the highway, and with slight traffic.
- It is not foregone that the unemployment diminishes in Angola, at least in a drastic way, in the next months.
- Most of the experts say that the probability of an occurring earthquake in the area of the Caurel is very small in an immediate future.

In the previous sentences we can recognize fuzzy predicates, fuzzy quantifiers and fuzzy probabilities. In this respect, other more conventional approaches, usually employees to represent knowledge, they lack of means to represent the meaning of the fuzzy concept efficiently. Thus, the models that are based in logical of first order, or those that are based on classic theories of the probability, don't allow us to manipulate the common sense knowledge correctly. The evident causes are the following:

- The derivate knowledge of the common sense is lexically imprecise.
- The derivate knowledge of the common sense is of non categorical nature.

On the other hand, the characteristics already studied of the fuzzy sets they give us hints on the way of proceeding, if what we want is to apply schemes of representation of the knowledge, and reasoning models, based on fuzzy logic:

- In fuzzy logic, the categorical reasoning is a particular case of the approximate reasoning
- In fuzzy logic everything is a degree problem
- Any logical system can be fuzzification.
- In fuzzy logic the knowledge should be interpreted as a collection of fuzzy restrictions that they operate about a collection of variables.
- In fuzzy logic, the reasoning problems and consequently the processes of inferences should be interpreted as propagations of the fuzzy restrictions mentioned in the previous paragraph.

This last point is of vital importance, and it deserves a brief explanation that will take us to the establishment of the so called “generalized modus ponens” as procedure

of basic inference in the fuzzy systems. How could we represent in a fuzzy system a declaration of the type “If x is A , then y is B ”, where A is a fuzzy subset of a referential U , B is a fuzzy subset of a referential V that can be equal or different to U , x is an element of U and y is an element of V ? The answer to this question is not unique, and several authors propose different solutions. In this respect, Zadeh proposes that the function of degree of belongs of the previous declaration can be calculated in the following way:

Declaration: If x is A , Then y is B

$$x \text{ is } A: f_A(x), x \in U$$

$$y \text{ is } B: f_B(y), y \in V$$

If x is A , Then y is B :

$$f_A \rightarrow B(x, y) = A' \mid + \mid B = \min\{1, 1 - f_A(x) + f_B(y)\}, x \in U, y \in V \quad (10)$$

Thus, we can introduce the well known inference mechanism as “modus ponens”, according to the one which:

If $A \rightarrow B$
and A
.....
Then B

The fuzzy systems use generalization of the “modus ponens” that we can represent in the following way:

If $A \rightarrow B$
and A
.....
Then B

where A resembles A but it is not A , and where B resembles B , but it is not B . This mechanism of inference is known with the name of “generalized modus ponens” and, always according to Zadeh, the corresponding expression to calculate $f_B(y)$ is the following:

$$f_B(y) = \sup_v [A' \mid + \mid B] \cap A, A \subset U, A \subset U, B \subset V, B \subset V \quad (11)$$

$$\text{for that } f_B(y) = \sup_v [\min[\min[1, 1 - f_A(x) + f_B(y)], f_A(x)]]], x \in U, y \in V$$

An example will contribute to clarify the process

Be $A \subset U, A \subset U, B \subset V, B \subset V$ where $U = V = \{1, 2, 3, 4\}$, and be:

$$f_A(x) = \left\{ \frac{0}{1} + \frac{0.6}{2} + \frac{1}{3} + \frac{0.5}{4} \right\} \quad (12)$$

$$f_A(x) = \left\{ \frac{0}{1} + \frac{0.2}{2} + \frac{0.6}{3} + \frac{1}{4} \right\} \quad (13)$$

$$f_B(x) = \left\{ \frac{0}{1} + \frac{1}{2} + \frac{0.6}{3} + \frac{0.2}{4} \right\} \quad (14)$$

Be also the following information:

- We know that: If x is A, Then y is B
- We know that: x is A

- (1) To find A'
- (2) To find $A' \mid + \mid B$ in $U \times V$
- (3) To evaluate $(A' \mid + \mid B) \cap A$ in $U \times V$
- (4) To find the expression that characterizes to B

Clearly this example is not more than the development “step to step” that allows us to characterize to the subset fuzzy B using the generalized modus ponens. This way (Tables 1 and 2):

$$A' \rightarrow f_{A'}(x) = \left\{ \frac{1}{1} + \frac{0.4}{2} + \frac{0}{3} + \frac{0.5}{4} \right\} \quad (15)$$

and finally:

$$B \rightarrow f_B(y) = \sup_v \left[(A' \mid + \mid B) \cap A \right], \text{ in } U \times V \quad (16)$$

$$f_B(y) = \left\{ \frac{0.5}{1} + \frac{1}{1} + \frac{1}{1} + \frac{0.7}{1} \right\} \text{ with } y \in V \quad (17)$$

Table 1. Generalized modus ponens

$$A' \mid + \mid B, \text{ in } U \times V = \text{---} \downarrow$$

	V			
U	1	2	3	4
1	1,0	1,0	1,0	1,0
2	0,4	1,0	1,0	0,6
3	0,0	1,0	0,6	0,2
4	0,5	1,0	1,0	0,7

Table 2. Generalized modus ponens

$$(A' \mid \mid B) \cap \text{ in } U \times V = \text{---} \downarrow$$

	V			
U	1	2	3	4
1	0,0	0,0	0,0	0,0
2	0,2	0,2	0,2	0,2
3	0,0	0,6	0,6	0,2
4	0,5	1,0	1,0	0,7

The generalized modus ponens is a first difference in the reasoning of the fuzzy systems, in relation to the reasoning in more classic and more conventional system, but also, we can also find other differences as much in representation as in reasoning that we summarize in the following points [6]:

- a) **Certainty:** In systems that they use bivalent logic the truth of a declaration could only have two values: The declaration is certain, or the declaration is false. On the contrary, in system multi valued, the truth of a declaration can be: an element of a finite group, an interval (example, [0,1]), or the Boole algebra. In fuzzy logic the truth of a declaration can be a partially orderly fuzzy subset, but it is usually assumed to existence of a diffuse subset of the interval [0,1], or said otherwise, a point of this interval. Thus, the denominated linguistic values of the truth of a declaration can be expressed by means of labels of the type: certain, very certain, not exactly certain,... that are labels corresponding to fuzzy subsets of the mentioned interval.
- b) **Predicates:** In bivalent systems the predicates are categorical, for example: mortal, pair, odd, higher than,... on the contrary, in fuzzy systems the predicates are, precisely, fuzzy, for example: high, soon, a lot bigger than,...
- c) **Modifiers:** In classic systems the only really utilized modifier is the negation NOT. In fuzzy systems there is a great variety of modifiers, for example: very, more or less, enough,... These modifiers are essential to generate the appropriate values of the linguistic variables involved in a process, for example: very young, not very old,...
- d) **Quantifiers:** In the classic systems there are only two quantifiers: universal and existential. On the contrary, in the fuzzy systems we find a great variety of quantifiers, for example: few, enough, usually, most,...
- e) **Probabilities:** In the classic logical systems, the probability is numeric. In the fuzzy systems the probability is expressed by means of labels linguists (fuzzy probabilities), of the type: commendable, not very probable, around 0.8,... The handling of such probabilities fuzzy should be make through of the fuzzy arithmetic call.
- f) **Possibilities:** Contrary to what happens with the classic logical system, the concept of possibility in the fuzzy systems is not bivalent. In fact, the equally as it happens to the probabilities, the possibilities can be treated as linguistic variables that adopt

values of the almost impossible, quite possible type,... The elements that we have just described, as basic components of the representation of the knowledge and of the fuzzy reasoning, they allow to be defined a wide variety in reasoning ways, among those that we will mention [9]:

- (1) Categorical reasoning: This reasoning type uses fuzzy declarations, but it doesn't use neither fuzzy quantifiers neither fuzzy probabilities. A simple example could be the following:

Carolina is a thin girl

Carolina is a very intelligent girl

Carolina is a thin and very intelligent girl

In this example the "thin" and "very intelligent" premises should be interpreted as fuzzy predicates. On the other hand, the fuzzy predicate of the conclusion is the conjunction of the previous premise

- (2) Syllogistic reasoning: Contrary to that described when speaking of the fuzzy categorical reasoning, the syllogistic reasoning produces inferences with premises that incorporated fuzzy quantifiers. A simple example could be the following:

Most of the Angolan people are black

Most of the black Angolan people are short

$(Most)^2$ of the Angolan people are black and short

In this case the fuzzy quantifier "Most" should be interpreted like a fuzzy proportion, and " $(Most)^2$ " is the square of "Most" in fuzzy arithmetic.

- (3) Dispositional Reasoning: In this reasoning type the premises are dispositions. The obtained conclusion is a maxim that should be interpreted as a command dispositional. A simple example could be the following:

To smoke a lot is usually cause of abundant cough

To avoid abundant cough, smoking a lot should be avoid

- (4) Qualitative reasoning: In fuzzy systems, the qualitative reasoning is defined as a reasoning way in which the input relationships and output of a system are represented by means of a fuzzy collection of rules of type IF-THEN, in those that the antecedents and the consequents include linguistic variables. This reasoning type is habitually employee in the applications of the fuzzy logic to the systems analysis and the control processes.

Actually, the application of the fuzzy sets to the intelligent systems is a topic of great interest in investigation. In all ways, although the theoretical bases of the fuzzy formalism are already quite clear, its application to systems of inferential nature finds problems that, nowadays, they continue without being resolved. Nevertheless that the fuzzy systems applied to control problems proportionate alternative solutions are of great brightness and elegance, in relation to the most traditional systems.

5 Conclusions

All the concepts seen up until now, they have allowed us to characterize to the fuzzy sets and to distinguish them of the ordinary sets. From the perspective of the artificial intelligence, the fuzzy model should allow us the representation of declarations. Any set, is fuzzy or is ordinary; it should be described in a convenient way. For obvious reasons, when we work with fuzzy sets we will prefer to use implicit descriptions or to use the functions of degree of belongs. In this last case, we will keep in mind that it gets lost the Boolean character of the function. The generalized modus ponens is a first difference in the reasoning of the fuzzy systems, in relation to the reasoning in more classic and more conventional systems. But, we can also find other differences summarized in the last part of this paper, as much in representation as in reasoning.

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A P2P Electricity Negotiation Agent Systems in Urban Smart Grids

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Abstract. Peer-to-Peer (P2P) energy trading (ET) is a paradigm of energy trading between consumers without intermediaries. This model of ET allows the commercialization of energy resources produced through renewable sources that do not need other local consumers. This energy trading between consumers is able to improve the local balance of energy generation and consumption. Currently, this paradigm is being evaluated to show the suitability of its application in today's society, significantly increasing the number of projects in this area worldwide. This article reviews the main models of application of this paradigm in smart cities, presenting the main characteristics of these approaches. This paper proposes an architectural model for P2P energy trading that solves the main deficiencies detected. The designed system allows the simulation of P2P processes using a novel negotiation model. This energy trading system is based on a Multi-Agent System (MAS) using the Robot Operating System (ROS). The system allows representing using independent agents each one of the zones that intervene in the process of negotiation of the energy of a city, being already representing a consumer's role or a producer's role of energy. The system has been tested on a model in which each zone uses real data about the role it simulates over a period of two and a half years. The preliminary results show how the energy trading market allows a smart city to evolve towards a high degree of sustainability.

Keywords: Energy trading · Energy negotiation · Smart cities · Sustainable cities · Multi-agent system

1 Introduction

In the last decades, the term smart city (SC) has been introduced to overcome the challenges caused by urbanization through cost-benefit decisions [3, 14]. A SC has complex systems which are created based on multiple connected subsystems *e.g.* Smart Grids (SGs) [1]. A SG provides a bi-directional communication data infrastructure to the agents of the energy system [2, 6]. Moreover, energy customers can play as proactive agents (producers) trading energy in urban distribution networks. There are several approaches for energy transaction -*e.g.* community-based, decentralised or distributed (P2P)- in distribution networks [12].

One of the main approaches that tend to apply MAS architecture is agreement technologies [9], where a series of participants (represented by individual agents) participate in a MAS to arrive at a solution that is optimal for all agents. To do this, they exchange a series of arguments in a negotiation process, defined in an ontology generally dependent on context to maximize the optimization of the negotiation result. Thus, different practical and theoretical approaches can be found in the literature that allows simulating the behavior of agents using agreement technologies [11].

Therefore, MASs are a suitable solution for large software systems whose participants are different entities with conflicting interests. Nevertheless, a specific development is necessary as there is no context-independent technology available that would be capable of adapting automatically to the problem. This paper presents a proposal in which a MAS has been developed using ROS. The developed system allows the interested entities to negotiate with each other, in this case, regarding energy efficiency.

The main novelty of this article lies in the application of a novel model of negotiation between ROS-based agents, which can be easily adapted to other contexts, unlike previous agent negotiation models. This negotiation model is based on an algorithm that can be scaled automatically, its computational cost is linear and it can be developed in any type of environment. The presented case study validates the feasibility of our proposal, as mentioned above, it focuses on taking information from a simulation on the consumption and generation of the energy of a SC and applying the algorithm to expose its advantages.

The next section of this article reviews different state-of-the-art researches related to our proposal. The third section describes the proposed system, detailing its architecture and modes of interaction. Then, the fourth section outlines the conducted case study and its results which have validated the functionality of the designed system. Finally, in Sect. 5, the conclusions and future lines of work are presented.

2 Related Works

Today, the importance of saving energy has led to research on multiple methods of energy optimization and changes in current trading models. It is crucial to design a system that would be highly efficient in the process of energy exchange between consumers to achieve the above objective. The following section reviews the current state of the art in this area.

2.1 Background on Smart City Energy Trading Platforms

One of the main objectives of Smart Cities is to become more sustainable through minimizing and optimizing the use of resources. Electricity is one of the most demanded resources because it is used in homes, factories, shopping centers, all these are found mainly in cities [7,8]. In this respect, one of the proposals that has aroused much interest in Europe is the process of buying and selling electricity (P2P model). This model proposes a paradigm shift in which the energy exchange between consumers helps them obtain energy at a lower price. This paradigm shift can be easily adapted thanks to the fact that the number of sensors deployed in cities continues to increase and therefore devices can connect even if the distance between them is large.

Several authors have put forward proposals that intend to make peer-to-peer a reality, nevertheless, these proposals have been designed to help producers maximize their profits. The higher income is mainly achieved through the elimination of the margin in the commercial agreement between the generator and the consumer; this is usually done by companies who commercialize and distribute energy. In this way, producers can offer energy at lower prices than those established by the market, favoring consumers, and still earning more money than if they sold it to the electricity market [13,16].

The industry is no stranger to these proposals and many companies have developed their platforms to enable a P2P exchange of energy between consumers. The Dutch company Vandebon¹ [16], allows consumers to buy energy directly from independent renewable energy producers (*e.g.* buying energy from farmers who own wind turbines or solar panels). Another platform of this kind is the British Open Utility platform² [13], which provides 30% benefits to its producers selling energy on the regulated market. GreenCom³ informs and encourages customers to consume electricity when green energy is available in the grid based on day-ahead forecasts. Community members earn credits for using green energy and shifting loads at times when more green electricity is available.

The Spanish startup Klenergy has developed the Pylon Network platform (⁴), which aims to eliminate the need for intermediaries in the purchase and sale of electricity from renewable sources. As a novelty in this P2P energy exchange platform, transactions are carried out using a new payment method, the Pylon-Coin

¹ Vandebon - <https://vandebron.nl/>.

² Open Utility platform - <https://www.openutility.com>.

³ GreenCom - <https://www.greencom-networks.com/>.

⁴ Pylon Network platform - <http://pylon-network.org/es/>.

(Blockchain technology). Another company that is using Blockchain for energy exchange in the P2P energy market is the LO3 energy company⁵. LO3 is carrying out a pilot project in Brooklyn, New York. The Brooklyn Microgrid⁶ allows users to buy and sell locally generated solar energy within their community. As shown in the study conducted by Mengelkamp et al. [10], the Brooklyn Microgrid is evaluated against the following seven components: i) Microgrid setup, ii) Grid connection, iii) Information system, iv) Market mechanism, v) Pricing mechanism, vi) Energy management trading system, and vii) Regulation.

However, some of the platforms found in the state of the art cannot be described as having genuine peer-to-peer processes because they only allow for unidirectional energy transmission. These platforms simply facilitate the contact between “unregulated” energy suppliers and their potential customers, rather than facilitating two-way energy traffic between network members.

For this reason, the platform proposed in this work allows any user with an intelligent meter to enter the market as a buyer or seller, configuring their energy purchase and sale tariffs *vis-à-vis* the suppliers of energy distributed through the platform. The intelligent meter allows its user to control how much energy to buy or sell at any given time. The platform will also provide knowledge on the profit that will come from investing in storage systems (if the price of energy fluctuates during the day, the user may save money by using electricity at the times when its price falls).

2.2 Robot Operating System (ROS)

ROS is a C++ and Python framework used to program MASs. Its name is very misleading [15] (since it is not an operating system). The purpose of ROS is to create reusable software capable of communicating with each other. Although this type of system has been used in robotic contexts for years and has proven to be quite useful, it has been seldom explored in other contexts. In this work, we have also explored new ways of using ROS. ROS has two main ways of communicating. The first way of communication is through a topic publisher/subscriber method where any node can publish through a topic and every subscriber node can receive the message. Every ROS topic has a unique name and an associated structure so messages do not need to be parsed. The second way of communication is a service method where a single node registers and other nodes can use the service (similar to a web service). Services also have a unique name and but their structure has two associated substructures, a request substructure, and a response substructure.

ROS might be seen as a centralized P2P communication system since communication is server-based (roscore). The server does not modify or alter the communication. There are several reasons for using ROS as the framework for SC development. The main reason for using ROS is that, sooner or later, robotic research and smart city research are going to merge. Since ROS has already

⁵ LO3 energy company - <https://lo3energy.com/>.

⁶ Brooklyn Microgrid - <http://brooklynmicrogrid.com/>.

become the main framework for robot development, it will be easier to merge SC technology with robot technology if we begin to develop SC using ROS.

3 System Overview

This section focuses on the technical details of the negotiation model proposed in this paper. It describes how the negotiation model has been accomplished. Moreover, it details the deployment of nodes using agents, to simulate different consumer profiles.

3.1 Architecture

Our proposal consists of different agents representing different city zones, each agent knows how much energy their zone has (or needs) but it does not know how much energy the other zones have (or need) so once the zone has (or needs) energy, the agent begins the negotiation process with the rest of the agents.

The multi-agent system will deploy and encourage nodes with surplus energy to act as energy vendors through the designed negotiation design. The multi-agent system has control nodes responsible for managing the balance between production and energy consumption. In case there are periods of surplus energy production, these control nodes communicate with the nodes of the production areas to cancel the production of energy and therefore no surplus energy is produced. Our simulation of the negotiation is semi-synchronous, this is because each node refreshes the measurement information at a similar time. However, a node may refresh its data and begins a negotiation using its new data before the other nodes have refreshed their data. Ideally, every node should be perfectly synchronized but this simulation method gives a more realistic scenario. In a non-simulated scenario, the data is supposed to be transmitted through post petitions in a web service from the set of sensors.

3.2 The Interaction of the Nodes

Each zone has an associated ROS node that represents the actions of the zone. Every node calculates the “power consumption net” as shown in Eq. 1.

$$Power_{net} = Power_{consumed} - Power_{generated} - Power_{gain.by.negotiation} \quad (1)$$

Each node publishes a ROS service named negotiation. As seen in Table 1 the service request structure is composed of an offer value that is a signed 64-bit float number, this represents the amount of the remaining “power consumption” (it is negative when representing generated power). The service response includes an acceptance value that represents how much from the offer can be traded and a status value that represents if the negotiation could be established. Each node generates an array that contains the name of the service that the other nodes provide avoiding adding their own service to the array.

Table 1. Negotiation structure

Request	
float64	offer
Response	
float64	acceptance
uint32	status
Constants	
uint32	<i>status_ok</i> = 0
uint32	<i>status_not_ok</i> = 1

Once a node receives new information about power generation and consumption, it calculates the power net. Then, the node sets the remaining power remaining to be equal to the power net. If the remaining power is different from 0, the node iterates over the service array to request the negotiation. Once a server is requested, the server checks if it is in the middle of another negotiation if that is the case, the server responds with a *status_non_ok* status. If the server is free to attend new negotiations, it will analyze if the offer and its remaining power have a different sign. If they do not have a different sign, it means that either both need power or both have extra power in which case the acceptance is a 0 and the status is *status_ok*. If the server finds that the signs are different, it calculates the highest possible amount of power that can be negotiated and returns it as the acceptance value. In that case, both the client and the server update their remaining power.

If the client receives a *status_non_ok* as a response, the client continues requesting negotiation to the remaining servers and once it finishes sending the requests to every server, it again asks the ones that returned *status_non_ok*. Notice that 2 nodes may be busy negotiating with the other node at the same time, leading to a death block. Because of this death block, the request process is repeated 10 times and then stopped assuming that the negotiation could not be established and hoping that the other node has not reached the 10th time, so the other node request can be accepted. The process of asking every node in the array to negotiate can be seen in Fig. 1 while the process of attending the petition is detailed in Fig. 2.

4 Case Study

A case study has been conducted to evaluate the system and demonstrate its superiority to previous, state-of-the-art energy trading models.

4.1 Simulation Set-Up

The mock-up performs simulations that allow comparing the Net Energy Consumption with Negotiation for the Net Energy Consumption without Negotiation. Each sensor generates its own data based on patterns of energy generation

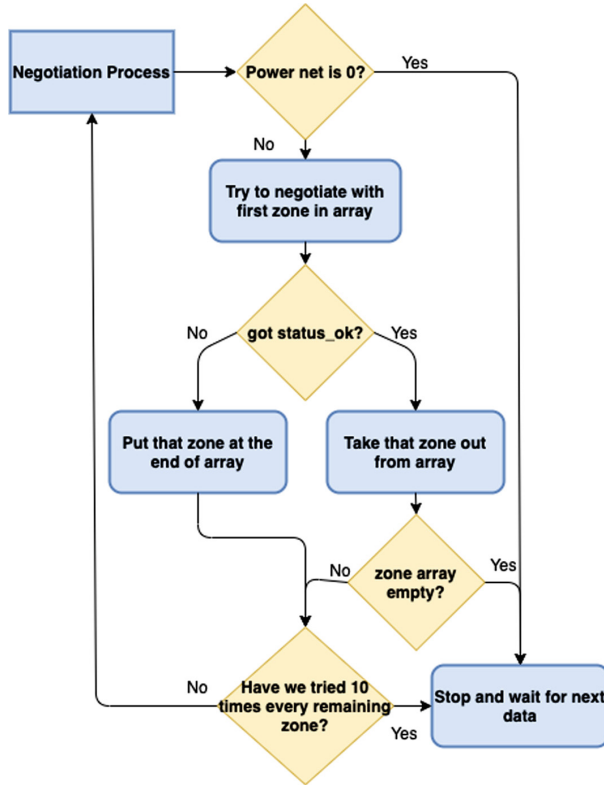


Fig. 1. Ask negotiation algorithm

and consumption empirically extracted after continuous observation of real areas similar to those described. The aim is to reliably reflect reality in a controlled environment that allows real-time measurements to be obtained and energy to be redistributed easily. For this study, the data was collected over a period of 60 hours (two and a half days), from the energy consumption simulations in urban zones. This process of extraction of patterns of generation and consumption is described in previous works [4,5]. We applied the algorithm to the generated data and could compare the amount of energy consumed in each zone with and without the negotiation process. In this case, we have simulated five zones of a smart city, as seen in Fig. 3.

Zone 1 was simulated assuming it is a residential, urban zone so it has several apartment buildings. Zone 2 is a residential zone in the suburbs whose consumption patterns are similar to zone 1. Zone 3 is a residential zone in the suburbs. Zone 4 is a business/commercial zone with some industrial buildings and skyscrapers in it. Zone 5 is an energy generation zone, here we have several wind generators, solar panels, and steam power plants.

4.2 Results

In Table 2, we can compare how much Net Power Consumed when negotiation took place and when it did not take place. We can also compare the amount of wasted power when negotiation took place and when it did not in Table 3. In this simulation we used data from a non-auto-sustainable city, because of this, the city must import from outside to meet its needs. This is not really important since the main goal of these simulations is to test the proposed algorithms. In Table 2 we can see how the power consumption diminish as each consuming zone's necessity is met by the generating zones. In the same way, Table 3 show

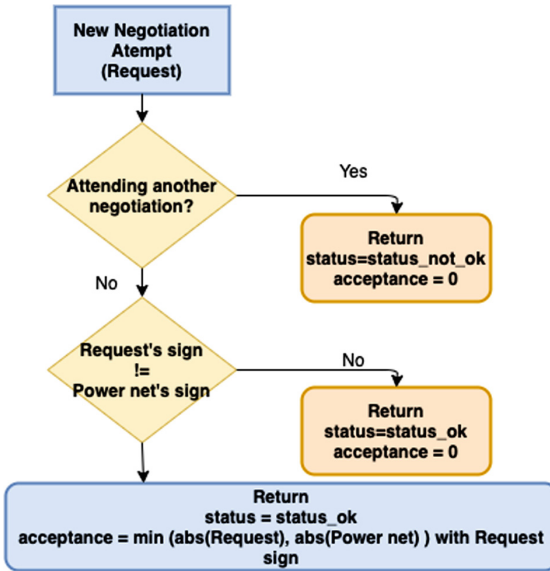


Fig. 2. Attending negotiation algorithm



Fig. 3. Mock up representation of the simulated smart city.

how each generating zone’s excess is being used by the consuming zones avoiding the excess to be wasted.

Table 2. The comparison of the amount of power consumed when the negotiating process took place and when it did not and the non-negotiating process.

	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Total
P.N. Con. without Neg. (kWh)	41,226.38	997.38	899.30	69,396.35	0	112,519.41
P.N. Con. with Neg. (kWh)	18,956.50	401.07	462.73	38,097.13	0	57,917.43
P. Con. saved (%)	54	60	49	45	0	49

Table 3. Power generation comparison between the negotiating process and the non-negotiating process.

	Zone 1	Zone 2	Zone 3	Zone 4	Zone 5	Total
P. Gen. wasted without Neg. (kWh)	0	0	5.60	0	57,033.42	57,039.02
P. Gen. wasted with Neg. (kWh)	0	0	0	0	2,437.01	2,437.01
P. Gen. saved from wasting (%)	0	0	100	0	95	96

5 Conclusions

This paper has presented a novel agent-based approach to energy negotiations between consumers and producers in a smart city. The use of a multi-agent system in this energy negotiation problem has been fundamental; by simulating different behaviors it has allowed us to find out about the interaction of users with our negotiation model. Real data have been used to simulate the functioning of the developed negotiation model, making it possible to model different roles. In addition, the system has mechanisms that allow it to extract weather forecast data for the next few days; in this way, it can simulate the negotiation in real-time. This is a key factor as it allows us to know the level of energy generated by the wind farm or by the solar panels, creating a highly precise simulation.

The case study has been carried out to evaluate our negotiation model, showing that the proposed solution makes a city sustainable, making it possible to supply the generated energy to the areas that demand it. The algorithm’s design enables the implementation of the model in any type of city, regardless of the number of zones involved and their role (consumer or producer). The algorithm developed for the energy trade between producers and consumers reduces the customers’ dependence on the energy supplied by trading companies by 49%. Thanks to the independence that our model gives to intermediaries, the price of energy reduces, benefiting both consumers and small producers (consumers have produced extra energy and can put it on sale in the market).

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Integration of IoT Technologies in the Maritime Industry

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Abstract. The transport and logistics sector, by its nature, requires an intensive and efficient exchange of data for effective management and decision-making. The maritime sector requires a special treatment due to the complex and heterogeneous environment. The rapid adoption of IoT technologies in the naval industry will facilitate the effective management of primary services, including vessel tracking, emissions control, predictive maintenance, safety and welfare. This article describes IoT solutions adapted to the current needs of the sector. The proposals are aligned with the SMARTSEA project, which aims to develop an interactive MSc course on Maritime and Surveyor ICT/IoT systems, helping complete the market void in technical and maintenance specialists generated by the prompt expansion of the Smart Maritime & Surveying industry.

Keywords: IoT · Smart maritime & surveying industry · Edge computing

1 Introduction

The Internet of Things (IoT) is a network of physical devices embedded with electronics, sensors and actuators, that enables inter-connectivity among devices and data exchange. Over the past five years, this new technology has grown rapidly [1], finding application in many sectors, such as energy, healthcare, transportation, industrial and security [2]. IoT is considered one of the leading gateway technologies to digital transformation. It is expected that by 2025, IoT reaches a potential market impact of USD 11.1 trillion [3].

Ships are crucial for the global transportation system, as 85% of the stock is carried by sea [4, 5]. In the maritime industry, the application of IoT technology enables shipping companies to connect their vessels in one platform, allowing data sharing with the entire corporate ecosystem, that stakeholders can exploit.

IoT systems are set to:

- improve the efficiency of the sector's activities,
- improve the transparency of the companies and institutions,
- increase the safety and well-being of workers on-board (e.g. telemedicine is very valuable in the case of accidents in remote locations),
- reduce inefficiencies, risks and costs (e.g. by reducing fuel consumption, optimising engine use or maintenance systems),
- minimise the environmental impact.

For the transportation and logistics sectors, which have always relied on exchanging decision-making data, the digitalization process has been clear, placing them ahead in the transition. However, the maritime industry, heavily anchored in traditional methodologies, is facing several obstacles. It operates in some of the most remote areas of the planet, where M2M interaction is complicated. Furthermore, the standardisation of IoT technologies and devices specially designed for this industry are relatively new.

Many institutions from the public and private sectors are making great efforts to facilitate its transition towards digitalization. The European Union, through its Horizon 2020, will allocate up to EUR 6.3 billion for research and development of ICT and IoT technologies [6, 7]. Sánchez-González et al. (2019) [8] compiled published work about the digitalization of the maritime industry focusing on eight domains: autonomous vehicles and robotics; artificial intelligence; big data; virtual reality, augmented and mixed reality; internet of things; the cloud and edge computing; digital security; and 3D printing and additive engineering. They conclude that the maritime sector is moving towards digitalisation at different rates for the domains aforementioned. Big data and AI have been widely studied, whereas robotics and IoT (which may have a significant impact on the sector) are underdeveloped.

In the private sector, large shipping industries are already investing in IoT techniques to optimize transparency, safety and reduce costs. Several SMEs offer IoT services focusing on ship tracking, emission control, predictive maintenance, safety and welfare. Here we highlight six services closely related to this study. They perform an analysis of vessel data supporting ship owners, crews, and agencies (e.g. coast guards, port authorities).

- *IoCurrents* [9] comprises an on-board mini-computer that collects and analyses data locally, a DataHub and a remote analytics cloud platform.
- *Green Sea Guard* [10] provides equipment for monitoring the ships' emissions, fuel evaluation and engine diagnostics.
- *Augury* [11] installs sensors that measure vibration, sound and temperature for the monitoring of machinery.
- *Parsyl* [12] installs sensors that track temperature, humidity, light, impact and GPS position of an individual pallet or package.
- *ZS Wellness* [13] has developed a health tracking system designed specifically for the maritime environment. The data collected can be used to create personalised wellness plans for individual crew members.

- *DanelecConnect* [14] is a ship-2-shore data automation solution which ensures timely, cost effective ship management.

In the transition towards digitalization, regulatory compliance is a key driver of adoption in the short term, but also a great chance for shipowners to see financial returns. There is a range of regulatory actions that govern the legal framework regarding ships and their operation [15]. One of the main jurisdictional rules is “The 1982 United Nations Conventions on the Law of the Sea (UNCLOS)”, which establishes rights and obligations concerning the ships [16]. The International Maritime Organisation (IMO) is the United Nations agency responsible for the safety and security of shipping and the prevention of atmospheric and marine pollution [17]. IMO’s most important treaties include: (i) The 1974 International Convention for the Safety of life at Sea (SOLAS) [18]; (ii) The 1973 International Convention for the Prevention of Pollution from Ships (MARPOL) [19]; and (iii) The 1972 Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) [20].

To operate ships efficiently and safely, access data from shipboard machinery and equipment is required. Navigational instruments normally use the IEC 61162 series of standards when transferring data, but access to other shipboard machinery and systems to collect data has not yet been standardised. Two relatively new ISO standards has been developed for this purpose. The ISO 19848:2018 establishes unified rules for developing machine and human identifiers and data structures for shipboard machinery and equipment; and the ISO 19847:2018 specifies requirements for performance, function, service and safety for the shipboard data server that stores data from shipboard machinery and equipment, and sends it off the ship.

“The Surveying & MARiTime internet of thingS EducAtion (SMARTSEA)” [21] is a project funded by the EU Erasmus+ Programme. It aims to develop an interactive MSc course on Maritime and Surveyor ICT/IoT systems, helping complete the market void in technical and maintenance specialists generated by the prompt expansion of the Smart Maritime & Surveying industry. Within the SMARTSEA framework, the remainder of this paper is structured as follow. Section 2 presents an IoT acquisition platform based on edge computing, including a few proposals for data management applied strategy along with the data-driven architecture. Section 3 defines an Augmented Reality design which provides experimental tools for training activities. Lastly, Sect. 4 summarises and concludes the paper.

2 Smart Maritime Platform Based on Edge Computing

The digitalization process of the maritime industry will allow Big data to support logistics management, decision making, incident management, alarms and process optimization. This section proposes a data acquisition platform based on edge computing.

2.1 Architecture for Data Collection and Management

The proposed platform comprises the following elements (Fig. 1):

- *Recovery node*: the collection and transmission of data from the IoT sensors of the underlying equipment and the environmental data are networked together, generally by a wireless protocol.
- *Edge gateway*: the edge gateway has the function of the data collector; providing computing, storage, network and other infrastructure resources. This collection enables making system decisions based on the results of the data analysis.
- *Cloud center*: employed for big data analysis and data mining, it is the key where models are performed with Artificial Intelligence (AI) methodologies from the data uploaded by the edge equipment.

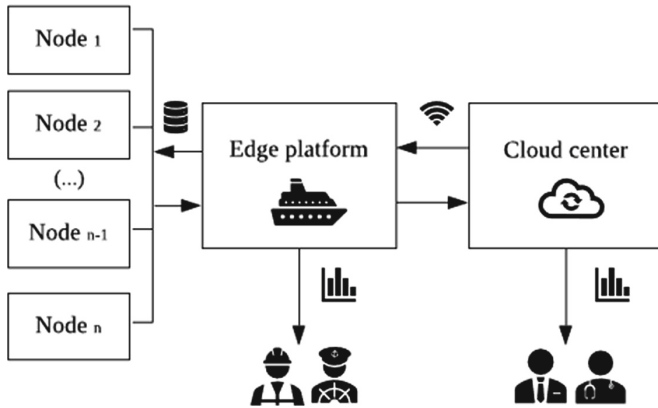


Fig. 1. An edge computing architecture for IoT data acquisition.

The IoT data collection is obtained based on the proposed edge computing platform. The system completes the analysis and storage functions of collected data on the edge side. It can also be combined with the cloud center to support more complex requirements. Adopting edge and central clouds to consider both timeliness and scale can be applied in a variety of industrial scenarios, so it is feasible in practice.

2.2 On-Board Edge Platform

The edge-computing platform will be anchored in the ship for the management and monitoring of the information collected by the sensors. Through the modules of the platform, the control of the different components will be performed by developing a layered architecture (Fig. 2). This architecture allows an incremental development of application and/or service management. Here, we proposed the following: (1) on-board management (e-health, load monitoring); (2) off-board management (eco-report, collision detection); (3) centralized incident management protocols; (4) Augmented Reality installation and management.

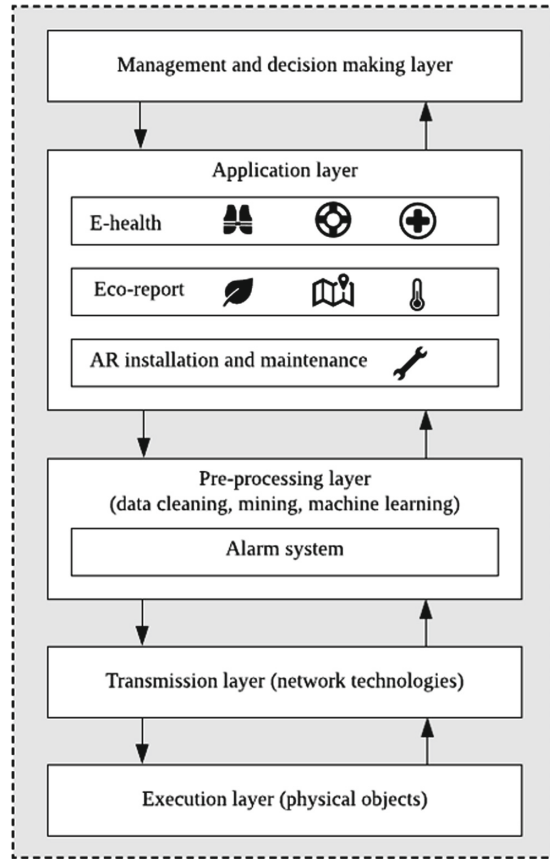


Fig. 2. Application levels for the architecture proposed. Adapted from Antão *et al.* [22].

2.3 Incident Management Models

Among the application levels mentioned in the previous section, the one proposed as incident detection systems is particularly revealing to handle variations in the acquisition of the data. Linear regression is one of the simplest algorithms employed in machine learning, and the one used in this paper.

Assuming a linear relationship between two physical quantities, the functional form can be written as $y_i = \alpha_0 x + \alpha_1 + \epsilon$ (the linear regression equation), where the constant α_0 and α_1 are the linear regression coefficients. Using this algorithm, we predict the scores on one variable from the scores on a second variable. However, the actual collected data does not always strictly satisfy the linear characteristics. The data points cannot be accurately placed on the straight line corresponding to the formula when drawing Fig. 3.

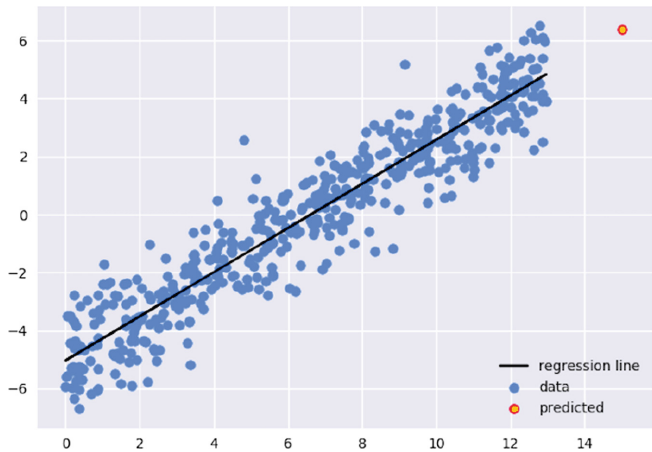


Fig. 3. Linear fitting for data incident management.

The difference between the actual value and the fitted value of y is defined as the fitting error or residual Δy_i , and the sum of squares of all fitting errors is added to obtain the sum of squares of errors. The best fitting line is also the line that minimizes the error sum of squares. Therefore, we can use the principle of finding the extreme to convert the problem of finding the best fit straight line into the problem of finding the smallest square of the error. A multiple regression analysis helped identify the significant independent variables.

A model with several independent variables could be reasonably accurate in predicting changes in the monitored data models in continuous value ranges; assuming that i -th the calibration data and the corresponding value on the fitted line is a quite simple method. We intended to implement the edge computing architecture with a

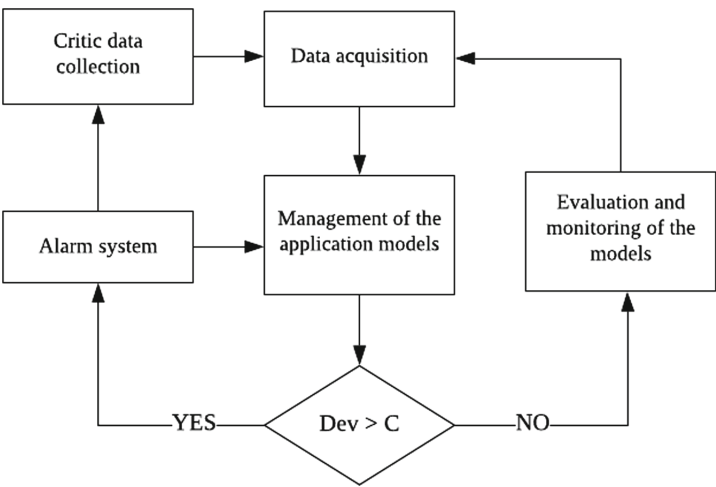


Fig. 4. Protocol for data acquisition process.

linear regression-based offline training displayed in the Fig. 4. IoT based system provides the availability and installation of an alert system interconnected easily.

3 Augmented Reality for Installation and Maintenance

In recent years, technologies that enhance or recreate real-world environments are increasingly influencing the world of the production industry. With Virtual Reality (VR), Augmented Reality (AR) and the combination of both, Mixed Reality (MR), it is possible to simulate almost any process carried out in the physical world [23]. The AR market is estimated to grow from USD 10.7 billion in 2019 to USD 72.7 billion by 2024 [24]. These data support the positive expectations about the future of these technologies. Training technicians to acquire new maintenance and assembly skills is crucial for various industries. Because these tasks can be very complicated, training technicians to efficiently perform new skills is challenging [25]. This sort of formation can be supported by AR, a powerful industrial training technology that directly links instructions on how to perform the service tasks to the machine parts that require processing. Because of the increasing complexity of maintenance duties, it is not enough to instruct the technicians in task execution. Instead, technicians must be trained in the underlying skills, sensorimotor and cognitive, that are necessary for the efficient acquisition and performance of new maintenance operations. Complex assembly and maintenance tasks in industrial environments are great domains for AR applications. The need for proper training and the access to large amounts of documentation are conditions making the use of AR techniques most promising. The basic idea of AR is to bring additional information as seamlessly as possible into the view of a user [26]. These facts illustrate the need for efficient training systems for maintenance and assembly skills that accelerate the acquisition of new maintenance procedures. Furthermore, these systems should improve the adjustment of the training process for new scenarios and enable the reuse of worthwhile existing training material.

In this context, we proposed to develop a novel concept and platform for multi-modal AR-based training of maintenance and assembly skills, which includes sub-skills training and the evaluation of the system. Because procedural skills are considered the most critical skills for maintenance and assembly operations, we focus on these skills and the appropriate methods for improving them. By providing spatially registered information on the task directly in the user's field of view, the system can guide the user through unfamiliar tasks and visualize information directly in the spatial context where it is relevant. In the framework of the SMARTSEA project, this tool will help the students learning how to install IoT devices, besides configuring the protocols required to link them to the node in the architecture presented.

4 Conclusions

More and more companies are offering IoT services in the maritime industry focusing on ship tracking, emission control, predictive maintenance, safety and welfare. Still, there is not a specialized university education that prepares technologists in this field.

There is a necessity to link IoT technologies to the maritime sector with the inclusion of edge computing architectures and related technologies. IoT technologies provide a live connection between the sea and the coast that stakeholders can exploit in decision-making while improving the efficiency of the sector's activities. The idea of developing specialized training in the integration of IoT technologies in the maritime industry arises from the need to digitalize this sector, one of the most anchored in traditional methods.

Acknowledgments. This research has been supported by the project “Surveying & MARiTime internet of thingS EducAtion (SMARTSEA)”, Reference: 612198-EPP-1-2019-1-ES-EPPKA2-KA, financed by the European Commission (Erasmus+: Higher Education - International Capacity Building).

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AIRUV: A Remotely Operated Underwater Vehicle with Artificial Intelligence Perspectives

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Abstract. Billions of physical devices and vehicles are nowadays enriched not only with electronics and sensors but also with intelligent software that enables more possibilities than just data collection and exchange. Moreover, over the last years, the Internet of Things (IoT) initiates a new era where the world will change deeply and decisively in many ways. In this context, both industry and research community attempts to merge engineering and artificial intelligence. This direction considers objects, vehicles and devices, as the driving force for autonomous IoT that enables intelligent management for crucial issues such as maritime monitoring. To this end, this article proposes a novel approach that combines theoretical and scientific knowledge, related to IoT and Artificial Intelligence to real-world needs as they are reported by an engineering perspective. The paper presents the first steps towards a model that will lead to an integrated, easy-to-replicate, water monitoring and information system in sea harbors.

Keywords: Underwater Vehicle · Artificial Intelligence · Internet of Things · Water monitoring system

1 Introduction

Over the last few years, the Internet of Things seems to be on the top of emerging IT technologies. Its main innovation consists of creating a world where everyone and everything, called Things, will be connected and able to communicate in a network. These Things, devices, services or even humans, will have intelligence and decision-making power that, among others, will let them cooperate to solve complex cases. There are already plenty of existing implemented IoT cases in a considerable domain range, such as smart living and healthcare [1–3]. Yet, there is another area that is expected to attract more attention in the near future; maritime monitoring.

Maritime environment of a harbor, especially if it is located in a relatively closed gulf, is environmentally burdened by numerous factors. All kinds of ships passing by on a daily basis like commercial ships, tankers, cargo ships, yachts, touristic activity ships, fishing ships etc., may be considered as significant pollutants of the harbor sea environment. Similarly, adjacent municipalities and industries are responsible of constant pollution (waste, dirt) as well as discharges from creeks, torrents and rivers. As a

special case, oil industry premises usually supply tankers through pipelines ending inside the harbor gulf, and where connections between supplied ships and the industry takes place. Pollution caused by all the aforementioned factors may significantly affect not only the sea water of the harbor but the entire ecosystem of the area as well, thus leading to degradation of the corresponding inshore environment and rendering constant and reliable monitoring of the harbor sea waters extremely important.

In this context, the proposed approach aims at combining a light-weighted ROUV (Remotely Operated Underwater Vehicle) equipped with a complete set of sensors in order to collect data and samples with a novel smart monitoring platform that will enable autonomous decision making/suggestion. Actually, the approach combines engineering (ROUV device) with Artificial Intelligence thought the smart monitoring IT system, realizing the vision of the IoT. The development of this informational monitoring platform will present, among others, environmental information about the sea waters as well as the state of the ROUV device itself. Access to them will be also provided, with gradation to various stakeholders and the public. Moreover, the data can be directly available to harbor or other managing entities.

The rest of the paper is organized as follows: The next section presents the Remotely Operated Underwater Vehicles with a discussion on the ROUV, the prototype device that was developed and used for the purposed of this research. Section 3 presents AIRUV monitoring platform and its contribution, demonstrating the added value of the approach. Section 4 discusses related work, and Sect. 5 concludes with final remarks and directions for future work.

2 Related Work

Many works have made implicit or explicit assumptions about the need of remotely operated underwater vehicles as far as it concerns maritime issues mainly focusing on navigation. Yet, there is still a lack regarding the combination of R.O.U.Vs and Artificial Intelligence in order to provide a complete, quite general-purpose monitoring system. For instance, authors in [14] survey the problem of navigation for autonomous underwater vehicles (AUVs), since navigation is critical for the safety and effectiveness of AUV missions but they do not take into account monitoring IT solutions like our approach. For the purposes of that study, the AUV systems that were described, regarding the navigation sensors and algorithms, included the HAUV [15], Seabed [16], Iver2 [17], ABE [18], and Nereus [19], AUVs and the Spray glider [20]. In [21], the authors suggest TurtleCam, a “smart” autonomous underwater vehicle for investigating behaviors and habitats of sea turtles. They used the TurtleCam system to simultaneously measure and observe leatherback turtle habitat and behavior, but their system is focused only on turtles.

3 Remotely Operated Vehicles for Underwater Monitoring

Although the necessity for environmental monitoring of sea water conditions in harbors has been recognized, usually most of the taken actions suffer from important drawbacks, ultimately rendering them inefficient. Particularly, from time to time, several management entities, local authorities and municipalities owing part of the harbors proceed to measurements of the sea water quality by means of expensive, sank in the sea environment, instruments. These gathered data in most cases are transmitted through telematics solutions to a base station for processing, aiming in facilitating initially pollution control and afterwards the decision making processes. High installation and operational cost, frequent cleaning requirements of the measuring stations, significant measurements inaccuracies due to high sea currents, lack of real time data transmission due to inefficient telematics services, are only few of the problems encountered that, explicitly or implicitly, make existing monitoring solutions non-operational. Therefore, scattered, standalone environmental measurement stations are installed in various places of the harbor gulfs taking static and unreliable measurements in general, being unreliable for decision making and practically offering poor or not at all monitoring of the harbors' maritime environment. The proposed approach aims at overcoming the lack of sufficient and efficient monitoring and therefore information of harbors' maritime environment, by achieving updated, real-time spatial environmental data. In the future, these data will be stored in a new data storing, processing and sharing platform that will allow stakeholders and the public to be informed through accessible web tools. A potential measuring system example based on Remotely Operated Underwater Vehicles is depicted below (see Fig. 1).

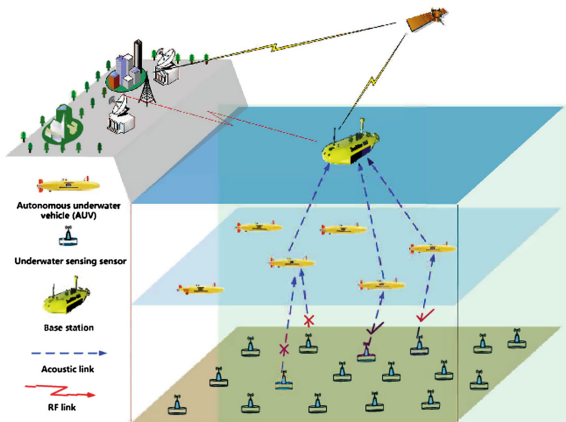


Fig. 1. Measuring system example [25]

Actually, Remotely Operated Underwater Vehicles (R.O.U.Vs) are commonly used for underwater routine or emergency monitoring [8–12]. Mainly they are used out of limits of autonomous diving. In most cases they are connected with surface through

cables in order to transmit data and/or image for a range for monitoring cases. Usually, such a vehicle is operated by a driver on sea surface. Over the last years, due to the growth of IoT, ROUVs are getting more and more attention. One of their main advantages is their operation ability in large depth and for almost unlimited times. The main applications of R.O.U.V.s include among others study of sea shore zone, oceanography, observation and support of diving operations, water cultivations, environmental monitoring including oil leakage, underwater archeology, harbor works, shipwrecks, surveillance of underwater cables and conductors, underwater monitoring – maintenance – reconstruction for ships, control and certification of underwater constructions, monitoring and maintenance of bridges, barriers, underwater video productions, education, photography as well as rescue operations support.

3.1 ROUV: The Remotely Operated Underwater Vehicle

The proposed approach, as far as, it concerns the measuring equipment for surveillance and maritime surveying proposes the use of a novel remotely operated underwater vehicle, called ROUV (see Fig. 2). This vehicle has been developed at the laboratory of Electric Vehicles and Automotive Electronics of the Department of Industrial Engineering and Management at the International Hellenic University. It is a small underwater vehicle, $457 \times 338 \times 254 \text{ mm}^3$ dimensions, weighting 10 kg on air. The ROUV is able to operate continuously for about 4 h as it is being powered from his on board Li-ion battery. In order to develop the latter, the Samsung INR18650-30Q Li-ion 18650 cells [13] have been selected, having as nominal specifications 3.6 V voltage, 3000 mAh capacity and 15 A maximum continuous discharge rate. In order for the overall battery to provide 14.8 V voltage and about 18 Ah capacity, 24 cells are connected in a 4S6P structure. This kind of structure having overall energy of 266.4 Wh, is able to provide 90 A nominal discharge current and 132 A maximum burst current to the ROUV electrical consumptions, values that are considered adequate for the operation of the ROUV. The battery pack's temperature is monitored through a simple NTC thermistor. The battery pack weighs about 1.15 kg while its dimensions are $75 \times 141 \times 55 \text{ mm}^3$, values adequate to allow its integration to a small ROUV.

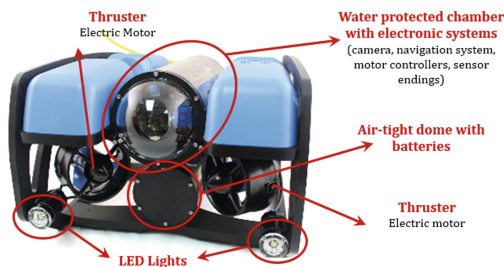


Fig. 2. The ROUV prototype

It supports a wired connection and operator control on surface (sea shore, boat, ship) even in large depths (100 m). A tether cable is used for the wired data transfer.

It also supports a variety of measurements with real time data storing and processing e.g. geographic positioning with GPS on surface and very small depths, water temperature (± 0.1 °C) for the thermocline control, pressure (± 2 mm $\dot{h}$ ± 0.2 mbar) for depth measurements. Moreover, it supports underwater photography and video taking (1080p HD video) recording with the additional aid of LED lighting (see Fig. 3). Additionally, it provides an open source software for control and navigation as well as a great expandability to other important features (various types of sensors, mechanical arm, underwater GPS). As for the future, the implementation team aims at using a robotic arm or a special suction device for the collection of underwater material.

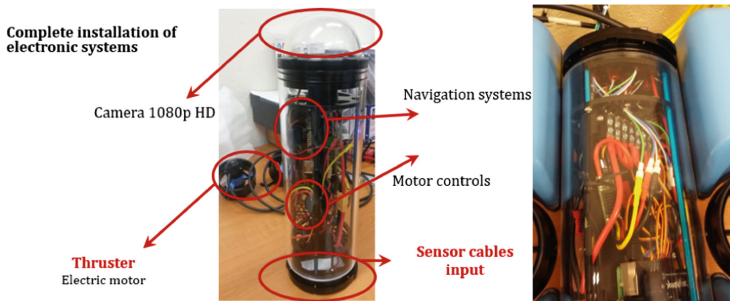


Fig. 3. The ROUV prototype electronic systems

Actually, the ROUV is equipped with a sufficient set of sensors, while the list will be extended, including diluted oxygen, ammonia/nitrate/nitrate salts, salinity/conductivity sensors that allow obtaining abundant and precise environmental data from sea waters at any time. The aim is to add a complete set of sensors for measurement of water quality. Specifically, three types of sensors are considered crucial for measuring purposes, namely diluted oxygen that is able to reveal the anoxic conditions (lack of oxygen), ammonia/nitric/nitrous salts that is able to reveal the extent of pollution and salinity/conductivity that is able to reveal changes in the composition of water. Moreover, in the AUVs [4–7], like ROUV, whose envisioned modes of operation include teleoperation and semi-autonomy, the addition of sonars or DVLs (Doppler Velocity Logs) in order to detect obstacles are also planned. Also, the ROUV will be enriched with an accelerometer (3-axis) for the estimation of the underwater movement of the vehicle, a special automated sucking device for the collection of water samples from specific depth or the collection of micro-material from the sea bottom, an underwater GPS addition for its exact underwater position and a wireless motion control and data transfer without the tether.

4 AIRUV Monitoring System

In order to proceed with an AI monitoring system that will use the advantages of the aforementioned ROUV device a few additions were proven necessary. First of all, a reliable communication system between the Measuring Station and a Ground Station is needed. This should consist of two links. The first, between the Measuring System and the Surface Boat which being an underwater link, can only be achieved by means of sound waves that will transfer low rate data recorded from the Measuring Station. Light technology underwater communication has also been referred, however it cannot be considered mature enough yet [26]. The second link established, based on radio waves, will allow data transfer from the Surface Boats to the Ground Station in a way that communication will be uninterruptable (direct transmission and GPRS). Moreover, for data processing purposes, we propose the development of an IoT analytics platform where the data collected from the Measuring Stations will be transmitted and stored. Our intention is the system to follow the lambda architecture principles with a batch, speed and serving layers.

This way, the data could be accessed with gradation to various stakeholders and the public. Moreover, the platform itself will be able to process data by conducting the appropriate reasoning, using AI techniques, which allow the system to proceed with alerts, suggestion or action upon the ROUV device and the state of harbor waters. To this end, the proposed approach will encompass at the next stage a WebGIS interface for managing and presenting the environmental data (maps, 2-D and 3-D graphs, matrices), including tools for statistical analysis, a public communication forum, a download section for researchers to be able to process the data and focused stakeholders participatory forums for discussion and exchange of opinions, an international stakeholders forum facilitating the cross-border exchange of knowledge and practices, an educational section (wiki-vironment) that will provide useful knowledge about the maritime environment of a harbor in an effort to increase public awareness.

However, the paper discusses the first steps towards this solution, focusing on the ROUV device and the underlying monitoring, measuring and decision making, methods used by the AIRUV (AI) platform. It is important to understand that the state of harbors, just like or even more than oceans, faces complex stressors and challenges. These include deterioration of water and water quality, the emergence and increased intensity of cases such as harmful algal blooms a microbiological and chemical contamination resulting from water usage and coastal runoff [23]. Hence, there is a need for system that will assist authorities to provide potential strategies that might mitigate this problem.

4.1 Measuring the Composition of Seawater

First of all, the intention of the proposed approach is to measure, using the ROUV, the composition of seawater in order to detect alarming or dangerous cases on the quality of (harbor) water. On average, chemical and physical compositions of seawater are quite similar in the oceans. Of course, in enclosed seas and areas closer to shore there are usually deviations in average water quality from those in open sea conditions. These variations are caused either by natural, usually cyclic, annual processes

involving marine growth and death or due to substantial discharge of nutrients or fertilizers at highly populated areas or cities, where adjacent coastal water quality may be lower than that in open sea.

Dissolved Oxygen

Dissolved Oxygen in water is necessary for (aerobic) life in water. Air is dissolved at the water surface and then transported into the water mass by turbulence and/or currents. Dissolved oxygen is usually reported in milligrams per liter (mg/L) or as a percent of air saturation. Based on the Henry's law, the dissolved oxygen is related to temperature and pressure.

$$DO = T \times p \quad (1)$$

where T is the water temperature and p is the partial pressure of oxygen at the water surface. The water temperature (T) is typically about 3 months behind the seasons. Its annual variation is responsible for cyclic changes in many parameters of seawater, such as diluted oxygen (DO). Moreover, it is also responsible in the variation in biological components, such as algae and bacteria [22]. Actually, the saturation concentration depends on temperature, pressure, and dissolved salts with a reversely proportional relationship.

In order to detect anoxic conditions, we have to compare the measuring findings with uniform conditions over a cross section. Hence, a simple oxygen balance for accepted conditions is given by:

$$Vol \frac{d}{dt}(DO) = kA(DO_m - DO) \quad (2)$$

where k is a dissociation rate coefficient. Usually, at harbors, there is a municipal and industrial discharge of organic and inorganic matter. This case is characterized through $BDO(t)$ or $CDO(t)$, represents (in mg O_2 /liter H_2O) biological and chemical oxygen demand, respectively. Hence, the degradation equation is represented as a first-order reaction:

$$\frac{d}{dt}(BDO(t)) = -k \cdot BDO(t) \quad (3)$$

Fluxes of Chemicals

Measuring and quantifying the fluxes of chemicals is important in order to study the extent of the water pollution at a harbor. To this end, ammonia, nitric, such as NH_4^+ , N_2O and NO_3^- , and key carbon species, such as CO_2 and CH_4 , are studied. Additionally, total dissolved inorganic carbon (CT), total alkalinity (TA), and carbonate ion concentration (CO_3^{2-}). In order to estimate pollution risk, we propose a form for

estimating the expected interval between exceedances of the value (x) of each measured chemicals. The relationship is given by the following equation:

$$T(x) = \lambda^{-1} / (1 - F(x)) \quad (4)$$

where λ is the sampling frequency and $F(x)$ is a dimensionless function related to $p(x)$, the probability density function for the variable x , given by

$$F(x) = \int_{-\infty}^x p(x) dx \quad (5)$$

The probability density function $p(x)$ of the continuous variable x has the following properties:

$$p(x) \geq 0, \quad \int_{-\infty}^{\infty} p(x) dx = 1 \quad \int_a^b p(x) dx = P(a < x < b), \quad (6)$$

where $P(a < x < b)$ is the probability of x falling between the values of a and b .

Additionally, in order to provide an estimation about future events related to water pollution we adopt the empirical simulation technique of [24]. It is based on the assumption that future events would be based on past appearances. It uses the rank of the ordered values m and the total number of observations N :

$$F(x) = m / (N + 1) \quad (7)$$

4.2 Monitoring the ROUV: Battery State of Charge

Of course, besides all sensor signals and data regularly monitored through the AI system, battery State of Charge (SoC) is also monitored as it is the most crucial parameter for the operation of the overall system, ROUV and sensing platform. Battery SoC is important as it is not only an indication of the remaining energy in the battery but also a parameter used for the estimation of the remaining operation time of the ROUV. Having in mind that the ROUV operates underwater and in a distance from its base, it is crucial not to run out of energy (shut down) in a large depth. The SoC of a battery at a time instant t is estimated by the following equation:

$$SoC(t) = SoC(t_0) - \frac{100}{Q_n} \int_{t_0}^t i(t) dt \quad (8)$$

where t_0 is the origin of time (start of operation), $i(t)$ is the discharging current and Q_n is the nominal capacity of the battery (18Ah in our case). The final SoC is calculated as a percentage of Q_n .

5 Conclusions

The paper argued that the recent growth of IoT enables new ways of transactions and management, which is vital but rather complicated. In this context, both the industry and research community attempt to merge engineering and artificial intelligence in order to provide novel approaches. The proposed approach focuses on maritime environments. It combines engineering (ROUV device) with Artificial Intelligence thought the smart monitoring IT system, realizing the vision of the IoT. To this end, a light-weighted ROUV (Remotely Operated Underwater Vehicle) equipped with a complete set of sensors is used in order to collect data and samples from sea harbors that can be processed and presented in a novel smart monitoring platform, enabling autonomous decision making/suggestion. As for future directions, our main interest is to provide even more capabilities for the ROUV by adding, among others, more sensors. On the other hand, we plan to focus on the development of the AIRUV IT platform in order to provide an easy-to-use digital interface that will allow interested parties to be informed or supported on decision making.

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Assessing the Current State of a Shipwreck Using an Autonomous Marine Robot: Szent Istvan Case Study

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Abstract. S.M.S. Szent Istvan, the only ship belonging to the Hungarian monarchy, met her end on June 10th 1918 shortly before dawn. It was sunk by Italian torpedo boats. On the 101st anniversary of this event the shipwreck was recorded for the first time by a multibeam sonar-mounted autonomous surface vehicle. The shipwreck has already suffered irreversible degradation of her steel and iron hull. Thus, the main objective of the bathymetric surveys was to assess the current state of the shipwreck and to set up a foundation which future monitoring operations could be built upon and compared with.

1 Introduction

Methods for recording and documenting underwater cultural heritage (UCH) sites have evolved significantly in the last two decades. The combined use of optical and acoustic technologies enables the provision of quality digital 3D reconstruction of large and complex underwater scenarios [1, 2]. These technologies in a non-intrusive manner, create the opportunity to study the UCH in the laboratories onshore [3]. Resulting digital reconstructions are often accepted for archaeological purposes, and in particular for documentation and monitoring activities, [4, 5].

One such UCH site is the shipwreck of SMS Szent István located approx. 8 nm off the coast of Premuda island in Croatia. SMS Szent István was an Austro-Hungarian battleship of the Tegetthoff class, constructed in Rijeka and Pula and completed in 1914. It was the only Austro-Hungarian ship to serve the Hungarian part of the monarchy. Its blueprints are given in Fig. 1a for reference in the remainder of this paper. It was named after the first Hungarian Christian king, Szent István (Stephen I. Saint) [6].

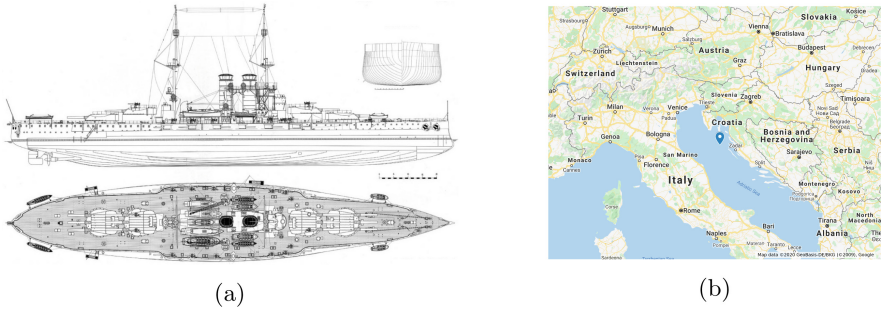
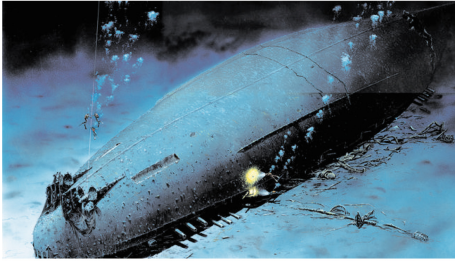


Fig. 1. (a) Blueprint of the cross section, side and top view of the SMS Szent Istvan, [6]. (b) Location of the UCH site SMS Szent Istvan shipwreck.

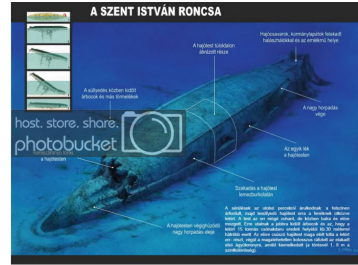
Shortly before dawn on June 10th 1918, the battleships Tegetthoff and Szent István were unexpectedly approached by the Italian torpedo boats 8 nm off Premuda Island, the location shown in Fig. 1b. The Italian torpedo boats MAS -15 and MAS -21 were on a mine search mission that night. MAS -21 attacks Tegetthoff, but torpedoes miss their target. Szent István received two direct torpedo hits from the boat MAS -15 at 3:31 in the morning. Italian boats manage to escape. This latest Austro-Hungarian ship turned and sank at 6:12 a.m., after three hours of agony, taking with it the lives of 89 sailors, mostly those who worked in the engine room [6].

The Szent Istvan wreck (deepest point at 68 meters) has been visited so far by many local and foreign divers. It lies inverted with the deck facing the bottom, with the cannons still facing left. Drawings of the shipwreck by Danijel Frka are given in Fig. 2. In Fig. 2a details of the southern side of the shipw's aft are shown, namely propellers, motor shafts, cannons, keel, as well as numerous fishing nets laying on the shipwreck and around it. A depression can be seen just above the cannons that was the consequence of ship's hull imploding the air trapped inside during the sinking. Moreover, from the shadow below the southern side of shipwreck's aft it can be deduced that the shipwreck is leaning a bit to its northern side, thus creating an opening under the ship's hull on its southern side. Details of the northern side of the torn bow, which broke during the sinking, are drawn in Fig. 2b.

The rest of this paper is organised as follows: the autonomous underwater vehicle and the sonar mounted onto it are described in detail in Sect. 2. Methodology of the underwater cultural heritage (UCH) site survey and data collection are presented in Sect. 3, while the results of bathymetric measurements are shown in Sect. 4. Lastly, the concluding and future work remarks are given in Sect. 5.



(a)



(b)

Fig. 2. Drawing of the (a) aft (b) bow of the SMS Szent Istvan shipwreck by Danijel Frka, [6].

2 Equipment

2.1 Autonomous Surface Vehicle

An autonomous surface vehicle (ASV) equipped with a Norbit WBMS 400/700 KHz multibeam sonar and accompanying Applanix navigation system along with a high-precision Trimble GPS antenna were also used to collect the acoustic data. This is one of the many application dependent versions of the so called Dynamic Positioning Platforms (PlaDyPos or H2Omni-X), called the Dynamic Bathymetric Imaging Platform (PlaDyBath), shown in the image below. The surface vehicle was developed by the Laboratory for Underwater Systems and Technologies (LABUST), Faculty of Electrical Engineering and Computing, University of Zagreb (UNIZG-FER), Croatia, and is used for a variety of applications from support to underwater archaeology [5], as a diving monitoring platform that allows navigation and monitoring of divers from the surface [7], as a communication router between underwater and aerial vehicles [8], used in ASV swarms for long-term underwater environment monitoring [9], mapping (obtaining photomosaic and bathymetry) of shallow water areas [2], and mine countermeasures [10].

The ASV is fully actuated with four thrusters that make up the X configuration. This configuration allows you to move horizontally under any orientation. The ASV has a diagonal length of 1m, is 0.35 m high and weighs about 30 kg with payload configuration in the experiments. The maximum speed in ideal conditions is 1 m/s. Such a configuration of the vessel is very suitable for research purposes due to its simple deployment process, robustness in real environmental conditions and low energy consumption, [7, 11].

However, for this particular application, the ASV has been redesigned into a catamaran shape for better hydrodynamic performance, shown in Fig. 3a. While PlaDyBath performs its tasks autonomously, the operator processes sonar data in low-resolution using Qimera software in order to monitor the quality of area coverage. As soon as the batteries are discharged, the vehicle is lifted up to the workboat, the batteries are changed, and high-quality sonar records are



Fig. 3. (a) Autonomous surface vehicle PlaDyBath with sonar mounted below, Trimble GPS antennae in the back and a WiFi antenna on the left. (b) ASV PlaDyBath during the monitoring mission.

transferred to the computer of the ASV operator. The ASV PlaDyBath has been used previously for surveying and 3D modelling of underwater cultural heritage (UCH) sites around Croatia, Italy, and Greece, as reported in [4] and [5].

2.2 Multibeam Sonar

The Norbit iWBMS multibeam sonar is the main sensor for ASV data acquisition, shown in Fig. 3b. Sonar is integrated with the latest GNSS-assisted inertial navigation system (Applanix SurfMaster), has 80kHz bandwidth, roll stabilization, an Ethernet interface and integrated sound speed measurement. The basic sonar features are 5–210° grip, adjustable measurement sector, 10 mm resolution, 256–512 beams, 200 kHz–700 kHz nominal frequency 400 kHz, range 0.2–275 m (160 m typical @ 400 kHz). Ping rate up to 60 Hz or adaptive, Resolution: longitudinal x transverse standard $0.9 \times 1.9^\circ$ @ 400 kHz and $0.5 \times 1.0^\circ$ @ 700 kHz.

3 Methodology of Acoustic Site Surveys

Based on the previous research by Croatian Conservation Institute [12], S.M.S Szent Istvan has already incurred irreversible degradation of her steel and iron hull. Such degradation will inevitably continue, threatening that artifacts could be forever lost and historical context destroyed. Finally, the wreck will completely disintegrate on the sea floor over time. To monitor the conservation status of the wreck the multibeam survey of the wreck was performed in June 2019. Survey had two goals: to provide reliable bathymetric model of the wreck and to measure the degradation status of the wreck. The parameter chosen to monitor the degradation status was the size of the gap between the ship hull and the seabed on the south side of the wreck.

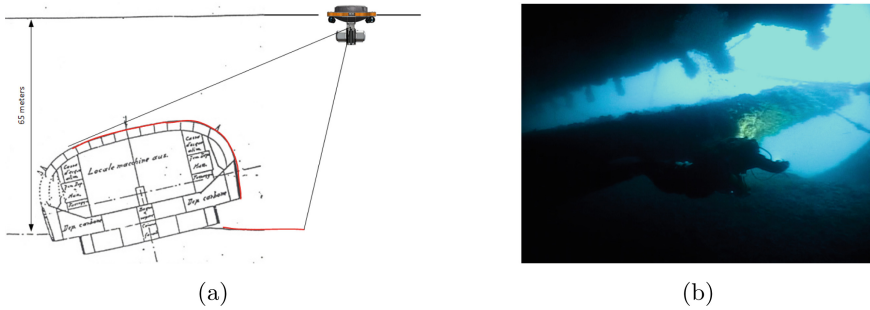


Fig. 4. (a) Methodology for measuring the Gap using ASV and the profiling multi-beam technology. Red line represents the sonified area. The image is conceptual, it does not respect the real proportions e.g. depth vs. hull size, nor sonar beam width and tilt angles. (b) Image representing the size of the gap between the ship hull and the seabed relative to diver. Image courtesy of Marino Brzac

As shown in Fig. 4a, the shipwreck lays upside down on the floor, leaning on its superstructure and resulting with the south side of the hull been lifted from the seabed. The size of the gap on the stern part of the ship is shown in Fig. 4b. Degradation of the steel hull underwater would eventually result in closing that gap until the hull completely collapses under its own weight. To measure the gap along the ship side, the profiling sonar carried by autonomous surface vehicle was used utilizing the methodology shown in Fig. 4a. It was necessary to design a mission for the ASV PlaDyBath, which runs 50 m from the south and north sides of the ship and parallel to the ship, as shown in Fig. 5 by yellow lines. The viewing angle of multibeam sonar was set to 60° , but the rays tilted to the left by 15° , as shown conceptually in Fig. 4a.

In total, an area of 200×75 m around the site was recorded by multibeam sonar, using standard lawnmower missions along, across, and from the sides of the wreck, as shown in Fig. 5 by white lines. In missions planned along and across the wreck to capture as much detail as possible, the sonar viewing angle was also set to 60° , but without ray tilting, and with adaptive ping frequency to ultimately obtain the highest quality shipwreck model from the sonar data. The navigational precision of the autonomous vehicle, and therefore the precision of geolocation of the 3D model reconstructed from sonar data, is of the order of 10 cm, which is more than sufficient for archaeological applications.

The missions planned in advance for the surface vehicle were in the form of transects spaced 25 m apart, thus providing sonar data with much redundancy and more detail in an important part of the shipwreck area. QPS Qimera software was used to reconstruct the bathymetric model from sonar and navigation data. A 0.5 m resolution was used for the bathymetric model because it is also a realistic physically achievable across-track resolution that the sonar mentioned above may have. Since each beam angle is approx. 1° , this would mean that the maximum physically achievable resolution corresponds to the beam footprint on the seabed, which is approx. $0.01 h = 0.65$ m, where h is the depth of the seabed at the site.

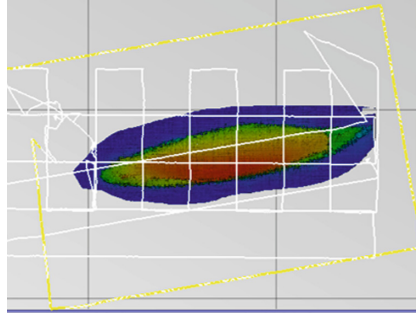


Fig. 5. Overlay of ASV PlaDyBath’s survey paths w.r.t. the top view of the bathymetric model of the shipwreck. (yellow) survey missions around the ship with tilted sonar beams to record the side of the ship (white) standard lawnmower missions along and across the wreck to capture the general morphology of the shipwreck.

4 Results of Bathymetric Surveys

The wreck length measured from the reconstructed 3D model created from the multibeam sonar data was 145 m, the wreck width was 28 m, and the bearing direction from the stern to the bow (bearing angle) was 79.4° . Size of the 3D model matches the real size of the ship very accurately. Precise coordinates of the stern and bow centers in the WGS84 system were also obtained.

Figure 6a shows the reconstructed bathymetry model of S.M.S. Szent Istvan shipwreck seen from its northern side. It is interesting to note how the sides of the shipwreck are very steep, almost vertical. This could be the consequence of a high number of outliers in the point cloud in these areas due to the fishing nets hanging all over the shipwreck. The ship’s propellers, motor shafts and the depression on its aft side are clearly visible in the model, as shown in Fig. 6b, which shows the aft of the ship from its southern side with a great similarity to the drawings of the shipwreck in Fig. 2a. Also, the torn bow part of the ship is shown in Fig. 6c, which is also identical to the drawings of Danijel Frka given in Fig. 2b.

Except for having made a 2.5D model of the shipwreck, which can be useful for presentation in museum exhibitions, the resulting point cloud of the model can be used to assess the change of the degradation state of shipwreck’s hull. With future monitoring missions and using software such as CloudCompare, the changes in shipwreck’s hull could be directly and easily tracked and documented.

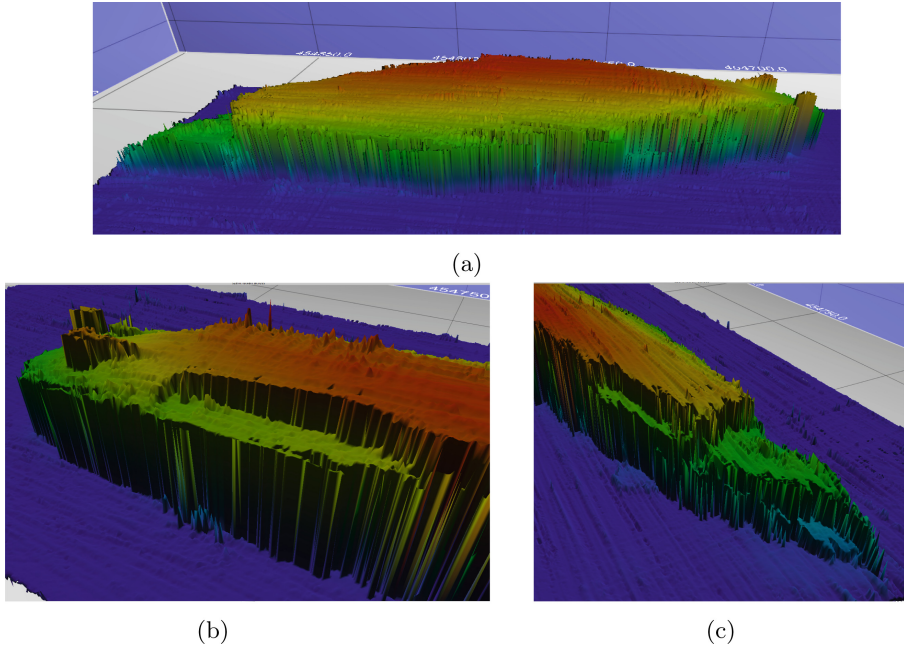


Fig. 6. (a) Bathymetry model of S.M.S. Szent Istvan seen from its northern side. (b) Bathymetry model details of the aft seen from its southern side: the depression in ship's hull, propellers, and motor shafts are clearly visible. (c) Bathymetry model details of the torn bow seen from its southern side.

4.1 Opening Under the Southern Side of the Shipwreck

Another interesting finding and confirmation of the reports we received from the Szent Istvan wreck divers is the opening below the south side of the ship. To record this with the multibeam sonar, its beams had to be tilted in order to catch the morphology of the opening. This made the sonar now able to record the sides of the ship that it could not detect when the sonar beam was not tilted to the side.

Since Qimera, which was used as surface reconstruction software from bathymetric data, reconstructs a 2.5D surface and pairs each (x, y) ordered pair in the horizontal plane to only one height z value, this opening could not be faithfully reconstructed in 3D. Instead, Fig. 7 shows 3 characteristic transverse profiles of the south side of the stern of the ship which clearly show an opening 4 m high, extending 100 m along the south side of the ship and entering an average of 3–4 m towards inside the ship. It is also interesting to note the outliers which were present in some of the ping returns, which are due to the nets laying all over the hull of the shipwreck and even over the above mentioned opening. This can be seen in Fig. 7b.

A proper 3D reconstruction of the shipwreck's side and the opening below it was not attainable, a plot of filtered pings in 3D is given in Fig. 7d. Depth

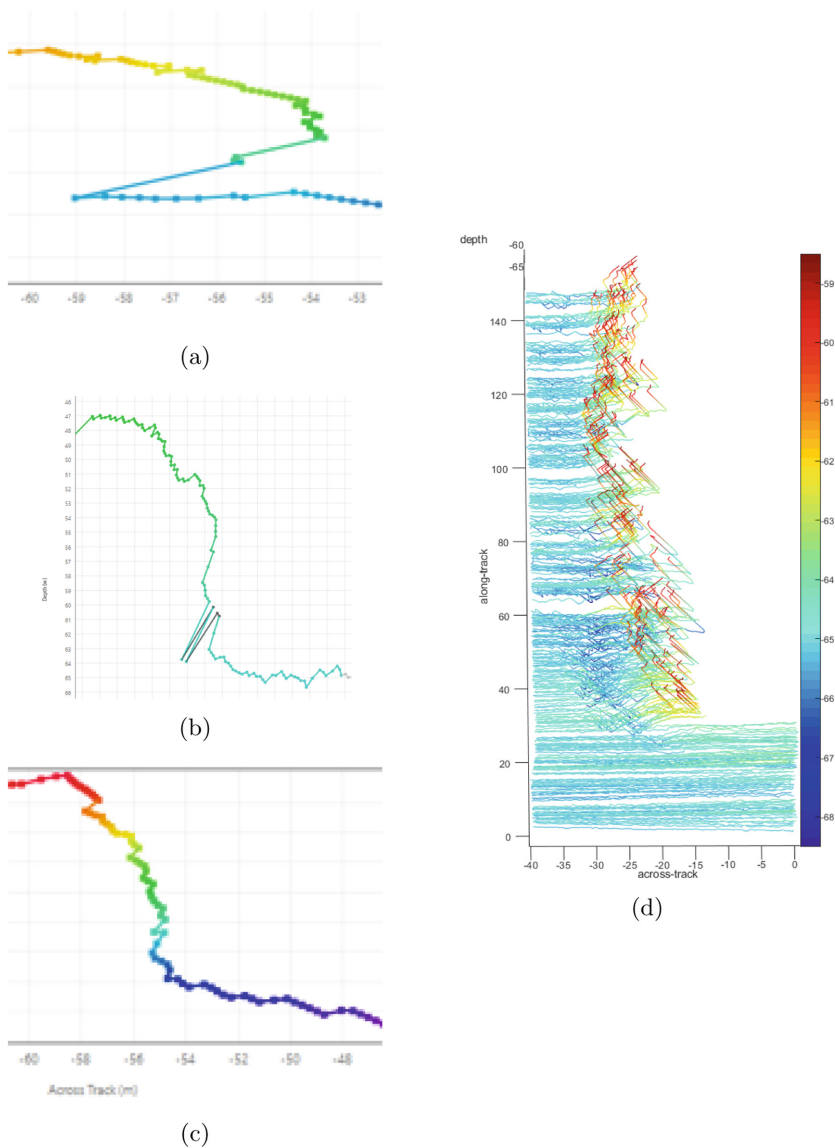


Fig. 7. Acoustic returns of the multibeam sonar: (a) inside the opening and the side of the ship, (b) the opening with outliers generated by the nets laying over the shipwreck’s hull (c) at the end of the opening, i.e. bow part of the shipwreck. (d) 3D line plot of the pings as a spatial representation of the opening below the southern side of the shipwreck.

range is clipped as $z \in [z_{sf}, z_{sf} + 10 \text{ m}]$, where z_{sf} is the depth at which the surrounding seafloor is, and is color-coded. Lower part of the visible shipwreck's hull represents the aft, and the top is the part of the ship closer to the bow, just where the opening ends. Moreover, south is left and north is right. The opening is clearly visible from this plot, as well as the depression left of the aft. This was probably a consequence of the ship hitting the seafloor and thus displacing the sediment.

Further analysis of the ping point cloud shown in Fig. 7d consisted of detecting and clustering points belonging to the seafloor and the shipwreck by thresholding depth values. The plot showing these two clustered point clouds is given in Fig. 8. The opening between these two structures is clearly visible as in some previous figures. However, this way the size of the opening can be assessed numerically, and the degradation level of the shipwreck's metal hull can be numerically represented through time with further monitoring missions.

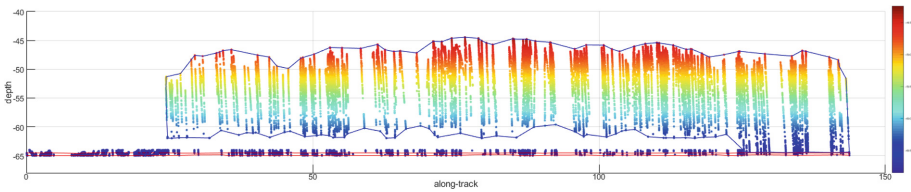


Fig. 8. Clustered and separated point clouds of the seabed and the shipwreck with the opening clearly visible in between. View is from the southern side of the shipwreck, so aft is on the left.

5 Conclusion

This paper presents the results of data analysis from bathymetric sonar mounted onto an ASV PlaDyBath. The survey was performed with two goals in mind: to provide complete and reliable bathymetric model of the wreck and to measure the degradation status of the wreck. The collected bathymetric sonar data provided a highly accurate georeferenced 3D model of the wreck. This model will support future surveys presenting the site on a big scale that would otherwise be difficult to perceive. The second goal, to establish the initial conditions (zero state) of the degradation status of the wreck for the surveys to come, was achieved by spatial measurement of the opening between the seabed and the ship hull. According to our results, opening stretches from the stern to the mid-forward part of the shipwreck in the length of 100 m with the width of 2 to 4 m. This paper shows that ASV equipped with multi-beam profiling sonar could represent the useful tool for mapping and evaluating the conservation status of the UCH site.

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**Special Session on Disruptive
Information and Communication
Technologies for Innovation and Digital
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Special Session on Disruptive Information and Communication Technologies for Innovation and Digital Transformation (DISRUPTIVE'20)

The workshop on Disruptive Information and Communication Technologies for Innovation and Digital transformation, organized under the scope of the DISRUPTIVE project, aims to discuss problems, challenges and benefits of using disruptive digital technologies, namely Internet of Things, Big data, cloud computing, multi-agent systems, machine learning, virtual and augmented reality, and collaborative robotics, to support the on-going digital transformation in society.

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InGenias: Women as a Precursor to Technological and Scientific Vocations

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Abstract. A very significant decline in technological and scientific vocations is being experienced in Europe, especially regarding the female sector. In this context, the InGenias Project was born, with the aim of making female university students and teachers protagonists of future technological vocations. They promote these vocations among secondary students showing a more social and human perspective of technology in society and how technology can improve the quality of life of citizens.

Keywords: STEM · Female careers · Technological vocations · Engineering · Secondary students · Disruptive technologies

1 Introduction

UNESCO in its latest report published in 2019 “Deciphering the code: the education of girls and women Science, Technology, Engineering and Mathematics (STEM)”, [1] has underlined the problem of the lack of female scientific and technical vocations. The Unesco Director (Irina Bokova) has warned that the lack of female representation in these sectors “*slows progress towards sustainable development*”, so “*we need to understand the obstacles that keep female students outside the STEM disciplines*” and “*we need to stimulate their interest from the early years, to combat stereotypes, to train teachers to encourage girls to choose STEM careers and to develop curricula that are gender sensitive and change ideas preconceived*”. In this way, since the enrollment of women in Technology and Engineering is becoming a challenge, many proposals in different institutions are carried out in many countries [2–6].

On the other hand, Spain is also currently suffering a significant decrease in Degrees related to Telecommunications and Engineering. Thus, the Official College of Telecommunications Engineers (COIT) in the last report (2017) reflected that Telecommunications Engineering is in full employment, since only 4.1% are actively seeking work [7]. Furthermore, according to a report promoted by the World Economic Forum [8], the professions with greater future projection are currently focused on technology and mathematics, followed by Engineering. This trend seems aligned with the Spanish reality. However, these data do not fit with the decline suffered by these Degrees. On the one hand, we believe that Telecommunications Engineering remains unknown to society, despite the development of electronics, communications networks,

Internet and telematic services. Even our own students are not aware of the potential of these sectors and the multiple fields where they can apply the acquired training, such as bioengineering, energy, security, defense, banking, automotive, etc. Besides, we also think that society perceives these studies as cold disciplines with little social orientation. All this joins that certain skills such as communication or social responsibility, are not directly developed in engineering profiles, but they are becoming key aspects for companies and society.

In this scenario, the InGenias Project was born in order to promote the acquisition of technological vocations among the youngest, trying to emphasize a more social and human perspective of these disciplines, supporting us in the presence of women in these areas through teachers and students of the Telecom School of the University of Valladolid (UVA). To enhance this initiative, we work with experts in scientific dissemination to detect communication problems that we must face, and therefore propose appropriate strategies, focusing on making women's presence visible in this sector. Besides, the project promotes the acquisition of transversal skills in university students, such as leadership, responsibility, communication and motivation. Indeed, we transform university students into a central element to encourage technological vocations among the youngest. On the other hand, InGenias responds to the needs of a society that is suffering a significant decrease in the number of technological vocations, especially in women. It is worth highlighting the potential of this proposal, as we propose a multidisciplinary work between different branches of knowledge. Finally, it is important to emphasize the great projection of this initiative because it can be extended to other technological-scientific contexts. Therefore, making visible the presence of women in technological and engineering sectors will help build a more diverse and plural educational context in gender.

2 Main Objectives and Methodology of InGenias

2.1 Objectives of the Project

InGenias is an innovation project led by female students and teachers of the Telecommunication Engineering School to be precursors of technological vocations and to increase the number of women who choose these studies. The communication strategy tries to combine two worlds: the university world and the school world. Both disparate but with a common characteristic: education. Therefore, the objectives to be achieved with this project are summarized as follows:

1. *To develop seminars on scientific and technological dissemination to train female teachers and students.* For this, the training team analyzed the most important communication problems of technological fields and designed appropriate and efficient dissemination strategies.
2. *To design oral presentations.* To provide an innovative perspective, we propose a talk format far from the traditional one so that teachers and students communicate their experiences with elevator speech strategies, so that in a few minutes they can arouse the interest and curiosity of their target audience, using simple language, close to the one school students use in their day-to-day. The students assume the

role of leaders to communicate their technological disciplines so they become a reference for the schools' students and a source of inspiration in which they are reflected. With this staging, a preferential place is given to the female presence in technological sectors in a natural way.

3. To develop experiments, prototypes and/or applications related to Telecommunications and Engineering. The more important property of the experiments/prototypes is that students can build them themselves and they can use them in their day-to-day. Then, we want to encourage the maker phenomenon or creative technology based on DIY (Do It Yourself) that is becoming much more than a trend among the youngest. However, we also want to emphasize the idea that technology can serve society so we want to develop experiments/prototypes with a high social and sustainability character.
4. To disseminate talks and prototypes in Secondary Schools. Dual teams formed by female students and teachers visit Secondary Schools to encourage technological vocations among teenagers, making women direct precursors of them.
5. To design a qualitative and quantitative study based on surveys to analyze the impact of the project. The provided feedback of this study will help us to improve the project. In addition, we want to go further so that surveys can throw information to detect possible gender biases and measure the level of knowledge of young people about technology and engineering.
6. To disseminate the project. A set of actions have been designed to publicize InGenias among students and members inside and outside the University. The results have been disseminated through digital channels, specifically twitter of the UCC (Scientific Culture Unit), the Science Park of the University of Valladolid and the E.T.S.I. Telecommunications [9].

2.2 Methodology of InGenias

The methodology and the steps followed in the Ingenias Project are summarized as:

1. The first step is to establish contact with Secondary Schools to offer the activity. Due to logistic issues and since we support the visit with experiments and prototypes, we initially restrict them to one city (Valladolid). This contact begins in September, to plan the visits from February to April.
2. Experts in scientific dissemination will help detect the main communication problems, and propose appropriate strategies, focusing especially on making visible the presence of women in our sectors. This training program will be elaborated for the dual teams who will attend Secondary Schools.
3. In parallel, the team of the project will work on the design and development of prototypes and experiments regarding Telecommunications and Technology.
4. Once the talks ("elevator pitch") and the experiments/prototypes are prepared, this new perspective of scientific communication will be disseminated among the youngest, visiting Secondary Schools (3rd and 4th courses, 13–15 years old). We think these courses are quite crucial since students especially lose interest in Technology and Engineering during these years.

3 Dissemination and Results of the InGenias Project

3.1 Dissemination of the Project

It should be noted that we have visited eight Secondary Schools in the first experience (academic course 2018–2019), attending 3rd and 4rd courses and reaching up to approximately 420 students. Figure 1 shows some pictures of the visits made by female teachers and students to the schools.



Fig. 1. Pictures made by the teams (teachers and students) in the Secondary Schools.

Furthermore, students and teachers have developed a set of experiments and applications related to Telecommunications/Engineering using low-cost technologies (Arduino, Rasberry pi) and emphasizing the social nature of technology, as it can be observed in Fig. 2. In this way, university students have shown some examples regarding disruptive technologies, so young students can be able to learn technological innovation that will make obsolete those that have being used until now (3D printing, Robotics, Virtual Reality, etc.). In fact, it has been designed a set of experiments such as a backpack alarm, a sensor presence to be connected to smartphones. However, now we are working on the design of prototypes for blind people. This will allow the

youngest to open new horizons, and that they can find in these sectors a future career, looking for other objectives and views different from those so far existing.

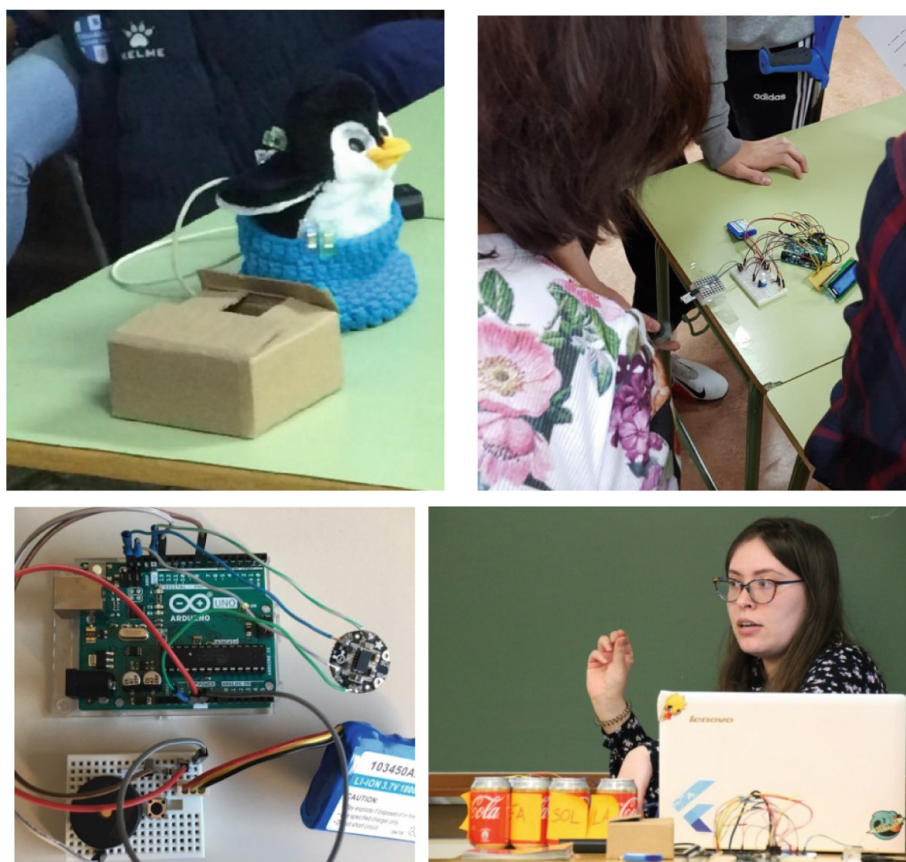


Fig. 2. Experiments developed by female students and teachers.

3.2 Results and Discussion

During the visits some surveys were launched to students and a qualitative and quantitative analysis of the results was carried out. These surveys were filled in by the students (420) that attended the activity schools and it was distributed during the last ten minutes. The survey consisted by a set of questions, some of multiple choice and others of open answer, both qualitative and quantitative, as it can be shown in Fig. 3.

These surveys aimed to obtain a feedback on the level of knowledge of students on related topics. Thus, Fig. 4 (a) and (b) show the level of prior knowledge of students according to age and gender with respect to Technology and Engineering. As can be seen, the level of knowledge of students in Engineering/Technology stands around 5.7 (on a scale of 0 to 10), but the levels in Telecommunications are lower, around 4.5.

1. Eres:

☐ Chico

☐ Chica

2. Edad:

3. ¿Te ha parecido interesante la actividad realizada? 1 (muy poco) y 10 (mucho)

1 2 3 4 5 6 7 8 9 10

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

4. ¿Antes de la actividad conocías algo de Ingeniería y/o Tecnología? 1 (muy poco) y 10 (mucho)

1 2 3 4 5 6 7 8 9 10

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

5. ¿Antes de la actividad conocías algo de la Ingeniería de Telecomunicaciones? 1 (muy poco) y 10 (mucho)

1 2 3 4 5 6 7 8 9 10

☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐ ☐

6. ¿Qué es lo que más te ha llamado la atención de la Ingeniería de Telecomunicaciones?

7. ¿Te podrías plantear ahora poder estudiar en un futuro Ingeniería de Telecomunicaciones?

☐ Sí

☐ No

8. (Si respondiste SÍ) ¿Por qué? (puedes elegir más de una respuesta)

☐ Porque es una carrera innovadora y dinámica

☐ Por tener muchas ramas en las que puedes trabajar

☐ Por su uso para poder mejorar la sociedad

☐ Por sus numerosas salidas laborales

☐ Otro:

9. (Si respondiste NO) ¿Por qué? (puedes elegir más de una respuesta)

☐ Porque me resulta poco atractiva para estudiar

☐ Porque me parece muy complicada

☐ Porque me gustan otras carreras diferentes

☐ Por su poca utilidad a nivel social

☐ Otro:

10. ¿Crees que hay que tener alguna cualidad especial para hacer una Ingeniería? ¿Cuál piensas?

11. Menciona a algún referente hombre o mujer en algún campo de la ingeniería y/o Tecnología.

Fig. 3. Example of the structure of the surveys.

Besides, knowledge levels remain more or less stable at all ages, except at 18 years old where they increase slightly. Regarding gender, it is observed how the degree of

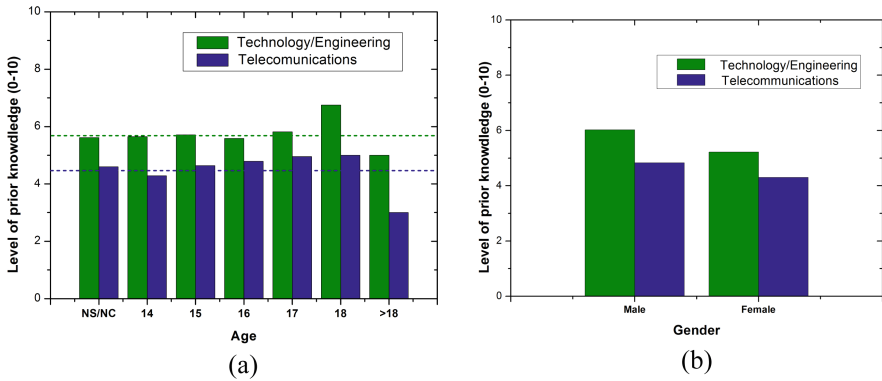


Fig. 4. Graphs that show the degree of prior knowledge of students in Technology/Engineering.

knowledge of girls is slightly below that of boys, both for Technology/Engineering and Telecommunications.

In addition, the study is focused on detecting possible gender biases when students opt for degrees related to Technology and Engineering. The preliminary analysis have shown that the number of boys who might consider studying Engineering Degrees is close to 70% while the number of girls is less than 50%, as it is observed in Fig. 5.

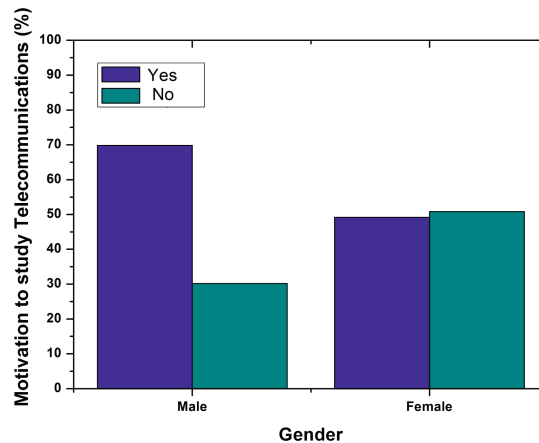


Fig. 5. Graphs that show the percentage of students who might consider studying a Telecommunications Degree after the visit.

If we analyze the options by which students may choose a Telecommunications/ Technological degree for their future, it is observed in Fig. 6 (a) that the option of being an innovative and dynamic degree (option A) together with the multiple fields that these degrees can cover in society (option B) are the most punctuated. On the contrary, it is observed that the least valued option is related to the number of job

opportunities (option D). If we go into analyzing this graph from the point of view of gender (Fig. 6 (b)), a balanced score is observed between both genders for options A and B. However, it is curious that Option C, which regards how technology can improve society, it is best scored by the female than by the male gender, 52% versus 34% respectively. In this way, it is also curious that option D, which regards to job opportunities, is best scored by the male than the female gender, although here the differences are smaller between both of them (35% versus 27%).

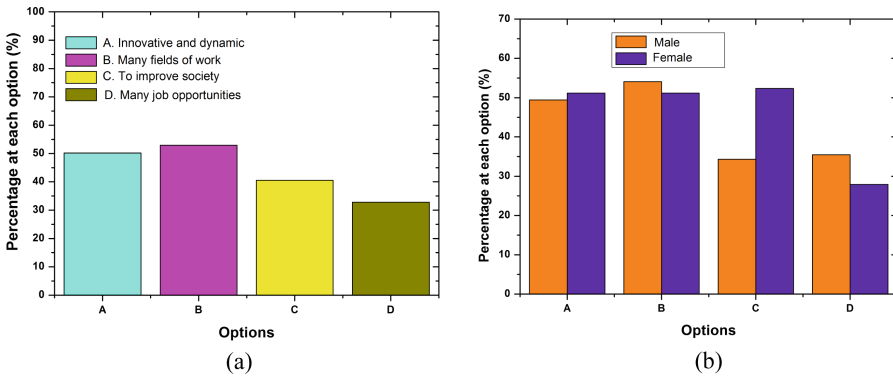


Fig. 6. Graphs showing the reasons why students could consider studying a Telecommunications Degree.

Finally, the results of the pilot experience have demonstrated that students do not know references in technological fields, especially women (Fig. 7 (a)). In fact, when secondary students were asked about possible reasons that drive young people away from studying degrees related to Technology and Engineering (Fig. 7 (b)) it should be noted that the social, educational and stereotypes components have impact in their answers, especially the social component with values close to 10%.

These results make us think on the one hand, that there is a strong ignorance on the part of young people from Engineering/Technological sectors and also they are not able to identify everyday referents, especially women referents. We have also observed that girls still perceive cultural and social biases as reasons not to opt for technological disciplines. As a consequence, we feel that there is a need to bring engineering and technology to the youngest through female referents with whom they can feel identified.

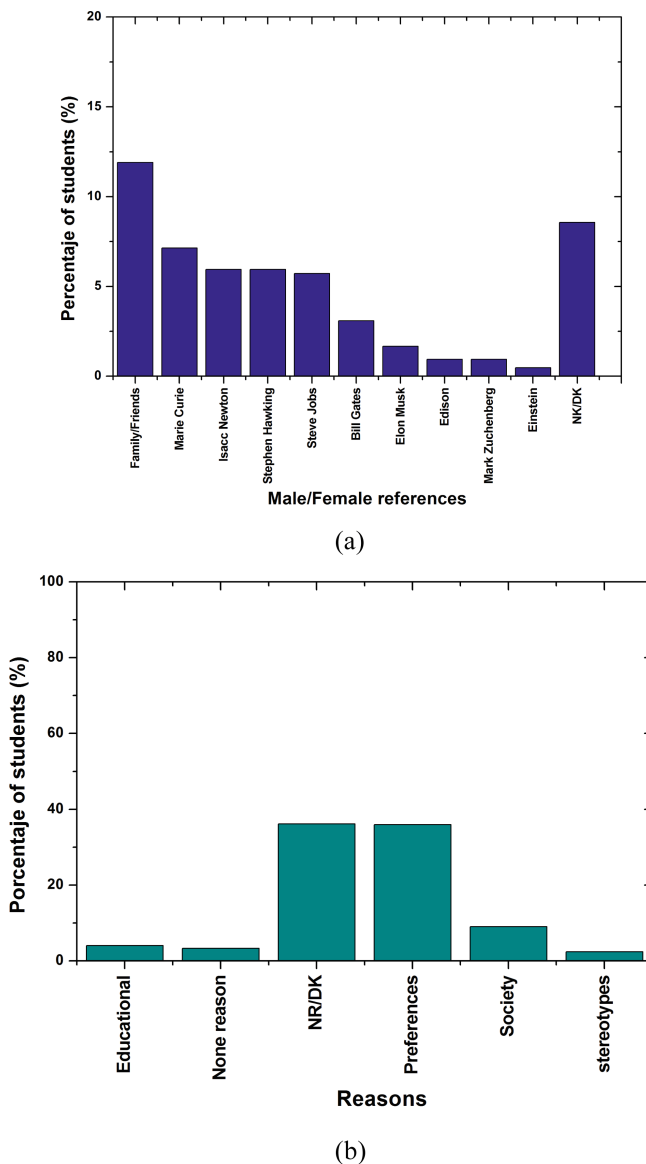


Fig. 7. Graphs that show the reasons why high school students believe that boys study more Engineering/Technology than girls

4 Conclusions

The first start-up of the InGenias Project has been quite successful among the visited Secondary Schools. In fact, a total of eight Secondary Schools have been visited, attending the classes of 3rd and 4rd courses (13–15 years old), around 420 students.

The visits took place between the months of February and April of 2019, since the Secondary Schools and the University teams had higher availability to carry out the activity during this period.

On the other hand, and as it has been commented before, all the qualitative and quantitative results obtained make us think that there is great ignorance among young people regarding Engineering/Technology careers. Even more, we have noticed that secondary students do not identify people who are technological references, especially women. We have also observed that girls still perceive cultural and social biases as reasons not to opt for technological or engineering disciplines. As a consequence, we think necessary to bring engineering and technology to young people but also emphasizing the female presence in these sectors. For all these reasons, we believe that to continue betting on the Project in successive academic years and to propose a more global and ambitious project will be very enriching for the transfer of technological and scientific knowledge to society.

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Contextual Adaptive Interfaces for Industry 4.0

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Abstract. Information technologies are intrinsically connected to the manufacturing processes, with more data generated each second. To efficiently operate machines, users must sort out information that is relevant to them in specific moments and contexts. In this paper, we propose an architecture that combines context – e.g. location, type of order, available assets, previous actions – with information established through user stereotypes.

Keywords: Context-aware applications · Adaptive interfaces · Industry 4.0

1 Introduction

With the advent of the Industry 4.0, information technologies have become an inescapable part of the manufacturing processes, generating and facilitating data to all parties involved. As systems grow in complexity, more expertise and data are required to handle them; and more information is necessary to properly learn how to use them [1, 2]. Selecting which information is important and relevant for a specific user in a given context is, therefore, a relevant step.

While an Enterprise Resource Planning system may have information regarding all possible tasks, there is the possibility not all of them can be executed on a given place or moment; therefore, for a more precise and adequate information filtering process, context and user data are both required. Additionally, the expertise of the user also comes into play: frequently, more experienced users are autonomous, while the less experienced require more guidance [3, 4].

User Modelling (UM) is frequently used in contexts where the system needs to adapt its behaviour to how it perceives the user. Systems and applications have, over the years, applied different approaches to how user information is stored and how it is affected through user-system interaction [5, 6]. For the UM to be able to fulfil its role as an adaptive force for the system, it requires a degree of ground information, i.e., an initial formalization of the assumed characteristics and preferences of its users. This information can be in one of two categories: (1) Domain Dependent Data (DDD) or

(2) Domain Independent Data (DID). DID features degrees of knowledge the system assumes its users would have regarding its application domain. Complementarily, DID includes psychological information about the users and their generic user profile.

Context Awareness (CA) approaches can be incorporated into a system in order to enhance user experience in different ways, such as the adaptation to different scenarios, by applying different algorithms and for personalized suggestions [7–9]. Regarding the adaptation to different scenarios, one can consider that context awareness can be relevant so that the right information can be delivered, in the right format and to the right user. E.g., this can be done through an intelligent and adaptative interface, in scenarios where the user can change places, change of status during a production process, among others. Operational information can be used as extra input for the algorithms used by the application or system in order to optimize their outcomes, potentially resulting in improved decision making, better planning, cost reduction and optimization of processes, among others.

Systems and application can gather contextual information through sensors (e.g. location of the user and/or the device, status, time, nearby locations, other devices, etc.) in order to filter, search and relay relevant information or services. Contextual information can be classified according to its complexity and how it can be combined with additional data. As such, primary context refers to raw data, captured directly by sensors and used independently from other known information sources. Secondary context, on the other hand, refers to the fusion of primary context sources for a more global appreciation of context [10].

The NIS Project (Núcleo Investigação Sistrade) consists in the development of solutions for usability, adapted to the Industry 4.0 challenges, by exploring the elaboration of methodologies and rules for the improved development of Human-Machine Interfaces and web-design of adaptative interfaces. In the research for augmented reality (AR) solutions for the industrial environment, we aim to present an architecture for such a system that will be able to consider historical information about the user (such as profile, expertise, among others) and combine it with contextual information regarding their location and previous tasks in order to determine how to best guide the user in the following tasks.

This paper is structured as follows: (1) Introduction, wherein the applicational context and the project were presented, (2) Context Awareness and User Profiling, providing a short review of the concepts involved in the project and specifying domain dependent and independent data to be used, (3) Proposal, wherein we present the architecture, adaptational model and technologies and (4) Conclusions.

2 Context Awareness and User Profile

The interface (graphical or otherwise) is the main interaction point between the user and any system. The information it shows must be able to adapt to each specific user, and its meaning and relevance will always be in regards to its context and location [11]. When the information is made available is also a relevant factor: the user must not be

overloaded with so much information they cannot make sense of it. Therefore, the user's context is an important factor when it comes to selecting the appropriate information to show [12].

There are several different approaches to implementing User Models, of which four categories can be extracted: knowledge-based, rule-based, behavioural and stereotyped-based [13, 14].

Knowledge-based approaches rely on the amount and quality of the information the user possesses over the application's domain and how it is obtained [5]. This information is generally forward by the users themselves through forms, enquiries or studies. Behaviour-based approaches, on the other hand, rely on the monitorization of the user while it performs tasks in the system [7], and different behaviours can be used for the formation of groups of users or heuristic extraction.

Models based on rules can be defined either automatically or semi-automatically through the use of machine learning algorithms or be formalized manually by domain experts. These rules can be used to infer new knowledge about the users, finding frequent application in scenarios where it is important to predict the users next move or anticipate possible errors.

Finally, stereotype-based approaches allow for the formation of possible groups of users through a relatively short number of relationships. They define, in a first step, which stereotypes are expected, attributing them a set of characteristics. These may include different layers for each user group, such as performance levels and expertise in their tasks [5, 13, 15].

2.1 Domain Dependent Data in Context-Awareness

Consider an AR system to be used in an assembly line, in which several tasks must be performed in sequence. Each task is therefore dependent on the one that precedes it and relies on different machinery. Additionally, if different products can be made on the same line, the same machinery can be used for a number of tasks. The system has support information regarding all possible actions that can be performed in the line but must decide which information to show the worker when a new order is being processed.

As such, when a given worker is at their workstation, the following contextual information can be captured: the location (and corresponding machinery), materials available and current order.

Information regarding available machinery can be used to establish which actions are possible in that area; i.e. by knowing the location of the worker, it is possible to filter the number of tasks that can be done by excluding all the actions that can only be done in different locations [11]. This gives us a subset of tasks and, therefore, a subset of support information that can be shown to the user. However, of these, not all are relevant: by knowing which product the worker is currently working on, it is possible to extrapolate a sequence of actions they can perform, filtering the support information further. This filtering process takes an even more relevant role when inserted in an AR environment where, on top of a risk of sensory overload to the user, there is a risk of complete occlusion of the environment the user is operating on. Impeding the user from properly concluding his tasks may have an unwanted impact either on security issues,

or operational and financial losses. As such, even the tasks that are relevant to the current context need to be properly evaluated as to justify its inclusion in the interface.

Additionally, not all support information is equally relevant; knowing more about the specific worker allows us to understand their ability and, therefore, how much support they need in performing their tasks, i.e., for the same location and tasks, different levels of support can be given. The importance of these needs to be assessed.

2.2 Evaluating Performance in User Profile

The system must know the characteristics and context of the user in order to be adaptable. Such characteristics depend not only on the user's stereotype but are also affected by other factors such as the user's performance. The system then must use information regarding actions – i.e., to possible tasks and sub-tasks – defined in the user's model in order to assess which contents are relevant to the current stereotype; data regarding performance is therefore used to adapt the content to the user's needs.

As the users execute their tasks, their behaviour will depend in the degree of confidence shown in performing their tasks: an experienced user is more likely to be more confident and to fulfil their tasks successfully, while a less experienced user may take a more cautious approach, needing more time and/or guidance [4, 16]. Keeping an historic record of past performances of a given user for each task is a simple way for the system to assess their confidence level and decide which information must be put in the forefront and which one is not particularly relevant. E.g., for an inexperienced user, it may be enough to provide a navigation system that allows the user to query about the task's general procedure. Exposing an inexperienced user to a higher amount of information may be a small price to pay when compared to a bad performance in a given task [4]. On the other hand, overloading an experienced user with huge amounts of information may do more harm than good, as they probably don't require the same amount of assistance [16]. Adaptation, therefore, happens through the omission or displaying of information based on the user's specific characteristics.

3 Proposal

Manufacturing Execution Systems and Enterprise Resource Planning systems hold and generate large amounts of data: more often than not, it is hard for users to establish which of it is relevant for their particular situation. By incorporating contextual information and the user's characteristics, we can develop algorithms that are able to give the users only the most relevant information according to their user profile. We propose that this information is delivered to the users through AR, with the aid of Hololens devices [17]. Which information is to be presented, and how, must be according to the context of the user and their personal needs.

The user's profile must therefore not only include the user's characteristics, but also temporal and environmental contexts, in a way that a broader, more dynamic model can be achieved in order to map the user's behavior in the different possible scenarios. I.e. the user's model will include information regarding preferences, interests, desires and/or needs.

3.1 Architecture

We will propose a stereotype-based approach, in which contextual information regarding the (i) user's expertise, (ii) their location, and (iii) available options will be used to determine which information will be available to the user at a given time.

Figure 1, below, illustrates how the adaptation mechanism depends on these different inputs:

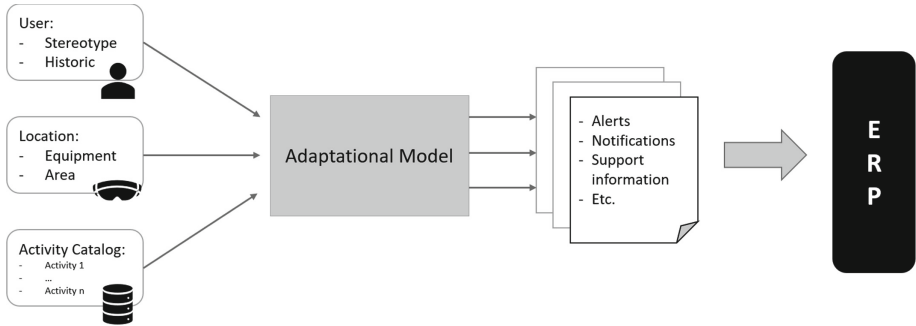


Fig. 1. System's main workflow diagram

User data, as previously mentioned, will identify the stereotype they belong to and, therefore, their characteristics. Additionally, it will also bring historical user data regarding past actions and the specific user's performance.

Information regarding the user's location will be supplied by a Hololens device, featuring data that identifies the space/room the user is currently in and identifying available devices.

Finally, we consider the existence of a catalogue containing all system's possible actions and helpful, support information regarding them that can be supplied to the users to help in the execution of said tasks.

3.2 Interaction and Adaptational Model

For any given moment and context, the goal of this Adaptational Model is to deliver only the most relevant information to the user. By searching through the catalogue containing all possible activities, the adaptational model will, on a first step, use location data to start the filtering process. As previously mentioned, this process will select a subset of information that is relevant to the location and usage of available assets.

As such, all the available contextual information can be used in a filtering pipeline, as illustrated in Fig. 2:

Considering that the filtering process may result in more than one task being possible under the current conditions, a ranking mechanism must be employed in order to assess which one is the most likely to be in execution. An expected activity flow and the user's action history can be used for this end. In a next step, information regarding

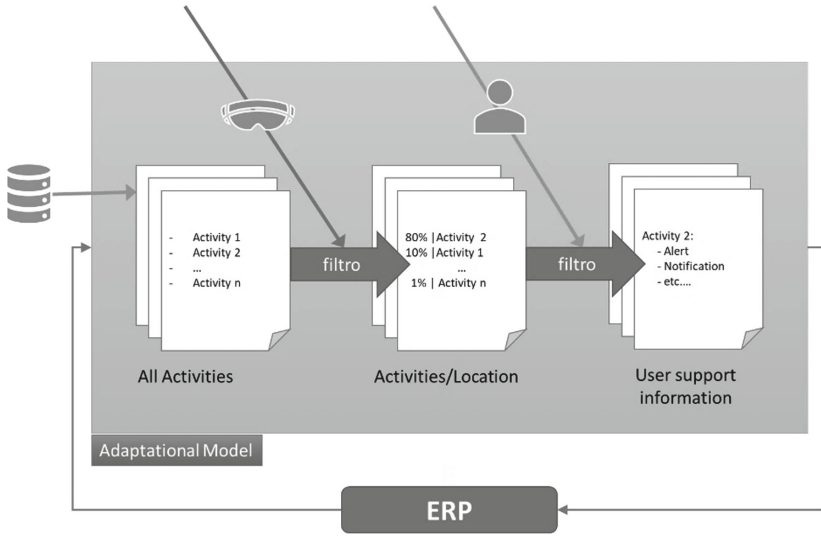


Fig. 2. Adaptational Model filtering process

performance will be used to assess which visual elements, such as menus, alerts or notifications, must be displayed either through the Hololens or other devices. However, identifying which task is currently being performed by a user is an uncertain and unprecise task. Even observing the user's recent activities may lead to a wrong conclusion, as the interpretation of their actions may not be an objective process. Bayesian Networks are commonly applied to scenarios where uncertainty is a factor [7, 18, 19], by providing a probabilistic model that allows reasoning over uncertain knowledge. In these acyclic graphs, each node represents a concept or a parameter, and each arch the direct, unidirectional influence between them. Each node has its own conditional probability table, quantifying the influence of the previous nodes over it.

In the context of this proposal, the Bayesian Networks define the relationships between activities, with each node being associated to one of the parameters identified in the User's Profile: be them the user's performance on a given tasks, or the steps that compose it.

For each user stereotype there's a Bayesian Network that represents all the tasks associated with it and relationships between them. Additionally, it includes relations between tasks that, while not exactly related to the stereotype, influence the tasks the user must perform. In Fig. 3, an illustrative subset of possible actions pertaining to Production Orders that can be executed by a Production Manager and their respective probabilities is shown:

Here, the nodes represent the possible tasks that can be performed by a Production Manager, how they influence each other and how likely the user is to perform them at a given moment. The probability tables are conditioned by previous interactions of the user with the system and evolve over time in order to better reflect their actual activities.

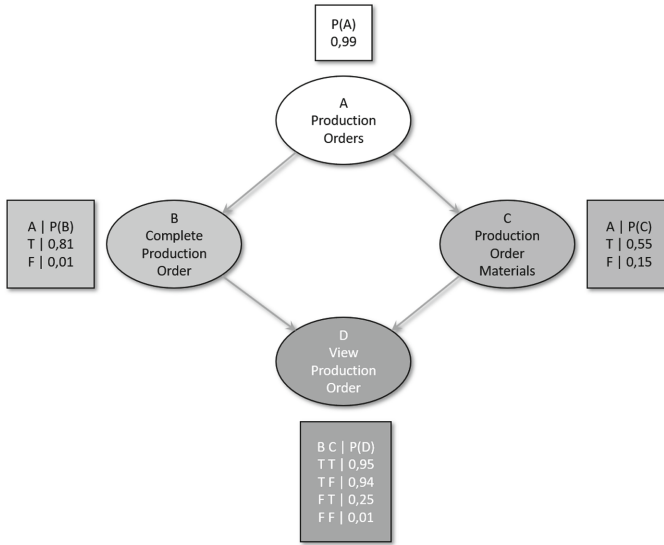


Fig. 3. Bayesian network of a Production Manager

4 Conclusions

How humans and machines interact is increasingly more relevant, as the informatization of services and industry grows; and systems that support the users as they perform their tasks, by adapting their interfaces to the person using them, become a necessity. Establishing a user model that encapsulates the expected behaviour and features of the users is only one of the pieces that will allow for such adaptive interfaces. In order to be as relevant as possible, other contextual information must be considered.

In order to show the user information regarding the tasks they must perform – which will depend on their location and available tools – it is necessary to combine context awareness technologies with user profiling information.

In this paper, we proposed an architecture that starts by modelling different user features through stereotypes, to which other contextual information is added, such as the location, available tools, previous tasks, among others. The combination of elements allows for the filtering of support information provided by MES and ERP systems, effectively showing the user the information that is most relevant not only in their current context, but more suited to their personal needs. The architecture presented is being implemented as three separate modules, including (1) the localization system, determining areas of interest in shop floor environment, (2) and the definition of the Bayesian network regarding the activities of the specified stereotypes. Following this implementation, the next step should be to use the outputs of the modules through the filtering pipeline in order to test the architecture in a proper industrial environment.

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OEE PRO: A Solution for Industry 4.0 in the Aeronautical Sector

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Abstract. We present a software tool and an architecture, called *oeo pro*, for the improvement of production and maintenance management in factories of the aeronautical industry. The tool, developed by the company Luce IT, allows you to capture large volumes of information from the different robots of the factory and from other sensors available in the shop floor. The software also processes and analyzes this information, calculating various indicators in order to obtain added value and optimize production. Through the use of this tool, reductions of 20% in lead time, 3% in the total product cost, 1.5% of the factory rate, and increments of 6% in the shop floor productive capacity have been achieved.

Keywords: Industry 4.0 · OEE · Software · Data capture · Big Data · Visualization · Aeronautical industry · Production · Maintenance

1 Introduction to Industry 4.0

Industry 4.0 can be seen as the fourth era of industrialization. It is based primarily on the integration of computing, communication and control (the so-called cyber-physical systems or CPS), and the analysis of large volumes of data (Big Data). Thus, there are several facilitators of Industry 4.0, including the Internet (as communications infrastructure), the Internet of Things paradigm, cloud computing, Big Data, artificial intelligence, robotics and human-machine interaction [1–3].

The introduction and coordinated use of all these technologies have as one of their main objectives the improvement of the efficiency of production processes, which translates into a reduction in operating costs. Therefore, it is essential to define a set of metrics or key performance indicators (KPI). Among these KPIs, the OEE (Overall Equipment Effectiveness) should be noted. It is an indicator of the productive efficiency of the industrial machinery. It is defined as the ratio between the time spent on producing products (which satisfy the quality requirements) to the scheduled production time, and it can be calculated as the product of three factors: availability, performance efficiency and quality rate [4]. Availability is the quotient between the planned production time, when the downtime due to breakdowns and machine configuration changes are discounted, and the planned production time. Performance efficiency is the

ideal time required to produce a product multiplied by the number of products made and divided by the actual runtime. Finally, the quality rate is the ratio between the number of products that meet the quality requirements and the total number of products produced [4]. In this way, the time wasted due to breakdowns and the need to configure the machines affects the availability, the micro-stops and the reduction of machine operating speed affects the performance efficiency, and the rejections of products due to insufficient quality (for various reasons) affects the quality rate, and all these events therefore affect the OEE [4]. However, to be able to calculate the OEE it is necessary to have real and reliable measurements of what the machines or robots do in a shop floor. Without that, any calculation you want to make to improve the use of machines, optimize operations, and improve maintenance, will not be trustworthy enough and will not be reflected in a true improvement of KPIs.

In this work, we present a software tool, called *oee pro* [5], for the improvement of production and maintenance management in factories of the aeronautical industry. The tool, developed by the company Luce IT, takes advantage of the enabling technologies of Industry 4.0 mentioned above, and allows capturing large volumes of information from the different robots of the shop floor, as well as from other available sensors in it. Then, it processes and analyzes this information using different methods (including artificial intelligence techniques) to calculate various KPIs, and to optimize maintenance and production processes.

2 Features of *oee pro*

oee pro is a software developed for factories of the aeronautical industry, which is capable of capturing, processing, analyzing and compacting more than 2 million daily events. Those events can be obtained from machines of different technologies and manufacturers, from external sensors, from information obtained from human operators, and through the interaction with other computer systems. The capture of this information allows the implementation of a disruptive performance analysis approach. It is based on a set of KPIs specifically designed for the aerospace sector, and it allows measuring the real added value of activities, relating them to processes, results and costs, and using them to optimize production, as well as for decision making to improve performance. The main features of *oee pro* are as follows:

- It is a production improvement and maintenance management software for Industry 4.0.
- It captures information in a non-intrusive way.
- It includes a unique information compaction algorithm that keeps valuable information.
- It is a robust and reliable system with self-check-state that constantly checks status of the various systems (connectors, sensors, databases, services, etc.) and implements automatic techniques to fix bad system operation.
- It has connectors with enterprise resource planning systems (ERP), material requirements planning systems (MRP), and manufacturing execution systems (MES) from main vendors.

- It allows to know the incidents in the activities carried out by the robots of the shop floor for the absolute control of the production.
- It implements artificial intelligence techniques, since it accumulates the knowledge of the shop floor and its technologies, avoiding processes, activities or environmental problems that incur wastes already detected, and predicts machine failures.
- It translates and visualizes all the information of the machine, the external sensors, the operators and the management systems in terms of added value and costs for the factory.
- It helps to standardize times in robot operations.

3 Architecture of *oeepro*

3.1 Software Modules

oeepro software which consists of 15 modules (Fig. 1). The basic modules of that architecture are *O3E Capture*, *O3E Data Box* and *O3E Dashboard*.

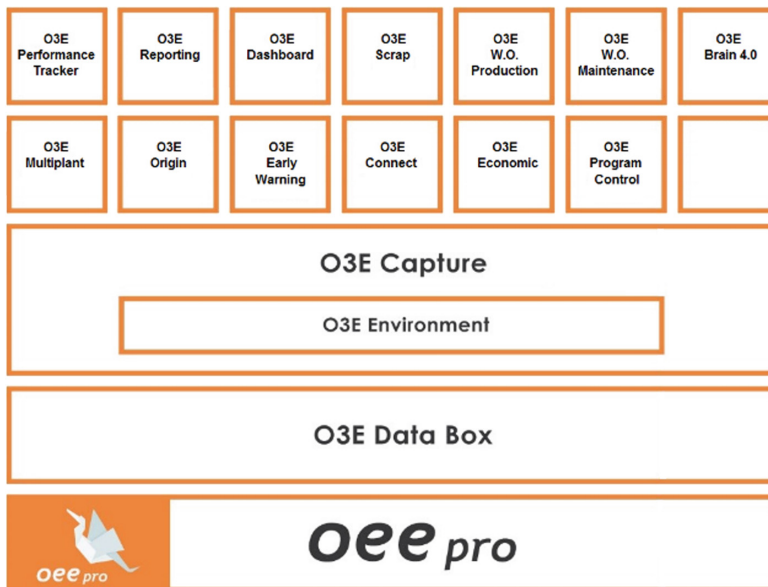


Fig. 1. *oeepro* modules.

- *O3E Capture* is responsible for the acquisition, in real time, of the signals coming from the robots of the shop floor (Fig. 2), performing this operation in a non-intrusive manner and without affecting the performance of the robots, nor the availability and reliability of them. In addition, relying on another element, called *O3E Environment*, it captures, through sensors and other means, external

parameters to the robot; in particular, environmental data like temperature and humidity. The *O3E Capture* module is also responsible for transforming signals into valuable information, cleaning and verifying their values. The module covers all activities of the normal operation of the robot (running, stopped, running at reduced speed), receiving signals from sensors, internal and external, and the incidents in the process (such as communication problems or stops caused by a fault). In addition, the stop detection system assigns the most common causes of stops automatically.

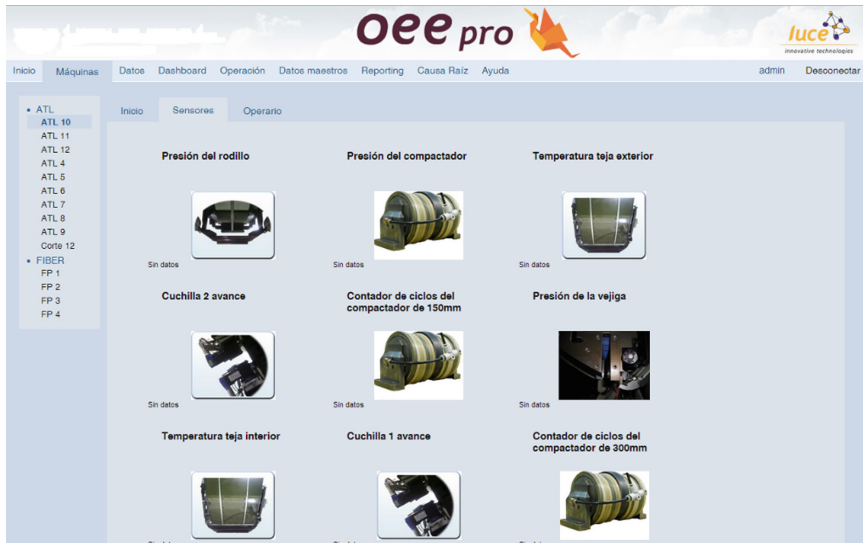


Fig. 2. Capture of data from different sensors of a robot.

- *O3E Data Box* is the core element which centralizes information. It stores, indexes and interprets the information, and generates business KPIs in real time. This module implements advanced analytical methods to process the information collected, and to find correlations in this information, as well as to automatically compact it. These steps are essential for the correct prediction of faults and other parameters. In addition, the module allows to generate an environmental map of the factory to monitor the real-time values of each probe in the shop floor, it allows to obtain in detail the activity of each robot (Fig. 3), and also allows the management and configuration of master data to enhance the flexibility of the system. This module, based on other modules shown in Fig. 1, provides intelligence to the system, learning the situations that are generating waste or that may do so in the future (and therefore optimizing the robot behavior), as well as the sequences of events that lead to breakdowns, generating alarms if there is a high probability that they will occur again. In addition, it implements a self-check-state procedure that verifies the correct operation of the system and implements actions to solve failures. Since the module centralizes the information collected, it includes mechanisms to

guarantee the integrity and security of the data against any physical error, and at the same time it offers high speed access and capacity.

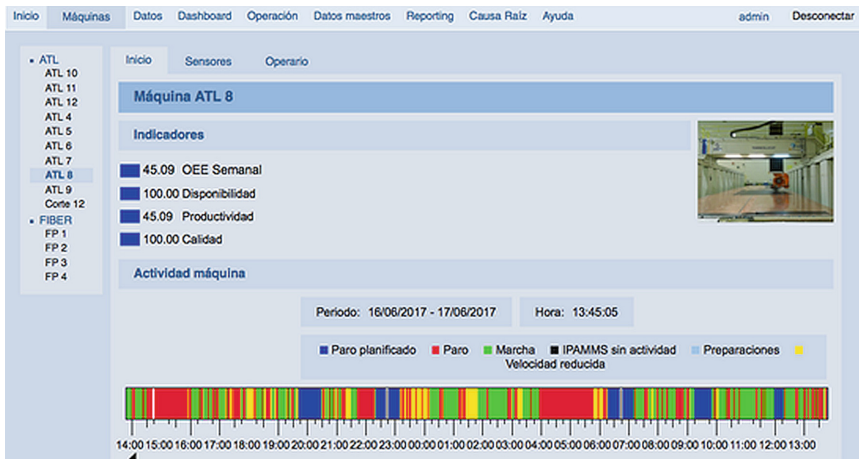


Fig. 3. Detail of the activity of an intelligent robot.

- The third basic module is *O3E Dashboard*. This module displays, in dashboards and executive reports, basic information for the management of the production, and about the prediction of failures. In this way, it provides information on the main metrics of production and maintenance to take immediate action when some KPI starts to be below the standard or when facing with a predicted breakdown. It is integrated into a screen very close to the place where the robot is located. Therefore, it also serves as a stimulus for operators to improve their work, controlling the evolution of the different KPIs.

Other modules of *oeo pro* are in charge, among other issues, of the management of work and maintenance orders, the calculation automatically and in real time of the evolution of the production, the monitoring of material consumption, the calculation of economic costs, the automatic sending of alerts and the generation of reports.

3.2 Development and Deployment Platform

The tool has been developed with the Java Enterprise Edition Seam framework, combined with Kepware (OPC server — Object linking and embedding for Process Control — that makes machine signals available for reading and interpretation) and .NET C # (services that collect the signals from the machine, provide processing semantics and insert them into the system database for further processing).

Although there are other possibilities to match client requirements, the platform can be deployed on the private cloud of Luce IT. It is equipped with real-time replication systems to a backup center to support the most serious incidents that could occur in the main data center, thus guaranteeing the achievement of a high service level agreement

(SLA). With regards to architecture management and incident management, 24×7 service monitoring is performed with real-time alerts. Moreover, the reactive resolution of incidents is combined with maintenance and anticipation operations in order to apply corrective measures before incidents occur. In addition, there is an ISO27001-certified information security management system.

4 Conclusions

oeepro is a solution for Industry 4.0 in the aeronautical sector that allows to improve production and manage maintenance more efficiently. The tool increases both productivity and product quality, it reduces production times and non-operational times due to lack of coordination and micro-stops, and also reduces costs of consumables and raw materials. In fact, through the use of this tool, reductions of 20% in lead time, 3% in the total product cost, 1.5% of the factory rate, and increments of 6% in the shop floor productive capacity have been achieved (these values being externally audited).

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Study Based on the Incidence of the Index of Economy and Digital Society (DESI) in the GDP of the Eurozone Economies

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Abstract. In the different developed societies, the different relationships between the so-called technological indicators and social development have been studied in the recent years. The aim of this work is to find a link between variables of the DESI (*digital economy and society index*) technology indicator and the GDP (*gross domestic product*) per capita for the current year. These relationships exist and are directly related to the use of certain services on the Internet by citizens and the integration of technology by companies.

Keywords: DESI · GDP · Technology · Digital transformation

1 Introduction

Over the past few years, companies in almost all industries have undertaken a number of initiatives to develop new digital technologies and explore the benefits of these [3]. The merger between the new digital technologies, called *Information and Communication Technologies (ICT)*, and traditional industrial production has largely generated what we know today as Industry 4.0 [1]. The concept of Industry 4.0 has allowed, among other things, to transform factories into intelligent environments where information, objects, and people are connected thanks to the convergence between the physical and virtual worlds through cyber-physical systems [4]. The process that has allowed progress towards Industry 4.0 has focused on the technological change that has been experienced especially in advanced economies during the last decades from the twentieth century until today. In those years, a transformation has been achieved, mainly linked to scientific progress, which has allowed a significant decrease of technological capital prices [2]. Despite of the advances that have taken place in economies such as Germany and the United States, during the last two decades some countries of the European Union have not been able to take advantage of all the benefits that the so-called digital revolution has been offering. Although the onset of the crisis in 2008 led to a reduction in spending on research and development due to some austerity fiscal policies [7, 8] and a reduction in business investment, from 2014 the situation has changed considerably. Specifically, the digitalization of certain economies is beginning to act as a growth engine [5], and from 2015 onwards, in cases such as Spain, it has been responsible for up to 30% of value-added growth [10, 11].

The digital transformation has not only been a growth engine in recent years in certain countries, such as Spain, but has also been a growth engine for the entire Eurozone. For this reason, the current work focuses on studying the relationship between the per capita GDP of the countries belonging to the Eurozone and the Digital Economy and Society Index (DESI) during the period 2015–2018.

1.1 The Digital Economy and Society Index (DESI)

The Digital Economy and Society Index (DESI) is an indicator that summarizes relevant indicators on Europe's digital performance and at the same time investigates the evolution of the European Union member states in terms of digital competitiveness. This index is produced annually by the European Commission [2]. The indicator in question is specifically broken down into 5 components and their main implications:

- **Connectivity:** Measures the deployment of broadband infrastructure and its quality. It is measured by 5 parameters: Fixed ADSL, Mobile ADSL, fast broadband, ultra-fast broadband, broadband price index. For example, access to fast and ultra-fast broadband services is a necessary condition for competitiveness.
- **Human Capital:** Measures the skills needed to take advantage of the possibilities offered by digital technology. It is measured by 2 indices: Internet users' skills, advanced skills and development.
- **Citizens' use of Internet services:** Represents a variety of online activities, such as the consumption of video calls of online content (videos, music, games, etc.), as well as online shopping and banking.
- **Integration of digital technology by businesses:** Measures the digitization of businesses and e-commerce. By adopting digital technologies, companies can improve efficiency, reduce costs and better serve customers and business partners.
- **Digital public services:** Measures the digitization of public services, focusing on e-government and e-health. Modernization and digitization of public services can generate efficiency gains for public administration, citizens and businesses alike (such as e-health and e-government).

2 Methodology

2.1 Population and Sample

Data for GDP per capita and GDP per capita of the previous year for each country in each time period have been obtained from Eurostat [3] and are expressed in millions of Euro at current prices. The data for the Digital Economy and Society Index (DESI) for each country in each time period has been obtained from the European Commission reports where the DESI index for each country is analyzed [2]. In our case we have chosen the countries related to the Eurozone, that is, we have data for 19 countries for which we have taken into account the time period 2015–2018.

2.2 Variables

In the following analysis, the variables selected are the following, so that we can observe, as anticipated, the impact of the DESI breakdown on GDP:

GDPPP = GDP per capita

GDPAN = GDP per capita of the previous year

desia = Connectivity

desib = Human Capital

desic = Use of internet services by citizens

desid = Business Integration of Digital Technology

desie = Digital public services

2.3 Estimation Techniques

For the model estimation we have used panel data, combining cross sections over several time periods. Specifically, we have followed the methodology of applying fixed effects, this being the most elementary and consistent methodology and the model to be estimated as in Eq. (1):

$$\log(\text{GDPPP}) = \beta_1 \log(\text{GDPAN}) + \beta_2 \log(\text{desia}) + \beta_3 \log(\text{desib}) + \beta_4 \log(\text{desic}) + \beta_5 \log(\text{desid}) + \beta_6 \log(\text{desie}) + u_{i,t} \quad (1)$$

The expression of the model has been elaborated by us and we have chosen to adopt logarithms as it allows us to understand the results more easily. In the model we take into account the GDP per capita of each country for the current and previous year considering the period 2015–2018 and all the disaggregated components of the DESI index during the same period. To estimate the model, we have used the R software, a free programming software, oriented to the statistical analysis, that through different libraries allows to design econometric models and to analyze them statistically.

3 Results

As a first step prior to the development of the study we have carried out an analysis of the correlations that can be seen in Table 1, where a relationship is shown clearly between GDP per capita of the previous year and GDP per capita of the current year. With regard to the rest of the correlations, it is understood as not high, the most significant relationship being that between the GDP per capita of the current year with regard to the desib variable, specifically at 0.62. Not observing a high correlation between the set of variables allows us to continue with the estimation of our model without the effect of multicollinearity that would be presented precisely by having a strong correlation between the explanatory variables of our model.

In code 1, we develop the model shown, where we can observe the clear and determined implication of the desic and desid variables in the GDP per capita indicator for the current year. In addition to these two implications, we can also observe the

Table 1. Correlation analysis.

	PIBPC	PIBAN	desia	desib	desic	desid	desie
PIBPC	1.000000	0.994836	0.481729	0.624592	0.299778	0.292481	-0.004682
PIBAN	0.994836	1.000000	0.482999	0.621272	0.298734	0.270716	-0.024390
desia	0.481729	0.482999	1.000000	0.577137	0.714044	0.485058	0.371132
desib	0.624592	0.621272	0.577137	1.000000	0.511841	0.432595	0.438125
desic	0.299778	0.298734	0.714044	0.511841	1.000000	0.259130	0.387760
desid	0.292481	0.270716	0.485058	0.432595	0.259130	1.000000	0.487801
desie	-0.004682	-0.024390	0.371132	0.438125	0.387760	0.487801	1.000000

Source: own elaboration

logical implication of the previous year's GDP per capita in the current year's GDP per capita, this implication being positive. It is important to point out that the implication of the DESI index variables is positive and is specifically in the use of Internet services by citizens and in the integration of digital technology by businesses.

Code 1. Regression Analysis.

```
Call:
plm(formula = log(PIBPC) ~ log(PIBAN) + log(desia) + log(desib) +
      log(desic) + log(desid) + log(desie), data = datest)

Balanced Panel: n = 19, T = 4, N = 76

Residuals:
    Min.      1st Qu.      Median      3rd Qu.      Max.
-0.06844738 -0.01367687  0.00052004  0.01000381  0.06409087

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
log(PIBAN)  1.3769e-01  4.8934e-02  2.8138  0.006938 **
log(desia) -2.1885e-02  6.9281e-02 -0.3159  0.753376
log(desib)  4.9375e-02  6.1260e-02  0.8060  0.423989
log(desic)  2.9505e-01  5.7769e-02  5.1073  4.918e-06 ***
log(desid)  2.7161e-01  4.3985e-02  6.1750  1.090e-07 ***
log(desie)  8.1522e-05  3.6255e-02  0.0022  0.998215
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:  0.20228
Residual Sum of Squares: 0.038294
R-Squared:  0.81069
Adj. R-Squared: 0.7216
F-statistic: 36.4002 on 6 and 51 DF, p-value: < 2.22e-16
```

Source: own elaboration

4 Conclusions

In developed economies, specifically in those economies with a clear technological base, different studies have been developed to really understand the implication that technological development has with indicators such as those related to the labor market.

According to the study carried out in this work, the implication between technology indicators and GDP per capita not only takes place but is also relevant specifically to aspects linked to the use of internet services by citizens and in the integration of digital technology by technology businesses. This result leads us to think clearly about the importance of technological development for a given country, since an increase in technology in different areas implies an improvement in per capita GDP. It would be interesting in future work to carry out a study of how the most developed countries have better technology and the percentages of investment in it. In addition to a comparison of how the adoption of technologies and their use in different parts of the economy affects the less developed countries of Europe compared to the major European powers, and to see if through the implementation of more technology in the less developed regions it helps a process of convergence between all European regions.

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Technology as a Lever for the Evolution and Recovery of the Financial and Construction Sectors in Spain

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Abstract. The economic crisis started in Spain between 2008 and 2009. During the previous years there had been a growth of the Spanish economy above 3% per year, combined with a reduction of the unemployment rate to values close to 8%, the lowest unemployment rate in the last two decades. This situation was seen by analysts and politicians as a perfect situation for the Spanish economy, which did not foreshadow the existence of the crisis that would follow, beyond a minority that warned of a bubble in the real estate sector and a possible slowdown in the economy. This study reviews the ratios and sectors that have suffered most from the consequences of the crisis and describes the need for the implementation of technology as levers to improve the productivity of these sectors.

Keywords: GDP · Spanish crisis · Subprime crisis · Financial sector · Construction sector

1 Introduction

The economic crisis suffered by Spain was not an isolated factor, as it was influenced by the subprime mortgage crisis in the United States which quickly moved to Europe and was aggravated by the real estate bubble suffered by Spain during those years [3]. The subprime crisis and the housing bubble were followed by the sovereign debt crisis, which once again had a major impact on Spain [1]. During those years, there were many cases of bankruptcy among Spanish companies, some of the most notable being the bankruptcy of Martinsa Fadesa and Bankia, both of which are related to the two sectors we are going to study in this article. These two bankruptcies would be only a part of the large number of bankruptcies and restructurings that occurred in the construction sector and the financial sector respectively. The impact in economic importance with respect to the GDP of these two sectors before the economic crisis and their subsequent decline during that crisis is one of the reasons why we have decided to choose these two sectors for our study. During the years of economic growth prior to the crisis there were two sectors, in addition to the hotel and catering industry, which grew rapidly and achieved extraordinary profits, these two sectors being construction and financial and insurance activities. In 2008, the construction sector contributed

11.30% of the Gross Added Value (GVA) of the GDP, and financial and insurance activities contributed 5.09% of the GVA [4].

Two of the most affected sectors during the crisis were those mentioned above, suffering large losses and a significant decrease in their contribution to GDP. They were the sectors with the highest number of companies in bankruptcy proceedings [4]. In 2018, construction contributed 6.23% of the GVA of the GDP while financial and insurance activities contributed 4.03% of the GVA of the GDP [4]. This reduction is very notable, since the current GDP is higher than that of 2008, but the contributions of the aforementioned sectors have not recovered. In the case of financial sector activities, and more specifically in banks, we can see that in addition to a deterioration in their importance in terms of their contribution to GDP, there is also a deterioration in their solvency, measured as the ratio between equity and assets, which went from 7.03% in 2004 to 6.03% in 2009, reaching a minimum of 5.94% in 2008 [1]. In absolute terms, the joint contribution to GDP in GVA of the construction sector and financial and insurance activities has decreased by 56,107 million euros between 2008 and 2018 [2]. The decrease in the level of importance in relation to the total GDP of the Spanish economy, both of the sector of financial and insurance activities and of the construction sector, has been the factor that has motivated the study that is presented below. We will focus on analyzing the differences in the solvency ratios in these two sectors in the period 2013–2018, coinciding with the exit from the crisis. We will use the solvency ratio as several studies have shown it to be a good indicator for measuring financial results when restructuring companies [8]. In addition, we will analyze the importance of the inclusion of technology in both sectors as a measure to improve their productivity and importance in the coming years.

2 Methodology

To obtain the data we have used the SABI (Iberian Balance Sheet Analysis System) database and for the data related to the contribution of the sectors to economic activity we have used the INE databases. We have selected companies corresponding to the construction sector and the financial and insurance activities sector, differentiating in turn between active and competitive companies in order to analyze their differences. The period chosen is from 2013 to 2018. However, we only have taken this period into account for active companies, since for companies in competition we consider the last year available before ceasing activity up to the previous 5 years. Once the companies for which we did not have data were eliminated, we were left with a total of 34 companies in tender in the financial and insurance activities sector, 948 companies in tender in the construction sector, 27,550 companies active in the financial and insurance activities sector, and 40,000 companies active in the construction sector. In order to obtain the average value of the ratios, the function of limited average was used, excluding from the sample the values belonging to the top 10% and the bottom 10% in order to eliminate the outliers that distorted the sample. Before analyzing the results, it should be pointed out that the values of the ratios obtained will be different, since the business activity of each sector is different in terms of how it operates. In addition to the study of the companies obtained through SABI, we have also carried out an

analysis of the main companies listed in both sectors on the Spanish stock exchange to see how they have evolved since the beginning of the crisis. We have focused on analyzing their price and market capitalization. The data for the analysis of the listed companies has been obtained through a tool for the analysis of the financial sector called Koyfin¹.

3 Results

The solvency ratio of the construction sector and the financial and insurance activities sector has been analyzed, distinguishing between active companies and those in competition. The solvency ratio is defined as assets divided by liabilities and denotes the ability to meet payment obligations. Ideally, this ratio should be above 1.5, although it will depend on the type of company which will be its optimum values. In any case, the ideal is that the assets are always higher than the liabilities, especially in the short term for the daily operation of the company (Table 1 and 2).

Figure 1 shows how the solvency ratios of companies in insolvency proceedings are deteriorating in both sectors, showing a trend towards difficulty in meeting their payment obligations, which would have led companies to enter into insolvency proceedings.

It can also be seen that the solvency ratio of the financial and insurance activities before entering into a tendering procedure is significantly lower than that of the construction sector, which maintains ratio levels above 2, above the ideal level.

However, if we analyze the EBITDA, we can see a progressive deterioration to negative values in the last year so that in the case of construction companies the declaration of insolvency by the company is due more to poor financial results than to a bad financing structure. Therefore, the solvency ratio would not be a good indicator to determine if a construction company is going to enter into bankruptcy proceedings (Tables 3 and 4).

Despite the fact that the period chosen for analysis (2013-2018) corresponds to the Spanish economy's exit from the crisis and high growth rates, many companies have been dragged down by poor results during the crisis period and have had to declare bankruptcy.

Figure 2 shows the differences between the solvency ratio for companies that are active in both sectors. As mentioned above, as in Fig. 1, there is a wide difference between the values of the solvency ratio for the construction sector and the financial and insurance activities. For construction companies the solvency ratio has similar values every year around values of 3.4 oscillating in a range of less than 0.2 points. These values denote stability in terms of their financial structure and maintain this stability throughout the years following the exit from the crisis. If we consider the companies listed on the IBEX 35, we can also see that from 2008 to date there have been significant losses in market capitalization and notable decreases in share prices in

¹ <https://www.koyfin.com/>.

Table 1. Financial and insurance activities (active companies).

Year	2013	2014	2015	2016	2017	2018
Ratio of solvency	59,77	56,89	60,83	62,70	56,53	59,24
EBITDA (thousand. Euros)	11,52	11,91	10,93	13,12	14,22	14,40

Source: own elaboration

Table 2. Financial and insurance activities (companies in bankruptcy).

	2013	2014	2015	2016	2017	2018
Ratio of solvency	3,03	1,67	2,18	1,10	1,23	0,92
EBITDA (thousand. Euros)	304,46	170,74	-757,68	-16647,73	-4720,44	-1501, 60

Source: own elaboration

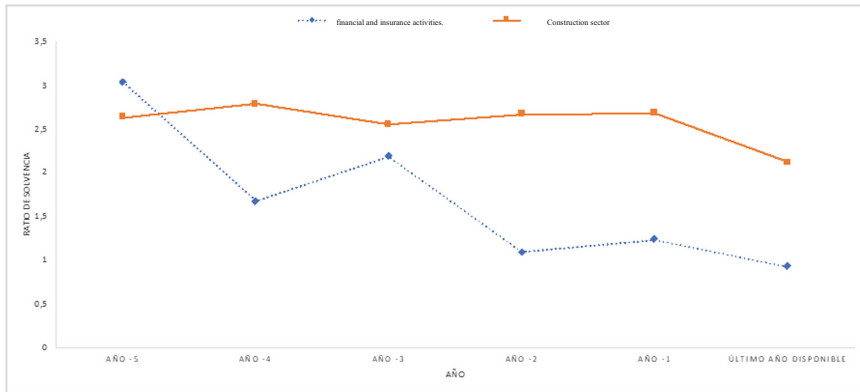


Fig. 1. Comparison of solvency ratios for companies in insolvency proceedings. Prepared by SABI. Source: own elaboration

Table 3. Construction sector (active companies).

Year	2013	2014	2015	2016	2017	2018
Ratio of solvency	3,34	3,38	3,33	3,31	3,33	3,49
EBITDA (thousand. Euros)	12,72	13,83	17,07	17,75	20,08	22,57

Source: own elaboration

Table 4. Construction sector (Companies in competition).

Year	2013	2014	2015	2016	2017	2018
Ratio of solvency	2,63	2,79	2,55	2,67	2,68	2,12
EBITDA (thousand. Euros)	106,74	107,62	74,85	47,79	14,83	-90,73

Source: own elaboration

both sectors. Due to the fact that these securities were very important in the composition of the IBEX in the pre-crisis period, the value of the index continues to be much lower than in 2007.

4 Technology as Levers for Recovery and Growth

The decrease in the importance of the two sectors studied in the Spanish economy, their difficulty in maintaining an adjusted financial structure, and their difficulty in increasing profits leads us to propose the need to introduce technology in their production process to try to solve the aforementioned problems. In the case of the financial and insurance activities sector, a very important digitalization process takes place [9], where the provision of more personalized services for each client through digital platforms such as computers or smartphones is becoming more important. This digitalization process combined with a reduction in staff and the number of offices is enabling the sector to reduce costs and increase productivity in recent years. For the construction sector, an increase in technology in its production process is necessary to increase productivity. In the case of Spain, there has been a drop-in productivity of around 0.5% between 1995-2015 measured in dollars per hour worked and per person employed [7]. To improve productivity in the industrial sector, it is necessary to introduce an element that has already been included in production processes in other sectors, automation and connectivity, giving way to an analysis of the benefits of what we call IoT [10]. In recent years we have seen the first steps in automation and robotization in the construction

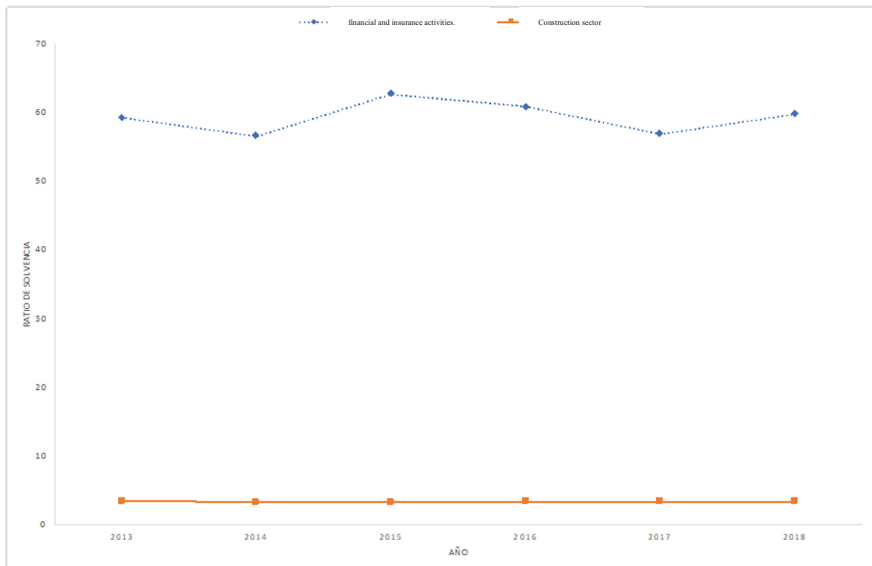


Fig. 2. Comparison of solvency ratios of active companies. Prepared by SABI. Source: own elaboration

sector, with innovations that in the coming years we will see if they result in improved productivity in the sector, since they allow for cost savings and increased efficiency.

5 Conclusions

In the sector of financial and insurance activities, solvency ratios are observed with values above 50 every year. However, there is greater variability in this sector as a complete transformation of the sector is taking place with changes in regulation and the appearance of new business models that do not allow it to have constant stability over the years. The high values of the solvency ratio in the financial sector, could be due to the fact that since the crisis there have been greater restrictions on the granting of credit and greater regulation of financial activities which has allowed them to adjust their balance sheets to obtain profits with greater financial stability. The decline in both the importance in the economy and the profits obtained in the sectors studied makes it necessary to adopt better technology in their productive processes that allows them to increase their productivity and income in order to continue developing an adequate economic activity in the future.

Acknowledgments. This work has been partially supported by the European Regional Development Fund (ERDF) through the Interreg Spain-Portugal V-A Program (POCTEP) under grant 0677_DISRUPTIVE_2_E (Intensifying the activity of Digital Innovation Hubs within the Poc-Tep region to boost the development of disruptive and last generation ICTs through cross-border cooperation).

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The Importance of Bankruptcy Prediction in the Advancement of Today's Businesses and Economies

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Abstract. The prediction of bankruptcy in companies is a problem that has concerned entrepreneurs, researchers and even governments for years, since detecting early signs that a company is going to enter bankruptcy involuntarily and being able to save it from that process, can help reduce the economic losses that bankruptcy entails, both in quantitative and qualitative terms. To try to avoid bankruptcy, it is very common to analyze the evolution of different financial ratios as it's been done in this article.

Keywords: Bankruptcy · Economic analysis · Company valuation

1 Introduction

The financial valuations of individual companies and companies grouped by sector have led to different studies on the valuation of investments at the government level and the geographical distribution of resources and companies, which have also been considered as factors to be analyzed [4], as well as territorial differences. Over the years, multiple methods have been and continue to be used to predict bankruptcy, as well as other calculations to maximize company profits [14]. As well as alternatives to succumbing to insolvency proceedings [6]. Numerous publications related to the prediction of financial behavior and evolution in companies have made use of econometric techniques [7], use of neural networks, discriminant analysis [1], or even machine learning techniques, the latter being the most recent and possible alluding to the large amount of data that companies are beginning to capture and store, which serve to develop economic models [8, 13]. In the study presented below, a series of financial variables have been established that show the situation of companies when they are active and when they are in competition, understanding companies that are in competition to be those that have entered the last year of activity available.

2 Methodology

The model set out below focuses on analyzing two financial ratios: the liquidity ratio and EBITDA (thousands of euros). These two parameters have been chosen because they are determining factors when it comes to understanding a company's solvency and

because they are parameters that allow us to visualize the most evident differences between companies with liquidity risks and those with low profits, allowing us to see which ones would be in a more delicate financial situation that could indicate the greater or lesser risk of going into insolvency. In order to carry out the study, a classification of companies by volume of operating income was made. Specifically, we have distinguished four groups according to operating income, taking the last available year as a reference:

Group (1) - Operating income from 0 to 1 million euro

Group (2) - Operating income from 1 to 5 million euros.

Group (3) - Operating income from 5 to 10 million euros.

Group (4) - Operating revenues of more than 10 million euros.

To select the companies, we used the SABI (Iberian Balance Sheet Analysis System) database, selected the range of operating income and generated a random sample. The number of companies chosen for each range is a total of one thousand companies per group. The sample of companies has been generated at random and therefore contains companies from all sectors. The data collected in the sample corresponds to the time period 2013–2018, calculating the annual average of each category of operating income. For the calculation of the average ratio values we have used a limited average function, choosing a percentage of 0.1, which means that we have eliminated the values of 10% from the top and 10% from the bottom. This has allowed us to eliminate extreme values or other values that could distort the sample. There is a sustained and gradual growth between 2013 and 2018 in all the ranges of companies' operating income, however, from an initial analysis of the data we perceive a difference in the period 2014–2015 in terms of the ratios and volumes of companies.

3 Method and Results

In the analysis carried out, it can be seen that companies, regardless of the volume of turnover, follow the same trend in terms of the growth of annual averages, something that could contradict Gibrat's law [5], a stochastic model proposed by the author in which the growth of companies is independent of size, but in our case it could not be fulfilled due to the elimination of extreme values. Therefore, the contribution of Singh A. and Whittington, in which they state that the decrease of the standard deviation with size is not as fast as if the companies were composed of subsidiary divisions that operate independently, could be assessed [12] (Tables 1 and 2).

Table 1. Operating income from 0 to 1 million euros. Source: own elaboration

Year	2013	2014	2015	2016	2017	2018
Liquidity ratio	1,96	2,02	2,16	2,26	2,41	2,56
EBITDA (thousand. Euros)	17,07	17,94	21,18	22,06	24,10	24,90

Table 2. Operating income from 1 to 5 million euros. Source: own elaboration

Year	2013	2014	2015	2016	2017	2018
Liquidity ratio	1,50	1,48	1,54	1,58	1,59	1,71
EBITDA (thousand. Euros)	93,04	109,48	129,39	143,48	156,19	166,85

In our study, companies with operating revenues of 0 to 1 million euros improved their liquidity ratio the most and companies with operating revenues above 10 million improved their EBITDA the most. This study [9] concluded the same results for companies in different sectors, where liquidity ratios were always higher on average than for smaller companies.

As can be seen in Fig. 1, the liquidity ratio is higher in companies with lower turnover. This is due to the fact that they are potentially growing companies and, coinciding with the recovery from the economic crisis, they have more facility to grow their business. On the other hand, consolidated companies have more stable liquidity ratios and hardly vary in the period studied. This is due to the fact that consolidated companies manage liquidity in a more stable manner, seeking similar values of liquidity each year, which allows them to develop their financial activity normally while adjusting the profitability of their business.

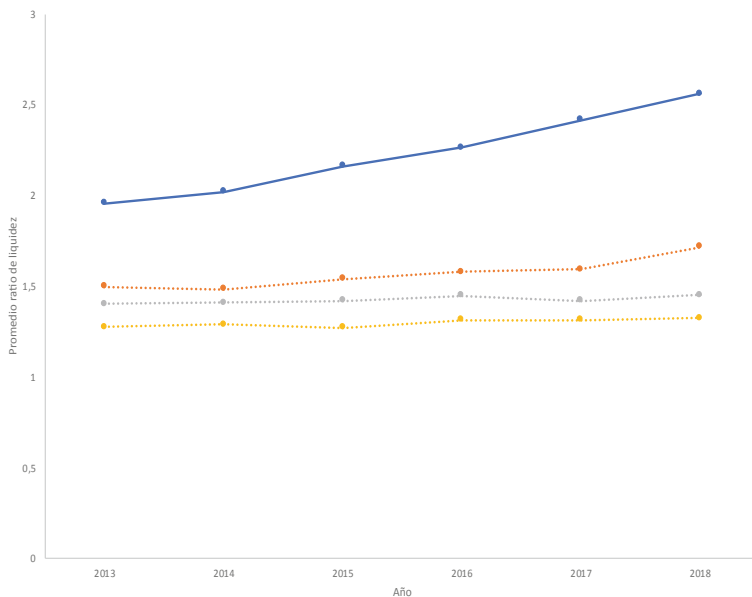
**Fig. 1.** Average liquidity ratio. Prepared by SABI. Source: own elaboration

Figure 2 shows the EBITDA for the different levels of operating income in the chosen period. It can be seen how the values of EBITDA follow a directly proportional relationship with the levels of operating income, with the higher operating income

having the highest EBITDA. If we analyze growth in relative terms, we can see that the type of companies that have increased their EBITDA most in the period studied are those with operating revenues of between 1 and 5 million euros. Specifically, their EBITDA has increased by 93.06% in that period.

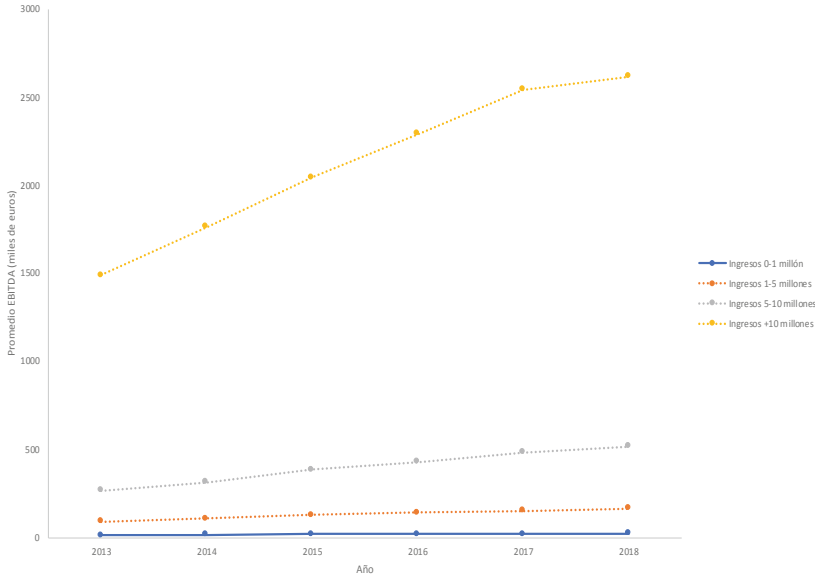


Fig. 2. Average EBITDA. Prepared by SABI Source: own elaboration

Figure 3 analyses the relationship between the liquidity ratio and EBITDA, distinguishing by level of operating income. It can be seen that the data are perfectly grouped by categories of operating income and each category has specific characteristics (Tables 3 and 4).

If we analyze it in terms of linear trends, we see how the trend changes from vertical in companies with an operating revenue volume between 0 and 1 million, to a horizontal trend in companies with an operating revenue volume over 10 million. There is a softening of the slope as operating revenues increase. This may indicate that there is a stabilization of the liquidity ratio as operating revenues and EBITDA increase, because companies with a higher volume of operating revenues and EBITDA are no longer looking so much for the company's growth but for its financial stability.

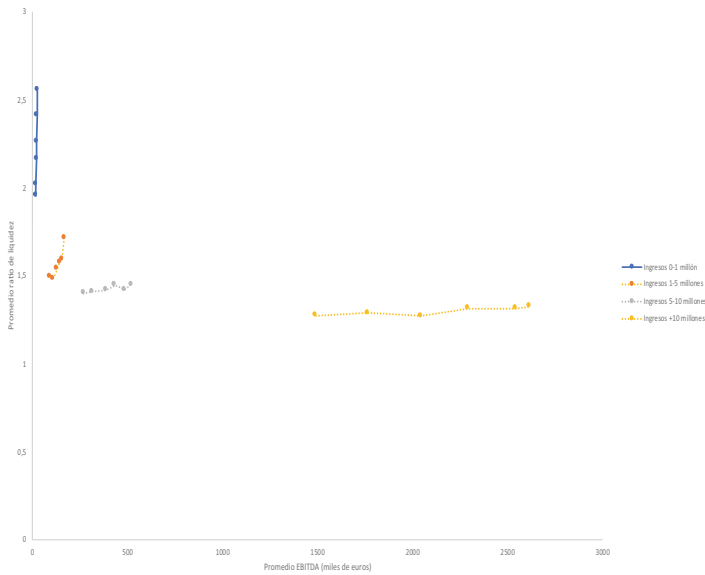


Fig. 3. Comparison of liquidity ratio vs. EBITDA Prepared by SABI. Source: own elaboration

Table 3. Operating income of 5 to 10 million euros. Source: own elaboration

Year	2013	2014	2015	2016	2017	2018
Liquidity ratio	1,40	1,41	1,42	1,45	1,42	1,45
EBITDA (thousand. Euros)	269,25	316,22	386,94	432,82	485,75	519,81

Table 4. Operating revenues in excess of 10 million euros. Source: own elaboration

Year	2013	2014	2015	2016	2017	2018
Liquidity ratio	1,27	1,29	1,27	1,31	1,31	1,33
EBITDA (thousand. Euros)	1489,54	1763,16	2046,04	2291,13	2542,43	2617,06

4 Conclusions

Once the results have been analyzed, we can draw the following conclusions in order of greatest to least importance:

As companies increase their EBITDA, there is a softening of the slope of the linear relationship between liquidity ratio and EBITDA. This is because as a company increases its EBITDA and is already in the maturity phase, it focuses on the financial stability of the company and on generating profits in a sustainable way.

- Analyzing the tables in Sect. 2 we have obtained that the greatest percentage change in EBITDA at all levels of operating income occurs either between 2013–2014 or between 2014–2015. After making a review we have associated this phenomenon to the fact that it is in these two periods when Spain manages for the first time since 2008 to achieve sustained growth in its economy. It was especially in these two periods that Spain achieved greater growth, and this allowed for the resumption of business activity and a more accelerated improvement in its economic situation. Subsequently, in the most recent years, the EBITDA of the companies continues to grow, but with a lower growth.
- - EBITDA is higher the higher the level of operating income of the companies. This relationship may be due to the fact that companies, as they increase their size and turnover, take advantage of economies of scale. This would allow them to lower production costs and increase their profits. On the other hand, the liquidity ratio is higher in companies with lower levels of operating income, since they are potentially growing companies and the conditions of the exit from the economic crisis from 2014 have facilitated the growth of their businesses.

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The Impact and Correlation of the Digital Transformation on GDP Growth in Different Regions Worldwide

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Abstract. Currently our society is experiencing a process of digital transformation worldwide, in 2016 the digital economy accounted for 22.5% of the world economy. The digital transformation has enabled the creation of new business models, the generation of opportunities and the maximization of efficiency in traditional companies that have wanted to reconvert their business model towards a new digital environment and the culture of data orientation. This document contains an analysis of how the adoption of digital technologies has a positive influence on the growth of the world economy as a whole, and particularly on the growth of some regions of the world.

Keywords: Digital transformation · Growth · Technology convergence first section

1 Introduction

Native digital companies have responded best to the digital transformation [1], maximizing profits and being more efficient with the new business models that have appeared in recent years [2, 3]. Despite the possibilities of interconnection that technological and digital advances allow, the digital transformation has not spread uniformly throughout the world, nor has it caused the same effects in all countries equally, as some have benefited more than others. One of the main reasons for this inequality is called the digital divide. The difference in access to the use of technology is called the digital divide, and it focuses particularly on the conditions and differences in access to the Internet [4] between different countries in the world. Due to the possibilities offered by the digital transformation and the differences between countries in terms of its implementation, at the last G20 summit, the digital transformation is an issue that has been included in the global agenda. It is expected to lead to more inclusive and sustainable prosperity for all countries in the world [5]. The fact that less digitally developed areas are missing out on the new capabilities offered by computing as well as, for example, artificial intelligence techniques that optimize industrial processes [6]. On the other hand, also techniques for reducing electricity consumption from which areas with fewer energy resources could benefit [7]. The reasons mentioned above, which have an international impact, lead us to analyze the inequality that exists around the benefits derived from the digital transformation in different parts of the world. To

this end, we will try to analyze the relationship between GDP and the digital adoption rate (DAI) in each continent.

2 Data and Methodology

This section analyses the relationship between GDP and the digital adoption rate (DAI), adding a GDP delay to its value in $t-2$. It will be analyzed whether there is a significant relationship between both variables and how this relationship is expressed on each continent. This study will allow us to visualize the effects that digital adoption has on each continent and how investment in digital adoption does not affect the growth of each region in the same way. The data for the study was obtained from the World Bank's database for both GDP and IAD. To show a better understanding of the study we will define the digital adoption rate (DAI). The Digital Adoption Index (DAI) is an index created by the World Bank to measure the adoption rate of technology in countries around the world. This index was introduced in "World Development Report 2016: Digital Dividends". What the authors pointed out in this report is that despite the great technological expansion that exists around the world, there is also great inequality in different areas [4]. The areas mentioned above are the benefits of technology, the quality of employment, and the ability of countries to participate in the global economy. In the case of this study, the model described below focuses on studying the benefits obtained through their influence on each country's GDP at the global level. GDP data are measured in 2010 \$ while data on the digital adoption rate are measured in 100 although they are usually measured in 1. Because of these differences, logarithms have been used in estimating the model. On the other hand, the digital adoption rate provides data for 178 countries, but due to lack of information we have had to eliminate Venezuela and Syria, so we are left with a total of 176 countries analyzed which we will group by continent. To analyze the relationship, we have defined a panel data model with fixed effects that can be defined as in Eq. (1):

$$\log(PIB_{i,t}) = \beta_1 \log(PIB_{i,t-2}) + \beta_2(DAI_{i,t}) + \varepsilon_{i,t} \quad (1)$$

Where $PIBi, t$ represents the GDP of country i in year t , $PIBi, t-2$ represents the GDP of country i in year $t-2$, and $DAIi, t$ represents the digital adoption rate (DAI) for country i in year t . We have considered only two time periods, $t = 2014$ and $t = 2016$, which are for which we have obtained the data.

The estimated model is a log-log model where an $x\%$ increase in one of the explanatory variables that are statistically significant implies an $x\%$ increase in the explained variable (*ceteris paribus*).

3 Results

The different results are detailed below. First, the entire data set has been analyzed, and then the sample has been categorized and classified by continent. For the dataset of all countries in the world (Code 1), the IAD has a coefficient of 0.36 and PIB_{t-2} has a

coefficient of 0.23. This means that a 1% increase in IAD for all countries combined would imply a 0.36% increase in PIB_t for all countries combined. Both variables are statistically significant at 5%, but the value of R^2 is nevertheless small ($R^2 = 0,40$). Although there is a relationship between the variables, R^2 indicates that there is a great dispersion in the data, and this may be due to the fact that there are great differences between the level of development of the different countries in the world. In obtaining these results on the levels of dispersion, we will study the results distinguishing by continents (Fig. 1).

```
> grun.dosvar <- plm(log(pib)~log(dai)+log(piban), data = DA)
> summary(grun.dosvar)
Oneway (individual) effect Within Model

Call:
plm(formula = log(pib) ~ log(dai) + log(piban), data = DA)

Balanced Panel: n = 176, T = 2, N = 352

Residuals:
    Min.    1st Qu.    Median    3rd Qu.     Max.
-0.160610 -0.015799  0.000000  0.015799  0.160610

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
log(dai)    0.361565    0.058383  6.1930 4.129e-09 ***
log(piban)  0.225316    0.064507  3.4929 0.0006058 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    0.65361
Residual Sum of Squares: 0.39003
R-Squared:                0.40326
Adj. R-Squared:          -0.20377
F-statistic: 58.7921 on 2 and 174 DF, p-value: < 2.22e-16
```

Fig. 1. (Code 1) World model regression analysis. Source: Own elaboration with R Software

For Europe (Code 2), IAD has a coefficient of 0.25 and PIB_{t-2} has a coefficient of 0.85. An increase of 1% in ICD for all countries as a whole would imply an increase of 0.25% in PIB_t for Europe as a whole. Both variables are statistically significant at a significance level of 5% and, unlike the model for all countries in the world, R^2 is higher ($R^2 = 0.59$). What we observe is that in this case the dispersion of data is less, that the IAD has less influence than in the previous case and that PIB_{t-2} has more influence. This may be because European countries already enjoy high levels of IAD and there is not as strong an increase in IAD as in other regions of the world, which would explain a smaller effect in predicting their level of GDP (Fig. 2).

```

> modeleuropa <- plm(log(pib)~log(piban)+log(dai), data = europa)
> summary(modeleuropa)
Oneway (individual) effect Within Model

Call:
plm(formula = log(pib) ~ log(piban) + log(dai), data = europa)

Balanced Panel: n = 41, T = 2, N = 82

Residuals:
    Min.      1st Qu.      Median      3rd Qu.      Max.
-8.5972e-02 -9.5042e-03  3.2803e-15  9.5042e-03  8.5972e-02

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
log(piban)   0.84743    0.16261   5.2113 6.417e-06 ***
log(dai)     0.24717    0.11311   2.1852  0.03495 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    0.10787
Residual Sum of Squares: 0.044121
R-Squared:               0.591
Adj. R-Squared:          0.15054
F-statistic: 28.1775 on 2 and 39 DF, p-value: 2.6827e-08

```

Fig. 2. (Code 2) Europe model regression analysis. Source: Own elaboration with R Software

For the case of Asia (Code 3), we found that IAD is the only statistically significant variable at 5%. The value of its coefficient is 0.56 and the estimate has a $R^2 = 0.68$, the highest in the study and the one that shows less dispersion of the data. This implies that an increase of 1% in IAD in Asian countries implies an increase of 0.56% in PIB_t for Asian countries. These results may be due to the fact that Asian countries, especially China, India and Indonesia, are experiencing high rates of economic growth combined with high rates of inclusion of technology among their population, which would make greater adoption of technology key to their growth (Fig. 3).

```

> modelasia <- plm(log(pib)~log(piban)+log(dai), data = asia)
> summary(modelasia)
Oneway (individual) effect Within Model

Call:
plm(formula = log(pib) ~ log(piban) + log(dai), data = asia)

Balanced Panel: n = 44, T = 2, N = 88

Residuals:
    Min.      1st Qu.      Median      3rd Qu.      Max.
-0.086438 -0.020466  0.000000  0.020466  0.086438

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
log(piban)   0.083785    0.088743   0.9441   0.3505
log(dai)     0.557991    0.082915   6.7297 3.545e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    0.20225
Residual Sum of Squares: 0.064912
R-Squared:               0.67905
Adj. R-Squared:          0.33518
F-statistic: 44.4309 on 2 and 42 DF, p-value: 4.3168e-11

```

Fig. 3. (Code 3) Asian model regression analysis. Source: Own elaboration with R Software

In the case of the Americas (Code 4), the opposite is true of Asia. In this case, only PIB_{t-2} is statistically significant at 5%, and has a coefficient of 0.77. This means that an increase of 1% in PIB_{t-2} means an increase of PIB_t by 0.77%. R^2 has a value of 0.61. In this case digital adoption would not be a statistically significant variable and therefore a variable that serves to predict PIB , and its value would be determined by PIB_{t-2} .

For the case of Africa (Code 5), none of the variables is statistically significant at 5% and in addition the value of R^2 is a very low value, namely 0.18. What this indicates to us is that none of the variables that we have used in the study are statistically significant to predict PIB_t and furthermore there is a great dispersion in the data. This is because Africa is a country with the largest number of underdeveloped countries in the world where there are other factors that can be more decisive for its economic growth than the adoption of technology, such as an improvement in human capital or the health conditions of its population (Figs. 4 and 5).

```
> modelamerica <- plm(log(pib)~log(piban)+log(dai), data = america)
> summary(modelamerica)
Oneway (individual) effect Within Model

Call:
plm(formula = log(pib) ~ log(piban) + log(dai), data = america)

Balanced Panel: n = 34, T = 2, N = 68

Residuals:
    Min.      1st Qu.      Median      3rd Qu.      Max.
-5.4251e-02 -1.2270e-02 -3.1134e-15  1.2270e-02  5.4251e-02

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
log(piban)   0.768875   0.151938   5.0604 1.667e-05 ***
log(dai)     -0.065026   0.111422  -0.5836   0.5636
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    0.066502
Residual Sum of Squares: 0.025963
R-Squared:               0.60959
Adj. R-Squared:          0.18257
F-statistic: 24.9821 on 2 and 32 DF, p-value: 2.9135e-07
```

Fig. 4. (Code 4) Regression analysis American model. Source: Own elaboration with R Software

```

> modelafrica <- plm(log(pib)~log(piban)+log(dai), data = africa)
> summary(modelafrica)
Oneway (individual) effect Within Model

Call:
plm(formula = log(pib) ~ log(piban) + log(dai), data = africa)

Balanced Panel: n = 46, T = 2, N = 92

Residuals:
    Min.    1st Qu.    Median    3rd Qu.    Max.
-0.160691 -0.021819  0.000000  0.021819  0.160691

Coefficients:
              Estimate Std. Error t-value Pr(>|t|)
log(piban)   0.13851    0.13689   1.0119  0.31712
log(dai)     0.27882    0.15125   1.8434  0.07201 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares:    0.23574
Residual Sum of Squares: 0.1936
R-Squared:    0.17874
Adj. R-Squared: -0.69852
F-statistic: 4.78803 on 2 and 44 DF, p-value: 0.01314

```

Fig. 5. (Code 5) Regression analysis African model. Source: Own elaboration with R Software

4 Conclusions

The adoption of technology has different impacts and contributes to growth in different ways depending on the region of the world being considered. In the model presented above, it can be seen how IAD is a statistically significant variable for economic growth globally as a whole (despite the wide dispersion of data), in Europe and Asia, it influences all regions positively. However, there are regions such as Africa and the Americas where it is not. As concluded in the “World Development Report 2016: Digital dividends”, not everyone has benefited equally from the expansion of technology around the world and the introduction of technology in companies, people and governments is not a measure that guarantees the economic growth of countries, and it is the inhabitants themselves who can develop strategies to promote digitalization in their countries as in the case of M-Pesa in Africa [8]. This study offers many lines of research, such as the study of the index by disaggregated components and how each one of them influences the growth of the different regions of the world. It could also be used to analyze similar models, but which include other variables such as human capital, expenditure on state GDP or expenditure on R&D.

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A Review of k-NN Algorithm Based on Classical and Quantum Machine Learning

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Abstract. Artificial intelligence algorithms, developed for traditional computing, based on Von Neumann's architecture, are slow and expensive in terms of computational resources. Quantum mechanics has opened up a new world of possibilities within this field, since, thanks to the basic properties of a quantum computer, a great degree of parallelism can be achieved in the execution of the quantum version of machine learning algorithms. In this paper, a study has been carried out on these properties and on the design of their quantum computing versions. More specifically, the study has been focused on the quantum version of the k-NN algorithm that allows to understand the fundamentals when transcribing classical machine learning algorithms into its quantum versions.

Keywords: Machine learning · Supervised learning · k-Nearest Neighbors · Quantum computing · Quantum k-NN

1 Introduction

Quantum mechanics is a theory in physics which describes nature at the smallest scales, including atomic and subatomic, and their interactions, as observable quantities. This theory arose to explain observations which could not be properly explained using classical physics. Some examples of first usage of concepts nowadays included in quantum mechanics include the Max Planck's solution to the ultraviolet catastrophe of the black body radiation [29] and the Albert Einstein's explanation of the photoelectric effect [7]. These results lead to suggest the hypothesis of the quantization of energy and that the interaction of an electromagnetic wave with matter occurs by means of elementary and indivisible processes.

In quantum mechanics, the state of a system at a given time is described by a complex wave function, or state vector in a complex vector space. This allows for

computing the probability of finding an electron in a particular region around the nucleus at a particular time. However, it cannot make a simultaneous prediction of related variables, such as position and momentum, to any precision. This is known as the Heisenberg's uncertainty principle which stated the inability to precisely locate the particle given its momentum and conversely [17].

The quantum mechanics has influenced other disciplines, such as chemistry, optics, electronics and information science. Richard Feynman and other authors have proposed that a quantum computer could perform simulations that are out of reach for regular computers [9]. The complexity of the algorithms can be different when executed in a quantum computer. Both problems, integer factorization and discrete log [42], are in polynomial complexity when using quantum computation, however, both are NP problems when using classical computation (although are suspected to not be NP-complete). Maybe because of the heavy computational load it needs, quantum computing is a emerging research area in the discipline of artificial intelligence and specifically applied to the machine learning [38].

This paper is organized as follows. Section 2 describes classical machine learning approaches and algorithms. Section 3 introduces background and terminology about the quantum computing and describe the mathematical background underlying in quantum physics and quantum computing, as well as the improvements obtained by applying quantum computing theories to machine learning algorithms and processes. Section 4 details how the K-Nearest Neighbors (KNN) algorithm is adapted to a quantum version. Finally, conclusions and future work are depicted in Sect. 5.

2 k-NN Algorithm Based on Classical Machine Learning

Artificial Intelligence (AI) aims to develop and use computer systems to reproduce the processes of human intelligence, necessary for learning, understanding, problem solving or decision making. AI is therefore a broad discipline that brings together several fields such as *natural language processing*, *expert systems* [8], *multi-agent systems* [1, 43], *recommender systems* [5, 12, 14], voice analysis and conversion to text (*speech to text* and *text to speech*) [22], *computer vision* [34], *planning systems* [30], *evolutionary computation*, *robotics* and, in the case of our study, *machine learning* [3, 4].

According to the different definitions of Artificial Intelligence by [36], *machine learning* falls into those definitions related to *thinking humanly* (related to cognitive systems). Thus, *machine learning* refers to the construction of computer programs that automatically improve their performance in a given task with experience. In machine learning, algorithms are used to parse and learn from data (*training data*). After that, the algorithms make predictions and make decisions about events in the real world. Machine learning algorithms can be subdivided into five main categories: *supervised learning*, *unsupervised learning*, *semi-supervised learning*, *ensemble learning* (also known as *integrated learning*), *deep learning* and *reinforcement learning*, although classification varies over time depending on the importance of each of the groups.

2.1 Supervised Learning

In the case of supervised learning, the machine has a supervisor who indicates an algorithm the result that should be obtained as output for a set of input data. That is, the machine learns about a *labeled dataset*. After learning, the algorithm is able to predict the output for a new combination of input data [40]. Thus, the basic objective of supervised learning is, given a relationship between input variables $\mathbf{X}$ and output variables $\mathbf{Y}$, to learn an objective function f that better maps these input variables to the output variables.

The usual in supervised learning will be that we have a labeled dataset and that we split that dataset in a *training dataset* so that our algorithm finds the relationship f and in a *test dataset* (eliminating the labels) to check the goodness of the function we have obtained with the selected algorithm (i.e., *cross-validation*). An example would be that in a set of image data we indicate for each one which ones correspond to “cats” and which ones correspond to “dog”. In this kind of case, where we expect to get a category on the way out, we are talking about *classification* problems. If, on the other hand, we expect to get a continuous numerical value at the output, for example, if we want to predict the value of an action over time, we are talking about a *regression* problem [13].

Regression algorithms are used to predict continuous numerical values rather than discrete variables or categories. This allows us to apply it to predict the value of shares, predict demand or sales volumes, such as electricity consumption [13], make medical diagnoses, and in general, any prediction in time series. Depending on the form of the function found, we find methods of *linear regression*, *polynomial regression* or more modern methods such as *lasso regression* or *ridge regression* (*Tikhonov regularization*) [15].

Classification algorithms are now widely used for applications such as spam filtering, sentiment analysis, language and similar document detection, handwritten character recognition, fraud detection or loan/risk evaluation. Among the most commonly used algorithms are *logistic regression* [31]; *decision trees* (such as ID3, CART, C4.5 or MDL), used in the diagnosis of diseases in medicine or in determining the granting of a loan, among many other possibilities [27]; *classification rules* techniques, such as PRISM; *Naive Bayes classifiers* [2016convergence]; *Support Vector Machines* [6]; or, focusing on the research of this work, the *K-Nearest Neighbors* algorithm [18].

k-Nearest Neighbors (k-NN) Algorithm. Like decision trees, the *k-NN* (*k-Nearest Neighbors*) [18] algorithm falls into what are known as *non-parametric learners*. That is, unlike *parametric learners*, this type of method does not require a predefined parametric function $\mathbf{Y} = f(\mathbf{X})$. This makes this type of algorithm suitable for those situations where the relationship between $\mathbf{X}$ and $\mathbf{Y}$ is too complex to be expressed as a linear model. In the k-NN algorithm, each instance is represented as a vector and to classify or make a prediction on an input data, the closest k points are taken and the average of their values is calculated, if we are working with continuous data, like the estimated value of a house, or its mode, if we are working with categorical data, like determining the breed

of a dog. The selection of the k hyperparameter is done by cross-validation, choosing the k that has the least error, on average, along the different iterations. This algorithm is used as a method for classification, as a fraud detection, as a method for regression, as a house price prediction, or to impute missing training data, imputing the average or mode of the neighbors instead of a missing value [19].

3 Quantum Computing in Machine Learning

Although it has not yet been possible to develop commercial hardware for the creation of quantum computers, we can design, albeit theoretically, the algorithms that will run on them [21]. One of these algorithms, possibly the most famous of them all, is Shor's algorithm [2]. Shor's algorithm shows that the factorization problem of very large numbers, for example those used in RSA keys, could be done in a time in the order of seconds. Currently, to factor a very large number, with computers based on the Von Neumann architecture, it would be necessary to wait a time in the order of years. That is why, nowadays, to secure the connections between devices through the Internet, RSA key pairs are used. With the implementation and commercialization of the first quantum computers, computer security as it is currently implemented, would become obsolete.

In the field of the machine learning, it has designed several algorithms too, like running subroutines of computationally expensive and the translation of stochastic methods into the language of quantum theory [39]. In [33] it has been proposed a Quantum Support Vector Machine(Q-SVM) for classification for Big Data. There it is detailed how a Q-SVM could be implemented in $O(\log NM)$ completion time, for both training and classification stages.

Other work assessing the capacity of Q-SVMs is [16]. It shows that when an SVM is used for a dataset with many features, the computation done by the kernel of the algorithm is very expensive. To solve that problem, it is proposed to use two methods of Q-SVMs that takes advantage of the exponentially large state space that characterizes quantum computing, through controllable entanglement and interference features. A similar approach is proposed in [37].

3.1 Quantum Computing: Background

In order for the design of algorithms within the field of quantum computing to improve performance with respect to their counterparts in classical Von Neumann's computing, quantum mechanical effects must be used. These effects are those that govern the subatomic world as the superposition of states, which says that a subatomic particle, until the time of its measurement, is in a multitude of states intermediate to the two typical states of operation in classical computing (0.1). Thanks to this characteristic it is possible to carry out any operation, the system allows to evaluate all the possibilities in only one step, that is to say, it carries out a parallel computation in a natural way; whereas classically, this evaluation process is carried out in sequential steps [2].

In classical computing, the smallest unit on which to store information is called a bit. A single bit can only store one value between two possible values: 0 or 1. Similarly, in quantum computing, the elementary unit is a qubit. The state of a qubit is determined by the composition of the probabilities it has of being, at the time of measurement its value, in one of its two possible states, $|0\rangle$ and $|1\rangle$.

The states in the q-bit are represented by vectors, belonging to a Hilbert's space L^2 [28]. Being its final value, at the time of measurement, determined by the sum of the probabilities it has of being in each of its possible states $|\psi\rangle = a_0|0\rangle + a_1|1\rangle$, where a_0 and a_1 are complex numbers that follow the normalization relation: $|a_0|^2 + |a_1|^2 = 1$. This means that a given qubit will have, when it is looked, a 0 value with a probability of $|a_0|^2$ and a value of 1 with a probability of $|a_1|^2$.

To work with qubits, it is necessary to use structures where they can be stored, the quantum registers. To represent the information stored in any quantum register, it is used the tensor product of the qubits that are stored in them [41]: $|\phi\rangle = |q_0\rangle \otimes |q_1\rangle$, where $|q_0\rangle$ and $|q_1\rangle$ are the qubits comprised in the quantum registry ϕ and the $\otimes$ is the tensor product symbol.

The state of the record, where it is shown the 4 basic states in which the 2 qubit register can be found, it is defined with $|\phi\rangle = a_0|00\rangle + a_1|01\rangle + a_3|10\rangle + a_4|11\rangle$, where the probabilities follow the rule $1 = |a_0|^2 + |a_1|^2 + \dots + |a_n|^2$.

In quantum computing, as in classical computing, it is necessary to design and implement circuits that allow the execution of an algorithm. In this case, quantum logic gates are used for the application of logic operations on sets of qubits. The main difference between classical and quantum logic gates is that classical logic gates work on a finite set of values, while quantum logic gates work on continuous data of the Hilbert space [45].

Although, due to the nature of qubits, it is thought that there can be infinite quantum logic gates, it has been demonstrated that by using only two universal logic gates, it is possible to perform any unitary transformation on a set of N qubits [10].

Thanks to the parallelism obtained from the application of the quantum physics theory, it is possible to reduce the execution time of the algorithms that are based on the classical computing based on Von Neumann's architecture, sequential by nature, to an order of logarithmic execution time in either both, supervised and non-supervised machine learning [20].

4 Quantum Algorithm for the K-Nearest Neighbors Classification

The importance of the k-NN algorithm lies in the fact that it is very often used as a subroutine in many machine learning models. For this reason, and as a search for its optimization, quantum versions of this algorithm have been designed, with a theoretic execution time on the order of $O(N)$ [35]. On the other hand, the

k-NN algorithm has a time complexity of $O(NxM)$ [32], where N is the training samples and M the number of dimensions in the dataset.

The basic operation of this algorithm is to measure distances between points in a data set. In this way, when it is intended to classify a sample, the distance between the sample to be classified and the rest of the training dataset is calculated. Examples of, simple and easy to implement, distance measurements algorithms are Euclidean, Manhattan and Hamming distance.

In order to tackle the problem of the computational cost of the k-NN execution, some authors have proposed a quantum version of the algorithm. Making use of the state superposition properties of quantum particles provides a great parallelism in its execution. This is due to the fact that each element of a quantum superposition state is operated simultaneously.

It is possible to demonstrate, although in a theoretical way, how making use of the quantum properties and mechanisms of the state superposition of the quantum particles, we are able to reduce the execution time of an algorithm like k-NN. While in traditional computing it is only possible to store, in a n bit register, a number of the set $\{0, 1, \dots, 2^n - 1\}$, in a n qubits register, it is possible to store that whole binary set of numbers with their corresponding probability $|c_i|^2$.

Thanks to the parallelism obtained by the properties offered by the quantum mechanics of theoretical quantum computers, the speed of execution of the quantum version of the algorithm can be improved. To design this adaptation, we can follow the next steps [35]:

- Prepare the dataset. Data characteristics must be converted to bit vectors, which are mapped for use in quantum computing. The data set can then be represented by N feature vectors $|v^p\rangle$, with $P = 1, 2, \dots, N$ and whose corresponding class is $c^p \in \{1, 2, \dots, l\}$, being able to be represented as $\{v_1^p \dots v_n^p, c^p\} \in \mathbb{H}_2^{\oplus n} \oplus \mathbb{H}_l$. Equation 1 shows the representation of the training set superposition.

$$|\tau\rangle = \frac{1}{\sqrt{N}} \sum_p \{v_1^p \dots v_n^p, c^p\} \quad (1)$$

- Take a sample and normalize it as a n -dimensional feature vector: $x_1 \dots x_n$. put in a register the unclassified quantum state and the τ training set in another register, with an ancillary qubit $|0\rangle$ in a third one. In Eq. 2 is shown the result of this step:

$$|\phi_0\rangle = \frac{1}{\sqrt{N}} \sum_p \{x_1 \dots x_n; v_1^p \dots v_n^p, c^p; 0\} \quad (2)$$

- Calculate the distance between the sample to be classified $\{x_1 \dots x_n\}$ and the training set $\{v_1^p \dots v_n^p, c^p\}$. Any distance metric can be used for this, e.g. in [35] it is used the Hamming distance metric. The result of the metric is then stored in the first register as $\{d_1^p \dots d_n^p\}$. Equation 3 shows the state of the registers.

$$|\phi_1\rangle = \frac{1}{\sqrt{N}} \sum_p \{d_1 \dots d_n; v_1^p \dots v_n^p, c^p; 0\} \quad (3)$$

- The training set of data, that are closer to the chosen sample than a t threshold value, are labeled in this step. The k data that is closer than the threshold is changed to the auxiliary qubit $|0\rangle$ to $|1\rangle$. In Eq. 4 it is formally defined this step:

$$|\phi_2\rangle = \frac{1}{\sqrt{N}} \left(\sum_{p \in \Omega} |d_1 \dots d_n; v_1^p \dots v_n^p, c^p; 1\rangle + \sum_{p \notin \Omega} |d_1 \dots d_n; v_1^p \dots v_n^p, c^p; 0\rangle \right) \quad (4)$$

where Ω determines the indices of the training data, whose distance to the sample is less than the threshold t .

- Finally, the labels with a $|1\rangle$ range in the auxiliary qubit are selected from the set of samples by means of a projection operator $\Gamma = \mathbb{I} \oplus |1\rangle\langle 1|$. Equation 5 shows how the samples have been selected, having only the vectors $|v^p\rangle$ that are k closest to the sample to classify:

$$|\phi_3\rangle = \Gamma|\phi_2\rangle = \alpha \sum_{p \in \Omega} |d_1 \dots d_n; v_1^p \dots v_n^p, c^p; 1\rangle \quad st. \sum |\alpha|^2 = 1 \quad (5)$$

being α the renormalized amplitude of each component in $|\phi_3\rangle$. Measuring c^p , we are able to determine now the category the sample x^p belongs to.

5 Conclusions and Future Work

In this work, we have made an introduction to the quantum mechanics and how they can be applied in a theoretical quantum computer. To make more concrete the study of this work, we have made a thorough study of machine learning algorithms, along with their classification, which will complement the study of the implementations of machine learning algorithms with quantum physics.

It has been shown some studies that have designed a quantum algorithm in the field of machine learning, obtaining in all of them, a great improvement in the theoretical execution time of their classical counterparts. In addition, it has been done a study of some of the flaws that have to be overcome in order to finally implement a quantum computer.

Finally, we have studied, in a minimalistic way, how it is designed the QKNN algorithm. Thanks to this study, the foundations have been laid on studies on other quantum algorithms and in the design of new ones. Also, it will help people that are starting to study this field, by having a good study of the art in this work.

This work will be continued in another study in which the implications of the quantum era will be shown, in the blockchain field [11]. The use of blockchain technology is one of the hottest topics in the world, since it brings an optimization and improvement to most of the current IoT systems [23–25]. In addition, and due to the high dependence that this technology has on the use of cryptographic keys, for the user's identification [26, 44], this technology will be seriously affected if Shor's algorithm is implemented in a commercialized quantum computer.

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Doctoral Consortium



Pragmatic Software Maintainability Management Using a Multi-agent System Working in Collaboration with the Development Team

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Abstract. This paper introduces an ongoing PhD work that focuses on the design of a multi-agent system that would assess software maintainability in collaboration with human developers. Evaluation of the maintainability is difficult to implement on actual industrial projects as specific criteria cannot be easily adapted to the context of a project, and metrics are not always relevant to the architectural pattern in use. The metrics can be bypassed without improving the quality of the source code. We have the ambition to tackle this problem through a more modular, flexible and dynamic assessment system that will ultimately evolve through a dialogic interaction with the development team. Our approach will first lay the foundations of a shared vocabulary between the artificial agents and the human team. We will further develop a multi-agent system first by introducing feedback from the human team and second by moving towards a real dialogic interaction.

Keywords: Software quality management · Distributed artificial intelligence · Explainability

1 Problem Statement

The evaluation of software quality, i.e. its conformity, maintainability and readability, is of crucial interest for all development activities. The main existing approaches fit into the three following categories:

- Static analysis of the code allows the application of many automatic controls based on existing or context-specific set of rules.

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- Automated dynamic testing controls the running software and make sure it behaves as expected.
- Manual peer reviews and tests often done by experts on the technology and/or the business of interest.

These approaches however have two limits as they are time-consuming and require contextual adaptation. In addition, developers often bypass them by artificially adapting the code and complying effortlessly to the project’s quality rules without respecting the “spirit of the rule”, which paradoxically degrades quality.

2 Related Work

Software quality is inherently multi-modal [4], and its evaluation has required the establishment of standards (e.g. ISO 9126) that have evolved considerably over time [6]. The literature also refers to software quality models, which defines a structured set of characteristics.

Here we focus more specifically on maintainability that can in turn be subdivided into readability, modifiability, stability and testability [2,4]. Using a maintainability model, maintainability can be evaluated using easily measurable metrics such as volume, cyclomatic complexity or code duplication [2]. Metrics are also a very active field of study that is based on empirical findings [8] but still needs to be confronted with industrial issues.

Moreover, objective metrics and subjective evaluations by developers do not systematically converge in the most complex cases [9]. The best alternative could be to mix both methods [9].

Another issue concerns the aggregation of metrics [3] to determine the maintainability of higher-level software components. Indeed, the effective maintenance of a software requires prioritizing the efforts of the development team. The results of such a model is significantly correlated with the experts’ evaluations. We state an agents-oriented approach could increase the extensibility and flexibility of the used model.

Finally, tests can also provide information regarding the quality of the code [1]. While the number of asserted objects within a unit test is positively correlated with complexity, a development team making minimalist assertions would perform artificially well on this criterion. There is a clear competition between test quality and production code quality. Several agents with different evaluation objectives could effectively manage this competition.

3 Hypothesis

We hypothesize that assessing maintainability with a modular, flexible and dynamic system in collaboration with the development team will greatly improve software quality, thus allowing effortless portability between projects.

4 Proposal

The objective of this PhD work is to develop a multi-agent system in order to evaluate and control the maintainability of a software artifact. To meet the industrial constraints of software engineering, this multi-agent system should be easy to implement, portable, flexible and explainable.

Ease of implementation is essential to limit the start-up costs of a project. The adoption and operation of agents into the development process would otherwise be too difficult and costly hence either delayed or impossible.

Portability will make it possible to take advantage of the achievements on similar projects, to pool costs and to increase the consistency of quality criteria across many projects.

Flexibility will allow agents to be easily and efficiently adapted over the course of the project's life, complying to the behaviors of different developers within the team.

Explainability will facilitate developer-agent interactions. The internal logic must be accessible and understandable in order for the operators to easily assess the relevance of the agents' feedback and correct it if needed.

5 Evaluation Plan

First, we will establish a common vocabulary shared between humans and artificial agents (or programs) to talk efficiently about maintainability. Works done on that matter [3,9] will provide an effective starting point.

Secondly, we will proceed with the development of a multi-agent system to evaluate maintainability with the help of classical metrics [2,4] and machine learning methods. This system will be evaluated in real conditions on an industrial project.

Third, we will introduce a feedback flow from the human development team, looking for quick convergence and effective portability of the multi-agent system. Moreover, the introduction of neural network agents to assess maintainability could help capture more complex causal relationships.

Finally, we wish to create a true dialogic interaction between agents and the human development team, an interaction that considers the different profiles, their experience and the context of the project. The latest works in explainability [5,7] will have to be harnessed to create a true meaningful and effective interaction.

6 Reflections

Through this work we hope to contribute to the establishment of a meaningful paradigm for small- and large-scale maintainability within a project or more broadly of the information system as a whole.

In addition, we wish to produce a methodology to implement a multi-agent system working in collaboration with a human team, providing a true dialogic interaction between artificial and human agents.

7 Conclusion

In the long term, we aim at developing an effective tool to interactively and continuously manage maintainability of large-scale projects.

To do so, we adopt a mixed approach of human-centered and iterative design that include both continuous feedback from development teams and the latest research from the scientific community.

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System Architecture Modelling Framework Applied to the Integration of Electric Vehicles in the Grid

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Abstract. A structured approach for the design of a model architecture is presented. From a simplified representation of the current situation, the aim is to implement regulations and test solutions in order to reach future climate targets. The challenge is to combine model accuracy to minimise uncertainty of the results on one side, and ease of implementation/plausibility of scenarios on the other side, which often decreases as systems complexify. A stage-wise model expansion is suggested to balance these two criteria. This approach is applied to the case of increased grid integration of electric vehicles (EVs) in Denmark.

Keywords: Modelling architecture · Stage-wise approach · EV grid integration

1 Introduction

The combination of electric vehicles (EVs) and smart grids is increasingly considered as a promising solution to solve both the decarbonisation of the transportation sector and the increasing grid power variability due to the uptake of renewable energies.

Indeed, EV batteries for passenger cars can have a storage capacity of up to 100 kWh [1], which represents approximately a 10-fold of the daily consumption of an average household in Denmark [2], and this storage capacity keeps increasing.

However, this large battery storage capacity also presents some challenges, as it puts a lot more stress on the existing grid infrastructure, particularly on the distribution level [3], quickly leading to grid overloads or voltage and frequency deviations.

In recent years, many academic research efforts focused on proposing innovative market frameworks to minimise the additional grid infrastructure costs of integrating EVs. While first models have been from a top-down coordinated control approach [4, 5], the increasing use of simulations and particularly agent-based modelling (ABM) simulations have recently encouraged market-regulated bottom-up approaches [6, 7].

A major flaw of the models presented in the previous paragraph, along with many other models, is the lack of consideration for the plausibility of the proposed scenarios, with focus staying on the accuracy of results and their effectiveness at achieving the desired climate goals. This work therefore proposes the setup of an architectural framework to ensure innovative solutions are tested in scenarios with minimised implementation risk/high plausibility. The approach is inspired by the business ecosystem modelling framework proposed in [8], which proposes a three-part methodology: architecture development, analysis of influential factors and simulation of proposed changes.

While the methodology is applied to the EV-grid integration case, the aim is to develop a method to test regulations and market mechanisms in the energy sector overall.

To present this work, this paper will first present the stage-wise approach of the model setup, before illustrating it with the specific case of electric vehicles. A final section links the proposed model architecture to an agent-based modelling simulation.

2 Method: The Stage-Wise Architecture

The modelling framework will first focus on a minimum viable ecosystem (MVE), where the main stakeholders at the level of interest are identified and their main interactions are defined. Interactions are separated into different types, which for now are categorised as product, data, information, regulatory and monetary flows. These different flow types relate to the interoperability layers defined in the Smart Grid Architecture Model Framework in [9], developed in an effort to standardise Smart Grid design at a European level. The more flow types a model contains, the more layers it covers in the framework, and therefore the more holistic the modelling approach becomes.

The outcome of this base-case scenario is compared to the fixed objectives within a certain timeframe. For Denmark, this would be a 70% reduction in carbon-equivalent emissions from 1990 to 2030 [10]. A sensitivity analysis on the model parameters is done to identify the key drivers within the system. Due to the limited amount of interactions in the base case, this reduces the number of parameters to a small amount.

In the next stage, the model is expanded to include more actors which are less directly involved but allow to model interactions in more detail. This allows to test whether the difference between the MVE results and the fixed objectives is simply due to modelling inaccuracy or not.

As a next step, a regulation agent is added to the model, to represent regulation updates from the government and regulatory authorities. The regulation agent is connected in priority to agents from the MVE, especially those with critical parameters identified in the first step. This ensures that impactful measures are focused on highly engaged stakeholders, which are more likely to react. The regulations should also prioritise individual agents over populations, as the latter have inevitable variability and are more likely to introduce uncertainty to the outcome of the regulations.

The next step introduces the different solutions to test within the constraints set by the regulation agent. The regulation agent therefore acts as a filter, quickly

disqualifying any solution not fitting the constraints or yielding poor results within the given constraints. If different solutions are tested simultaneously, they should each focus on different parts of the system and avoid interacting with each other, as this increases the uncertainty of their output and reduces the likelihood of being implemented.

3 The Architecture for EV Grid Integration

The stage-wise approach is illustrated in Fig. 1 for the EV integration case.

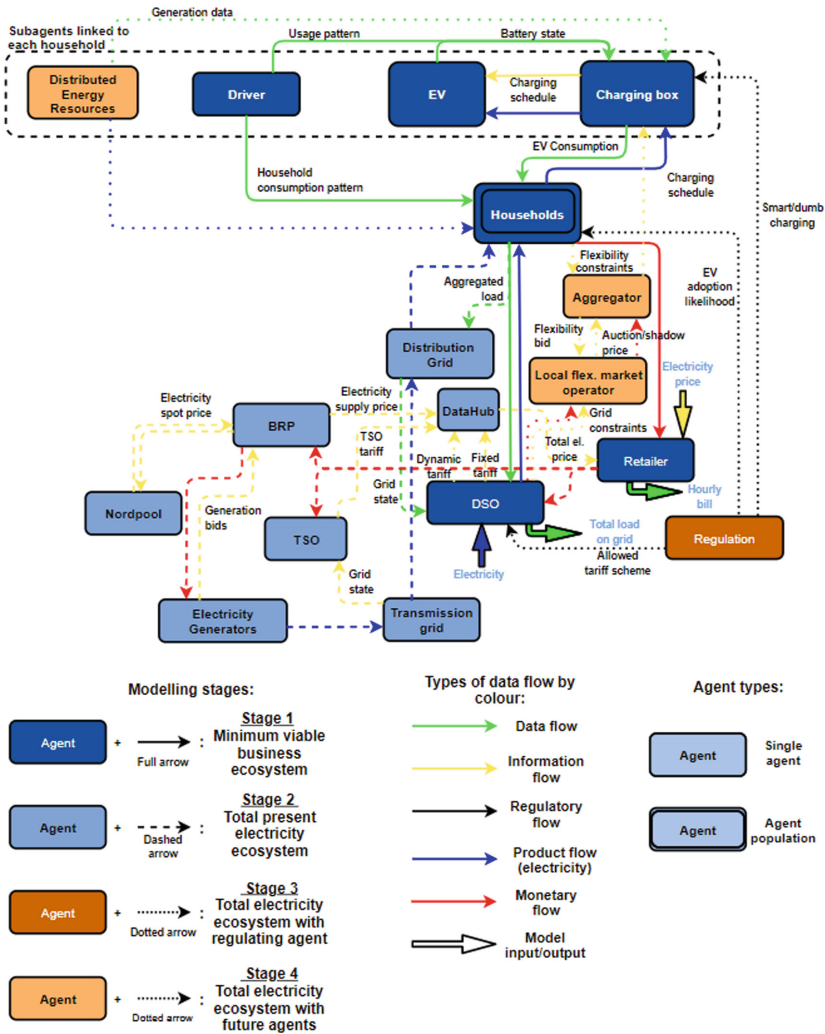


Fig. 1. Model architecture for EV integration case

In the MVE (dark blue actors with full arrows), most drawn-out flows are data flows. This basic ecosystem mainly relies on actors communicating the output of their actions, without taking the action of other actors into account. As more stakeholders are added, more information is exchanged. Information flows occur as stakeholders' actions become more dependent on others' behavior and more data processing is needed. Examples are charging scheduling, day-ahead price tariffs, forecasts...

The regulations implemented in stage 3 only focus on the MVE agents. In stage 4, two solutions are shown: the addition of distributed energy resources and the introduction of a flexibility market. Each solution acts on different parts of the system: the first focuses mainly on data and electricity flows; the latter focuses more on information and financial flows. This avoids too much interactions between the proposed solutions.

4 Future Work

This model will be implemented using ABM simulations in AnyLogic. Once the first and second stages are implemented and validated, the regulations proposed in the third stage need to be based on a thorough literature review of Denmark's regulation history. The market mechanisms and solutions tested in stage four should undergo a first screening process based on plausibility and accuracy criteria before being added in AnyLogic. Scenarios will be run in increasing order of complexity and decreasing order of plausibility.

5 Conclusion

The adequate system integration of a proposed solution follows a four-step approach: identification of stakeholders with highest invested interest and influence, grouped into an MVE; extension of the ecosystem to peripheral players to enrich the type of agent interactions; introduction of a regulation agent focusing in priority on the MVE for high-plausibility and low uncertainty regulations; testing of proposed solutions in the model, while avoiding the overlap of solutions in the ecosystem.

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Hierarchical Coalition Formation in Multi-agent Systems

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Abstract. Coalition formation has been addressed in multi-agent systems for a long time. Quite surprisingly, hierarchical structures, that often naturally occur in the real-world, have barely been investigated. I propose to consider organisational hierarchies of coalition structures. Both the complexity as well as suitable algorithms have to be investigated for this more general setting.

Keywords: Hierarchical coalition formation · Multi-agent systems

1 Introduction

Hierarchies naturally occur or are used in companies, agencies, or organisations in the real-world. For instance in rescue or disaster response operations, or in military operations, we have a high-stake environment, in which deliberation to reach a consensus is impractical, and therefore hierarchical command structures are defined [5]. Besides hierarchies, I also introduce constraints and allow overlapping coalitions: This helps to combine a horizontal dimension (e.g., interaction graph) with a vertical dimension (e.g., a hierarchy) to represent a robust organisational structure.

2 Related Work

A simple approach to address constraints in the coalition formation process is by considering *interaction graphs* [1, 8], where agents are represented by vertices. The edges between them establish a relation (e.g., trust, synergy, etc). Another well-defined framework for representing constraints is Constrained Coalition Formation (CCF) [6]. The basic form of CCF defines the permitted sizes for coalitions as well as the agents that should be together (i.e., positive constraints) and the ones who must not be together in the same coalition (i.e., negative constraints).

The negative constraints from CCF are introduced in a valuation structure [4] in the form of *pivotal agents*, which are then evaluated by a valuation function in a particular way. Pivotal agents are said to be incompatible and should stay in different coalitions. Apart from those features, none of the previous approaches that take constraints into account deal with overlaps between coalitions [3].

3 Hierarchical Coalition Formation

I introduce a new Game where I look for a *hierarchy of coalition structures* as a solution. This hierarchical setting is important for many scenarios in which a coalition structure is strongly related with different levels of a hierarchy.

As an example, suppose a school is organising an excursion to an environmental park. Two buses were rented for taking kids from the school to the park and arrive at different sides of the park. At the park, four trucks, two for each bus, will take the kids for sightseeing to different areas. Five teachers are responsible for the kids, students cannot stay without supervision, and two particular students need a special care of at least two teachers for the whole time. If we split the teachers in groups of three and two alongside with the students, those who require special care could get into any bus. However, if they get into the bus with only two professors, at the truck division they would later receive attention of only one professor, which should be avoided. So a solution is a sequence of coalitions satisfying certain conditions.

3.1 The Hierarchical Game

As usual, a coalition C is a set of agents, $C \subseteq A$, $A = \{1, \dots, n\}$ and a coalition structure CS is given by a partition of the set of agents ($C, C' \in CS : C \cap C' = \emptyset, \bigcup CS = A$). The value of a coalition structure CS is given by $v(CS) = \sum_{C \in CS} v(C) : v : 2^A \mapsto \mathbb{R}$. The classical coalition structure graph $\mathcal{G}$ has been extensively studied and leads to a natural hierarchy: $CS \prec CS'$ iff for all $C' \in CS'$ there is a $C \in CS$ such that $C' \subseteq C$ (reflecting a lattice structure). So this is the usual definition of a characteristic function game (cfg) $\langle A, v \rangle$.¹

While in the classical case, one is interested to find in $\mathcal{G}$ the CS with the highest value (depicted in red in Fig. 1), I am interested to find the best *hierarchical* CS , which corresponds to a path in $\mathcal{G}$ (the green path in Fig. 1). The value of this path is then the sum of the values of its CS 's. My excursion example can be modelled by choosing the function v suitably.

Definition 1. A *Hierarchical Coalition Structure HCS* over a cfg $\langle A, v \rangle$ is any list of coalition structures $HCS = (CS_1, \dots, CS_h)$, where $CS_i \prec CS_{i+1}$. h is called the *length* of HCS . The set of all hierarchical coalition structures over A of fixed length h is denoted by Π_h^{CS} . I also call $\langle A, v, h \rangle$ a *hierarchical game*.

The value of HCS is $v(HCS) = \sum_{CS \in HCS} v(CS)$.

¹ Note that I can choose to work with any form of the game (i.e., disjoint coalitions, with overlaps, or externalities [7]) because I am only interested in the relation between coalition structures.

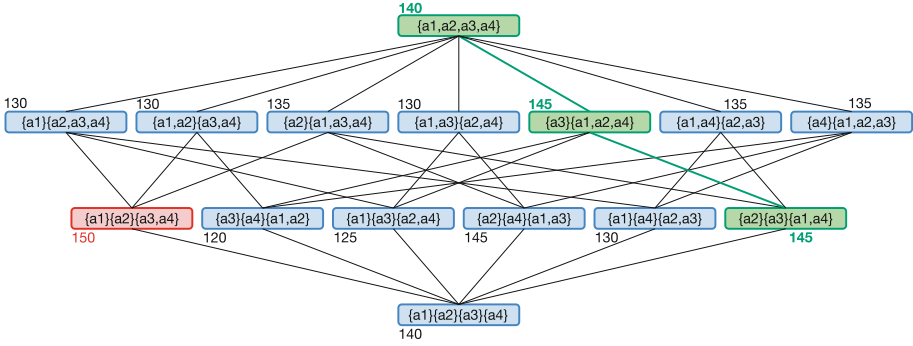


Fig. 1. The classical Coalition Structure Graph $n = 4$. The optimal CS is depicted in red, with a value of 150. The green path represents the optimal HCS for $h = 3$.

The aim is to find an HCS that maximises the overall social welfare of the system, given h : $\operatorname{argmax}_{HCS \in \Pi^{CS}} v(HCS)$, where $|HCS| = h$. The final outcome of this new game is a pair $(HCS, \mathbf{X})$ where HCS is the optimal HCS and $\mathbf{X}$ is the payoff matrix that will distribute the value of each coalition in each hierarchy level among its members. Figure 1 shows an example of a hierarchical game $\mathcal{G} = \langle \{a1, a2, a3, a4\}, v, 3 \rangle$. The optimal CS is $\{\{a1\}, \{a2\}, \{a3, a4\}\}$. However, the optimal HCS structure is depicted in green. Here, the (classical) optimal CS is not even a member of the optimal *hierarchical* CS .

3.2 Complex Coalition Formation

Coalition formation is an interesting technique and the number of possible coalitions is huge. However, many real-world problems impose constraints on potential coalitions [5] and solutions of those problems can be modelled by looking for sequences of coalition structures, not just a single one (my excursion example is exactly of this kind). Using hierarchical games, I can achieve this and address two dimensions in this formation process: horizontal (i.e., constraints and relations between agents) and vertical (i.e., hierarchical structure) dimensions.

When dealing with constraints, I have to modify Definition 1. Firstly, allowing an overlapping coalition formation game [3], secondly, considering a list of hierarchy levels $H = (c_1, \dots, c_h)$ in which h is the number of levels in the hierarchy. Each constraint c consists of one or more tuples, each containing:

Formation Structure: Taking into account constraints to be applied over each *individual coalition*. In particular, an interaction graph in which coalitions are induced over it [8], a set of permitted sizes for coalitions [6], and a set of set of agents that must stay together (i.e., an extension of [4]).

Combination Structure: Where constraints are defined over *coalition structures*. I fix the permitted sizes for a coalition structure [6] and for an overlap between any two coalitions. In addition, I fix a vector that indicates how many coalitions each agent can participate [3].

Valuation Structure: Following the idea proposed in [4].

In the vertical dimension, I combine coalition structures and do not consider them in isolation as in the classical form of the game. Consequently, the outcome in this setting is the same as defined in Sect. 3.1.

4 Evaluation Plan

The proposed hierarchical game requires a formal definition and complexity analysis for finding optimal coalition structures, in particular for games with externalities. I also need to propose and evaluate algorithms that receive inputs, as the ones defined in Sect. 3.2, and output a hierarchical relation of coalition structures. Any potential algorithms have to be compared against classical algorithms for the simplified case. A potential algorithm should be able to address those versions of game, provided that it receives the correct inputs (e.g., no overlaps between coalitions to evaluate against disjoint coalitions). The final topic that will benefit the Multi-Agent System (MAS) community is the integration of my hierarchical approach into a MAS framework. For example, considering the JaCaMo framework [2], I can receive as input for my algorithm a *MOISE* structural specification.

5 Reflections

The main contributions expected from this thesis are: (i) an attempt to apply hierarchical structure into coalition formation process; and (ii) an algorithm, to be developed, that solves a variation of the hierarchical problem that considers constraints in games with overlaps. I aim to integrate the developed technique into a MAS platform (e.g., JaCaMo platform [2]) so it could be used to solve real-world problems by MAS developers.

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An Intelligent Platform for the Management of Underwater Cultural Heritage

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Abstract. Documentation and management of underwater cultural heritage are crucial to preserving humankind's history. This research proposal aims to develop a smart platform that facilitates the conservation and management of underwater cultural heritage, contributing to the generation and transfer of knowledge in the fields of Edge Computing, Intelligent Models and Virtual Organisations.

Keywords: Internet of Things · Artificial intelligence · Edge computing · Underwater cultural heritage

1 Introduction

Documentation and management of underwater cultural heritage are crucial to preserving humankind's history and tangible testimonies while securing its accessibility to present and future generations. Precautionary conservation solutions, starting from the evaluation and analysis of the state of cultural heritage until the restoration activities, still imply high costs and are difficult to implement in marine environments.

The Internet of Things (IoT) is a network of physical “smart” devices embedded with electronics, software, sensors and actuators, that allows interconnectivity among devices and data exchange. This new technology has grown rapidly [1], finding applications in several sectors [2] (e.g. energy, healthcare, industrial, IT and networks, security and public safety and transportation). The European Union, through its Horizon 2020 programme, will allocate up to EUR 6.3 billion for research and development of ICT and IoT technologies [3, 4]. It is expected that by 2025, IoT will reach a potential market impact of USD 11.1 trillion [5].

This research proposal aims to develop an intelligent platform that allows the inclusion of Artificial Intelligence (AI) algorithms and models [6–21] for the conservation, restoration and management of underwater cultural heritage. It will have the capacity to combine information stored in databases with data acquired in real-time [22–34]. To solve critical systems, we will design an architecture that facilitates the integration of intelligent algorithms capable of managing data and information in real-time, responding in execution time, and that have backup mechanisms [35–54]. Besides, we will attempt to introduce new algorithms based on automatic learning techniques that help to create intelligent systems. The platform will use the power of a

cloud for decision making and the flexibility to distribute intelligence to the edge of the network [55–64].

The “Technological Consortium TO develop sustainability of underwater Cultural heritage (TECTONIC)” is a project funded by the EU Horizon 2020 Programme [65]. It aims to implement, improve and assess innovative materials, techniques, tools and methodologies for the conservation, restoration and management of the underwater cultural heritage from selected pilot sites. The intelligent platform proposed here will facilitate the unification of all the technology (3D reconstruction of pilot sites and a decision support tool for cultural heritage risk assessment) developed by the TECTONIC project consortium.

2 Conclusions

Over the past five years the Internet of Things (IoT) technology has grown rapidly, finding applications in several sectors. It is considered one of the leading gateway technologies to digital transformation.

This work aims to develop an intelligent platform for the conservation, restoration and management of underwater cultural heritage, within the framework of the TECTONIC project. This research work will analyse intelligent systems that allow decentralised decision-making, reducing network latency, facilitating the process in case of communication failures and increasing security. The platform will have greater autonomy and scalability, facilitating its integration with other systems.

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Analysis of Self-presentation and Self-verification of the Users on Twitter

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Abstract. From the cave paintings of early humans to the popularity of social media these days, it is understandable that for human beings self-expressing is a crucial need for all of us and social media has made it easier by posting content and express our thoughts about the other's contents. Due to this urge and the competition that has been occurred by the concept of social media, the definition of so many entities has been changed. A good example can be beauty. Due to this issue, the ratio in plastic surgeries and editing the photos has been increased a lot. The interesting challenge is understanding how people present themselves and define their reality. Also, how to make a computer understand these behaviors and then categorize these users. In this paper, we are going to focus on twitter users and will introduce the useful features to feed the algorithm with, to make the analysis of the behavior of the users possible and have a better understanding of the self-presentation of the users on Twitter.

Keywords: Social media analysis · User behavior mining · Feature extraction · Analytics · Self-presentation · Self-verification

1 Introduction to Social Media

These days by the advancement of technology [2–22], the Internet is getting more popular day by day, therefore using social media is getting more popular. People spend a lot of time on social media to read about others and share information about their life and thoughts. In fact, on the one hand, they are using social media to express themselves and get validation¹ by others. This self-presentation creates a competition in which everybody is trying to get more and more verification from others so they try harder and share more information and eventually they may reshape their reality by emphasizing positive parts that they want to share and ignoring the negative ones. In the next subsection, we will define the concept of self-presentation and self-verification which are some of the challenges in the analysis of social media.

¹ On social media verification is defined by receiving more likes, comments, retweets, views, etc.

2 Self-presentation and Self-verification, and Proposed Method

Self-Presentation defined by the way people perceive themselves whether in real life or on social media. Whether they are alone, or they are with others. The behavior, when they are alone, maybe similar or in contrast to the behavior around other people or on social media. This complicated behavior is because each person's comfort zone is different. Comfort zone is a real or virtual place that an individual is not worried about the judgment or criticism of others and it will be defined by two zones of privacy and familiarity. The difference between these two zones is, in the zone of privacy the individual feels safe and secure being alone without being worried about the judgment of others and in the zone of familiarity the individual has the same feeling of safety and trust but by being around other people. The familiarity zone usually is made by friends and family who accept the individuals the way they are.

Self-presentation is a set of methods [20–30] and strategies that we use selectively to apply in situations to shape an enhance or change our self-image to others which can be conscious or unconscious. In other words, it is the process of self-manipulation in front of the new person. To achieve ingratiation by behaving in a certain way to be liked or satisfy other's expectations. Or to achieve self-promotion by talking highly about themselves and boosting their positive points regardless of the failures and negative points. And it needs a tactic for doing it properly.

However, Self-Verification is the manner of acting individuals to shape the thoughts of society about the way they perceive other people.

Due to [1] people use three strategies to create their self-verifying social world. The first one is "Opportunity structures", as finding a place in society that satisfies their needs [31–40]. For example, they make a relationship with ones who can confirm their experience and self-view and end relationships that won't receive self-verification. The second one is "Identity cues" it contains systematic communication of self-views to others. People tend to show signs and marks about who they are and communicate with others from them. Like their beliefs, clothes, attitude, income amount, etc. [2, 41–46]. The more negative feedback they get from their actions, the more unwanted and depressed they become.

The final strategy is "Seeing", including three stages of information processing: attention, recall, and interpretation. People with positive self-views spend more time examining evaluations that they expected to be positive and true. But people with negative self-views spend more time examining evaluations that they want them to be negative. Also, Positive-minded people remember more positive statements than negative minded people. People like to interpret information for their self-esteem level. They want to look for matters in their relationships [47–49].

As the proposed method, we suggest studying the behavior of people on Twitter and extract features indicating users' behavior. These features can be extracted from the content (Tweets), the user's profile, or the information extracted from the graph of relationships [49–59]. After behavior mining, we are aiming to map these behaviors to the set of defined fuzzy strategies. This research aims to define the convenient strategy

that users are using to present themselves on Twitter and later try the methodology for other social media and aggregate the results altogether.

3 Conclusion and Future Work

In this paper, we proposed to investigate the content that users share on social media and the relationships a user is having with other users and use the behavior mining methods to define the self-presentation and self-verification of the users. For further studies, we want to find the interest of each category and use recommender systems to promote possible offers that are more interesting for the specific category.

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An Energy-Aware Dynamic Resource Management Technique Using Deep Q-Learning Algorithm and Joint VM and Container Consolidation Approach for Green Computing in Cloud Data Centers

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Abstract. These days by a high increase in the amount of computation and big data gathering and analysis, everybody needs more resources. Buying more computational and storage resources are so expensive. However, cloud computing solved this problem by providing a “pay as you go” plans therefor, users will only pay for resources that they used. However, using this technology has its challenges. One of them is resource management, which is focusing on the methodologies of dedicating resources to the users with the minimum of waste. In this paper, we propose a novel energy-aware resource management technique, using the concepts of both joint VM and container consolidation approach and deep Q-Learning algorithm for green computing in cloud data centers in order to minimize the waste of resources, migration rate, and energy.

Keywords: Cloud computing · Consolidation · Containerization · Data center · Energy consumption · Resource management · Dynamic resource management · Deep Q-Learning

1 Introduction

There are four types of cloud service models, including infrastructure as a service (IaaS), platform as a service (PaaS), software as a service (SaaS), and Container as a Service (CaaS) [1]. Our target in this research work is the CaaS environment. The CaaS principle is that make the resources and process isolated from the residuum of the system. As containers can be utilized denser in comparison with virtual machines, they increase the need for efficient cloud resource utilization. Containerization, as a light-weight remedy, is applied for management and positioning the portable packages and applications on a large number of servers [2–10].

Consolidation is one of the most significant solutions for dynamic resource management in cloud data centers, which take advantage of virtualization technology [11–

15]. Consolidation occurs by live migration VMs and containers in cloud data centers. In this approach, several virtual machines and containers are filled in the minimum number of physical machines due to power off or turn the unoccupied hosts status to sleep mode to minimize energy consumption, which is a significant concern for the cloud providers [3, 16–25].

Previous researchers have applied segregated virtual machine and container consolidation. Also, the authors in [1] verified that container consolidation is more energy efficient than VM consolidation. As the novelty approach in this paper, we focus on joint VM and container consolidation and show that this solution is more energy-efficient than the separate consolidation of VMs and containers [26–32]. Because in some cases, if containers migrate, there will be less costly than the VMs or vice versa; for that reason, this selection should be perspicacious. To evaluate our work, we consider some criteria such as SLA violation, number of container migrations, and the number of virtual machine migrations with the energy consumption in our research [33–42, 67].

In this paper, we propose to use Deep Q-Learning Algorithm to decide whether containers should be migrated or VMs intelligently.

The authors in [4] have presented a task scheduling and resource allocation model based on Hybrid Ant Colony Optimization and Deep Reinforcement Learning. Also, their targets were to belittle the overall task completion time and enhance the utilization of idle resources. They have validated their proposed work by the CloudSim simulator [10, 43–48].

The authors in [5] have explored Reinforcement Learning (RL) solutions for controlling the horizontal and vertical elasticity of container-based applications. They have considered the increment of compatibility to various workloads as their purpose. Also, they have demonstrated their proposed capability through simulations and prototype-based [10, 49–56] experiments.

The researchers in [6] have investigated the task scheduling of fog based IoT applications. Also, they have focused on minimizing long-term service delays and computation costs under the resource along with deadline constraints. To intend for this problem, they have applied the reinforcement learning approach and have presented a Double Deep Q-Learning (DDQL)-based scheduling algorithm using the target network and experience replay techniques [56–62]. Finally, they have validated that their proposed algorithm outperforms other states of the arts.

This paper organized as follows: In Sect. 2, we propose our method, which is the deep Q-Learning. In Sect. 3, the conclusion of the work has been presented.

2 Proposed Method

Reinforcement Learning algorithms are the group of the algorithms that help us build very intelligent models and robots which can do a specific task only by themselves, without any superior knowledge of the environment, through a trial and error in a feedback loop, Fig. 1. The agent takes action, apply it to the environment, and see how good its action was, by the value of the reward function [7].

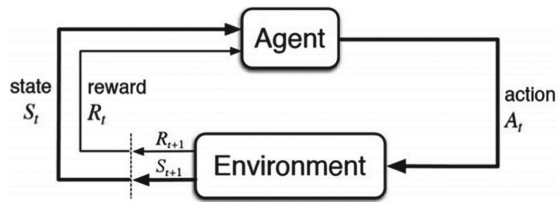


Fig. 1. Markov decision process reinforcement learning [3]

We propose to apply the Deep Q-Learning algorithm because Q-learning is aiming to move towards the maximization of the reward that, in the Q-Learning algorithm it is defined as a summation of immediate reward and a part of future reward. So, in each situation agent will make a decision based on these two. Doing this procedure for a while will lead to obtaining reaching the optimum policy, which in this problem is defined as how we should manage the resources and in which situation decide to move the VMs, containers, or the combination of them.

3 Conclusion

In this paper, we are aiming to solve resource management problems dynamically by finding the optimum policy for dedicating resources [63–66] to the clients, so it consumes the least amount of power, and the waste of hardware resources will be the least, therefore, the rate of migration will be minimized. We are using Deep Q-Learning algorithm because this algorithm does not need a big knowledge about the environment and the model will learn through a trial and error process, so it is great to be used in cases that we can map the problem to a Markov decision problem and we do not have great knowledge about the environment. Due to the dynamic characteristic of clouds, there is not too much knowledge available about what exactly happens in the cloud; therefore, this algorithm will work properly and will obtain the optimum policy to optimize the usage of energy, waste of hardware resources and migration rate.

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Deep Tech and Artificial Intelligence for Worker Safety in Robotic Manufacturing Environments

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Abstract. The main objective of the work is to prevent hazardous situations for workers in automated and robotized agile-production environments by means of a new open-source FIWARE-based Smart Factory Service. New system will implement Deep Learning models based on Convolutional Neural Networks, Deep Reinforcement Learning (Double Deep Q Networks) and Recurrent Neural Networks for the detection of anomalous behavior patterns and the prediction of the workers' level of attention, fatigue and distraction. For this purpose, FIWARE IoT Agents will gather video, audio and ECG data coming from camera and microphone arrays deployed in the environment, as well as ECG incorporated in wearables.

Keywords: Internet of Things · Convolutional Neural Networks · Recurrent Neural Networks · Deep Reinforcement Learning · Industry 4.0

1 Introduction

According to Eurostat, in 2016, the manufacturing sector employed 30.4 million people in the EU-28, working in more than 2.1 million enterprises with a turnover of 7,418,942 million EUR [1–7]. In terms of the number of employees, the largest sectors are the manufacture of food products, the manufacture of fabricated metal products (except machinery and equipment), followed by the manufacture of machinery and equipment [8–12]. According to the European Commission and the International Federation of Robots (IFR), by 2020, it is expected that more than 3 million industrial robots will be in use around the world [5, 13–23]. In 2025, investment in industrial robots will grow 10% per year in the 25 largest exporter nations, compared to the 2–3% growth in recent years. Globally, in 2017, the average of industrial robotic density was 74 robot units per 10,000 employees [17, 24–32]. The average robot density in Europe is 99. Out of the 21 countries with an above-average robotic density, 14 are in the EU [33–36].

In 2017, in the EU-28, there were just over 3.3 million non-fatal accidents that resulted in at least 4 calendar days of absence from work and 3 552 fatal accidents. Within the EU-28, the construction, transportation and storage, manufacturing, and agriculture, forestry and fishing sectors together accounted for around two thirds

(65.2%) of all fatal accidents at work and more than two fifths (43.6%) of all non-fatal accidents at work. The manufacturing industry has the highest rate of non-fatal accidents (18.7%) and the third-highest rate of fatal accidents (14.0%) [37–42].

According to the US Department of Labor, in addition to their social costs, workplace injuries and illnesses have a major impact on an employer's bottom line. It has been estimated that US employers pay almost 1 billion USD per week for direct workers' compensation costs alone. In this sense, the cost of fatigue is of increasing concern to organizations across the globe as fatigue-related accidents and losses are extremely high [43–49]. It is observed that for every 1 USD spent on direct costs like medication, insurance claims, absenteeism, or property damages, there is almost double or sometimes triple the amount spent on indirect costs like presenteeism due to lost productivity, decreased cognition, near-miss accidents, delays in production and sometimes even loss of end customers [50–54].

Considering all these figures, there is a growing trend in investment in solutions that detect fatigue and reduce accidents in increasingly robotic manufacturing environments. There are models based on Deep Learning that have been designed for fatigue detection from video [7, 52, 55–58] or ECG [59–63]. Nonetheless, none of them use Deep Reinforcement Learning and Recurrent Neural Networks to analyze sentiment in the tone of the worker when answering to the question generated by a synthesizer. In this sense, it will be performed a research in Convolutional Neural Networks [64–67], Reinforcement Learning [68–70], sentiment analysis [55] predictive algorithms and IoT oriented to Industry 4.0 [71–73]. Therefore, an innovative Smart Manufacturing Service will be implemented to detect fatigue and distractions from video (Convolutional Neural Networks), speech (Double Deep Q-Learning and Recurrent Neural Networks) and ECG (Recurrent Neural Networks). FIWARE-ready IIoT devices will be utilized for gathering context data [74, 75].

2 Conclusions

This research will be focused on research and develop a new Smart Factory Service will use Deep Learning techniques to monitor workers and detect/predict their level of attention, fatigue and possible distractions, thus preventing potential hazardous situations that can provoke injuries. This Smart Factory Service will use Convolutional Neural Networks, Deep Reinforcement Learning and Recurrent Neural Networks to detect anomalous behavior patterns from video, audio and ECG data, gathered by appropriate FIWARE-ready IIoT devices, including ambient camera/microphone arrays, and ECG in wearables.

To obtain those functionalities, the Smart Factory Service will be based on FIWARE and standard service enablers, including IoT Agents for data ingestion from sensors; Kurento Generic Enabler for video/audio processing in real-time; Cygnus Generic Enabler for historical IoT/video/audio data management; FogFlow for the execution of Deep Learning techniques using Edge servers; OPC-UA and FIROS Agents for production line and robot deactivation in hazardous situations; as well as Perseo Generic Enabler for alarm activation in situation that involve risk. Therefore, the Smart Factory Service will take advantage of existing FIWARE components and it will

be possible to reutilize it in new Industry 4.0 scenarios where workers deal with potentially hazardous situations because of their interaction with automated and robotized production lines.

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Virtual Agent Societies to Provide Solutions to an Investment Problem

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Abstract. This research uses the paradigm of Virtual Agent Societies (multi-agent systems) to model and automate the administration of investment portfolios, equipped with sentiment analysis to propose more optimal solutions adapted to market requirements. The agents act and collaborate with a certain autonomy and intelligence to provide solutions to an investment problem. This design is made by different artificial intelligence techniques capable of extracting knowledge and anticipating the appearance of anomalous situations.

Keywords: Virtual agent societies · Artificial intelligence · Big data

1 Introduction

Processing data from different heterogeneous sources, optimizing multiple aspects of information retrieval, risk profiling, asset allocation, decision making and user recommendations on the investment portfolio, is a challenge. Such data sources include human knowledge, open data, social networks, etc. The integration of these data sources is a goal, but the main objective of the research is to process that information to achieve the highest system performance and thus develop more optimal solutions, which would benefit the investor's portfolio. The design of a platform under the paradigm of virtual societies of agents [18–30] is proposed. The agents act and collaborate with a certain autonomy and intelligence to provide solutions to an investment problem. Figure 1 shows a general scheme describing the interaction of virtual agent organizations (VAOs). The VAOs, in order to fulfil their objectives, develop learning strategies [19–58] through the fulfilment of roles and rules in order to implement new functionalities resulting from their own learning process.

- Risk profiling and asset allocation VAO. It relies on the evolutionary intelligence VAO to manage the portfolio by applying the necessary algorithms.
- Evolutionary intelligence VAO. It applies machine learning algorithms, evolutionary algorithms and data clustering. It also performs the training, testing and tuning phase of the learning to be achieved as requested by the other VAOs.
- Big Data Management VAO. Performs data acquisition, data merging, data source monitoring and data analysis.

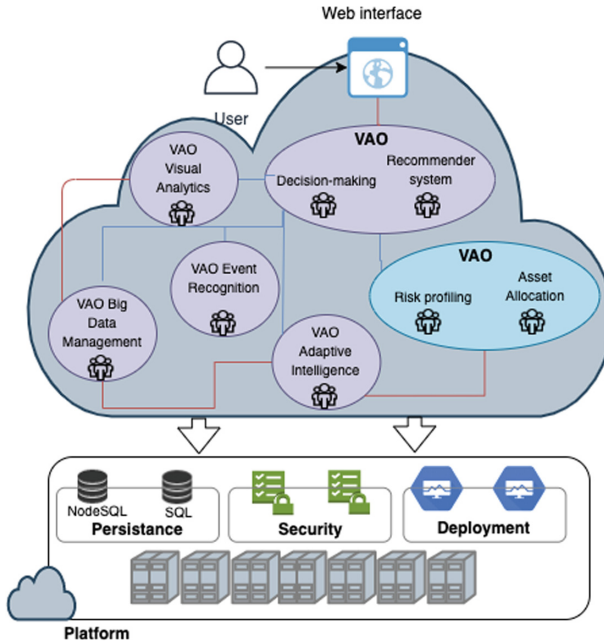


Fig. 1. Flowchart of virtual agent organizations

- Event recognition VAO. It has the role of monitoring/surveillance in the business environment to trigger events or alarms in atypical situations in the market, for which a readjustment of the portfolio or investment is needed.
- Visual analysis VAO. It displays the data and results. It allows the user to extract knowledge from the interaction with the visualizations.

2 Conclusion

The platform proposed will be designed under the paradigm of Virtual Agent Societies to provide solutions to an investment problem in the business area, finding actions, behavior patterns and establishing more efficient investment portfolio programs. Virtual Agents Organizations have been identified as the most suitable technology for automatic knowledge management [1–17], due to their robustness and autonomy. VAOs establish the level of abstraction needed to communicate with the environment and interact with other VAOs cooperatively for problem solving []. This design is made by different artificial intelligence techniques [] capable of extracting knowledge and anticipating the appearance of anomalous situations, detecting them or even proposing different solutions and improvements to the system in a completely autonomous manner.

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Proposing to Use Artificial Neural Networks for NoSQL Attack Detection

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Abstract. Relationships databases have enjoyed a certain boom in software worlds until now. These days, with the rise of modern applications, unstructured data production, traditional databases do not completely meet the needs of all systems. Regarding these issues, NOSQL databases have been developed and are a good alternative. But security aspects stay behind. Injection attacks are the most serious class of web attacks that are not taken seriously in NoSQL.

This paper presents a Neural Network model approach for NoSQL injection. This method attempts to use the best and most effective features to identify an injection. The features used are divided into two categories, the first one based on the content of the request, and the second one independent of the request meta parameters. In order to detect attack payloads features, we work on character level analysis to obtain malicious rate of user inputs. The results demonstrate that our model has detected more attack payloads compare with models that work black list approach in keyword level.

Keywords: Security · Attack detection · NoSQL injection · Big data · Feature extraction · Deep learning · Artificial neural network

1 Introduction

These days, there are new technologies such as IoT, enterprise systems, health care systems that due to generate big data on a large scale. NoSQL is one approach that helps applications deal with huge data. NoSQL technologies are new which security aspect are not considered seriously. Thing is NoSQL is not only technologies that use to handle big data. Modern applications use several API and microservices to process big data. This new variety of technologies that helps to increase attack surfaces. we are now faced with a large area of concepts and technologies. In this work, we are going to investigate security of data. There are two major areas to consider when reviewing Data security. The first one is storage and another is data transfer. There are several storage and transfer security issues, in this paper, we will highlight injection problems in the NoSQL storage database. In practical there are some NoSQL major security challenges the real work attacks come from four major reasons:

Lacking Consideration of Security Life Cycle Model. The security errors have a wave species on applications. It means if we have one functional fault, we can fix it by changing code in one part of the code. But for example, if we have SQL injection

security vulnerability it means we have this problem in all of the code. Therefore, it is important to consider the best security practice in the design phase. Thing is nowadays companies are interested in providing user-friendly and high-performance applications. based on these goals' developers will work on functionality and performance more. For example, Developers usually use prepared simple code in every technology that maybe that code was not provided well in the best security practice. It is because, NoSQL technologies are new and they aren't developed for many years like relational DB SQL languages. The amazing thing is Most of developer has heard about SQL Injection and they know the one of best solution is parametrize query. But the problem is don't have enough knowledge in NoSQL technologies. As we know the main problem in every injection attacker are inject special characters in query and change the execution result.

Increase the Attack Surface by Using Third-Parties: As we mentioned before, Modern web applications include the number of prepared tools, plugins, and Protocols. web developers need components and APIs that improve UX, APIs, features and etc. they can develop them manually and spend a lot of time on these, but they usually select existing third-party solutions. There are a lot of plugins that implemented for NoSQL documents and Those plugins are mostly free or community-built projects. therefore, they cannot guarantee they are secured from common attacks. In addition, Developers usually install tools or plugin at once and usually leave it by default configuration. Here there are two scenarios, first the plugin is popular community-built projects that needs regular update and if developers don't care about regular updates them. So even if developer programming in a strict secure way, plugin vulnerability will open doors for the attacker. Base on this reason, applications will target to many *Zero-day* vulnerabilities. Second one is using unpopular third-party plugins that contains malicious code. For example, developers usually use containers to encapsulating everything or take less time to install NoSQL databases. Thing is some docker images are tempered by hackers and it will be getting worse if developers use default docker configuration and run a docker as the root user. defaults to running containers using the root user. When a Docker file doesn't specify a USER, it defaults to executing the container using the root user. So, it maybe Couse to malicious code in images run as root privilege and due to our security issue for data storage.

Regarding mentioned issues, we need some services to monitor networks to detect malicious activity. IDS (Intrusion detection system) are the most common solution. IDS models have been an active research area. There are a lot of approaches implemented to detect injection base on log mining. One primary problem with some IDS is they need to define an especial log format to create features. Generating special log format due to overhead in system. One way to overcome these issues is to use general resources that will be found in all of the servers.

In this work we are going to use the most effective features for our model that will be found in all of the servers. In other words, we want to use the available resources and don't implement any approach to generate the required data.

To collecting, data we should understand NoSQL injection attacks concepts. NoSQL injection attacks are kind of injection attacks. As we know, every technology has special characters that reserved for it. Attackers can send malicious input to applications and do serious damage to systems.

In above example, attacker have changed SQL query result by using special characters injection. In NoSQL injection attacks the scenario is same. For example, MongoDB expects input in json array format:

```
find({'username':'admin'})
```

And PHP library uses associate arrays for query:

```
$param['username']='admin'

User.find($param)
```

If hacker send following user input as HTTP request and use MongoDB NoSQL reserved character:

[http://example.com/profile.php?username\[\\$ne\]='admin'](http://example.com/profile.php?username[$ne]='admin')

The result is:

```
User.find({'username':{'$ne':'admin'}})
```

So, attacker can inject and change the NoSQL query result. One way to detect most of attacks is investigate user inputs. Therefore, our important material in this work is user input. To obtain these goals we used the apache access.log file. As we know apache and Nginx are the most common web servers for Linux. Access log format of these two both stored in the combined log format [13]. So if we implement parser base one of them, we almost have a parser for another one, too. The additional advantage of using access.log is that it results in apache or Nginx log system and we can find this in a lot of servers.

Another important advantage of using access.log is, it contains user input in requested URL. As mentioned before, user inputs are one of important data that can consider it. Moradpoor [17] employing Neural Network for detection of SQL Injection attacks, proposed model is based on Assigned vectors to the SQLI attack keywords such as SELECT, DELETE, FROM, TABLE, etc. for example, following attack payload contains keywords For URLs:

```
AND updatexml(rand(), concat(0x3a, (SELECT concat(CHAR(126), data_info,
CHAR(126)) FROM data_table.data_column LIMIT data_offset,1)),null)-
```

The problem is these methods is kind of black list approach and attacker usually use encoding methods to bypass the black list. In following attack payload attacker convert all SQL keywords by CHAR() function.

```
Declare @cmd as varchar(3000);Set @cmd = (CHAR(101)+CHAR(120)+ CHAR(101)+
CHAR(99)+CHAR(32)+CHAR(109)+CHAR(97)+CHAR(115)+CHAR(116)+CHAR(101)+CHAR(114)
+CHAR(46)+CHAR(46)+CHAR(120)+CHAR(112)+CHAR(95)+CHAR(99)+CHAR(109)+CHAR(100)
+CHAR(115)+CHAR(104)+CHAR(101)+CHAR(108)+CHAR(108)+CHAR(32)+CHAR(39)+CHAR(10
0)+CHAR(105)+CHAR(114)+CHAR(39)+CHAR(59));EXEC(@cmd);-
```

As we can see the special character like single quote, prates are important part of this attack payloads that attacker have to use them. Base on this concept, we have selected character base analyze. The schematic of architecture our approach is shown in Fig. 1.

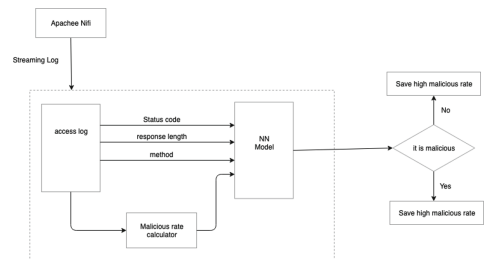
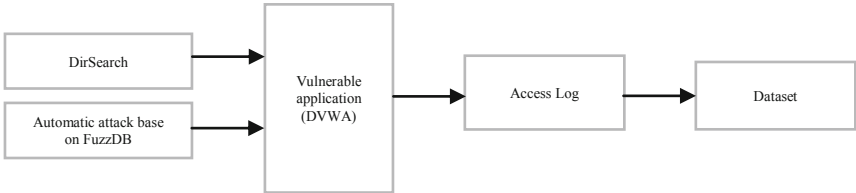


Fig. 1. NN model for NoSQL injection attacks

In rest of this section we will give details of proposed approach. To identify the NoSQL injections, a model must first be created. The intended model is a machine-based model. Due to the lack of specific datasets for this purpose. we need provide dataset base on access log. Regarding this goal we provided following scenario to generate dataset.



We have extended vulnerable application that called DVWA to add page that is vulnerable NoSQL injection to it. [14] After it we crawl web site with Dirsearch tool [13] it is simple command line to brute force website. Also, we visit all of website pages and Then we labelled these requests as normal request.

To Label NoSQL attacks we used FuzzDB attack payloads. This database will provide payload for automatic attack. Also, we exploited web page manually, too. These requests are labelled as NoSQL attacks. We provide some automatic test for SQL Injection and XSS attacks to increase the likelihood compare with real attacks. After it, we provided train and test dataset to train model and tune parameter model.

After creating model, the logs which provided by apache Nifi will send to identifier system. The operations are as follows:

- 1 The first step is to collect the logs. For this purpose, apache access.log are used. There are two method to collect logs: first method is Directory path: we can just put access.log path for example /var/log/apache2/access.log, and second one: Remote

link: there are several web services such as S3 that are used to store data or logs. We provided this method to receive access.log remotely too.

- 2 The next step is to prepare the received log. This is a preparation to extract the features of the received log. For this purpose, the URL query string is separated and the malicious rate is calculated. The other features for the log are: HTTP status code, HTTP response length, HTTP request method
- 3 At this step, the features extracted from the log are given to the model. And the model provided with these features specifies the type of request.
- 4 Finally, according to the type detected for the log, the system alert security issue base on tolerance rate.

2 Conclusion

An important outcome of this research is to provide an automatic structured for detecting NoSQL injects from user requests. in this paper based on log file we extract several features. The features can divide two classes, the first one is based on the content of the request query, and the other one independent of the request meta parameters. in the content-based feature, we estimate a new metric that depends on the user inputs. in the others we used server response and we expect by these features achieved an accurate System. In result Our model helps to detect more attacker bypass techniques and ignore to report false positive alarms.

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Predictive Maintenance Proposal for Server Infrastructures

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Abstract. Predictive maintenance has played a key role in industry complexes for several years. The prediction of upcoming errors has the potential to increase the average productivity of any enterprise and to avoid losing opportunities due to a partial shut-down of the production system. Such ideas can be applied to large-scale server infrastructures achieving a fully automatic system which warns the administrators of a potential shut-down before it takes place. Mathematical algorithms and statistical tools can be used to model the standard behaviour of the system and, when an anomaly is detected, to warn the system operator. Furthermore, machine learning can also be used to model such a behaviour and to identify the most likely cause of the anomaly.

Keywords: Predictive maintenance · Mathematical tools · Servers

1 Introduction

Predictive maintenance is a series of actions techniques designed with the objective of detecting possible failures and defects of machinery in the early stages to prevent these failures from manifesting themselves as a larger failure during its operation, preventing them from causing emergency shutdowns and downtime, causing negative financial impact [1–9]. The equipment is intervened without any symptoms of having problems. The great capacity of current server infrastructures to register detailed logs from its systems make it possible to develop a early-warning system which sends out an alert before there is any system shut-down [10–14]. Furthermore, the system can provide a list of the most likely items to have caused the incidence and of the most representative logs of the incidence. In our work, both ideas are implemented and the mentioned list is generated by the RAKE (Rapid Automatic Keyword Extraction) algorithm [15–19]. This system is feed in real-time by the new logs of the systems and is designed to ensure that any ab-normal behaviour of the infrastructure is detected, therefore focusing on minimising type II error. This article tackles server infrastructure shutdown prediction using both mathematical and machine leaning algorithms, which is a novelty approach in this field [20–28].

In this paper, a system meant to predict future incidences in servers is provided. In our approach, machine learning is used in order to identify the keywords of a future (or past) shutdown, and mathematical algorithms are used to analyse the probability of a future coming incidences, sending an alert when the probability is above a set threshold. The incidence is therefore identified both in place and time through the integration of the previous techniques, by knowing the keywords of the anomaly (place of the incidence) and the predicted hour (time of the incidence).

2 Conclusion

This work proposes a hybrid algorithm of artificial intelligence using servers logs as input. The algorithm has been applied to the data collected in a big company's log-gatherer in a real-life scenario. The main goal of the proposed system is to create a system which warns its administrators of a imminent incidence automatically. The most significant results obtained in this work are listed below.

This paper provides a novel, machine learning-based and self-diagnosing algorithm, which allows to avoid future incidences in the servers by detecting anomalies and letting administrators know of probable future failures, as well as of its most likely causes. In our future work, we will fine-tune the keyword extraction method and to increase the performance of the predictor.

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